



Assetto Corsa EVO

Game Version : **0.7.1**

Assets Type : **Cars**

Subtype : **Car Physics Assets**



Preface

This document is an independent, community-driven technical reference for the vehicle physics assets of **Assetto Corsa EVO**. It is **not** an official publication of Kunos Simulazioni, nor is it endorsed, reviewed, or maintained by the studio. Assetto Corsa EVO and related marks remain the property of their respective owners.

The material herein was compiled through careful inspection of shipped physics assets, comparative analysis across vehicle configurations, and reasoned interpretation of field names, value ranges, and observed behavior. Where the engine's internal implementation is not publicly specified, units, roles and relationships are **inferred**. They should therefore be treated as working hypotheses - useful for modding, research, and discussion - rather than as authoritative studio specifications.

Readers should expect that some entries may contain inaccuracies, incomplete coverage, or interpretations that later evidence revises. Game updates, newly examined assets, and community verification may all require corrections. This reference is published so that knowledge of otherwise opaque systems can be shared, challenged, and improved openly.

Use this guide as a structured starting point. Cross-check critical parameters against in-game behaviour and primary asset data before relying on any single conclusion. Contributions that refine descriptions, confirm units, or document previously unmapped fields are welcome and strengthen the resource for everyone.

Reporting Corrections

If you find an error, an incomplete field, or a more convincing interpretation of a unit or description, please report it on GitHub using the **Documentation correction** issue template :

<https://github.com/CorsaClub/ace-documentation/issues/new?template=documentation-correction.yml>

Include the section (for example `ig.TYRE`), the field ID when relevant, the current text, your proposed correction, and the evidence behind it (asset values, game version, telemetry, or comparative checks). Maintainers will review reports and update the reference accordingly. If you already have a ready-made edit, a Pull Request is equally welcome.

1. Brake System [.brakesystem]	13
A. Description	13
I. General Description	13
II. Area of Influence / Impact on Vehicle Dynamics	13
III. Key Architecture & Data Fields Explained	13
1 - Base Mechanical & Physical Parameters	13
2 - Dynamic Control Units (EBB & Steer Brake)	14
3 - Control Stages & Algorithmic Logic	14
4 - EBB Modes	14
B. Schema	15
C. Measurement Units & Descriptions	16
D. Example data	17
I. Chosen Cars for Example	17
II. Example	18
Alfa Romeo Giulia GTAm	18
Lancia Delta HF Integrale EVO II (slug : ks_lancia_delta_hf_integrale_evo_ii)	19
Ferrari 296 GT3	19
Ferrari SF25	20
2. Brakes [.brakes]	23
A. Description	23
I. General Description	23
II. Area of Influence / Impact on Vehicle Dynamics	23
III. Key Architecture & Data Fields Explained	23
1 - Thermal Modeling & Heat Dissipation	23
2 - Dimensions, Wear, and Degradation (M M = Per Millimeter)	24
3 - Perf Curve (Friction Coefficient vs. Temperature)	24
IV. Interpretation of Asset Implementation & Data Profiles	24
Profile A: Split Front/Rear Axle Configurations (e.g., Vintage/Road Cars)	24
Profile B: Single Shared Compound Profiles & Endurance Pads (e.g., Racing GT3)	25
Reading the Performance Curve Trend	25
B. Schema	25
C. Measurement Units & Descriptions	26
D. Example data	28
I. Chosen Brakes for Example	28

II. Example	28
Vintage Road [Front]	28
Vintage Road [Rear]	29
Racing GT3 [Pad 2]	30
3. Car Data [.car]	32
A. Description	32
I. General Description	32
II. Area of Influence / Impact on Vehicle Dynamics	32
III. Key Architecture & Data Fields Explained	32
1 - Global Mass & Inertia Properties	32
2 - Dimensions, Track and Alignement Coordinates	33
3 - Fuel Management & Consumables	33
IV. Short Interpretation of Asset Implementation	33
B. Schema	34
C. Measurement Units & Descriptions	43
D. Example data	58
I. Chosen Car Data for Example	58
II. Example	58
Ferrari 296 GTB	58
Audi R8 LMS GT3 Evo II	67
Renault 5 GT Turbo	75
4. Car Engine [.carengine]	79
A. Description	79
I. General Description	79
II. Area of Influence / Impact on Vehicle Dynamics	79
III. Key Architecture & Data Fields Explained	79
1 - Power Generation & Power Curves	79
2 - Rotational Dynamics & Throttle Response	80
3 - Aspiration, Turbocharging & Thermal Behavior	80
IV. Short Interpretation of Asset Implementation	80
B. Schema	81
C. Measurement Units & Descriptions	82
D. Example data	86
I. Chosen Car Engine for Example	86

II. Example	86
Alpine A290 b	86
Ferrari SF 25	87
Chevrolet Camaro ZL1	91
Datsun 240z Fairlady	92
5. Car Setup [.carsetup]	94
A. Description	94
I. General Description	94
II. Area of Influence / Impact on Vehicle Dynamics	94
III. Key Architecture & Data Fields Explained	94
1 - Tyres (The Contact Patch)	94
2 - Aerodynamics (The Airflow Platform)	95
3 - Suspension Geometry & Rates	95
4 - Dampers (Transient Shock Absorption)	95
5 - Drivetrain & Differential	95
IV. Interpretation of Setup Configuration Strategies	96
B. Schema	96
C. Measurement Units & Descriptions	97
D. Example data	101
I. Chosen Car Engine for Example	101
II. Example	101
Audi Sport Quattro	101
Alfa Romeo Junior	104
Ferrari 488 Challenge Evo [preset : safe_1]	106
6. Car Setup Limits [.carsetuplimits]	110
A. Description	110
I. General Description	110
II. Area of Influence / Impact on Vehicle Dynamics	110
III. Key Architecture & Data Fields Explained	110
1 - The Anatomy of a Parameter Limit Object	110
2 - Core Categories of Application	111
IV. Interpretation of Setup Limits Strategies	111
B. Schema	112
C. Measurement Units & Descriptions	121

D. Example data	124
I. Chosen Car Engine for Example	124
II. Example	124
BMW M4 CSL	124
Lamborghini Countach	145
7. Car Setup Units [.carsetupunits]	166
A. Description	166
I. General Description	166
II. Area of Influence / Impact on Vehicle Dynamics	166
III. Key Architecture & Data Fields Explained	166
1 - Wheel and Tyre Telemetry Alignment	166
2 - Suspension Components & Kinematics	167
3 - Aerodynamics & Aerostatic Clearance	167
IV. Interpretation of Setup Units Strategies	167
B. Schema	167
C. Measurement Units & Descriptions	169
D. Example data	172
I. Chosen Car Engine for Example	172
II. Example	172
Setup Units	172
8. Car Tuning Parts [.tuningpart]	176
A. Description	176
I. General Description	176
II. Area of Influence / Impact on Vehicle Dynamics	176
III. Key Architecture & Data Fields Explained	176
1 - Tuning Part Core Definitions	176
2 - Component Path Overrides	177
IV. Interpretation of Tuning Part Strategies	177
B. Schema	177
C. Measurement Units & Descriptions	183
D. Example data	192
I. Chosen Cars for Example	192
II. Example	192
Toyota Supra MK IV	192

<i>Datsun 240z Fairlady</i>	193
<i>Porsche 992 GT3 Cup</i>	195
9. Car Electronics [.carelectronics]	196
A. Description	196
I. General Description	196
II. Area of Influence / Impact on Vehicle Dynamics	196
III. Key Architecture & Data Fields Explained	196
1 - Traction Control (TC) Logic Matrix	196
2 - Anti-lock Braking System (ABS) Map Settings	197
IV. Interpretation of Tuning Part Strategies	197
B. Schema	197
C. Measurement Units & Descriptions	198
D. Example data	201
I. Chosen Cars for Example	201
II. Example	202
<i>Lamborghini Huracan ST Evo 2</i>	202
<i>Maserati MC20 GT2</i>	206
<i>Porsche 992 GT3 Cup</i>	209
10. Clutch [.clutch]	212
A. Description	212
I. General Description	212
II. Area of Influence / Impact on Vehicle Dynamics	212
III. Key Architecture & Data Fields Explained	212
1 - base mechanical & Intertial Parameters	212
2 - Autoclutch Profile Parameters	213
IV. Interpretation of Tuning Part Strategies	213
B. Schema	213
C. Measurement Units & Descriptions	214
D. Example data	215
I. Chosen Cars for Example	215
II. Example	215
<i>Caterham 485 CSR</i>	215
<i>Ferrari F2004</i>	215
<i>Volkswagen Golf GTI mk8</i>	216

11. Coilover [.coilover]	217
A. Description	217
I. General Description	217
II. Area of Influence / Impact on Vehicle Dynamics	217
III. Key Architecture & Data Fields Explained	217
1 - Wheel Rate & Spring Parameters	217
2 - Damper Profile Parameters	218
3 - Alignments & Geometry Fields	218
IV. Interpretation of Tuning Part Strategies	219
B. Schema	219
C. Measurement Units & Descriptions	220
D. Example data	223
I. Chosen Cars for Example	223
II. Example	223
Caterham 485 CSR	223
Alpine A110s	225
Dallara EXP	226
12. Damper Curves [.dampercurves]	229
A. Description	229
I. General Description	229
II. Area of Influence / Impact on Vehicle Dynamics	229
III. Key Architecture & Data Fields Explained	229
1 - Wheel Rate & Spring Parameters	229
2 - Damper Profile Parameters	229
3 - Alignments & Geometry Fields	229
IV. Interpretation of Tuning Part Strategies	229
B. Schema	229
C. Measurement Units & Descriptions	229
D. Example data	230
I. Chosen Cars for Example	230
II. Example	230
Ford - GT3 Dampers	230
Penske	231
Porsche Cayman Dampers	233

13. Drivetrain [.drivetrain]	235
A. Description	235
I. General Description	235
II. Area of Influence / Impact on Vehicle Dynamics	235
III. Key Architecture & Data Fields Explained	235
1 - Wheel Rate & Spring Parameters	235
2 - Damper Profile Parameters	235
3 - Alignments & Geometry Fields	235
IV. Interpretation of Tuning Part Strategies	235
B. Schema	235
C. Measurement Units & Descriptions	238
D. Example data	241
I. Chosen Cars for Example	241
II. Example	241
Audi RS3 Sportback	241
Ferrari F40 LM	245
Abarth 695 Biposto	247
14. Gearbox [.gearbox]	249
A. Description	249
I. General Description	249
II. Area of Influence / Impact on Vehicle Dynamics	249
III. Key Architecture & Data Fields Explained	249
1 - Wheel Rate & Spring Parameters	249
2 - Damper Profile Parameters	249
3 - Alignments & Geometry Fields	249
IV. Interpretation of Tuning Part Strategies	249
B. Schema	249
C. Measurement Units & Descriptions	250
D. Example data	252
I. Chosen Cars for Example	252
II. Example	252
Porsche 718 Cayman GT4 CS MR	252
Alpine A290 b	254
Renault 5 GT Turbo	254

15. General [.generalcar]	256
A. Description	256
I. General Description	256
II. Area of Influence / Impact on Vehicle Dynamics	256
III. Key Architecture & Data Fields Explained	256
1 - Wheel Rate & Spring Parameters	256
2 - Damper Profile Parameters	256
3 - Alignments & Geometry Fields	256
IV. Interpretation of Tuning Part Strategies	256
B. Schema	256
C. Measurement Units & Descriptions	257
D. Example data	258
I. Chosen Cars for Example	258
16. Surface 3D [.surface3d]	259
A. Description	259
I. General Description	259
II. Area of Influence / Impact on Vehicle Dynamics	259
III. Key Architecture & Data Fields Explained	259
1 - Wheel Rate & Spring Parameters	259
2 - Damper Profile Parameters	259
3 - Alignments & Geometry Fields	259
IV. Interpretation of Tuning Part Strategies	259
B. Schema	259
C. Measurement Units & Descriptions	260
D. Example data	261
I. Chosen Cars for Example	261
II. Example	261
Mercedes AMG GT2	261
Audi R8 LMS GT3 Evo 2	263
Dallara Stradale Coupe	266
17. Suspension [.suspension]	268
A. Description	268
I. General Description	268
II. Area of Influence / Impact on Vehicle Dynamics	268

<i>III. Key Architecture & Data Fields Explained</i>	268
1 - Wheel Rate & Spring Parameters	268
2 - Damper Profile Parameters	268
3 - Alignments & Geometry Fields	268
<i>IV. Interpretation of Tuning Part Strategies</i>	268
B. Schema	268
C. Measurement Units & Descriptions	270
D. Example data	276
<i>I. Chosen Cars for Example</i>	276
<i>II. Example</i>	276
Volkswagen Golf GTI Mk1	276
Honda S2000 AP1	277
Porsche 992 GT3 R Rennport	279
18. Turbo [.turbo]	281
A. Description	281
<i>I. General Description</i>	281
<i>II. Area of Influence / Impact on Vehicle Dynamics</i>	281
<i>III. Key Architecture & Data Fields Explained</i>	281
1 - Wheel Rate & Spring Parameters	281
2 - Damper Profile Parameters	281
3 - Alignments & Geometry Fields	281
<i>IV. Interpretation of Tuning Part Strategies</i>	281
B. Schema	281
C. Measurement Units & Descriptions	282
D. Example data	283
<i>I. Chosen Cars for Example</i>	283
<i>II. Example</i>	283
Peugeot 205 T16	283
Chevrolet Camaro ZL1	284
Toyota Supra MKIV	284
19. Tyre [.tyre]	286
A. Description	286
<i>I. General Description</i>	286
<i>II. Area of Influence / Impact on Vehicle Dynamics</i>	286

<i>III. Key Architecture & Data Fields Explained</i>	286
1 - Wheel Rate & Spring Parameters	286
2 - Damper Profile Parameters	286
3 - Alignments & Geometry Fields	286
<i>IV. Interpretation of Tuning Part Strategies</i>	286
B. Schema	286
C. Measurement Units & Descriptions	289
D. Example data	297
<i>I. Chosen Tyres for Example</i>	297
<i>II. Example</i>	297
<i>Eco Size : 165 - 60 - 12</i>	297
<i>Vintage Size : 195 - 60 - 15</i>	299
<i>F1 2025 [Wet] Size : 305 - 720 - 18</i>	301
20. Wing [.wing]	305
A. Description	305
<i>I. General Description</i>	305
<i>II. Area of Influence / Impact on Vehicle Dynamics</i>	305
<i>III. Key Architecture & Data Fields Explained</i>	305
1 - Wheel Rate & Spring Parameters	305
2 - Damper Profile Parameters	305
3 - Alignments & Geometry Fields	305
<i>IV. Interpretation of Tuning Part Strategies</i>	305
B. Schema	305
C. Measurement Units & Descriptions	306
D. Example data	308
<i>I. Chosen Tyres for Example</i>	308
<i>II. Example</i>	309
<i>Audi RS6 Avant</i>	309
<i>Lotus Emira</i>	310
<i>Ferrari SF-25</i>	312

1. Brake System [.brakesystem]

A. Description

I. General Description

The `BRAKESYSTEM` asset is the core configuration file that defines the entire physical, mechanical, and electronic behavior of a vehicle's braking system within the simulation engine.

Rather than just setting a raw stopping power value, this asset acts as a central hub. It governs how braking force is distributed between the front and rear axles (static and dynamic bias), how secondary systems like the handbrake behave, and how advanced modern electronic driving aids—such as Electronic Brake Balance (EBB) and brake-based torque vectoring systems—alter braking pressure in real time based on telemetry.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters configured in this asset directly impact how the vehicle handles during one of the most critical phases of driving:

- **Stopping Distance (Braking Power):** Dictated by the total available torque, defining how quickly the vehicle can decelerate.
- **Braking Stability (Brake Bias):** Controls the balance of force. Too much front bias leads to understeer or front-wheel lockup; too much rear bias causes instability, potentially throwing the car into a spin when braking heavily into a corner.
- **Cornering Agility & Trajectory Control:** Through dynamic controllers, the system can modulate brake pressure on individual wheels or axles based on steering angle, lateral G-forces, or wheel slip, acting like a performance Stability Control (ESP) or electronic torque vectoring system.
- **Energy Recovery Systems (Hybrid/EV Integration):** Variables linked to the ERS (Energy Recovery System) interact with regenerative braking, managing how hydraulic pressure scales alongside electric motor resistance (`ERSCOASTTORQUE`) and battery state-of-charge.

III. Key Architecture & Data Fields Explained

The schema is organized into three main categories: **Mechanical Constants**, **Electronic Subsystems**, and **Algorithmic Logic (Stages)**.

1 - BASE MECHANICAL & PHYSICAL PARAMETERS

- **Total Torque:** The absolute maximum braking torque (usually in *Nm*) applied across the entire vehicle when the brake pedal is depressed 100%.
- **Front Bias:** The baseline, static percentage of braking force sent to the front axle (*e.g. 0.60 = 60% front, 40% rear*).
- **Hand Brake Torque:** Dedicated torque value applied exclusively by the handbrake (typically impacting only the rear wheels).

- **Has Cockpit Bias & Bias Step:** Determines whether the driver can adjust the brake balance on the fly from inside the cockpit, and by what percentage increment each button press alters the bias (e.g. 0.5%).
- **Front/Rear Compound Path:** File paths linking the brakes to external thermal and friction physics models (governing brake wear, fade, and optimal operating temperature windows).

2 - DYNAMIC CONTROL UNITS (EBB & STEER BRAKE)

Advanced vehicles use automated controllers to alter braking performance based on real-time physics data:

1. **Controller EBB / Controllers EBB:** Electronic Brake Balance controllers. They continuously shift the brake balance forward or backward depending on live driving conditions.
2. **Steer Brake Controller:** An agile-handling system that applies micro-braking forces to specific wheels based on steering inputs (`STEERDEG`) or yaw rates (`STEERYAWDELTA`). It mimics modern electronic differentials or stability management.
3. **Torque Controller EBB:** Dynamically scales total braking torque output to prevent global lockups or optimize regenerative braking efficiency.

3 - CONTROL STAGES & ALGORITHMIC LOGIC

Every dynamic controller processes data through one or multiple **Stages**. Each stage functions as an automated logical loop:

- **Input Var (Input Variable):** The live telemetry channel the game reads (e.g., `BRAKE` pedal pressure, wheel `SLIPRATIO`, lateral forces `LATG`, or suspension/weight transfer data like `LOADSPREADLF`).
- **LUT (Look-Up Table):** A path to a external `.CURVE` file. This data curve acts as an instruction set: *"If the input variable is X, apply a modifier value of Y."*
- **Combinator Mode:** Dictates how this specific stage interacts with others in the pipeline (`ADD` to sum the values together, or `MULT` to multiply them).
- **Filter Gain:** A smoothing coefficient that dampens rapid telemetry spikes, preventing jittery or unnaturally sudden braking corrections.
- **Up / Down Limit:** Standard clamping values ensuring the controller output stays within safe, physically realistic boundaries.

4 - EBB MODES

`EBBDISABLED`: Electronic brake distribution is off. The car relies entirely on its static physical brake bias.

`EBBINTERNAL`: Uses hardcoded engine-level logic to distribute braking.

`EBBDYNAMICCONTROLLERABSOLUTE` / `RELATIVE`: Fully activates the custom look-up tables and stages defined in the asset to modulate balance absolutely or relative to the base setup.

B. Schema

- 1. Total Torque : float
- 2. Front Bias : float
- 3. Hand Brake Torque : float
- 4. Has Cockpit Bias : boolean
- 5. Bias Step : float
- 6. Front Compound Path [x] : float | can have multiple Front Compound Path
- 7. Rear Compound path [x] : float | can have multiple Rear Compound Path
- 8. Controller EBB : object with an array of stages within
 - 8a. Name : string
 - 8b. Stages [x] : object | Controller EBB can have multiple stages
 - 8b1. Input Var : enum
 - 8b2. Combinator Mode : enum
 - 8b3. Lut : string - path
 - 8b4. Filter Gain : float
 - 8b5. Up Limit : float
 - 8b6. Down Limit : float
 - 8b7. Current Value : float
 - 8b8. Const Value : float
- 9. Controllers EBB [x] : object with an array of stages within
 - 8a. Name : string
 - 8b. Stages [x] : object | Controllers EBB [x] can have multiple stages
 - 8b1. Input Var : enum
 - 8b2. Combinator Mode : enum
 - 8b3. Lut : string - path
 - 8b4. Filter Gain : float
 - 8b5. Up Limit : float
 - 8b6. Down Limit : float
 - 8b7. Current Value : float
 - 8b8. Const Value : float
- 10. Steer Brake Controller : object with an array of stages within
 - 8a. Name : string
 - 8b. Stages [x] : object | Steer Brake Controller can have multiple stages
 - 8b1. Input Var : enum
 - 8b2. Combinator Mode : enum
 - 8b3. Lut : string - path
 - 8b4. Filter Gain : float
 - 8b5. Up Limit : float
 - 8b6. Down Limit : float
 - 8b7. Current Value : float
 - 8b8. Const Value : float
- 11. Torque Controller EBB : object with an array of stages within
 - 8a. Name : string
 - 8b. Stages [x] : object | Torque Controller EBB can have multiple stages
 - 8b1. Input Var : enum
 - 8b2. Combinator Mode : enum
 - 8b3. Lut : string - path

- 8b4. Filter Gain : float
- 8b5. Up Limit : float
- 8b6. Down Limit : float
- 8b7. Current Value : float
- 8b8. Const Value : float
- 12. EBB Mode : enum
- 13. EBB Front Multiplier : float
- 14. EBB Min Speed : float

Enum List - Brake System

Enum	Values
Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
CombinatorMode	UndefinedMode, Add, Mult
EBB Mode	ebbDisabled, ebbInternal, ebbDynamicControllerAbsolute, ebbDynamicControllerRelative

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Total Torque	Nm (Newton-meters)	Defines the maximum total braking torque available for the entire vehicle.
2.	Front Bias	% (Percentage / Distribution ratio)	Sets the percentage of total braking torque allocated to the front axle (e.g., 0.60 = 60%).
3.	Hand Brake Torque	Nm (Newton-meters)	Specifies the maximum braking torque applied by pulling the mechanical handbrake.
4.	Has Cockpit Bias	None (Boolean : True / False)	Toggles whether the driver can adjust the brake balance manually from inside the cockpit while driving.
5.	Bias Step	% (Percentage increment)	The step size by which the brake bias changes per click when adjusted (e.g., 0.005 for 0.5% increments).
6.	Front Compound Path	None (File path)	Points to the physics/lookup file defining the friction and thermal properties of the front brake pads.

ID	Name	Unit of Measurement	Description
7.	Rear Compound Path	None (File path)	Points to the physics/lookup file defining the friction and thermal properties of the rear brake pads.
8a.	Name	None (String)	An internal label or identifier for the specific brake controller or modifier stage.
8b1.	Input Var	None (Telemetry enum)	The telemetry variable used as the trigger input for this controller (e.g., <i>SteerDEG</i> , <i>Brake</i> , <i>Speed</i>).
8b2.	Combinator Mode	None (Math enum : Add / Mult)	Determines how this stage's output combines with the baseline value (Addition or Multiplication).
8b3.	Lut	None (.curve file path)	References a Look-Up Table curve file that maps the raw input value to a specific output factor.
8b4.	Filter Gain	Coefficient (Smoothing multiplier)	Controls the input signal smoothing filter; acts as a dampener to prevent sudden spikes in controller application.
8b5.	Up Limit	Depends on the input variable	The upper bounding limit for the input signal, clamping values that exceed this threshold.
8b6.	Down Limit	Depends on the input variable	The lower bounding limit for the input signal, clamping values that fall below this threshold.
8b7.	Current Value	Depends on the input variable	The current, real-time value processed by the controller during simulation.
8b8.	Const Value	Depends on the input variable	A fallback fallback constant value used if no dynamic input curve or telemetry is active.
12.	EBB Mode	None (Mode enum)	Selects the operational logic profile for the Electronic Brake Balance system.
13.	EBB Front Multiplier	Coefficient (Scaling factor)	Scaling factor that dynamically adjusts the front brake bias authority under EBB intervention.
14.	EBB Min Speed	Km/h or m/s * (Spped threshold)	The minimum vehicle speed below which the Electronic Brake Balance system deactivates.

D. Example data

I. Chosen Cars for Example

- Alfa Romeo Giulia GTAm (slug : ks_alfa_romeo_giulia_gtam)
- Lancia Delta HF Integrale EVO II (slug : ks_lancia_delta_hf_integrale_evo_ii)

- Ferrari 296 GT3 (slug : ks_ferrari_296_gt3)
- Ferrari SF25 (slug : ks_ferrari_sf_25)

II. Example

Alfa Romeo Giulia GTAm

```
1. Total Torque : 4100.00000
2. Front Bias : 0.75000
3. Hand Brake Torque : 200.00000
4. Has Cockpit Bias : false
5. Bias Step : 0.50000
6. Front Compound Path : None
7. Rear Compound path : None
8. Controller EBB
  8a. Name : None
  8b. Stages 1
    8b1. Input Var : LoadSpreadLF
    8b2. Combinator Mode : Add
    8b3. Lut :
content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_0.curve
    8b4. Filter Gain : 0.01000
    8b5. Up Limit : 1.00000
    8b6. Down Limit : 0.00000
    8b7. Current Value : 0.00000
    8b8. Const Value : 0.00000
  8b. Stages 2
    8b1. Input Var : SlipAngleRearMAX
    8b2. Combinator Mode : Mult
    8b3. Lut :
content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_1.curve
    8b4. Filter Gain : 0.95000
    8b5. Up Limit : 1.20000
    8b6. Down Limit : 0.00000
    8b7. Current Value : 0.00000
    8b8. Const Value : 0.00000
  8b. Stages 3
    8b1. Input Var : Brake
    8b2. Combinator Mode : Mult
    8b3. Lut :
content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_2.curve
    8b4. Filter Gain : 0.95000
    8b5. Up Limit : 1.20000
    8b6. Down Limit : 0.00000
    8b7. Current Value : 0.00000
    8b8. Const Value : 0.00000
9. Controllers EBB [x] : None
10. Steer Brake Controller : None
11. Troque Controller EBB : None
12. EBB Mode : ebbDisabled
13. EBB Front Multiplier 1.10000
14. EBB Min Speed : 0.00000
```

Lancia Delta HF Integrale EVO II (slug : ks_lancia_delta_hf_integrale_evo_ii)

- 1. Total Torque : 2800.00000
- 2. Front Bias : 0.78000
- 3. Hand Brake Torque : 1300.00000
- 4. Has Cockpit Bias : false
- 5. Bias Step : 0.00000
- 6. Front Compound Path : None
- 7. Rear Compound path : None
- 8. Controller EBB
 - 8a. Name : None
 - 8b. Stages 1
 - 8b1. Input Var : LoadSpreadLF
 - 8b2. Combinator Mode : Add
 - 8b3. Lut :

content\cars\ks_lancia_delta_hf_integrale_evo_ii\data\ctrl_ebbCONTROLLER_0.curve

- 8b4. Filter Gain : 0.01000
- 8b5. Up Limit : 1.00000
- 8b6. Down Limit : 0.00000
- 8b7. Current Value : 0.00000
- 8b8. Const Value : 0.00000
- 8b. Stages 2
 - 8b1. Input Var : Brake
 - 8b2. Combinator Mode : Mult
 - 8b3. Lut :

content\cars\ks_lancia_delta_hf_integrale_evo_ii\data\ctrl_ebbCONTROLLER_1.curve

- 8b4. Filter Gain : 0.95000
- 8b5. Up Limit : 1.20000
- 8b6. Down Limit : 0.00000
- 8b7. Current Value : 0.00000
- 8b8. Const Value : 0.00000
- 9. Controllers EBB [x] : None
- 10. Steer Brake Controller : None
- 11. Troque Controller EBB : None
- 12. EBB Mode : ebbDisabled
- 13. EBB Front Multiplier 1.20000
- 14. EBB Min Speed : 0.00000

Ferrari 296 GT3

- 1. Total Torque : 4300.00000
- 2. Front Bias : 65.00000
- 3. Hand Brake Torque : 200.00000
- 4. Has Cockpit Bias : true
- 5. Bias Step : 0.20000
- 6. Front Compound Path 1 :

content\cars\common_phsx\brakes\racing\racing_GT3_pad2.brakes

- 7. Rear Compound path 2 :

content\cars\common_phsx\brakes\racing\racing_GT3_pad2.brakes

- 8. Controllers EBB [x] : None
- 9. Controller EBB : None

- 10. Steer Brake Controller : None
- 11. Troque Controller EBB : None
- 12. EBB Mode : ebbDisabled
- 13. EBB Front Multiplier 0.00000
- 14. EBB Min Speed : 0.00000

Ferrari SF25

- 1. Total Torque : 5500.00000
- 2. Front Bias : 54.00000
- 3. Hand Brake Torque : 0.00000
- 4. Has Cockpit Bias : true
- 5. Bias Step : 0.10000
- 6. Front Compound Path : None
- 7. Rear Compound path : None
- 8. Controller EBB : None
- 9. Controllers EBB 1
 - 8a. Name : Mig0
 - 8b. Stages 1
 - 8b1. Input Var : Brake
 - 8b2. Combinator Mode : Add
 - 8b3. Lut :

content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_0.c
urve

- 8b4. Filter Gain : 0.90000
- 8b5. Up Limit : 10.00000
- 8b6. Down Limit : 0.00000
- 8b7. Current Value : 0.00000
- 8b8. Const Value : 0.00000
- 8b. Stages 2
 - 8b1. Input Var : ErsCoastTorque
 - 8b2. Combinator Mode : Add
 - 8b3. Lut :

content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve

- 8b4. Filter Gain : 0.90000
- 8b5. Up Limit : 10.00000
- 8b6. Down Limit : 0.00000
- 8b7. Current Value : 0.00000
- 8b8. Const Value : 0.00000
- 8. Controllers EBB 2
 - 8a. Name : Mig2
 - 8b. Stages 1
 - 8b1. Input Var : Brake
 - 8b2. Combinator Mode : Add
 - 8b3. Lut :

content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_2.c
urve

- 8b4. Filter Gain : 0.90000
- 8b5. Up Limit : 10.00000
- 8b6. Down Limit : 0.00000
- 8b7. Current Value : 0.00000
- 8b8. Const Value : 0.00000
- 8b. Stages 2

```

| | | 8b1. Input Var : ErsCoastTorque
| | | 8b2. Combinator Mode : Add
| | | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
| | | 8b4. Filter Gain : 0.90000
| | | 8b5. Up Limit : 10.00000
| | | 8b6. Down Limit : 0.00000
| | | 8b7. Current Value : 0.00000
| | | 8b8. Const Value : 0.00000
- 8. Controllers EBB 3
| | | 8a. Name : Mig4
| | | 8b. Stages 1
| | | | 8b1. Input Var : Brake
| | | | 8b2. Combinator Mode : Add
| | | | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_4.c
urve
| | | 8b4. Filter Gain : 0.90000
| | | 8b5. Up Limit : 10.00000
| | | 8b6. Down Limit : 0.00000
| | | 8b7. Current Value : 0.00000
| | | 8b8. Const Value : 0.00000
- 8b. Stages 2
| | | | 8b1. Input Var : ErsCoastTorque
| | | | 8b2. Combinator Mode : Add
| | | | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
| | | 8b4. Filter Gain : 0.90000
| | | 8b5. Up Limit : 10.00000
| | | 8b6. Down Limit : 0.00000
| | | 8b7. Current Value : 0.00000
| | | 8b8. Const Value : 0.00000
- 8. Controllers EBB 4
| | | 8a. Name : Mig6
| | | 8b. Stages 1
| | | | 8b1. Input Var : Brake
| | | | 8b2. Combinator Mode : Add
| | | | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_6.c
urve
| | | 8b4. Filter Gain : 0.90000
| | | 8b5. Up Limit : 10.00000
| | | 8b6. Down Limit : 0.00000
| | | 8b7. Current Value : 0.00000
| | | 8b8. Const Value : 0.00000
- 8b. Stages 2
| | | | 8b1. Input Var : ErsCoastTorque
| | | | 8b2. Combinator Mode : Add
| | | | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
| | | 8b4. Filter Gain : 0.90000
| | | 8b5. Up Limit : 10.00000
| | | 8b6. Down Limit : 0.00000

```

```
| | | 8b7. Current Value : 0.00000
| | | 8b8. Const Value : 0.00000
- 8. Controllers EBB 5
| | | 8a. Name : Mig8
| | | 8b. Stages 1
| | | | 8b1. Input Var : Brake
| | | | 8b2. Combinator Mode : Add
| | | | 8b3. Lut :
```

content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_8.c
urve

```
| | | 8b4. Filter Gain : 0.90000
| | | 8b5. Up Limit : 10.00000
| | | 8b6. Down Limit : 0.00000
| | | 8b7. Current Value : 0.00000
| | | 8b8. Const Value : 0.00000
| | | 8b. Stages 2
| | | | 8b1. Input Var : ErsCoastTorque
| | | | 8b2. Combinator Mode : Add
| | | | 8b3. Lut :
```

content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve

```
| | | 8b4. Filter Gain : 0.90000
| | | 8b5. Up Limit : 10.00000
| | | 8b6. Down Limit : 0.00000
| | | 8b7. Current Value : 0.00000
| | | 8b8. Const Value : 0.00000
- 10. Steer Brake Controller : None
- 11. Troque Controller EBB : None
- 12. EBB Mode : ebbDynamicControllerRelative
- 13. EBB Front Multiplier 0.00000
- 14. EBB Min Speed : 0.00000
```

2. Brakes [.brakes]

A. Description

I. General Description

The **Brakes** asset defines the microscopic physical properties, thermal dynamics, and wear characteristics of the individual hardware components (disks/rotors and pads) on a specific wheel or axle.

While the previous **BRAKESYSTEM** asset manages macroscopic commands, bias, and electronics, the **BRAKES** asset handles the actual physics of friction and heat. It simulates how brake pads grab the disks, how heat builds up during intense deceleration, how ambient airflow and rain cool the components down, and how the braking performance degrades due to overheating (fade) or physical wear over time.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters in this file dictate the endurance, consistency, and tactile feel of the brakes under racing conditions:

- **Brake Fade & Optimal Temperature Window:** Defines how temperature fluctuations change the friction coefficient. Brakes that are too cold or too hot will lose stopping power, drastically altering stopping distances.
- **Thermal Management & Cooling Strategy:** Controls how fast the brakes heat up and shed energy. This directly influences how hard a driver can push lap after lap, and dictates if cooling ducts need adjustment.
- **Brake Wear & Lifespan:** Determines the consumption rate of pads and disks, which is vital for endurance racing strategy where brake swaps might be necessary.
- **Weather Adaptability:** Governs how environmental factors like rain disrupt both the cooling rate and the friction surface.

III. Key Architecture & Data Fields Explained

The data in this schema is structured into three fundamental categories: **Thermal Modeling**, **Wear & Dimensions**, and the **Friction Performance Curve**.

1 - THERMAL MODELING & HEAT DISSIPATION

- **Cool Transfer & Cool Speed Factor:** Base coefficients for convective cooling. **COOL TRANSFER** sets the ambient cooling rate, while **COOL SPEED FACTOR** dictates how much faster the brakes cool as vehicle velocity increases (forcing more air through the brake ducts).
- **Rain Cool Factor:** Modifies the cooling rate when the track is wet, as water spraying onto the brakes drastically increases heat dissipation.

- **Emissivity:** The radiant heat efficiency of the brake material, simulating how much heat energy is emitted as infrared radiation (crucial at glowing, high-temperature states).
- **Surface:** The physical surface area of the brake assembly exposed to airflow.
- **Thermal Capacity & Core Thermal Capacity:** The amount of heat energy required to raise the temperature of the brake surface and the internal "core" mass. Higher capacity means the brakes heat up more slowly but take longer to cool down.
- **Thermal Conductivity & Conduction Thickness:** Controls the rate of heat transfer from the friction surface into the core of the brake disk.

2 - DIMENSIONS, WEAR, AND DEGRADATION (M M = PER MILLIMETER)

- **Disk / Pad Thickness:** The initial, brand-new thickness of the brake disk and pads (typically in millimeters).
- **Disk / Pad Consumption Rate:** The speed at which material is worn away relative to friction and heat cycles.
- **T Reference Wear:** The reference temperature at which normal wear calculations occur. Exceeding this temperature usually spikes the wear exponentially.
- **Perf Decrease M M:** The percentage drop in braking performance or friction for every millimeter of pad/disk material lost to wear.
- **Mu Reduction M M:** The literal decrease of the friction coefficient (μ) as the pads and disks get thinner.
- **Area Reduction M M:** The reduction of effective pad contact surface area as the components wear down.
- **Gamma Correction M M:** A scaling factor that non-linearly adjusts how wear alters the brake feel and performance curves over time.

3 - PERF CURVE (FRICTION COEFFICIENT VS. TEMPERATURE)

This is a look-up table mapping Temperature (X-axis in °C) to Performance Efficiency (Y-axis where 1.000 = 100% target friction). It dictates the "sweet spot" of the brake compound.

IV. Interpretation of Asset Implementation & Data Profiles

Depending on the vehicle category and simulation depth, the deployment of .BRAKES files generally follows one of two implementation profiles:

PROFILE A: SPLIT FRONT/REAR AXLE CONFIGURATIONS (E.G., VINTAGE/ROAD CARS)

In setups where front and rear hardware are separated into distinct files, the simulation treats the axles as independent thermal entities.

- **Axle Balancing:** Developers can mirror the properties exactly (as seen in some classic vehicles where front and rear drums or basic disks share structural dimensions) or modify them to reflect different rotor sizes.
- **Thermal Synchronization:** Separate files allow the front brakes to face realistic, intense thermal loading (due to forward weight transfer) while the rear brakes are modeled with lower heat accumulation or distinct cooling properties.

PROFILE B: SINGLE SHARED COMPOUND PROFILES & ENDURANCE PADS (E.G., RACING GT3)

For modern racing environments—often denoted by specific naming conventions like `_PADI`, `_PAD2` (representing alternative endurance, sprint, or wet-weather friction compounds)—the simulation frequently utilizes a single master `.BRAKES` asset applied globally across the vehicle.

- **Compound Swaps:** In this scenario, switching files changes the fundamental chemistry of the pads vehicle-wide. For example, a `_PAD2` compound might represent a long-distance endurance pad characterized by lower consumption rates (`PAD CONSUMPTION RATE`) and a wider, flatter thermal efficiency window, albeit at the cost of a slightly lower maximum friction coefficient (μ).
- **Unified Thermal Performance:** A shared file applies identical material limits (such as `T REFERENCE WEAR` and `THERMAL CAPACITY`) globally, leaving the volumetric difference in heat handling to be modulated purely by the car's distinct front/rear brake duct settings.

READING THE PERFORMANCE CURVE TREND

The performance curve look-up table translates directly to driver feel and pedal consistency on track:

- **The Bite Threshold (0°C - 200°C):** A high efficiency value ($0.90+$) at low temperatures indicates a street-friendly or versatile compound that works instantly without requiring a warm-up phase. Modern carbon-ceramic or pure racing pads will often show a massive dip here ($0.40 - 0.60$), requiring aggressive warming tactics.
- **The Sweet Spot (300°C - 600°C):** The plateau where the curve hits its absolute maximum (1.000). A wider plateau indicates an easy-to-manage brake system that provides a consistent stopping distance over a broad operational window.
- **Thermal Fade Zone (700°C+):** The rate at which the curve drops at extreme temperatures defines the severity of the brake fade. A gradual taper allows the driver to feel the car losing stopping power safely, while a sharp drop-off causes catastrophic brake failure under heavy racing conditions.

B. Schema

- 1. Cool Transfer : float
- 2. Torque K : float
- 3. Cool Speed Factor : float
- 4. Rain Cool Factor : float
- 5. Emissivity : float

- 6. Surface : float
- 7. Thermal Capacity : float
- 8. Core Thermal Capacity : float
- 9. Thermal Conductivity : float
- 10. Conduction Thickness : float
- 11. Disk Consumption Rate : float
- 12. Pad Consumption Rate : float
- 13. Disk Thickness : float
- 14. Pad Thickness : float
- 15. Perf Decrease M M : float
- 16. Gamma Correction M M : float
- 17. Mu Reduction M M : float
- 18. Area Reduction M M : float
- 19. T Reference Wear : float
- 20. Perf Curve : string – path

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Cool Transfer	Coefficient (W/K)	Base ambient cooling rate; defines how fast the brakes shed heat to the surrounding air when the vehicle is completely stationary.
2.	Torque Conversion / K	Dimensionless Factor	Mechanical torque conversion multiplier; scales how efficiently hydraulic line pressure translates into raw stopping torque.
3.	Cool Speed Factor	Coefficient	Airflow cooling multiplier; dictates the linear or non-linear scaling of heat dissipation as the vehicle speed increases.
4.	Rain Cool Factor	Dimensionless Ratio	Wet-weather cooling modifier; amplifies the global heat dissipation rate to simulate rain and track water spray striking the brake assembly.
5.	Emissivity	Dimensionless (0.0 - 1.0)	Thermal radiation efficiency (ϵ); defines how effectively the brake material radiates infrared heat, becoming highly critical at glowing, high-temperature states.
6.	Surface	m^2 (Square meters)	Total physical exposed surface area of the brake disc/assembly; larger surfaces naturally facilitate faster convective cooling.
7.	Thermal Capacity	J/K or J/°C	Heat storage capacity of the thin outer friction surface layer; determines how quickly the contact zone heats up under friction.

ID	Name	Unit of Measurement	Description
8.	Core Thermal Capacity	J/K or J/°C	Heat storage capacity of the internal "core" mass of the brake disc; serves as the main thermal reservoir absorbing energy from the surface.
9.	Thermal Conductivity	W/(m-K) or Coefficient	Internal heat transfer rate; controls how fast heat energy moves from the hot outer friction surface into the cooler internal core.
10.	Conduction Thickness	m (Meters) or mm	Physical distance/depth representing the boundary layer for internal heat transfer between the surface friction node and the core mass.
11.	Disk Consumption Rate	Wear Coefficient	Material wear rate of the brake disc/rotor, tracking physical thickness reduction relative to temperature and kinetic energy absorption cycles.
12.	Pad Consumption Rate	Wear Coefficient	Material wear rate of the brake pads, tracking friction compound depletion relative to temperature and kinetic energy absorption cycles.
13.	Disk Thickness	mm (Millimeters)	The initial, brand-new physical structural thickness of the brake disc/rotor.
14.	Pad Thickness	mm (Millimeters)	The initial, brand-new physical thickness of the wearable friction material layer on the brake pad.
15.	Perf Decrease M M	% / mm (mm^{-1})	Global percentage drop in overall braking efficiency and biting performance for every single millimeter of total pad/disk material lost to wear.
16.	Gamma Correction M M	Dimensionless Exponent	Non-linear scaling factor; progressively adjusts how material wear non-linearly alters pedal feedback, compliance, and friction behavior over time.
17.	Mu Reduction M M	$\Delta\mu/mm$	The literal linear reduction of the raw friction coefficient (μ) applied as the pads and discs become thinner.
18.	Area Reduction M M	m^2/mm or $\%/mm$	The progressive reduction of the effective pad-to-disc contact surface area as the pad material shaves down.

ID	Name	Unit of Measurement	Description
19.	T Reference Wear	°C (Degress Celsius)	Threshold reference temperature; defines the thermal boundary above which component wear rates begin to spike exponentially.
20.	Perf Curve	None (File path)	File path pointing to an external .CURVE look-up table mapping raw operating temperature (X-axis in °C) to friction coefficient efficiency (Y-axis).

D. Example data

I. Chosen Brakes for Example

- Vintage Road Front (slug : vintage_road_front) / Vintage Road Rear (vintage_road_rear)
- Racing GT3 [Pad 2] (slug : racing_gt3_pad2)

II. Example

Vintage Road [Front]

```

1. Cool Transfer : 1.30000
2. Torque K : 0.70000
3. Cool Speed Factor : 1.50000
4. Rain Cool Factor : 0.80000
5. Emissivity : 0.70000
6. Surface : 0.20000
7. Thermal Capacity : 100.00000
8. Core Thermal Capacity : 1600.00000
9. Thermal Conductivity : 250.00000
10. Conduction Thickness : 0.00500
11. Disk Consumption Rate : 0.18000
12. Pad Consumption Rate : 0.19800
13. Disk Thickness : 32.00000
14. Pad Thickness : 29.00000
15. Perf Decrease M M : 15.00000
16. Gamma Correction M M : 1.50000
17. Mu Reduction M M : 0.01000
18. Area Reduction M M : 0.16000
19. T Reference Wear : 600.00000
20. Perf Curve :
content\cars\common_phsx\brakes\vintage\tcurve_vintage_front.curve

```

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P0	0.000	0.900
P1	100.000	0.950

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P2	500.000	0.980
P3	600.000	1.000
P4	800.000	0.980
P5	900.000	0.950
P6	1000.000	0.850
P7	1200.000	0.800

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

Vintage Road [Rear]

```

1. Cool Transfer : 1.30000
2. Torque K : 0.70000
3. Cool Speed Factor : 1.50000
4. Rain Cool Factor : 0.80000
5. Emissivity : 0.70000
6. Surface : 0.20000
7. Thermal Capacity : 100.00000
8. Core Thermal Capacity : 1600.00000
9. Thermal Conductivity : 250.00000
10. Conduction Thickness : 0.00500
11. Disk Consumption Rate : 0.18000
12. Pad Consumption Rate : 0.19800
13. Disk Thickness : 32.00000
14. Pad Thickness : 29.00000
15. Perf Decrease M M : 15.00000
16. Gamma Correction M M : 1.50000
17. Mu Reduction M M : 0.01000
18. Area Reduction M M : 0.16000
19. T Reference Wear : 600.00000
20. Perf Curve :
content\cars\common_phsx\brakes\vintage\tcurve_vintage_rear.curve

```

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P0	0.000	0.900
P1	100.000	0.950
P2	500.000	0.980
P3	600.000	1.000
P4	800.000	0.980
P5	900.000	0.950
P6	1000.000	0.850

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P7	1200.000	0.800

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

Racing GT3 [Pad 2]

- 1. Cool Transfer : 0.90000
- 2. Torque K : 0.70000
- 3. Cool Speed Factor : 1.60000
- 4. Rain Cool Factor : 0.80000
- 5. Emissivity : 0.70000
- 6. Surface : 0.40000
- 7. Thermal Capacity : 100.00000
- 8. Core Thermal Capacity : 1600.00000
- 9. Thermal Conductivity : 250.00000
- 10. Conduction Thickness : 0.00500
- 11. Disk Consumption Rate : 0.01050
- 12. Pad Consumption Rate : 0.02100
- 13. Disk Thickness : 32.00000
- 14. Pad Thickness : 29.00000
- 15. Perf Decrease M M : 13.00000
- 16. Gamma Correction M M : 1.00000
- 17. Mu Reduction M M : 0.01000
- 18. Area Reduction M M : 0.08000
- 19. T Reference Wear : 500.00000
- 20. Perf Curve :

content\cars\common_phsx\brakes\racing\tcurve_racing_GT3_pad2

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P0	-84.050	0.774
P1	-0.905	0.839
P2	39.477	0.914
P3	88.596	0.952
P4	121.867	0.967
P5	156.821	0.974
P6	203.872	0.978
P7	252.036	0.980
P8	303.284	0.980
P9	365.345	0.980
P10	380.803	0.979
P11	443.598	0.977

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P12	526.801	0.973
P13	599.594	0.966
P14	722.684	0.945
P15	781.584	0.936
P16	849.036	0.923
P17	908.993	0.898
P18	972.698	0.854
P19	1047.645	0.734
P20	1245.218	0.567

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

3. Car Data [.car]

A. Description

I. General Description

The **Car Data** asset is the primary foundational master file for a vehicle's physics package in the simulation engine. If **BrakeSystem** and **Brakes** represent localized hardware components, **Car Data** represents the physical carcass, weight distribution, structural constraints, and regulatory rules of the entire vehicle.

It acts as the structural blueprint that holds all other modular components (engine, tires, suspension, and brakes) together by defining how the global mass moves, how the center of gravity shifts, and how the physical dimensions interact with the environment.

II. Area of Influence / Impact on Vehicle Dynamics

The properties configured within the **Car Data** asset dictate the base handling DNA and fundamental physical limits of the vehicle before any specific aerodynamic downforce or mechanical grip is added:

- **Weight Distribution & Inertia:** Governs the vehicle's resistance to rotational changes (pitch, roll, and yaw). It directly dictates how quick the car changes direction and how much weight transfers under braking, acceleration, or cornering.
- **Dimensional Footprint (Track & Wheelbase):** Determines the baseline mechanical stability. A wider track increases cornering stability, while a longer wheelbase stabilizes the car at high speeds but reduces low-speed agility.
- **Driver Environment & Ergonomics:** Configures the driver's perspective (camera placement, cockpit nodes) and controls the operational boundaries of steering inputs.
- **Fuel Weight Dynamics:** Simulates dynamic mass changes over a race stint. As fuel burns off, the overall vehicle weight drops and the center of gravity shifts, drastically altering handling characteristics over time.

III. Key Architecture & Data Fields Explained

The parameters in this schema are universally categorized into **Mass & Inertia**, **Dimensions & Offsets**, and **Fuel & Regulation Logistics**.

1 - GLOBAL MASS & INERTIA PROPERTIES

- **Total Mass / Dry Weight:** The baseline mass of the vehicle without fuel, driver, or fluids (expressed in **KG**). This is the fundamental variable for $F=MA$ calculations.
- **Inertia (Pitch, Roll, Yaw Vectors):** Critical tensors (I_x, I_y, I_z) defining how mass is distributed away from the center of gravity. High polar inertia means a car resists spinning but is hard to catch once it breaks traction.

- **Center of Gravity Height (CG Height):** The vertical position of the balance point. Lower CG decreases body roll and load transfer across tires, significantly increasing overall cornering potential.
- **Weight Bias (Front/Rear %):** The static balance of the car at rest (e.g. 45/55 for mid-engine layouts). It underpins the car's natural handling bias (understeer vs. oversteer tendency).

2 - DIMENSIONS, TRACK AND ALIGNEMENT COORDINATES

- **Wheelbase:** The linear distance between the center lines of the front and rear axles.
- **Track Width (Front/Rear):** The lateral distance between the center points of the left and right tires on an axle.
- **Graphics / Collision Mesh Offsets:** Coordinates that align the structural 3D model and physical hitbox with the theoretical center of gravity, ensuring visual impacts and ground scraping clip accurately.
- **Steer Lock / Rack Ratio:** Max steering angle of the front wheels and the corresponding ratio to the steering wheel column. Defines the vehicle's turning circle and quickness of steering inputs.

3 - FUEL MANAGEMENT & CONSUMABLES

- **Fuel Tank Capacity:** The maximum volume of fuel the car can physically hold (typically in Liters or Gallons).
- **Fuel Tank Position (X, Y, Z Coordinates):** The exact physical location of the fuel payload inside the chassis. Crucial for calculating how the center of gravity and weight balance shift as the fuel burns down.
- **Driver Mass Toggle:** Dictates whether the weight of an average driver (e.g., standard regulatory 75 KG - 80 KG) is automatically appended to the physical mass calculations.

IV. Short Interpretation of Asset Implementation

When analyzing a vehicle's physics using the `CAR DATA` profile, you can immediately predict its handling characteristics through basic parameter archetypes:

- **The Prototype/Open-Wheel Profile:** Characterized by ultra-low total mass (< 800 KG), a wide track width, and an incredibly low Center of Gravity Height. The inertia tensors are compressed closely to the center, yielding instantaneous directional response.
- **The GT3/Production Profile:** Features higher total mass (1200 KG - 1400 KG), higher CG positions, and a sprawling wheelbase. These vehicles experience heavy weight transfer, demanding careful management of pitch and roll stiffness elsewhere in the setup.
- **The Endurance Strategy Vector:** By looking at the Fuel Tank Capacity paired with its structural 3D coordinates, you can see how heavily a full stint will punish tire wear. If the tank is behind the rear axle, a full fuel load will temporarily shift the weight bias

significantly rearward, introducing entry understeer that slowly fades into snappier oversteer as the tank empties.

B. Schema

- 1. Screen Name : **string**
- 2. General : **object**
 - 2a. Screen Name : **string**
 - 2b. Total Mass : **float**
 - 2c. Tank Position : **x, y, z float**
 - 2d. Fuel : **float**
 - 2e. Max Fuel : **float**
 - 2f. Efficiency : **float**
 - 2g. Kg Per Liter : **float**
 - 2h. Body Box Sizes : **x, y, z float**
 - 2i. Pickup Front Height : **float**
 - 2j. Pickup Rear Height : **float**
 - 2k. Check Rules : **boolean**
 - 2l. Minimum Height : **float**
 - 2m. Torsional Stiffness : **float**
 - 2n. Torsional Damping : **float**
 - 2o. Body Mesh Offset : **object**
 - 2o1. Position : **x, y, z float**
 - 2o2. Rotation : **x, y, z, float**
 - 2o3. Scale : **x, y, z float**
- 3. General Path : **string – path**
- 4. Suspensions : **object**
 - 4a. Wheel Base : **float**
 - 4b. Longitudinal Cg Location : **float**
 - 4c. Base Y Front : **float**
 - 4d. Base Y Rear : **float**
 - 4e. Track Front : **float**
 - 4f. Track Rear : **float**
 - 4g. Damage : **object**
 - 4g1. Min Velocity : **float**
 - 4g2. Gain : **float**
 - 4g3. Max Damage : **float**
 - 4g4. Debug Log : **boolean**
 - 4h. Coilover Front path : **string – path**
 - 4i. Coilover Rear Path : **string – path**
 - 4j. Front Suspension Path : **string – path**
 - 4k. Rear Suspension Path : **string – path**
 - 4l. Heavy Springs [x] : **object** | Can have multiple Heavy Springs,
where x can be 1, 2, 3, ...
 - 4l1. Spring Rate : **float**
 - 4l2. Progressive Sprint Rate : **float**
 - 4l3. Bump Stop Up : **object**
 - 4l3a. Range : **float**
 - 4l3b. Reference : **float**
 - 4l3c. Force : **float**
 - 4l3d. Gamma : **float**
 - 4l3e. Length : **float**
 - 4l3f. Damping : **float**
 - 4l4. Bump Stop Down : **object**

- 4l4a. Range : float
- 4l4b. Reference : float
- 4l4c. Force : float
- 4l4d. Gamma : float
- 4l4e. Length : float
- 4l4f. Damping : float
- 4l5. Collar Position : float
- 4l6. Damper : object
 - 4l6a. Fast : object
 - 4l6a1. Bump : float
 - 4l6a2. Rebound : float
 - 4l6b. Slow : object
 - 4l6b1. Bump : float
 - 4l6b2. Rebound : float
 - 4l6c. Fast Threshold Bump : float
 - 4l6d. Fast Threshold Rebound : float
 - 4l6e. Cooling Surface : float
 - 4l6f. Nominal Force : float
 - 4l6g. Min Stress Fatigue : float
 - 4l6h. Max Stress Fatigue : float
 - 4l6i. Thermal Capacity : float
 - 4l6j. Heat Transfer Coefficient : float
 - 4l6k. Lut List : string - path
 - 4l6l. Damper Lut Scale : string - path
- 4l7. Helper K : float
- 4l8. Helper Range : float
- 4l9. Rod Controllers : object with an array of stages within
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple

stages

- 4l9b1. Input Var : enum
- 4l9b2. Combinator Mode : enum
- 4l9b3. Lut : string - path
- 4l9b4. Filter Gain : float
- 4l9b5. Up Limit : float
- 4l9b6. Down Limit : float
- 4l9b7. Current Value : float
- 4l9b8. Const Value : float

- 4m. Arb Front : object
 - 4m1. Stiffness : float
 - 4m2. Controller : object
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple

stages

- 4l9b1. Input Var : enum
- 4l9b2. Combinator Mode : enum
- 4l9b3. Lut : string - path
- 4l9b4. Filter Gain : float
- 4l9b5. Up Limit : float
- 4l9b6. Down Limit : float
- 4l9b7. Current Value : float
- 4l9b8. Const Value : float

- 4n. Arb Rear : object
 - 4n1. Stiffness : float
 - 4n2. Controller : object
 - 4l9a. Name : string

```

| | | | 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
| | | | | 4l9b1. Input Var : enum
| | | | | 4l9b2. Combinator Mode : enum
| | | | | 4l9b3. Lut : string - path
| | | | | 4l9b4. Filter Gain : float
| | | | | 4l9b5. Up Limit : float
| | | | | 4l9b6. Down Limit : float
| | | | | 4l9b7. Current Value : float
| | | | | 4l9b8. Const Value : float
| | | | - 4o. Flex Bar Front : object
| | | | | 4o1. Stiffness : float
| | | | | 4o2. Controller : object
| | | | | 4l9a. Name : string
| | | | | 4l9b. Stages [x] : float | Rod Controllers can have multiple
stages
| | | | | 4l9b1. Input Var : enum
| | | | | 4l9b2. Combinator Mode : enum
| | | | | 4l9b3. Lut : string - path
| | | | | 4l9b4. Filter Gain : float
| | | | | 4l9b5. Up Limit : float
| | | | | 4l9b6. Down Limit : float
| | | | | 4l9b7. Current Value : float
| | | | | 4l9b8. Const Value : float
| | | | - 4p. Flex Bar Rear : object
| | | | | 4p1. Stiffness : float
| | | | | 4p2. Controller : object with an array of stages within
| | | | | 4l9a. Name : string
| | | | | 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
| | | | | 4l9b1. Input Var : enum
| | | | | 4l9b2. Combinator Mode : enum
| | | | | 4l9b3. Lut : string - path
| | | | | 4l9b4. Filter Gain : float
| | | | | 4l9b5. Up Limit : float
| | | | | 4l9b6. Down Limit : float
| | | | | 4l9b7. Current Value : float
| | | | | 4l9b8. Const Value : float
| | | | - 4q. Dampers Controller : object
| | | | | 4q1. Wheel Gain : float
| | | | | 4q2. Heave Gain : float
| | | | | 4q3. Pitch Gain : float
| | | | | 4q4. Roll Gain : float
| | | | | 4q5. Front : object
| | | | | | 4q5a. Base : object
| | | | | | | 4q5a1. Type : enum
| | | | | | 4q5b. Heave : object
| | | | | | | 4q5a1. Type : enum
| | | | | | 4q5c. Roll : object
| | | | | | | 4q5a1. Type : enum
| | | | | | 4q5d. Pitch : object
| | | | | | | 4q5a1. Type : enum
| | | | | | 4q5e. Max : object
| | | | | | | 4q5a1. Type : enum
| | | | | | 4q5f. Min : object
| | | | | | | 4q5a1. Type : enum

```

- 4q6. Rear : **object**
 - 4q5a. Base : **object**
 - └ 4q5a1. Type : **enum**
 - 4q5b. Heave : **object**
 - └ 4q5a1. Type : **enum**
 - 4q5c. Roll : **object**
 - └ 4q5a1. Type : **enum**
 - 4q5d. Pitch : **object**
 - └ 4q5a1. Type : **enum**
 - 4q5e. Max : **object**
 - └ 4q5a1. Type : **enum**
 - 4q5f. Min : **object**
 - └ 4q5a1. Type : **enum**
- 4r. Has Dampers Cockpit Settings : **boolean**
- 5. Drivetrain Path : **string - path**
- 6. Gearbox Path : **string - path**
- 7. Clutch Path : **string - path**
- 8. Engine Path : **string - path**
- 9. Brakes Path : **string - path**
- 10. Steering System : **object**
 - 10a. Four W S Controllers : **object with an array of stages within**
 - 4l9a. Name : **string**
 - 4l9b. Stages [x] : **object** | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : **enum**
 - 4l9b2. Combinator Mode : **enum**
 - 4l9b3. Lut : **string - path**
 - 4l9b4. Filter Gain : **float**
 - 4l9b5. Up Limit : **float**
 - 4l9b6. Down Limit : **float**
 - 4l9b7. Current Value : **float**
 - 4l9b8. Const Value : **float**
- 11. Electronics : **object**
 - 11a. T C : **object**
 - 11a1. Has T C2 : **boolean**
 - 11a2. Frequency Hz : **float**
 - 11a3. Min Speed Kmh : **float**
 - 11a4. Gear Change Time : **float**
 - 11a5. Min Cut Level : **float**
 - 11a6. Max Cut Level : **float**
 - 11a7. Settings [x] : **object** | T C can have multiple Settings
 - 11a7a. Min Slip Ratio : **float**
 - 11a7b. Max Slip Ratio : **float**
 - 11a7c. Ref Slip Angle Deg : **float**
 - 11a7d. Engine Cut Level : **float**
 - 11a7e. Angular A C Cgain : **float**
 - 11a7f. Oversteer Gain : **float**
 - 11a7g. Slip Angle Activation Deg : **float**
 - 11b. A B S : **object**
 - 11b1. Settings [x] : **object** | A B S can have multiple Settings
 - 11b1a. Min Slip Ratio : **float**
 - 11b1b. Max Slip Ratio : **float**
 - 11b1c. Ref Slip Angle Deg : **float**
 - 11b1d. Cut Level : **float**
 - 11b1f. Max Torque Variation : **float**
 - Frequency : **float**

- Channels : integer
- Min Speed Kmh : float
- 11c. E D L : object
 - 11c1. Active : boolean
 - 11c2. Brake Torque Power : float
 - 11c3. Brake Torque Coast : float
 - 11c4. Dead Zone Coast : float
 - 11c5. Dead Zone Power : float
 - 11c6. Max Spin Power : float
 - 11c7. Max Spin Coaster : float
 - 11c8. Min Speed : float
- 11d. E S P : object
 - 11d1. Frequency Hz : float
 - 11d2. Min Speed Kmh : float
 - 11d3. Settings [x] : object | E S P can have multiple settings
 - 11d3a. Gain : float
 - 11d3b. Steer Gain : float
 - 11d3c. Min Steer Gain : float
 - 11d3d. Steer Gain Max Speed : float
 - 11d3e. Oversteer Gain : float
 - 11d3f. Understeer Gain : float
 - 11d3g. Max Slip Ratio : float
 - 11d3h. Dead Zone : float
 - 11d3i. Filter Gain : float
 - 11d3j. Brake Perc : float
 - 11d3k. Brake Perc Activation : float
- 12. Electronics Path : string - path
- 13. Controls : object
 - 13a. Ff Mult : float
 - 13b. Steer Lock : float
 - 13c. Steer Ratio : float
 - 13d. Linear Steer Rod Ratio : float
 - 13e. Steer Assist : float
- 14. Box Colliders [x] : object | can have multiple Box Colliders
 - 14a. Center : x, y, z float
 - 14b. Size : x, y, z float
 - 14c. Pitch Rotation Deg : float
- 15. Front Tyre Compounds [x] : string - path | can have multiple Front Tyre Compounds path
- 16. Rear Tyre Compounds [x] : string - path | can have multiple Rear Tyre Compounds path
- 17. Aero : object
 - 17a. Slip Gain Multiple : float
 - 17b. Speed Factor Mult : float
 - 17c. Downforces [x] : object | can have multiple Downforces
 - 17c1. Position : x, y, z float
 - 17c2. Cl Gain : float
 - 17c3. Cd Gain : float
 - 17c4. Yaw Gain : float
 - 17c5. Drag Per Cool Transfer : float
 - 17c6. Damage C L [x] : string | can have multiple Damage C L
 - 17c7. Damage C D [x] : string | can have multiple Damage C D
 - 17c8. Downforce Controllers [x] : object | can have multiple Downforce Controllers
 - 17c8a. Combinator Mode : enum
 - 17c8b. Input : enum

- 17c8c. Filter : float
- 17c8d. Up Limit : float
- 17c8e. Down Limit : float
- 17c8f. Lut : string – path
- 17c9. Lift Per Front Angle : float
- 17c10. Lift Per Rear Angle : float
- 17c11. Drag Per Front Angle : float
- 17c12. Drag Per Rear Angle : float
- 17c13. Default Front Angle : float
- 17c14. Default Rear Angle : float
- 17d. Front Lift : string – path
- 17e. Rear Lift : string – path
- 17f. Drag : string – path
- 17g. Wings Path [x] : string – path | can have multiple Wings Path
- 18. Drs : object
 - 18a. Ignore Zones : boolean
 - 18b. Limit G : float
 - 18c. Wing Connections [x] : object | can have multiple Wing

Connections

- 18c1. Mode : enum
- 18c2. Connected Wing : integer
- 18c3. Effect : float
- 18c4. Angle : float
- 19. Ers : object
 - 19a. Torque Lut : string – path
 - 19b. Coast Lut : string – path
 - 19c. Battery Charge Kj : float
 - 19d. Has Button Override : boolean
 - 19e. Max Kj Per Lap : float
 - 19f. Max Charge Kj Per Lap : float
 - 19g. Heat Charge Perc : float
 - 19h. Heat Power Kw : float
 - 19i. Default Power Controller Index : integer
 - 19j. Power Controllers Front [x] : object | can have multiple Power

Controllers Front

- 4l9a. Name : string
- 4l9b. Stages [x] : object | Rod Controllers can have multiple

stages

- 4l9b1. Input Var : enum
- 4l9b2. Combinator Mode : enum
- 4l9b3. Lut : string – path
- 4l9b4. Filter Gain : float
- 4l9b5. Up Limit : float
- 4l9b6. Down Limit : float
- 4l9b7. Current Value : float
- 4l9b8. Const Value : float
- 19k. Power Controllers Rear [x] : object | can have multiple Power

Controllers Rear

- 4l9a. Name : string
- 4l9b. Stages [x] : object | Rod Controllers can have multiple

stages

- 4l9b1. Input Var : enum
- 4l9b2. Combinator Mode : enum
- 4l9b3. Lut : string – path
- 4l9b4. Filter Gain : float
- 4l9b5. Up Limit : float

- 4l9b6. Down Limit : float
 - 4l9b7. Current Value : float
 - 4l9b8. Const Value : float
- 19l. Brake Rear Correction : float
- 19m. Has Cockpit Controls : boolean
- 19n. Cockpit Controls : object
 - 19n1. Delivery Profile : boolean
 - 19n2. Mgu H Mode : boolean
 - 19n3. Recovery : boolean
- 19o. Has Front Motors : boolean
- 19p. Front Motors : object
 - 19p1. Torque Lut : string - path
 - 19p2 Discharge Time : float
 - 19p3. Torque Vectoring Bias : float
- 20. Setup Limits : string path
- 21. Collider Mesh : string path
- 22. Body Mesh Offset : object
 - 22a. Position : x, y, z float
 - 22b. Rotation: x, y, z float
 - 22c. Scale : x, y, z float
- 23. Stock Setup : string path
- 24. Ai Setup : string path
- 25. Wet Setup : string path
- 26. Performance Modes [x] : object | can have multiple Performance

Modes

- 26a. Performance Mode Name : string
- 26b. Electronics Settings : object
 - 26b1. Tc1 : float
 - 26b2. Tc2 : float
 - 26b3. Abs : float
 - 26b4. Esc : float
 - 26b5. Ebb : float
 - 26b6. Engine Map : float
 - 26b7. Telemetry laps To Record : float
 - 26b8. Turbo Boost Lv : float
 - 26b9. Ers Deployment Map : float
 - 26b10. Ers Recharge Lv : float
 - 26b11. Ers Heat Charging : float
- 26c. Brakes Settings : object
 - 26c1. Front Bias : float
 - 26c2. Torque Multiplier : float
 - 26c3. Brake Ducts [x] : float | can have multiple Brake Ducts
- 26d. Damper Settings [x] : object | can have multiple Damper

Settings

- 26d1. Slow Bump : float
- 26d2. Fast Bump : float
- 26d3. Slow Rebound : float
- 26d4. Fast Rebound : float
- 26e. Differential Data : object
 - 26e1. Type : enum
 - 26e2. Power : float
 - 26e3. Coast : float
 - 26e4. Preload : float
 - 26e5. Front Share : float
 - 26e6. Torque Bias Ratio Power : float
 - 26e7. Torque Bias Ratio Coast : float

- 26e8. Thermal Capacity : float
- 26e9. Surface : float
- 26e10. Heat Transfer Coeff : float
- 26e11. Wear Factor : float
- 26e12. Friction Reduction With T : float
- 26e13. Friction Ref T : float
- 26f. Four W D Differentials : object
 - 26f1. Front Diff : object
 - 26e1. Type : enum
 - 26e2. Power : float
 - 26e3. Coast : float
 - 26e4. Preload : float
 - 26e5. Front Share : float
 - 26e6. Torque Bias Ratio Power : float
 - 26e7. Torque Bias Ratio Coast : float
 - 26e8. Thermal Capacity : float
 - 26e9. Surface : float
 - 26e10. Heat Transfer Coeff : float
 - 26e11. Wear Factor : float
 - 26e12. Friction Reduction With T : float
 - 26e13. Friction Ref T : float
 - 26f2. Center Diff : object
 - 26e1. Type : enum
 - 26e2. Power : float
 - 26e3. Coast : float
 - 26e4. Preload : float
 - 26e5. Front Share : float
 - 26e6. Torque Bias Ratio Power : float
 - 26e7. Torque Bias Ratio Coast : float
 - 26e8. Thermal Capacity : float
 - 26e9. Surface : float
 - 26e10. Heat Transfer Coeff : float
 - 26e11. Wear Factor : float
 - 26e12. Friction Reduction With T : float
 - 26e13. Friction Ref T : float
 - 26f3. Rear Diff : object
 - 26e1. Type : enum
 - 26e2. Power : float
 - 26e3. Coast : float
 - 26e4. Preload : float
 - 26e5. Front Share : float
 - 26e6. Torque Bias Ratio Power : float
 - 26e7. Torque Bias Ratio Coast : float
 - 26e8. Thermal Capacity : float
 - 26e9. Surface : float
 - 26e10. Heat Transfer Coeff : float
 - 26e11. Wear Factor : float
 - 26e12. Friction Reduction With T : float
 - 26e13. Friction Ref T : float
- 26g. Front Lock Controllers : object
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : enum
 - 4l9b2. Combinator Mode : enum
 - 4l9b3. Lut : string - path

- 4l9b4. Filter Gain : float
 - 4l9b5. Up Limit : float
 - 4l9b6. Down Limit : float
 - 4l9b7. Current Value : float
 - 4l9b8. Const Value : float
 - 26h. Center Lock Controllers : object
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : enum
 - 4l9b2. Combinator Mode : enum
 - 4l9b3. Lut : string – path
 - 4l9b4. Filter Gain : float
 - 4l9b5. Up Limit : float
 - 4l9b6. Down Limit : float
 - 4l9b7. Current Value : float
 - 4l9b8. Const Value : float
 - 26i. Rear Lock Controllers : object
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : enum
 - 4l9b2. Combinator Mode : enum
 - 4l9b3. Lut : string – path
 - 4l9b4. Filter Gain : float
 - 4l9b5. Up Limit : float
 - 4l9b6. Down Limit : float
 - 4l9b7. Current Value : float
 - 4l9b8. Const Value : float
 - 26j. Awd Clutches [x] : object | can have multiple Awd Clutches
 - 26j1. Position : integer
 - 26j2. Preload : float
 - 26k. Turbo Controllers [x] : object with an array of stages within | can have multiple Turbo Controllers
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : enum
 - 4l9b2. Combinator Mode : enum
 - 4l9b3. Lut : string – path
 - 4l9b4. Filter Gain : float
 - 4l9b5. Up Limit : float
 - 4l9b6. Down Limit : float
 - 4l9b7. Current Value : float
 - 4l9b8. Const Value : float
 - 26l. Turbo Settings : object
 - 26l1. Boost Lv : float
 - 27. Ai Car Data : string – path
 - 28. mm : integer

Enum list - Car Engine

Id	Enum	Values
4l9b1	Input Var	UndefinedInput, Brake, Gas, LatG, Long, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
4l9b2	CombinatorMode	UndefinedMode, Add, Mult
4q5a1	Type	<None>, Poly3, Poly5, Piece Wise, Damper Lut Data
18c1	Mode	UseEffect, UseAngle
26e1	Type	LSD, Spool, Torsen, EpicyclicTorsen, EpicyclicLSD, TorqueVectoring

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Screen Name	None (String)	Display label for the vehicle profile in editor or UI menus; does not affect physics simulation.
2a.	Screen Name	None (String)	Human-readable name shown for the General block (mass, fuel, body envelope) in tooling interfaces.
2b.	Total Mass	kg (Kilograms)	Baseline vehicle mass including driver and fluids as configured; fundamental input for $F=ma$ weight transfer and inertia calculations.
2c.	Tank Position	m (Meters, X / Y / Z)	3D coordinates of the fuel tank centroid relative to the vehicle reference frame; governs how CG and weight bias shift as fuel burns.
2d.	Fuel	L (Liters)	Current fuel volume at session start; dynamically reduces mass and alters balance during the stint.
2e.	Max Fuel	L (Liters)	Maximum fuel tank capacity; defines the upper bound for fuel load and endurance strategy calculations.
2f.	Efficiency	Dimensionless ratio	Fuel consumption efficiency modifier applied to engine fuel burn calculations.

ID	Name	Unit of Measurement	Description
2g.	Kg Per Liter	kg/L (Kilograms per Liter)	Fuel density conversion factor; translates volumetric fuel (liters) into mass (kg) for physics weight updates.
2h.	Body Box Sizes	m (Meters, X / Y / Z)	Axis-aligned bounding box dimensions of the chassis body envelope; used for regulatory checks and collision approximation.
2i.	Pickup Front Height	m (Meters)	Vertical pickup/reference height at the front axle; aligns suspension hardpoints and ground contact geometry.
2j.	Pickup Rear Height	m (Meters)	Vertical pickup/reference height at the rear axle; aligns suspension hardpoints and ground contact geometry.
2k.	Check Rules	None (Boolean : True / False)	Enables automated regulatory rule validation (e.g., minimum ride height) against series constraints.
2l.	Minimum Height	m (Meters)	Regulatory minimum ride-height threshold enforced when Check Rules is active.
2m.	Torsional Stiffness	Nm/rad or N/m (Torsional spring rate)	Chassis torsional spring rate; resists twist between front and rear axles under asymmetric loading.
2n.	Torsional Damping	Nm·s/rad (Torsional damping)	Chassis torsional damping coefficient; dissipates oscillatory twist energy in the body structure.
2o1.	Position	m (Meters, X / Y / Z)	Translation offset applied to the visual/collision body mesh relative to the physics origin (General > Body Mesh Offset).
2o2.	Rotation	deg or rad (X / Y / Z)	Euler rotation offset applied to the body mesh for visual/collision alignment (General > Body Mesh Offset).
2o3.	Scale	Dimensionless (X / Y / Z)	Non-uniform scale factor applied to the body mesh offset block for model alignment.
3.	General Path	None (File path)	External reference to a .GENERAL.CAR asset overriding or extending global mass/inertia parameters.
4a.	Wheel Base	m (Meters)	Longitudinal distance between front and rear axle centerlines; stabilizes high-speed behavior and affects rotation agility.

ID	Name	Unit of Measurement	Description
4b.	Longitudinal Cg Location	Ratio (0.0 - 1.0)	Longitudinal center-of-gravity position along the wheelbase as a fraction from front axle (e.g., 0.45 = slightly front-biased).
4c.	Base Y Front	m (Meters)	Reference vertical suspension hub height at the front axle in the chassis coordinate system.
4d.	Base Y Rear	m (Meters)	Reference vertical suspension hub height at the rear axle in the chassis coordinate system.
4e.	Track Front	m (Meters)	Lateral distance between left and right front tire centerlines; wider track increases roll stability.
4f.	Track Rear	m (Meters)	Lateral distance between left and right rear tire centerlines.
4g1.	Min Velocity	m/s or km/h (Speed threshold)	Minimum impact velocity required before suspension/chassis damage accumulation begins.
4g2.	Gain	Dimensionless coefficient	Damage accumulation rate multiplier per unit of impact energy above the minimum velocity threshold.
4g3.	Max Damage	Ratio (0.0 - 1.0)	Upper cap on cumulative suspension/chassis damage before performance degradation saturates.
4g4.	Debug Log	None (Boolean : True / False)	Enables verbose logging of suspension damage events for development and tuning diagnostics.
4h.	Coilover Front Path	None (File path)	Path to the front axle .coilover asset defining spring rates, damper curves, and bump-stop geometry.
4i.	Coilover Rear Path	None (File path)	Path to the rear axle .COILOVER asset.
4j.	Front Suspension Path	None (File path)	Path to the front .SUSPENSION asset defining kinematics, wishbone geometry, and camber curves.
4k.	Rear Suspension path	None (File path)	Path to the rear .SUSPENSION asset.
4l1.	Spring Rate	N/m (Newtons per meter)	Linear spring stiffness of the heavy-duty auxiliary spring element (used on trucks, off-road, or dual-rate setups).
4l2.	Progressive Spring Rate	N/m ² or N/m (Progressive rate)	Secondary progressive spring rate; stiffness rises non-linearly as compression increases.

ID	Name	Unit of Measurement	Description
4I3a.	Range	m (Meters)	Operational travel range of the upper bump-stop before full engagement (Heavy Springs > Bump Stop Up).
4I3b	Reference	m (Meters)	Reference suspension position at which the upper bump-stop force model begins scaling.
4I3c.	Force	N (Newtons)	Peak resistive force delivered by the upper bump-stop at maximum compression.
4I3d.	Gamma	Dimensionless exponent	Non-linear shaping exponent for upper bump-stop force vs. compression curve.
4I3e.	Length	m (Meters)	Physical contact length of the upper bump-stop element.
4I3f.	Damping	N·s/m (Damping coefficient)	Hydraulic or rubber damping applied during upper bump-stop engagement.
4I4a.	Range	m (Meters)	Operational travel range of the lower (rebound) bump-stop (Heavy Springs > Bump Stop Down).
4I4b.	Reference	m (Meters)	Reference position at which the lower bump-stop force model activates.
4I4c.	Force	N (Newtons)	Peak resistive force delivered by the lower bump-stop at maximum extension.
4I4d.	Gamma	Dimensionless exponent	Non-linear shaping exponent for lower bump-stop force curve.
4I4e.	Length	m (Meters)	Physical contact length of the lower bump-stop element.
4I4f.	Damping	N·s/m (Damping coefficient)	Damping applied during lower bump-stop engagement on rebound.
4I5.	Collar Position	m (Meters)	Adjustable spring perch/collar position controlling static ride height and preload on the heavy spring.
4I6a1.	Bump	N·s/m (Damping rate)	Fast damper channel bump (compression) damping rate within the Heavy Springs embedded damper model.
4I6a2.	Rebound	N·s/m (Damping rate)	Fast damper channel rebound (extension) damping rate.
4I6b1.	Bump	N·s/m (Damping rate)	Slow damper channel bump damping rate for low-velocity suspension movements.

ID	Name	Unit of Measurement	Description
4I6b2.	Rebound	N·s/m (Damping rate)	Slow damper channel rebound damping rate.
4I6c.	Fast Threshold Bump	m/s (Velocity threshold)	Suspension compression velocity above which the fast bump damper circuit engages.
4I6d.	Fast Threshold Rebound	m/s (Velocity threshold)	Suspension extension velocity above which the fast rebound damper circuit engages.
4I6e.	Cooling Surface	m ² (Square meters)	Exposed damper body surface area used for thermal dissipation modeling.
4I6f.	Nominal Force	N (Newtons)	Reference force level for damper nominal operating point and fatigue calculations.
4I6g.	Min Stress Fatigue	Pa or N (Stress threshold)	Lower bound stress level below which damper fatigue accumulation is negligible.
4I6h.	Mas Stress Fatigue	Pa or N (Stress threshold)	Upper bound stress level at which damper fatigue reaches maximum degradation rate.
4I6i.	Thermal Capacity	J/K or J/°C	Heat storage capacity of the damper assembly; controls temperature rise under repeated high-energy cycles.
4I6j.	Heat Transfer Coefficient	W/(m ² ·K) (Coefficient)	Convective heat transfer rate from the damper body to ambient airflow.
4I6k.	Lut List	None (File path)	Path to a damper force-velocity look-up table list defining non-linear damping behavior.
4I6l.	Damper Lut Scale	None (File path)	Scaling curve applied to the damper LUT output for fine-tuning peak forces.
4I7.	Helper K	N/m (Spring rate)	Stiffness of the helper/tender spring assisting the main spring at low compression.
4I8.	Helper Range	m (Meters)	Travel range over which the helper spring is active before the main spring fully engages.
4I9a.	Name	None (String)	Internal identifier for a dynamic controller block. Reused across Rod Controllers, ARB/Flex Bar controllers, Steering, ERS, diff-lock, and turbo controllers.
4I9b1.	Input Var	None (Telemetry enum)	Live telemetry channel used as controller input (e.g., Brake, Gas, Speed, SlipRatio, SteerDEG, ErsChargeLevel).

ID	Name	Unit of Measurement	Description
4l9b2.	Combinator Mode	None (Math enum : Add / Mult)	How this stage output combines with prior stages: additive offset or multiplicative scaling.
4l9b3.	Lut	None (.CURVE file path)	Look-up table mapping the input variable to an output modifier factor.
4l9b4.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter coefficient dampening rapid telemetry spikes for smooth controller response.
4l9b5.	Up Limit	Depends on input variable	Upper clamp on the processed input signal before LUT evaluation.
4l9b6.	Down Limit	Depends on input variable	Lower clamp on the processed input signal before LUT evaluation.
4l9b7.	Current Value	Depends on input variable	Runtime value of the controller stage output during simulation (debug/telemetry).
4l9b8.	Const Value	Depends on input variable	Fallback constant output when no dynamic input or LUT is active.
4m1.	Stiffness	Nm/deg or N/m (Anti-roll bar rate)	Torsional stiffness of the front anti-roll bar resisting body roll in cornering.
4n1.	Stiffness	Nm/deg or N/m (Anti-roll bar rate)	Torsional stiffness of the rear anti-roll bar.
4o1.	Stiffness	Nm/deg or N/m (Flex bar rate)	Stiffness of the front flex/secondary torsion bar (compliance element in complex suspension layouts).
4p1.	Stiffness	Nm/deg or N/m (Flex bar rate)	Stiffness of the rear flex/secondary torsion bar.
4q1.	Wheel Gain	Dimensionless coefficient	Global scaling gain for per-wheel damper authority in the active damper controller.
4q2.	Heave Gain	Dimensionless coefficient	Gain applied to heave-mode (vertical chassis) damper control signals.
4q3.	Pitch Gain	Dimensionless coefficient	Gain applied to pitch-mode damper control (nose dive / squat compensation).
4q4.	Roll Gain	Dimensionless coefficient	Gain applied to roll-mode damper control (lateral weight transfer management).
4q5a1.	Type	None (Enum : none / Poly3 / Poly5 / PieceWise / DamperLutData)	Damper control curve type for each motion mode (Base, Heave, Roll, Pitch, Max, Min) on front/rear axles.

ID	Name	Unit of Measurement	Description
4r.	Has Dampers Cockpit Settings	None (Boolean : True / False)	Allows the driver to adjust damper settings from cockpit controls during a session.
5.	Drivetrain Path	None (File path)	Path to the .DRIVETRAIN asset defining differential type, gear ratios, and torque split logic.
6.	Gearbox Path	None (File path)	Path to the .GEARBOX asset defining gear ratios, shift times, and auto-blip behavior.
7.	Clutch Path	None (File path)	Path to the .CLUTCH asset defining clutch friction, inertia, and autoclutch engagement.
8.	Engine Path	None (File path)	Path to the .CARENGINE asset defining torque curves, turbo, and engine braking.
9.	Brakes Path	None (File path)	Path to the .BRAKESYSTEM asset defining total braking torque, bias, and EBB controllers.
10a.	Four W S Controllers	None (Controller object)	Four-wheel steering dynamic controller array; modulates rear steer angle via staged LUT pipelines (uses 419 stage fields).
11a1.	Has T C2	None (Boolean : True / False)	Enables a secondary traction-control map (TC2) for dual-stage intervention logic.
11a2.	Frequency Hz	Hz (Hertz)	Control loop update frequency of the traction-control system.
11a3.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum vehicle speed below which traction control deactivates.
11a4.	Gear Change Time	s (Seconds)	Temporary TC suppression window duration during gear changes to avoid false interventions.
11a5.	Min Cut Level	Ratio (0.0 - 1.0)	Minimum engine torque cut level applied at the lowest TC intervention step.
11a6.	Max Cut Level	Ratio (0.0 - 1.0)	Maximum engine torque cut level at the most aggressive TC step.
11a7a.	Min Slip Ratio	Dimensionless ratio	Lower slip-ratio threshold at which TC begins monitoring wheel spin for this settings map.
11a7b.	Max Slip Ratio	Dimensionless ratio	Upper slip-ratio threshold triggering full TC intervention for this settings map.
11a7c.	Ref Slip Angle Deg	deg (Degrees)	Reference rear slip angle used to scale TC aggression relative to yaw behavior.

ID	Name	Unit of Measurement	Description
11a7d.	Engine Cut Level	Ratio (0.0 - 1.0)	Engine torque reduction factor applied when slip thresholds are exceeded.
11a7e.	Angular A C Cgain	Dimensionless gain	Yaw-rate / angular acceleration gain for TC correction authority.
11a7f.	Oversteer Gain	Dimensionless gain	Additional TC sensitivity multiplier when oversteer is detected.
11a7g.	Slip Angle Activation Deg	deg (Degrees)	Minimum slip angle required before this TC map becomes active.
11b1a.	Min Slip Ratio	Dimensionless ratio	Lower wheel slip threshold for ABS monitoring on this settings map.
11b1b.	Max Slip Ratio	Dimensionless ratio	Upper slip threshold triggering maximum ABS pressure modulation.
11b1c.	Ref Slip Angle Deg	deg (Degrees)	Reference slip angle for ABS yaw-stability cross-correlation.
11b1d.	Cut Level	Ratio (0.0 - 1.0)	Brake pressure reduction factor applied during ABS pulsing.
11b1f.	Max Torque Variation	Nm or ratio	Maximum allowed brake torque fluctuation per ABS cycle on this map.
11b2.	Frequency	Hz (Hertz)	ABS control loop pulsing frequency.
11b3.	Channels	None (Integer)	Number of independent ABS control channels (typically one per wheel or per axle).
11b4.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum speed below which ABS intervention is disabled.
11c1.	Active	None (Boolean : True / False)	Master enable for Electronic Differential Lock (EDL) brake-based torque transfer.
11c2.	Brake Torque Power	Nm (Newton-meters)	Brake torque applied to the spinning wheel under power (on-throttle) to emulate a locking diff.
11c3.	Brake Torque Coast	Nm (Newton-meters)	Brake torque applied during coasting to synchronize wheel speeds.
11c4.	Dead Zone Coast	Ratio or rad/s	Speed-difference deadband below which coast EDL does not intervene.
11c5.	Dead Zone Power	Ratio or rad/s	Speed-difference deadband below which power EDL does not intervene.

ID	Name	Unit of Measurement	Description
11c6.	Max Spin Power	rad/s or ratio	Maximum allowed wheel spin differential under power before EDL fully engages.
11c7.	Max Spin Coaster	rad/s or ratio	Maximum allowed wheel spin differential on coast before EDL engages.
11c8.	Min Speed	m/s or km/h (Speed threshold)	Minimum vehicle speed required for EDL operation.
11d1.	Frequency Hz	Hz (Hertz)	Electronic Stability Program control loop update frequency.
11d2.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum speed below which ESP deactivates.
11d3a.	Gain	Dimensionless gain	Global ESP intervention authority multiplier for this settings map.
11d3b.	Steer Gain	Dimensionless gain	Steering-angle sensitivity gain for ESP yaw correction.
11d3c.	Min Steer Gain	Dimensionless gain	Minimum steering gain applied at low speeds within ESP logic.
11d3d.	Steer Gain Max Speed	km/h (Kilometers per hour)	Speed at which steering gain reaches its maximum ESP authority.
11d3e.	Oversteer Gain	Dimensionless gain	Corrective gain when rear slip (oversteer) is detected.
11d3f.	Understeer Gain	Dimensionless gain	Corrective gain when front slip (understeer) is detected.
11d3g.	Max Slip Ratio	Dimensionless gain	Maximum wheel slip ratio before ESP applies full brake/engine intervention.
11d3h.	Dead Zone	deg or ratio	Yaw/steer error deadband where ESP remains passive.
11d3i.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter on ESP sensor inputs to prevent oscillatory corrections.
11d3j.	Brake Perc	Ratio (0.0 - 1.0)	Percentage of available brake pressure ESP may apply for stability correction.
11d3k.	Brake Perc Activation	Ratio (0.0 - 1.0)	Slip/yaw threshold fraction at which ESP begins applying brake intervention.
12.	Electronics Path	None (File path)	External reference to a standalone .CARELECTRONICS asset overriding inline TC/ABS/ESP definitions.
13a.	Ff Mult	Dimensionless multiplier	Force-feedback strength multiplier scaling steering wheel torque output to the driver.

ID	Name	Unit of Measurement	Description
13b.	Steer Lock	deg (Degrees)	Maximum steering wheel rotation angle from lock to lock.
13c.	Steer Ratio	Ratio (:1)	Steering rack ratio converting steering wheel angle to front wheel angle.
13d.	Linear Steer Rod Ratio	M/rad or dimensionless	Linear rack displacement per unit steering input; fine-tunes low-speed steering response.
13e.	Steer Assist	Dimensionless multiplier	Power-steering assist strength scaling self-aligning torque feedback.
14a.	Center	m (Meters, X / Y / Z)	Center position of an axis-aligned box collider used for simplified body collision detection.
14b.	Size	m (Meters, X / Y / Z)	Full dimensions (width, height, length) of the box collider.
14c.	Pitch Rotation Deg	deg (Degrees)	Pitch-axis rotation applied to orient the box collider relative to the chassis.
15.	Front Tyre Compounds	None (File path)	Path(s) to front axle .tyre compound assets; multiple entries allow compound selection.
16.	Rear Type Compounds	None (File path)	Path(s) to rear axle .tyre compound assets.
17a.	Slip Gain Multiple	Dimensionless multiplier	Global scaling factor for aerodynamic slip-stream and yaw-sensitivity effects.
17b.	Speed Factor Mult	Dimensionless multiplier	Multiplier applied to speed-dependent aero force calculations (downforce/drag scaling).
17c1.	Position	m (Meters, X / Y / Z)	Application point of an aerodynamic downforce/lift element on the chassis.
17c2.	Cl Gain	Dimensionless coefficient	Lift coefficient gain multiplier for this downforce element.
17c3.	Cd Gain	Dimensionless coefficient	Drag coefficient gain multiplier for this downforce element.
17c4.	Yaw Gain	Dimensionless coefficient	Sensitivity of aero forces to vehicle yaw angle (side-force generation).
17c5.	Drag Per Cool Transfer	Coefficient	Drag penalty factor linked to coolant or brake-duct thermal transfer (aero-thermal coupling).
17c6.	Damage C L	None (String / curve reference)	Damage-state modifier curve for lift coefficient degradation on this aero element.

ID	Name	Unit of Measurement	Description
17c7.	Damage C D	None (String / curve reference)	Damage-state modifier curve for drag coefficient degradation.
17c8a.	Combinator Mode	None (Math enum : Add / Mult)	How downforce controller stage output combines with baseline aero coefficients.
17c8b.	Input	None (Telemetry enum)	Telemetry input driving dynamic downforce modulation (speed, ride height, DRS state, etc.).
17c8c.	Filter	Coefficient (Smoothing multiplier)	Input smoothing filter for downforce controller stages.
17c8d.	Up Limit	Depends on input variable	Upper clamp on downforce controller input.
17c8e.	Down Limit	Depends on input variable	Lower clamp on downforce controller input.
17c8f.	Lut	None (.curve file path)	Look-up table mapping controller input to aero coefficient modifier.
17c9.	Lift Per Front Angle	Coefficient per deg	Rate of lift change per degree of front ride-height / pitch angle.
17c10.	Lift Per Rear Angle	Coefficient per deg	Rate of lift change per degree of rear ride-height / pitch angle.
17c11.	Drag Per Front Angle	Coefficient per deg	Rate of drag change per degree of front body angle.
17c12.	Drag Per Rear Angle	Coefficient per deg	Rate of drag change per degree of rear body angle.
17c13.	Default Front Angle	deg (Degrees)	Baseline front body angle reference for aero angle-dependent calculations.
17c14.	Default Rear Angle	deg (Degrees)	Baseline rear body angle reference for aero angle-dependent calculations.
17d.	Front Lift	None (File path)	Path to a .SURFACE3D front lift aero map defining Cl vs. ride height / speed.
17e.	Rear Lift	None (File path)	Path to a .SURFACE3D rear lift aero map.
17f.	Drag	None (File path)	Path to a .SURFACE3D drag map defining Cd vs. speed and configuration.
17g.	Wings Path	None (File path)	Path(s) to .WING assets defining adjustable aerodynamic devices (splitters, wings, diffusers).
18a.	Ignore Zones	None (Boolean : True / False)	When true, DRS activation ignores track zone restrictions (development override).

ID	Name	Unit of Measurement	Description
18b.	Limit G	G (Lateral G-force)	Maximum lateral G beyond which DRS automatically deactivates for safety.
18c1.	Mode	None (Enum : UseEffect / UseAngle)	DRS wing connection mode: apply a coefficient effect or a physical angle offset.
18c2.	Connected Wing	None (Integer index)	Index of the wing element in the Wings Path array controlled by this DRS connection.
18c3.	Effect	Dimensionless coefficient delta	Aerodynamic coefficient change applied when DRS is open (Mode = UseEffect).
18c4.	Angle	deg (Degrees)	Physical wing angle offset applied when DRS is open (Mode = UseAngle).
19a.	Torque Lut	None (.curve file path)	Look-up table mapping deployment input to electric motor assist torque (kW/Nm).
19b.	Coast Lut	None (.curve file path)	Look-up table defining regenerative braking / coast recovery torque.
19c.	Battery Charge Kj	kJ (Kilojoules)	Total energy storage capacity of the hybrid battery pack.
19d.	Has Button Override	None (Boolean : True / False)	Allows driver button override of automatic ERS deployment strategy.
19e.	Max Kj Per Lap	kJ/lap (Kilojoules per lap)	Regulatory or strategic cap on ERS energy deployment per lap.
19f.	Max Charge Kj Per Lap	kJ/lap (Kilojoules per lap)	Maximum energy that can be recovered into the battery per lap.
19g.	Heat Charge Perc	Ratio (0.0 - 1.0)	Fraction of waste heat energy convertible into battery charge (MGU-H simulation).
19h.	Heat Power Kw	kW (Kilowatts)	Thermal energy recovery power ceiling from exhaust/heat sources.
19i.	Default Power Controller Index	None (Integer)	Index of the default ERS deployment map selected at session start.
19j.	Power Controllers Front	None (Controller array)	Array of front-axle ERS deployment controllers; each entry uses the shared 4I9 stage fields (Name + Stages).
19k.	Power Controllers Rear	None (Controller array)	Array of rear-axle ERS deployment controllers; each entry uses the shared 4I9 stage fields.

ID	Name	Unit of Measurement	Description
19l.	Brake Rear Correction	Ratio or Nm	Rear brake torque correction factor when ERS regenerative braking is active.
19m.	Has Cockpit Controls	None (Boolean : True / False)	Exposes ERS deployment/ recovery controls to the driver in the cockpit UI.
19n1.	Delivery Profile	None (Boolean)	Cockpit toggle for ERS power delivery profile selection.
19n2.	Mgu H Mode	None (Boolean)	Cockpit toggle for MGU-H harvesting mode.
19n3.	Recovery	None (Boolean)	Cockpit toggle for regenerative recovery aggressiveness.
19o.	Has Front Motors	None (Boolean : True / False)	Enables front-axle electric motor(s) for hybrid or torque-vectoring applications.
19p1.	Torque Lut	None (.curve file path)	Torque delivery curve for front-axle electric motor(s).
19p2.	Discharge Time	s (Seconds)	Time constant governing front motor energy discharge rate from the battery.
19p3.	Torque Vectoring Bias	Ratio (-1.0 - 1.0)	Left/right torque distribution bias for front-axle electric torque vectoring.
20.	Setup Limits	None (File path)	Path to .CARSETUPLIMITS asset defining min/max/step bounds for garage setup sliders.
21.	Collider Mesh	None (File path)	Path to the detailed collision mesh used for body contact and scraping detection.
22a.	Position	m (Meters, X / Y / Z)	Root-level body mesh translation offset (distinct from General 2o block).
22b.	Rotation	deg or rad (X / Y / Z)	Root-level body mesh rotation offset.
22c.	Scale	Dimensionless (X / Y / Z)	Root-level body mesh scale factor.
23.	Stock Setup	None (File path)	Path to the default .CARSETUP preset loaded for new sessions.
24.	AI Setup	None (File path)	Path to the AI-specific .CARSETUP preset used by computer-controlled opponents.
25.	Wet Setup	None (File path)	Path to the wet-weather .CARSETUP preset (tyre pressures, wing angles, etc.).
26a.	Performance Mode Name	None (String)	Label for a performance mode preset (e.g., Qualifying, Race, Wet, Safe).

ID	Name	Unit of Measurement	Description
26b1.	Tc1	None (Map index / level)	Traction Control map level stored in this performance mode.
26b2.	Tc2	None (Map index / level)	Secondary TC map level for dual-TC systems.
26b3.	Abs	None (Map index / level)	ABS intervention map level for this performance mode.
26b4.	Esc	None (Map index / level)	ESP/stability control map level for this performance mode.
26b5.	Ebb	None (Map index / level)	Electronic Brake Balance mode/level for this performance mode.
26b6.	Engine Map	None (Map index)	Engine power map selection (fuel/power trade-off).
26b7.	Telemetry Laps To Record	None (integer / lap number)	Number of laps of telemetry data to retain when this mode is active.
26b8.	Turbo Boost Lv	bar or level index	Turbo boost level preset for this performance mode.
26b9.	Ers Deployment Map	None (Map index)	Turbo boost level preset for this performance mode.
26b10.	Ers Recharge Lv	None (Level index)	ERS energy recovery aggressiveness level.
26b11.	Ers Heat Charging	None (Level index)	MGU-H heat-to-battery charging level.
26c1.	Front Bias	Ratio (0.0 - 1.0)	Brake balance front percentage stored in this performance mode.
26c2.	Torque Multiplier	Dimensionless multiplier	Global braking torque scaling factor for this mode.
26c3.	Brake Ducts	Ratio (0.0 - 1.0)	Brake cooling duct opening level(s); multiple entries for front/rear ducts.
26d1.	Slow Bump	Clicks or N·s/m	Slow-speed compression damper setting for this performance mode.
26d2.	Fast Bump	Clicks or N·s/m	Fast-speed compression damper setting.
26d3.	Slow Rebound	Clicks or N·s/m	Slow-speed rebound damper setting.
26d4.	Fast Rebound	Clicks or N·s/m	Fast-speed rebound damper setting.
26e1.	Type	None (Enum : LSD / Spool / Torsen / EpicyclicTorsen / EpicyclicLSD / TorqueVectoring)	Differential mechanism type governing torque split behavior.
26e2.	Power	Nm or ratio	Differential lock/acceleration sensitivity under power (on-throttle).

ID	Name	Unit of Measurement	Description
26e3.	Coast	Nm or ratio	Differential lock sensitivity on coast (off-throttle).
26e4.	Preload	Nm (Newton-meters)	Static preload torque required before differential plates begin slipping.
26e5.	Front Share	Ratio (0.0 - 1.0)	Fraction of total driveline torque directed to the front axle (AWD/ 4WD).
26e6.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Torque bias ratio between axles or sides under power.
26e7.	Torque Bias Ratio Coast	Ratio	Torque bias ratio between axles or sides on coast.
26e8.	Thermal Capacity	J/K or J/°C	Heat storage capacity of differential components for thermal wear modeling.
26e9.	Surface	m ² (Square meters)	Exposed surface area of differential housing for cooling calculations.
26e10.	Heat Transfer Coef	W/(m ² ·K) (Coefficient)	Heat dissipation rate from differential to ambient airflow.
26e11.	Wear Factor	Dimensionless coefficient	Rate at which differential friction surfaces degrade under load and temperature.
26e12.	Friction Reduction With T	Ratio/°C	Linear friction reduction per degree of temperature rise in the differential.
26e13.	Friction Ref T	°C (Degrees Celsius)	Reference temperature at which nominal differential friction is defined.
26f1.	Front Diff	None (Differential object)	Front axle differential block; contains 26e1–26e13 fields for AWD/4WD layouts.
26f2.	Center Diff	None (Differential object)	Center differential block for torque split between front and rear axles.
26f3.	Rear Diff	None (Differential object)	Rear axle differential block.
26g.	Front Lock Controllers	None (Controller object)	Dynamic controller modulating front differential lock under telemetry conditions.
26h.	Center Lock Controllers	None (Controller object)	Dynamic controller modulating center differential lock.
26i.	Rear Lock Controllers	None (Controller object)	Dynamic controller modulating rear differential lock.
26j1.	Position	None (Integer index)	Index position of an AWD clutch pack in the driveline layout.

ID	Name	Unit of Measurement	Description
26j2.	Preload	Nm (Newton-meters)	Static preload torque on this AWD clutch before slip occurs.
26k.	Turbo Controllers	None (Controller object)	Dynamic controller array modulating turbo boost via staged LUT pipelines (uses 419 stage fields).
26l1.	Boost Lv	bar or level index	Turbo boost level preset within this performance mode.
27.	Ai Car Data	None (File path)	Path to an AI-specific .car data override tuning opponent behavior without altering player physics.
28.	mm	None (Integer)	Internal unit-scaling or precision flag used by the editor toolchain (millimeter reference mode).

D. Example data

I. Chosen Car Data for Example

- Ferrari 296 GTB (slug : ks_ferrari_286_gtb)
- Audi R8 LMS GT3 Evo II (slug : ks_audi_r8_lms_gt3_evo_2)
- Renault 5 GT Turbo (slug : ks_renault_5_gt_turbo)

II. Example

Ferrari 296 GTB

```

1. Screen Name : None
2. General
  2a. Screen Name : Ferrari 296 GTB
  2b. Total Mass : 1750.00000
  2c. Tank Position : 0.00000, -0.16715, -0.32949
  2d. Fuel : 60.00000
  2e. Max Fuel : 120.00000
  2f. Efficiency : 0.00000
  2g. Kg Per Liter : 0.75500
  2h. Body Box Sizes : 2.00000, 0.70000, 4.22149
  2i. Pickup Front Height : -0.35200
  2j. Pickup Rear Height : -0.35200
  2k. Check Rules : false
  2l. Minimum Height : 0.00000
  2m. Torsional Stiffness : 30000.00000
  2n. Torsional Damping : 500.00000
  2o. Body Mesh Offset
    2o1. Position : 0.000, 0.000, 0.000
    2o2. Rotation : 0.000, 0.000, 0.000
    2o3. Scale : 0.000, 0.000, 0.000

```

```
- 3. General Path : None
- 4. Suspensions
  - 4a. Wheel Base : 2.60000
  - 4b. Longitudinal Cg Location : 0.39500
  - 4c. Base Y Front : -0.11000
  - 4d. Base Y Rear : -0.08200
  - 4e. Track Front : 1.72600
  - 4f. Track Rear : 1.71000
  - 4g. Damage
    - 4g1. Min Velocity : 40.00000
    - 4g2. Gain : 0.00040
    - 4g3. Max Damage : 0.05000
    - 4g4. Debug Log : true
  - 4h. Coilover Front path :
    content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_front.coilover
  - 4i. Coilover Rear Path :
    content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_rear.coilover
  - 4j. Front Suspension Path :
    content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_front.suspension
  - 4k. Rear Suspension Path :
    content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_rear.suspension
  - 4l. Heavy Springs : None
  - 4m. Arb Front
    - 4m1. Stiffness : 26000.00000
    - 4m2. Controller
      - 4l9a. Name : None
      - 4l9b. Stages : None
  - 4n. Arb Rear
    - 4n1. Stiffness : 20000.00000
    - 4n2. Controller
      - 4l9a. Name : None
      - 4l9b. Stages : None
  - 4o. Flex Bar Front
    - 4o1. Stiffness : 0.00000
    - 4o2. Controller
      - 4l9a. Name : None
      - 4l9b. Stages : None
  - 4p. Flex Bar Rear
    - 4p1. Stiffness 0.00000
    - 4p2. Controller
      - 4l9a. Name : None
      - 4l9b. Stages : None
  - 4q. Dampers Controller : None
  - 4r. Has Dampers Cockpit Settings : false
- 5. Drivetrain Path :
  content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.drivetrain
- 6. Gearbox Path :
  content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.gearbox
- 7. Clutch Path :
  content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.clutch
- 8. Engine Path :
  content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.carengine
- 9. Brakes Path :
  content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.brakesystem
- 10. Steering System
  - 10a. Four W S Controllers
```

- 4l9a. Name : None
- 4l9b. Stages : None
- 11. Electronics
 - 11a. T C
 - 11a1. Has T C2 : false
 - 11a2. Frequency Hz : 333.00000
 - 11a3. Min Speed Kmh : 35.00000
 - 11a4. Gear Change Time : 0.0800
 - 11a5. Min Cut Level : 8.00000
 - 11a6. Max Cut Level : 0.90000
 - 11a7. Settings 1
 - 11a7a. Min Slip Ratio : 0.00000
 - 11a7b. Max Slip Ratio : 0.00000
 - 11a7c. Ref Slip Angle Deg : 0.00000
 - 11a7d. Engine Cut Level : 0.00000
 - 11a7e. Angular A C Cgain : 0.00000
 - 11a7f. Oversteer Gain : 0.00000
 - 11a7g. Slip Angle Activation Deg : 0.00000
 - 11a7. Settings 2
 - 11a7a. Min Slip Ratio : 0.09000
 - 11a7b. Max Slip Ratio : 0.35000
 - 11a7c. Ref Slip Angle Deg : 15.00000
 - 11a7d. Engine Cut Level : 1.80000
 - 11a7e. Angular A C Cgain : 4.50000
 - 11a7f. Oversteer Gain : 3.00000
 - 11a7g. Slip Angle Activation Deg : 6.00000
 - 11a7. Settings 3
 - 11a7a. Min Slip Ratio : 0.07000
 - 11a7b. Max Slip Ratio : 0.20000
 - 11a7c. Ref Slip Angle Deg : 12.00000
 - 11a7d. Engine Cut Level : 1.50000
 - 11a7e. Angular A C Cgain : 6.00000
 - 11a7f. Oversteer Gain : 5.00000
 - 11a7g. Slip Angle Activation Deg : 5.00000
 - 11a7. Settings 4
 - 11a7a. Min Slip Ratio : 0.04500
 - 11a7b. Max Slip Ratio : 0.15000
 - 11a7c. Ref Slip Angle Deg : 7.00000
 - 11a7d. Engine Cut Level : 1.0000
 - 11a7e. Angular A C Cgain : 10.00000
 - 11a7f. Oversteer Gain : 7.00000
 - 11a7g. Slip Angle Activation Deg : 3.00000
 - 11b. A B S
 - 11b1. Settings 1
 - 11b1a. Min Slip Ratio : -1.00000
 - 11b1b. Max Slip Ratio : -1.00000
 - 11b1c. Ref Slip Angle Deg : 0.00000
 - 11b1d. Cut Level : 0.00000
 - 11b1f. Max Torque Variation : 0.70000
 - 11b1. Settings 2
 - 11b1a. Min Slip Ratio : 0.07000
 - 11b1b. Max Slip Ratio : 0.12000
 - 11b1c. Ref Slip Angle Deg : 9.00000
 - 11b1d. Cut Level : 0.05000
 - 11b1f. Max Torque Variation : 1.00000

- Frequency : 40.00000
- Channels : 4
- Min Speed Kmh : 0.00000
- 11c. E D L : None
- 11d. E S P
 - 11d1. Frequency Hz : 100.00000
 - 11d2. Min Speed Kmh : 20.00000
 - 11d3. Settings 1
 - 11d3a. Gain : 0.00000
 - 11d3b. Steer Gain : 0.00000
 - 11d3c. Min Steer Gain : 0.00000
 - 11d3d. Steer Gain Max Speed : 0.00000
 - 11d3e. Oversteer Gain : 0.00000
 - 11d3f. Understeer Gain : 0.00000
 - 11d3g. Max Slip Ratio : 0.00000
 - 11d3h. Dead Zone : 0.00000
 - 11d3i. Filter Gain : 0.00000
 - 11d3j. Brake Perc : 0.00000
 - 11d3k. Brake Perc Activation : 0.00000
 - 11d3. Settings 2
 - 11d3a. Gain : 0.50000
 - 11d3b. Steer Gain : 0.50000
 - 11d3c. Min Steer Gain : 0.50000
 - 11d3d. Steer Gain Max Speed : 0.00000
 - 11d3e. Oversteer Gain : 0.50000
 - 11d3f. Understeer Gain : 0.50000
 - 11d3g. Max Slip Ratio : 0.19000
 - 11d3h. Dead Zone : 0.30000
 - 11d3i. Filter Gain : 0.95000
 - 11d3j. Brake Perc : 0.10000
 - 11d3k. Brake Perc Activation : 0.90000
- 12. Electronics Path : None
- 13. Controls
 - 13a. Ff Mult : 1.10000
 - 13b. Steer Lock : 342.00000
 - 13c. Steer Ratio : 12.00000
 - 13d. Linear Steer Rod Ratio : 0.00200
 - 13e. Steer Assist : 1.00000
- 14. Box Colliders 1
 - 14a. Center : 0.00000, -0.26000, 0.33000
 - 14b. Size : 1.78000, 0.15000, 3.94000
 - 14c. Pitch Rotation Deg : 0.200000
- 15. Front Tyre Compounds 1 :
content\cars\common_phsx\tyres\hypercar\hypercar_245_35_20.tyre
- 16. Rear Tyre Compounds 1 :
content\cars\common_phsx\tyres\hypercar\hypercar_305_35_20.tyre
- 17. Aero
 - 17a. Slip Gain Multiple : 1.00000
 - 17b. Speed Factor Mult : 2.00000
 - 17c. Downforces : None
 - 17d. Front Lift : None
 - 17e. Rear Lift : None
 - 17f. Drag : None
 - 17g. Wings Path 1 :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb0.wing

```

| L 17g. Wings Path 2 :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb1.wing
| L 17g. Wings Path 3 :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb2.wing
| L 17g. Wings Path 4 :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb3.wing
- 18. Drs : None
- 19. Ers
|   - 19a. Torque Lut :
content\cars\ks_ferrari_296_gtb\data\kers\KERS_TORQUE.curve
|   - 19b. Coast Lut :
content\cars\ks_ferrari_296_gtb\data\kers\KERS_TORQUE.curve
|   - 19c. Battery Charge Kj : 26820.00000
|   - 19d. Has Button Override : false
|   - 19e. Max Kj Per Lap : 0.00000
|   - 19f. Max Charge Kj Per Lap : 0.00000
|   - 19g. Heat Charge Perc : 0.00000
|   - 19h. Heat Power Kw : 0.00000
|   - 19i. Default Power Controller Index : 0
|   - 19j. Power Controllers Front : None
|   - 19k. Power Controllers Rear 1
|       - 4l9a. Name : MAP Q
|       - 4l9b. Stages 1
|           - 4l9b1. Input Var : Gas
|           - 4l9b2. Combinator Mode : Add
|           - 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_Q_CONTROLLER_G
AS.curve
|       - 4l9b4. Filter Gain : 0.96000
|       - 4l9b5. Up Limit : 1.00000
|       - 4l9b6. Down Limit : -1.00000
|       - 4l9b7. Current Value : 0.00000
|       - 4l9b8. Const Value : 0.00000
|       - 4l9b. Stages 2
|           - 4l9b1. Input Var : Gear
|           - 4l9b2. Combinator Mode : Mult
|           - 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
|       - 4l9b4. Filter Gain : 0.96000
|       - 4l9b5. Up Limit : 1.00000
|       - 4l9b6. Down Limit : -1.00000
|       - 4l9b7. Current Value : 0.00000
|       - 4l9b8. Const Value : 0.00000
|       - 4l9b. Stages 3
|           - 4l9b1. Input Var : SlipRatioRearAVG
|           - 4l9b2. Combinator Mode : Mult
|           - 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
|       - 4l9b4. Filter Gain : 0.96000
|       - 4l9b5. Up Limit : 1.00000
|       - 4l9b6. Down Limit : -1.00000
|       - 4l9b7. Current Value : 0.00000
|       - 4l9b8. Const Value : 0.00000
|       - 4l9b. Stages 4

```

```

| | | | 4l9b1. Input Var : RPMS
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 5
| | | | 4l9b1. Input Var : Gas
| | | | 4l9b2. Combinator Mode : Add
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHROTTLE_coast.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 6
| | | | 4l9b1. Input Var : Brake
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 19k. Power Controllers Rear 2
| | | | 4l9a. Name : MAP P
| | | | 4l9b. Stages 1
| | | | 4l9b1. Input Var : Gas
| | | | 4l9b2. Combinator Mode : Add
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_P_CONTROLLER_G
AS.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 2
| | | | 4l9b1. Input Var : Gear
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000

```

```

| | | | - 4l9b. Stages 3
| | | | | - 4l9b1. Input Var : SlipRatioRearAVG
| | | | | - 4l9b2. Combinator Mode : Mult
| | | | | - 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
| | | | | - 4l9b4. Filter Gain : 0.96000
| | | | | - 4l9b5. Up Limit : 1.00000
| | | | | - 4l9b6. Down Limit : -1.00000
| | | | | - 4l9b7. Current Value : 0.00000
| | | | | - 4l9b8. Const Value : 0.00000
| | | | | - 4l9b. Stages 4
| | | | | | - 4l9b1. Input Var : RPMs
| | | | | | - 4l9b2. Combinator Mode : Mult
| | | | | | - 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
| | | | | - 4l9b4. Filter Gain : 0.96000
| | | | | - 4l9b5. Up Limit : 1.00000
| | | | | - 4l9b6. Down Limit : -1.00000
| | | | | - 4l9b7. Current Value : 0.00000
| | | | | - 4l9b8. Const Value : 0.00000
| | | | | - 4l9b. Stages 5
| | | | | | - 4l9b1. Input Var : Gas
| | | | | | - 4l9b2. Combinator Mode : Add
| | | | | | - 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHROTTLE_coast.
curve
| | | | | - 4l9b4. Filter Gain : 0.96000
| | | | | - 4l9b5. Up Limit : 1.00000
| | | | | - 4l9b6. Down Limit : -1.00000
| | | | | - 4l9b7. Current Value : 0.00000
| | | | | - 4l9b8. Const Value : 0.00000
| | | | | - 4l9b. Stages 6
| | | | | | - 4l9b1. Input Var : Brake
| | | | | | - 4l9b2. Combinator Mode : Mult
| | | | | | - 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
| | | | | - 4l9b4. Filter Gain : 0.96000
| | | | | - 4l9b5. Up Limit : 1.00000
| | | | | - 4l9b6. Down Limit : -1.00000
| | | | | - 4l9b7. Current Value : 0.00000
| | | | | - 4l9b8. Const Value : 0.00000
| | | | | - 19k. Power Controllers Rear 3
| | | | | | - 4l9a. Name : MAP H
| | | | | | - 4l9b. Stages 1
| | | | | | | - 4l9b1. Input Var : Gas
| | | | | | | - 4l9b2. Combinator Mode : Add
| | | | | | | - 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_H_CONTROLLER_G
AS.curve
| | | | | - 4l9b4. Filter Gain : 0.96000
| | | | | - 4l9b5. Up Limit : 1.00000
| | | | | - 4l9b6. Down Limit : -1.00000
| | | | | - 4l9b7. Current Value : 0.00000

```



```

| | | L 4l9b8. Const Value : 0.00000
| | | - 4l9b. Stages 2
| | | | 4l9b1. Input Var : Gear
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | - 4l9b. Stages 3
| | | | 4l9b1. Input Var : SlipRatioRearAVG
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | - 4l9b. Stages 4
| | | | 4l9b1. Input Var : RPMS
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | - 4l9b. Stages 5
| | | | 4l9b1. Input Var : Gas
| | | | 4l9b2. Combinator Mode : Add
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHROTTLE_coast.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | - 4l9b. Stages 6
| | | | 4l9b1. Input Var : Brake
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000

```

- 19l. Brake Rear Correction : 5.00000
- 19m. Has Cockpit Controls : false
- 19n. Cockpit Controls
 - 19n1. Delivery Profile : false
 - 19n2. Mgu H Mode : false
 - 19n3. Recovery : false
- 19o. Has Front Motors : false
- 19p. Front Motors
 - 19p1. Torque Lut : None
 - 19p2 Discharge Time : 0.00000
 - 19p3. Torque Vectoring Bias : 0.00000
- 20. Setup Limits :
content\cars\ks_ferrari_296_gtb\data\setup\limits_f296gtb.carsetuplimits
- 21. Collider Mesh :
content\cars\ks_ferrari_296_gtb\collider\ferrari_296_gtb_collider.mesh
- 22 Body Mesh Offset
 - 22a. Position : 0.000, -0.482, 0.247
 - 22b. Rotation: 0.100, 0.000, 0.000
 - 22c. Scale : 1.000, 1.000, 1.000
- 23. Stock Setup : content/cars/
ks_ferrari_296_gtb\data\setup\f296gtb_stock.carsetup
- 24. Ai Setup : None
- 25. Wet Setup : None
- 26. Performance Modes 1
 - 26a. Performance Mode Name : QUAL
 - 26b. Electronics Settings
 - 26b1. Tc1 : 2.00000
 - 26b2. Tc2 : 1.00000
 - 26b3. Abs : 1.00000
 - 26b4. Esc : 0.00000
 - 26b5. Ebb : 0.000
 - 26b6. Engine Map : 0.00000
 - 26b7. Telemetry laps To Record : 0.00000
 - 26b8. Turbo Boost Lv : 0.00000
 - 26b9. Ers Deployment Map : 0.000
 - 26b10. Ers Recharge Lv : 100.000
 - 26b11. Ers Heat Charging : 0.000
 - 26c. Brakes Settings : None
 - 26d. Damper Settings : None
 - 26e. Differential Data : None
 - 26f. Four W D Differentials : None
 - 26g. Front Lock Controllers : None
 - 26h. Center Lock Controllers : None
 - 26i. Rear Lock Controllers : None
 - 26j. Awd Clutches : None
 - 26k. Turbo Controllers : None
 - 26l. Turbo Settings : None
- 26. Performance Modes 2
 - 26a. Performance Mode Name : PERF
 - 26b. Electronics Settings
 - 26b1. Tc1 : 2.00000
 - 26b2. Tc2 : 1.00000
 - 26b3. Abs : 1.00000
 - 26b4. Esc : 1.00000
 - 26b5. Ebb : 0.000
 - 26b6. Engine Map : 0.00000

- 26b7. Telemetry laps To Record : 0.00000
- 26b8. Turbo Boost Lv : 0.00000
- 26b9. Ers Deployment Map : 1.000
- 26b10. Ers Recharge Lv : 100.000
- 26b11. Ers Heat Charging : 0.000
- 26c. Brakes Settings : None
- 26d. Damper Settings : None
- 26e. Differential Data : None
- 26f. Four W D Differentials : None
- 26g. Front Lock Controllers : None
- 26h. Center Lock Controllers : None
- 26i. Rear Lock Controllers : None
- 26j. Awd Clutches : None
- 26k. Turbo Controllers : None
- 26l. Turbo Settings : None
- 26. Performance Modes 3
 - 26a. Performance Mode Name : HYBRID
 - 26b. Electronics Settings
 - 26b1. Tc1 : 2.00000
 - 26b2. Tc2 : 1.00000
 - 26b3. Abs : 1.00000
 - 26b4. Esc : 1.00000
 - 26b5. Ebb : 0.000
 - 26b6. Engine Map : 1.00000
 - 26b7. Telemetry laps To Record : 0.00000
 - 26b8. Turbo Boost Lv : 0.00000
 - 26b9. Ers Deployment Map : 2.000
 - 26b10. Ers Recharge Lv : 100.000
 - 26b11. Ers Heat Charging : 0.000
 - 26c. Brakes Settings : None
 - 26d. Damper Settings : None
 - 26e. Differential Data : None
 - 26f. Four W D Differentials : None
 - 26g. Front Lock Controllers : None
 - 26h. Center Lock Controllers : None
 - 26i. Rear Lock Controllers : None
 - 26j. Awd Clutches : None
 - 26k. Turbo Controllers : None
 - 26l. Turbo Settings : None
- 27. Ai Car Data :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.aicardata
- 28. mm : 1

Audi R8 LMS GT3 Evo II

- 1. Screen Name : Audi R8 LMS GT3 Evo II
- 2. General
 - 2a. Screen Name : Audi R8 LMS GT3 Evo II
 - 2b. Total Mass : 1355
 - 2c. Tank Position : 0.00000, -0.02900, -0.25600
 - 2d. Fuel : 0.00000
 - 2e. Max Fuel : 120.00000
 - 2f. Efficiency : 0.40000

- 2g. Kg Per Liter : 0.75500
- 2h. Body Box Sizes : 1.98000, 1.19000, 4.46000
- 2i. Pickup Front Height : -0.34900
- 2j. Pickup Rear Height : -0.31800
- 2k. Check Rules : true
- 2l. Minimum Height : 450.00000
- 2m. Torsional Stiffness : 40000.00000
- 2n. Torsional Damping : 400.00000
- 2o. Body Mesh Offset
 - 2o1. Position : 0.000, 0.000, 0.000
 - 2o2. Rotation : 0.000, 0.000, 0.000
 - 2o3. Scale : 0.000, 0.000, 0.000
- 3. General Path : None
- 4. Suspensions
 - 4a. Wheel Base : 2.70000
 - 4b. Longitudinal Cg Location : 0.42000
 - 4c. Base Y Front : -0.03500
 - 4d. Base Y Rear : -0.03500
 - 4e. Track Front : 1.66700
 - 4f. Track Rear : 1.67000
 - 4g. Damage
 - 4g1. Min Velocity : 40.00000
 - 4g2. Gain : 0.00040
 - 4g3. Max Damage : 0.05000
 - 4g4. Debug Log : true
 - 4h. Coilover Front path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_front.coilover
 - 4i. Coilover Rear Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_rear.coilover
 - 4j. Front Suspension Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_front.suspension
 - 4k. Rear Suspension Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_rear.suspension
 - 4l. Heavy Springs : None
 - 4m. Arb Front
 - 4m1. Stiffness : 56800.00000
 - 4m2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
 - 4n. Arb Rear
 - 4n1. Stiffness : 39900.00000
 - 4n2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
 - 4o. Flex Bar Front
 - 4o1. Stiffness : 0.00000
 - 4o2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
 - 4p. Flex Bar Rear
 - 4p1. Stiffness : 0.00000
 - 4p2. Controller

- 4l9a. Name : None
 - 4l9b. Stages : None
 - 4q. Dampers Controller : None
 - 4r. Has Dampers Cockpit Settings : false
- 5. Drivetrain Path :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.drivetrain
- 6. Gearbox Path :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.gearbox
- 7. Clutch Path :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.clutch
- 8. Engine Path :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.carengine
- 9. Brakes Path :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.brakesystem
- 10. Steering System
 - 10a. Four W S Controllers
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 11. Electronics
 - 11a. T C
 - 11a1. Has T C2 : true
 - 11a2. Frequency Hz : 333.00000
 - 11a3. Min Speed Kmh : 30.00000
 - 11a4. Gear Change Time : 0.07000
 - 11a5. Min Cut Level : 0.00000
 - 11a6. Max Cut Level : 6.00000
 - 11a7. Settings 1
 - 11a7a. Min Slip Ratio : 0.00000
 - 11a7b. Max Slip Ratio : 0.00000
 - 11a7c. Ref Slip Angle Deg : 0.00000
 - 11a7d. Engine Cut Level : 0.00000
 - 11a7e. Angular A C Cgain : 0.00000
 - 11a7f. Oversteer Gain : 0.00000
 - 11a7g. Slip Angle Activation Deg : 0.00000
 - 11a7. Settings 2
 - 11a7a. Min Slip Ratio : 0.18000
 - 11a7b. Max Slip Ratio : 0.35000
 - 11a7c. Ref Slip Angle Deg : 15.00000
 - 11a7d. Engine Cut Level : 1.50000
 - 11a7e. Angular A C Cgain : 1.00000
 - 11a7f. Oversteer Gain : 1.00000
 - 11a7g. Slip Angle Activation Deg : 5.00000
 - 11a7. Settings 3
 - 11a7a. Min Slip Ratio : 0.15000
 - 11a7b. Max Slip Ratio : 0.30000
 - 11a7c. Ref Slip Angle Deg : 13.00000
 - 11a7d. Engine Cut Level : 1.50000
 - 11a7e. Angular A C Cgain : 1.50000
 - 11a7f. Oversteer Gain : 1.50000
 - 11a7g. Slip Angle Activation Deg : 5.00000
 - 11a7. Settings 4

- 11a7a. Min Slip Ratio : 0.12000
- 11a7b. Max Slip Ratio : 0.28000
- 11a7c. Ref Slip Angle Deg : 11.00000
- 11a7d. Engine Cut Level : 1.25000
- 11a7e. Angular A C Cgain : 2.00000
- 11a7f. Oversteer Gain : 2.00000
- 11a7g. Slip Angle Activation Deg : 4.50000
- 11a7. Settings 5
 - 11a7a. Min Slip Ratio : 0.10000
 - 11a7b. Max Slip Ratio : 0.25000
 - 11a7c. Ref Slip Angle Deg : 10.00000
 - 11a7d. Engine Cut Level : 1.00000
 - 11a7e. Angular A C Cgain : 2.50000
 - 11a7f. Oversteer Gain : 2.50000
 - 11a7g. Slip Angle Activation Deg : 4.00000
- 11a7. Settings 6
 - 11a7a. Min Slip Ratio : 0.08000
 - 11a7b. Max Slip Ratio : 0.22000
 - 11a7c. Ref Slip Angle Deg : 8.50000
 - 11a7d. Engine Cut Level : 1.00000
 - 11a7e. Angular A C Cgain : 3.00000
 - 11a7f. Oversteer Gain : 3.00000
 - 11a7g. Slip Angle Activation Deg : 4.00000
- 11a7. Settings 7
 - 11a7a. Min Slip Ratio : 0.07000
 - 11a7b. Max Slip Ratio : 0.22000
 - 11a7c. Ref Slip Angle Deg : 7.50000
 - 11a7d. Engine Cut Level : 1.00000
 - 11a7e. Angular A C Cgain : 3.50000
 - 11a7f. Oversteer Gain : 3.50000
 - 11a7g. Slip Angle Activation Deg : 3.50000
- 11a7. Settings 8
 - 11a7a. Min Slip Ratio : 0.06000
 - 11a7b. Max Slip Ratio : 0.22000
 - 11a7c. Ref Slip Angle Deg : 7.00000
 - 11a7d. Engine Cut Level : 0.50000
 - 11a7e. Angular A C Cgain : 4.00000
 - 11a7f. Oversteer Gain : 4.00000
 - 11a7g. Slip Angle Activation Deg : 3.00000
- 11a7. Settings 9
 - 11a7a. Min Slip Ratio : 0.05000
 - 11a7b. Max Slip Ratio : 0.19000
 - 11a7c. Ref Slip Angle Deg : 7.00000
 - 11a7d. Engine Cut Level : 0.45000
 - 11a7e. Angular A C Cgain : 5.00000
 - 11a7f. Oversteer Gain : 5.00000
 - 11a7g. Slip Angle Activation Deg : 3.00000
- 11a7. Settings 10
 - 11a7a. Min Slip Ratio : 0.05000
 - 11a7b. Max Slip Ratio : 0.17000
 - 11a7c. Ref Slip Angle Deg : 7.00000
 - 11a7d. Engine Cut Level : 0.40000
 - 11a7e. Angular A C Cgain : 7.50000
 - 11a7f. Oversteer Gain : 6.00000
 - 11a7g. Slip Angle Activation Deg : 3.00000
- 11a7. Settings 11

- 11a7a. Min Slip Ratio : 0.04000
- 11a7b. Max Slip Ratio : 0.15000
- 11a7c. Ref Slip Angle Deg : 6.00000
- 11a7d. Engine Cut Level : 0.35000
- 11a7e. Angular A C Cgain : 8.00000
- 11a7f. Oversteer Gain : 6.50000
- 11a7g. Slip Angle Activation Deg : 2.50000
- 11a7. Settings 12
 - 11a7a. Min Slip Ratio : 0.03000
 - 11a7b. Max Slip Ratio : 0.15000
 - 11a7c. Ref Slip Angle Deg : 6.00000
 - 11a7d. Engine Cut Level : 0.30000
 - 11a7e. Angular A C Cgain : 8.00000
 - 11a7f. Oversteer Gain : 7.00000
 - 11a7g. Slip Angle Activation Deg : 2.50000
- 11a7. Settings 13
 - 11a7a. Min Slip Ratio : 0.03000
 - 11a7b. Max Slip Ratio : 0.14000
 - 11a7c. Ref Slip Angle Deg : 5.00000
 - 11a7d. Engine Cut Level : 0.30000
 - 11a7e. Angular A C Cgain : 8.50000
 - 11a7f. Oversteer Gain : 7.50000
 - 11a7g. Slip Angle Activation Deg : 3.00000
- 11b. A B S
 - 11b1. Settings 1
 - 11b1a. Min Slip Ratio : -1.00000
 - 11b1b. Max Slip Ratio : -1.00000
 - 11b1c. Ref Slip Angle Deg : 0.00000
 - 11b1d. Cut Level : 0.00000
 - 11b1f. Max Torque Variation : 0.00000
 - 11b1. Settings 2
 - 11b1a. Min Slip Ratio : 0.12000
 - 11b1b. Max Slip Ratio : 0.14000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
 - 11b1. Settings 3
 - 11b1a. Min Slip Ratio : 0.11000
 - 11b1b. Max Slip Ratio : 0.14000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
 - 11b1. Settings 4
 - 11b1a. Min Slip Ratio : 0.10000
 - 11b1b. Max Slip Ratio : 0.12000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
 - 11b1. Settings 5
 - 11b1a. Min Slip Ratio : 0.08000
 - 11b1b. Max Slip Ratio : 0.10000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
 - 11b1. Settings 6
 - 11b1a. Min Slip Ratio : 0.07000

- 11b1b. Max Slip Ratio : 0.08000
- 11b1c. Ref Slip Angle Deg : 7.00000
- 11b1d. Cut Level : 0.20000
- 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 7
 - 11b1a. Min Slip Ratio : 0.06000
 - 11b1b. Max Slip Ratio : 0.08000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 8
 - 11b1a. Min Slip Ratio : 0.05000
 - 11b1b. Max Slip Ratio : 0.07000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 9
 - 11b1a. Min Slip Ratio : 0.05000
 - 11b1b. Max Slip Ratio : 0.06000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 10
 - 11b1a. Min Slip Ratio : 0.04000
 - 11b1b. Max Slip Ratio : 0.05000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 11
 - 11b1a. Min Slip Ratio : 0.02500
 - 11b1b. Max Slip Ratio : 0.03500
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 12
 - 11b1a. Min Slip Ratio : 0.01000
 - 11b1b. Max Slip Ratio : 0.02000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- Frequency : 40.00000
- Channels : 4
- Min Speed Kmh : 20.00000
- 11c. E D L : None
- 11d. E S P
 - 11d1. Frequency Hz : 0.00000
 - 11d2. Min Speed Kmh : 0.00000
 - 11d3. Settings : None
- 12. Electronics Path : None
- 13. Controls
 - 13a. Ff Mult : 1.50000
 - 13b. Steer Lock : 240.00000
 - 13c. Steer Ratio : 14.30000
 - 13d. Linear Steer Rod Ratio : 0.00235
 - 13e. Steer Assist : 1.00000
- 14. Box Colliders 1

- 14a. Center : 0.00000, -0.23500, 0.30000
- 14b. Size : 1.78000, 0.20000, 4.07000
- 14c. Pitch Rotation Deg : 0.65000
- 15. Front Tyre Compounds 1 :
content\cars\common_phsx\tyres\racing_slicks\slick_325_680_18.tyre
- 16. Rear Tyre Compounds 1 :
content\cars\common_phsx\tyres\racing_slicks\slick_325_705_18.tyre
- 17. Aero
 - 17a. Slip Gain Multiple : 1.00000
 - 17b. Speed Factor Mult : 2.00000
 - 17c. Downforces 1
 - 17c1. Position : 0.82140, -0.34900, 1.57000
 - 17c2. Cl Gain : 1.00000
 - 17c3. Cd Gain : 1.00000
 - 17c4. Yaw Gain : 0.00000
 - 17c5. Drag Per Cool Transfer : 0.01500
 - 17c6. Damage C L 1 : 0.01000
 - 17c6. Damage C L 2 : 0.00500
 - 17c6. Damage C L 3 : 0.00300
 - 17c6. Damage C L 4 : 0.00300
 - 17c7. Damage C D 1 : 0.01500
 - 17c7. Damage C D 2 : 0.00500
 - 17c7. Damage C D 3 : 0.00300
 - 17c7. Damage C D 4 : 0.00300
 - 17c8. Downforce Controllers : None
 - 17c9. Lift Per Front Angle : 0.00000
 - 17c10. Lift Per Rear Angle : 0.00000
 - 17c11. Drag Per Front Angle : 0.00000
 - 17c12. Drag Per Rear Angle : 0.00000
 - 17c13. Default Front Angle : 0.00000
 - 17c14. Default Rear Angle : 0.00000
 - 17c. Downforces 2
 - 17c1. Position : 0.82140, -0.34900, 1.57000
 - 17c2. Cl Gain : 1.00000
 - 17c3. Cd Gain : 1.00000
 - 17c4. Yaw Gain : 0.00000
 - 17c5. Drag Per Cool Transfer : 0.01500
 - 17c6. Damage C L 1 : 0.01000
 - 17c6. Damage C L 2 : 0.00500
 - 17c6. Damage C L 3 : 0.00300
 - 17c6. Damage C L 4 : 0.00300
 - 17c7. Damage C D 1 : 0.01500
 - 17c7. Damage C D 2 : 0.00500
 - 17c7. Damage C D 3 : 0.00300
 - 17c7. Damage C D 4 : 0.00300
 - 17c8. Downforce Controllers : None
 - 17c9. Lift Per Front Angle : 0.00000
 - 17c10. Lift Per Rear Angle : 0.00000
 - 17c11. Drag Per Front Angle : 0.00000
 - 17c12. Drag Per Rear Angle : 0.00000
 - 17c13. Default Front Angle : 0.00000
 - 17c14. Default Rear Angle : 0.00000
 - 17c. Downforces 3
 - 17c1. Position : 0.82140, -0.31800, -1.13500
 - 17c2. Cl Gain : 1.00000
 - 17c3. Cd Gain : 1.00000

- 17c4. Yaw Gain : -0.10000
- 17c5. Drag Per Cool Transfer : 0.01000
- 17c6. Damage C L 1 : 0.00500
- 17c6. Damage C L 2 : 0.01000
- 17c6. Damage C L 3 : 0.00300
- 17c6. Damage C L 4 : 0.00300
- 17c7. Damage C D 1 : 0.00500
- 17c7. Damage C D 2 : 0.01500
- 17c7. Damage C D 3 : 0.00300
- 17c7. Damage C D 4 : 0.00300
- 17c8. Downforce Controllers : None
- 17c9. Lift Per Front Angle : 0.00000
- 17c10. Lift Per Rear Angle : 0.00000
- 17c11. Drag Per Front Angle : 0.00000
- 17c12. Drag Per Rear Angle : 0.00000
- 17c13. Default Front Angle : 0.00000
- 17c14. Default Rear Angle : 0.00000
- 17c. Downforces 4
 - 17c1. Position : 0.82140, -0.31800, -1.13500
 - 17c2. Cl Gain : 1.00000
 - 17c3. Cd Gain : 1.00000
 - 17c4. Yaw Gain : -0.10000
 - 17c5. Drag Per Cool Transfer : 0.01000
 - 17c6. Damage C L 1 : 0.00500
 - 17c6. Damage C L 2 : 0.01000
 - 17c6. Damage C L 3 : 0.00300
 - 17c6. Damage C L 4 : 0.00300
 - 17c7. Damage C D 1 : 0.00500
 - 17c7. Damage C D 2 : 0.01500
 - 17c7. Damage C D 3 : 0.00300
 - 17c7. Damage C D 4 : 0.00300
 - 17c8. Downforce Controllers : None
 - 17c9. Lift Per Front Angle : 0.00000
 - 17c10. Lift Per Rear Angle : 0.00000
 - 17c11. Drag Per Front Angle : 0.00000
 - 17c12. Drag Per Rear Angle : 0.00000
 - 17c13. Default Front Angle : 0.00000
 - 17c14. Default Rear Angle : 0.00000
- 17d. Front Lift :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\frontczmap.surface3d
- 17e. Rear Lift :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\rearczmap.surface3d
- 17f. Drag :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\cxmap.surface3d
- 17g. Wings Path 1 :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_rear_modifier.wing
- 17g. Wings Path 2 :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_front_modifier.wing
- 18. Drs : None
- 19. Ers : None
- 20. Setup Limits :
 - content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_front_modifier.wing

- 21. Collider Mesh :
content\cars\ks_audi_r8_lms_gt3_evo_2\collider\audi_r8_gt3_evo2_collider.mesh
- 22. Body Mesh Offset
 - 22a. Position : 0.000, -0.419, 0.230
 - 22b. Rotation: 0.650, 0.000, 0.000
 - 22c. Scale : 1.000, 1.000, 1.000
- 23. Stock Setup :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\setup\ks_audi_r8_lms_gt3_evo_2_safe.carsetup
- 24. Ai Setup : None
- 25. Wet Setup :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\setup\ks_audi_r8_lms_gt3_evo_2_wet.carsetup
- 26. Performance Modes : None
- 27. Ai Car Data :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.aicardata
- 28. mm : 1

Renault 5 GT Turbo

- 1. Screen Name : None
- 2. General
 - 2a. Screen Name : Renault 5 GT Turbo
 - 2b. Total Mass : 910.00000
 - 2c. Tank Position : 0.00000, -0.20000, -1.56750
 - 2d. Fuel : 30.00000
 - 2e. Max Fuel : 50.00000
 - 2f. Efficiency : 0.00000
 - 2g. Kg Per Liter : 0.75500
 - 2h. Body Box Sizes : 1.60000, 1.10000, 3.30000
 - 2i. Pickup Front Height : -0.49000
 - 2j. Pickup Rear Height : -0.48000
 - 2k. Check Rules : false
 - 2l. Minimum Height : 0.00000
 - 2m. Torsional Stiffness : 11000.00000
 - 2n. Torsional Damping : 50.00000
 - 2o. Body Mesh Offset
 - 2o1. Position : 0.000, 0.000, 0.000
 - 2o2. Rotation : 0.000, 0.000, 0.000
 - 2o3. Scale : 0.000, 0.000, 0.000
- 3. General Path : None
- 4. Suspensions :
 - 4a. Wheel Base : 2.40700
 - 4b. Longitudinal Cg Location : 0.64000
 - 4c. Base Y Front : -0.42000
 - 4d. Base Y Rear : -0.43000
 - 4e. Track Front : 1.34000
 - 4f. Track Rear : 1.31500
 - 4g. Damage
 - 4g1. Min Velocity : 40.00000
 - 4g2. Gain : 0.00040
 - 4g3. Max Damage : 0.05000

```

| | L 4g4. Debug Log : true
| | 4h. Coilover Front path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_front.coil
over
| | 4i. Coilover Rear Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_rear.coilo
ver
| | 4j. Front Suspension Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_front.susp
ension
| | 4k. Rear Suspension Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_rear.coilo
ver
| | 4l. Heavy Springs : None
| | 4m. Arb Front
| | | 4m1. Stiffness : 8000.00000
| | | 4m2. Controller
| | | | 4l9a. Name : None
| | | | 4l9b. Stages : None
| | 4n. Arb Rear
| | | 4n1. Stiffness : 6000.00000
| | | 4n2. Controller
| | | | 4l9a. Name : None
| | | | 4l9b. Stages : None
| | 4o. Flex Bar Front
| | | 4o1. Stiffness : 0.00000
| | | 4o2. Controller
| | | | 4l9a. Name : None
| | | | 4l9b. Stages : None
| | 4p. Flex Bar Rear
| | | 4p1. Stiffness : 10000.00000
| | | 4p2. Controller
| | | | 4l9a. Name : None
| | | | 4l9b. Stages : None
| | 4q. Dampers Controller : None
| | 4r. Has Dampers Cockpit Settings : false
| 5. Drivetrain Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.drivetrain
| 6. Gearbox Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.gearbox
| 7. Clutch Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.clutch
| 8. Engine Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.carengine
| 9. Brakes Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.brakesystem
| 10. Steering System
| | 10a. Four W S Controllers
| | | 4l9a. Name : None
| | | 4l9b. Stages : None
| 11. Electronics
| | 11a. T C
| | | 11a1. Has T C2 : false
| | | 11a2. Frequency Hz : 150.00000
| | | 11a3. Min Speed Kmh : 20.00000
| | | 11a4. Gear Change Time : 0.08000

```

- 11a5. Min Cut Level : 10.00000
- 11a6. Max Cut Level : 1.00000
- 11a7. Settings 1 :
 - 11a7a. Min Slip Ratio : -1.00000
 - 11a7b. Max Slip Ratio : -1.00000
 - 11a7c. Ref Slip Angle Deg : 0.00000
 - 11a7d. Engine Cut Level : 0.00000
 - 11a7e. Angular A C Cgain : 0.00000
 - 11a7f. Oversteer Gain : 0.00000
 - 11a7g. Slip Angle Activation Deg : 0.00000
- 11b. A B S
 - 11b1. Settings 1
 - 11b1a. Min Slip Ratio : -1.00000
 - 11b1b. Max Slip Ratio : -1.00000
 - 11b1c. Ref Slip Angle Deg : 0.00000
 - 11b1d. Cut Level : 0.00000
 - 11b1f. Max Torque Variation : 0.70000
 - 11b1. Settings 2
 - 11b1a. Min Slip Ratio : 0.06500
 - 11b1b. Max Slip Ratio : 0.12000
 - 11b1c. Ref Slip Angle Deg : 8.00000
 - 11b1d. Cut Level : 0.40000
 - 11b1f. Max Torque Variation : 0.75000
 - Frequency : 40.00000
 - Channels : 4
 - Min Speed Kmh : 0.00000
- 11c. E D L
 - 11c1. Active : true
 - 11c2. Brake Torque Power : 40.00000
 - 11c3. Brake Torque Coast : 200.00000
 - 11c4. Dead Zone Coast : 0.05000
 - 11c5. Dead Zone Power : 0.10000
 - 11c6. Max Spin Power : 0.40000
 - 11c7. Max Spin Coaster : 0.20000
 - 11c8. Min Speed : 0.00000
- 11d. E S P
 - 11d1. Frequency Hz : 0.00000
 - 11d2. Min Speed Kmh : 0.00000
 - 11d3. Settings : None
- 12. Electronics Path : None
- 13. Controls
 - 13a. Ff Mult : 2.90000
 - 13b. Steer Lock : 630.00000
 - 13c. Steer Ratio : -19.60000
 - 13d. Linear Steer Rod Ratio : 0.00220
 - 13e. Steer Assist : 1.00000
- 14. Box Colliders 1
 - 14a. Center : 0.00000, -0.24000, -0.29000
 - 14b. Size : 1.40000, 0.10000, 3.23000
 - 14c. Pitch Rotation Deg : 0.00000
- 15. Front Tyre Compounds 1 :
 - content\cars\common_phsx\tyres\road\road_195_55_13.tyre
- 16. Rear Tyre Compounds 1 :
 - content\cars\common_phsx\tyres\road\road_195_55_13.tyre
- 17. Aero
 - 17a. Slip Gain Multiple : 1.00000

```
| 17b. Speed Factor Mult : 2.00000
| 17c. Downforces : None
| 17d. Front Lift : None
| 17e. Rear Lift : None
| 17f. Drag : None
| 17g. Wings Path 1 :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo0.wing
| 18. Drs : None
| 19. Ers : None
| 20. Setup Limits :
content\cars\ks_renault_5_gt_turbo\data\setup\renault_5_gt_turbo.carsetuplimits
| 21. Collider Mesh :
content\cars\ks_renault_5_gt_turbo\collider\renault_5_gt_turbo_collider.mesh
| 22. Body Mesh Offset
| 22a. Position : 0.000, -0.632, -0.340
| 22b. Rotation : 0.700, 0.000, 0.000
| 22c. Scale : 1.000, 1.000, 1.000
| 23. Stock Setup :
content\cars\ks_renault_5_gt_turbo\data\setup\renault_5_gt_turbo.carsetup
| 24. Ai Setup : None
| 25. Wet Setup : None
| 26. Performance Modes : None
| 27. Ai Car Data :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo.aicardata
| 28. mm : 1
```

4. Car Engine [.carengine]

A. Description

I. General Description

The **Car Engine** asset is the heart of the vehicle's propulsion system within the simulation engine. While previous assets focused on chassis dimensions (**CAR DATA**) or deceleration (**BRAKE SYSTEM**), the **CAR ENGINE** file dictates how the vehicle generates mechanical energy.

It maps the complete thermodynamic and rotational behavior of the internal combustion engine (ICE), electric motor, or hybrid powertrain. It calculates how much raw power is produced at any given RPM, how the engine responds to the driver's throttle inputs, how it builds or sheds rotational speed (inertia), and how its performance is impacted by thermal limits and turbocharging.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters configured in the **CAR ENGINE** asset heavily influence the vehicle's straight-line speed, drivability, and endurance:

- **Acceleration & Top Speed:** Driven entirely by the torque output across the rev range. The shape of the power curve dictates how fast the car pulls out of corners and its absolute top speed on straights.
- **Traction Control & Drivability:** How smoothly or aggressively the engine delivers torque when the driver steps on the gas pedal. A snappy engine makes throttle modulation difficult, increasing wheelspin.
- **Engine Braking & Corner Entry:** When the driver lifts off the throttle, internal engine friction and compression create a drag torque (**ENGINE BRAKE**). This slows down the driven wheels, acting as a natural brake balance shift toward the drive axle.
- **Fuel Consumption & Thermal Management:** Dictates the fuel burn rate relative to RPM and throttle positioning, directly affecting pit-stop strategy. It also manages engine temperatures, where overheating can cause component failure or power loss.

III. Key Architecture & Data Fields Explained

The data within a **CAR ENGINE** schema is typically divided into **Power Generation**, **Rotational Dynamics**, and **Turbo/Aspiration Systems**.

1 - POWER GENERATION & POWER CURVES

- **Torque Curve / Power Curve:** The foundational lookup table mapping **Engine RPM (X-axis)** to **Torque Output (Y-axis, usually in Nm)**. The engine's raw horsepower is directly calculated from this relationship.

- **Max RPM / Limiter:** The absolute maximum rotational speed the engine can achieve before a fuel/ignition cut-off is triggered to prevent damage.
- **Idle RPM:** The baseline rotational speed at which the engine runs when there is zero driver throttle input.

2 - ROTATIONAL DYNAMICS & THROTTLE RESPONSE

- **Engine Inertia:** The rotational mass of the flywheel, crankshaft, and pistons. A low inertia value allows the engine to rev up and drop RPMs incredibly fast (like a Formula 1 car), while high inertia creates a slower, heavier revving characteristic (like a heavy V8 or diesel truck).
- **Engine Brake Torque:** The negative torque (resistance) generated by the engine when the throttle is fully released (0%). Crucial for modulating car balance during weight transfers.
- **Throttle Map / Curve:** A lookup table correlating physical pedal position to actual throttle plate opening. This can be linear, aggressive (for immediate bite), or progressive (for easier throttle control in wet weather).

3 - ASPIRATION, TURBOCHARGING & THERMAL BEHAVIOR

- **Turbo Boost / Wastegate Settings:** Configures turbocharger behavior, including maximum boost pressure, spool-up time (turbo lag), and wastegate limits.
- **Fuel Consumption Factor:** The efficiency coefficient of fuel burn per unit of power generated, used to compute real-time fuel depletion.
- **Optimal Temperature & Damage Thresholds:** The ideal operational thermal window for water and oil. Running the engine above these critical thresholds triggers progressive performance degradation or catastrophic engine failure.

IV. Short Interpretation of Asset Implementation

When analyzing an engine's physics profile, the data layout reveals the distinct mechanical identity of the powertrain:

- **The High-Revving Naturally Aspirated Profile (e.g., V10 GT3 / Vintage Race Cars):** Characterized by an ultra-low **ENGINE INERTIA** and a linear **TORQUE CURVE** that peaks very high up in the rev range (8000+ RPM). These engines have a sharp, instantaneous response to the throttle and feature high **ENGINE BRAKE TORQUE**, meaning lifting off the gas sharply stabilizes or upsets the rear axle depending on vehicle balance.
- **The Modern Turbocharged Profile (e.g., Modern GT3 / Le Mans Hypercars):** Features a flat, wide plateau in the **TORQUE CURVE** across the mid-range RPMs rather than a steep peak. It includes complex turbocharger data blocks defining spool rates. The throttle map is often configured progressively to mask "turbo kick"—the sudden surge of torque that happens when boost pressure builds up—ensuring the tires aren't instantly overwhelmed.
- **The Hybrid/EV Assist Profile:** Includes overlaying curves where an electric motor delivers instant maximum torque at 0 RPM, seamlessly tapering off as the internal combustion engine reaches its optimal power band higher up.

B. Schema

- 1. Engine Type : **enum**
- 2. Inertia : **float**
- 3. Power Curve : **string - path**
- 4. Coast Curve : **string - path**
- 5. Maps **[x]** : **object** | **can have multiple Maps**
 - 5a. Type : **enum**
 - 5b. Power Mult : **float**
 - 5c. Consumption Mult : **float**
 - 5d. Throttle Response Curve : **string - path**
 - 5e. Throttle Gain K R P M : **float**
 - 5f. Throttle Ref R P M Move : **float**
 - 5g. Throttle Lag Up : **float**
 - 5h. Throttle Lag Dn : **float**
 - 5i. Throttling Factor : **float**
- 6. Minimum : **integer**
- 7. Limiter : **integer**
- 8. Limiter Cycles : **integer**
- 9. Throttle Response Curve : **string - path**
- 10. Throttle Lag Up : **float**
- 11. Throttle Lag Dn : **float**
- 12. Throttle Rev Chocking : **float**
- 13. Ignition Time S : **float**
- 14. Starter Engine Torque : **float**
- 15. Start E C U Assist : **object**
 - 15a. Rpm Range : **float**
 - 15b. Gain : **float**
 - 15c. Speed Range K H M : **float**
 - 15d. Rpm Limiter : **float**
 - 15e. Limiter Cycles : **integer**
 - 15f. Use Clutch : **boolean**
 - 15g. Slip Ratio Target : **float**
 - 15h. Clutch Gain : **float**
 - 15i. Gas Gain : **float**
- 16. Turbo Controllers **[x]** : **object with an array of stages within** | **can have multiple Turbo Controllers**
 - 16a. Name : **string**
 - 16b. Stages **[x]** | **object** | **can have multiple Stage**
 - 16b1. Input Var : **enum**
 - 16b2. Combinator Mode : **enum**
 - 16b3. Lut : **string - path**
 - 16b4. Filter Gain : **float**
 - 16b5. Up Limit : **float**
 - 16b6. Down Limit : **float**
 - 16b7. Current Value : **float**
 - 16b8. Const Value : **float**
- 17. Waste Gate Controllers **[x]** : **object with an array of stages within** | **can have multiple Waste Gate Controllers**
 - 16a. Name : **string**
 - 16b. Stages **[x]** | **object** | **can have multiple Stage**
 - 16b1. Input Var : **enum**
 - 16b2. Combinator Mode : **enum**
 - 16b3. Lut : **string - path**

- 16b4. Filter Gain : float
- 16b5. Up Limit : float
- 16b6. Down Limit : float
- 16b7. Current Value : float
- 16b8. Const Value : float
- 18. Max Turbo Boost : float
- 19. Bov Threshold : float
- 20. Turbos To Load [x] : string - path | can have multiple Turbos To Load
- 21. Battery Data : object
 - 21a. Capacity Kwh : float
 - 21b. Dischage Efficiency : float
 - 21c. Charge Efficiency : float
 - 21d. Temp Eff Loss Per Deg : float
 - 21e. Convection K : float
 - 21f. Convection Forced K : float
 - 21g. Thermal Capacity : float
 - 21h. Max Temp : float
 - 21i. Min Temp : float

Enum - Car Engine

1	Engine Type	motor, combustion
5a	Type	gamma, cervone
16b1	Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
16b2	CombinatorMode	UndefinedMode, Add, Mult

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Engine Type	None (Enum : motor / combustion)	Selects the propulsion model: motor for electric-only powertrains, combustion for internal-combustion (ICE) or hybrid ICE behavior.
2.	Inertia	kg·m ² (Kilogram square meters)	Rotational inertia of the engine assembly (crankshaft, flywheel, pistons). Low values yield fast revving (F1); high values create sluggish RPM changes (heavy V8).

ID	Name	Unit of Measurement	Description
3.	Power Curve	None (.CURVE file path)	Primary look-up table mapping engine RPM (X-axis) to torque output in Nm (Y-axis); the foundation of all acceleration and top-speed calculations.
4.	Coast Curve	None (.CURVE file path)	Look-up table defining negative engine torque (engine braking) as a function of RPM when throttle is fully released (0%).
5.	Maps	None (Object array)	Array of selectable engine maps (power modes); each entry defines power/consumption multipliers and per-map throttle behavior.
5a.	Type	None (Enum : gamma / cervone)	Engine map blending/interpolation profile type governing how map parameters transition between operating points.
5b.	Power Mult	Dimensionless multiplier	Scalar applied to the base power/torque curve output for this engine map (e.g., 0.82 = 82% power for economy mode).
5d.	Throttle Response Curve	None (.CURVE file path)	Per-map look-up table mapping pedal position to effective throttle plate opening (linear, aggressive, progressive, wet).
5e.	Throttle Gain K RPM	Dimensionless gain	RPM-dependent gain scaling throttle response sensitivity; modulates how aggressively the engine reacts at different rev ranges.
5f.	Throttle Ref RPM Move	RPM (Revolutions per minute)	Reference RPM delta threshold triggering throttle gain adjustments during rapid RPM changes.
5g.	Throttle Lap Up	S (Seconds) or ratio	Time constant or smoothing factor for throttle opening response; higher values soften pedal tip-in.
5h.	Throttle Lag Dn	S (Seconds) or ratio	Time constant or smoothing factor for throttle closing response; higher values delay lift-off engine braking onset.
5i.	Throttling Factor	Dimensionless coefficient	Global throttle shaping factor applied within this engine map for fine-tuning drivability.
6.	Minimum	RPM (Revolutions per minute)	Idle RPM — the baseline engine speed maintained at zero throttle input.
7.	Limiter	RPM (Revolutions per minute)	Maximum engine RPM before the rev limiter cuts fuel/ignition to protect the engine.

ID	Name	Unit of Measurement	Description
8.	Limiter Cycles	None (Integer)	Number of soft-cut cycles the rev limiter performs before hard limiting; affects limiter feel and sound.
9.	Throttle Response Curve	None (.CURVE file path)	Global (non-map-specific) throttle pedal-to-plate response curve used when no per-map override is active.
10.	Throttle Lag Up	s (Seconds) or ratio	Global throttle opening lag/ smoothing applied across all maps unless overridden per map.
11.	Throttle Lag Dn	s (Seconds) or ratio	Global throttle closing lag/ smoothing applied across all maps unless overridden per map.
12.	Throttle Rev Chocking	Dimensionless coefficient	Rev-choking factor limiting RPM rise under partial throttle; simulates intake restriction or anti-lag behavior.
13.	Ignition Time S	s (Seconds) or ratio	Ignition timing delay or startup ignition sequence duration; affects initial fire-up and low-RPM response.
14.	Starter Engine Torque	Nm (Newton-meters)	Cranking torque delivered by the starter motor to bring the engine from rest to idle RPM.
15a.	Rpm Range	RPM (Revolutions per minute)	RPM window within which the start/launch assist system actively modulates torque and clutch slip.
15b.	Gain	Dimensionless gain	Authority multiplier of the start ECU assist torque and slip control loop.
15c.	Speed Range KMH	km/h (Kilometers per hour)	Vehicle speed range over which launch assist remains active before disengaging.
15d.	Rpm Limiter	RPM (Revolutions per minute)	Temporary RPM ceiling enforced during launch assist to prevent over-rev on clutch dump.
15e.	Limiter Cycles	None (Integer)	Number of rev-limiter cycles permitted during launch assist before hard cut.
15f.	Use Clutch	None (Boolean : True / False)	Enables clutch-slip-based launch control logic instead of pure throttle modulation.
15g.	Slip Ratio Target	Dimensionless ratio	Target wheel slip ratio maintained by launch assist for optimal traction off the line.

ID	Name	Unit of Measurement	Description
15h.	Clutch Gain	Dimensionless gain	Clutch engagement modulation gain within the launch assist controller.
15i.	Gas Gain	Dimensionless gain	Throttle input scaling gain applied during launch assist for traction management.
16.	Turbo Controllers	None (Controller array)	Array of dynamic turbo boost controllers; each entry contains a Name and Stages pipeline (fields 16a, 16b1–16b8).
16a.	Name	None (String)	Internal identifier for a turbo or wastegate controller block.
16b1.	Input Var	None (Telemetry enum)	Telemetry channel driving the controller stage (commonly RPMS, Gas, or Boost pressure).
16b2.	Combinator Mode	None (Math enum : Add / Mult)	How this stage output combines with prior stages: additive offset or multiplicative scaling.
16b3.	Lut	None (.CURVE file path)	Look-up table mapping the input variable to a turbo boost or wastegate modifier.
16b4.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter coefficient smoothing rapid input spikes for stable boost control.
16b5.	Up Limit	Depends on input variable	Upper clamp on the processed controller input signal.
16b6.	Down Limit	Depends on input variable	Lower clamp on the processed controller input signal.
16b7.	Current Value	Depends on input variable	Runtime output value of the controller stage during simulation.
16b8.	Const Value	Depends on input variable	Fallback constant output when no dynamic input or LUT is active.
17.	Waste Gate Controllers	None (Controller array)	Array of wastegate dynamic controllers; reuses the same 16a Name and 16b1–16b8 stage field structure as Turbo Controllers.
18.	Max Turbo Boost	bar (bar gauge pressure)	Maximum allowable turbocharger boost pressure ceiling; hard limit for forced-induction output.
19.	Bov Threshold	bar or ratio	Blow-off valve (BOV) activation threshold; pressure differential at which excess boost is vented.
20.	Turbos To Load	None (.TURBO file path)	Path(s) to .turbo assets defining turbocharger spool dynamics, compressor maps, and wastegate physics.

ID	Name	Unit of Measurement	Description
21a.	Capacity Kwh	kWh (Kilowatt-hours)	Total electrical energy storage capacity of the traction battery pack.
21b.	Discharge Efficiency	Ratio (0.0 - 1.0)	Energy conversion efficiency from battery to motor during discharge (e.g., 0.80 = 80%).
21c.	Charge Efficiency	Ratio (0.0 - 1.0)	Energy conversion efficiency from regenerative braking or charging back into the battery.
21d.	Temp Eff Loss Per Deg	Ratio/°C (Per degree Celsius)	Efficiency loss per degree of battery temperature deviation from optimal operating point.
21e.	Convection K	W/(m²·K) or coefficient	Natural convection heat transfer coefficient for passive battery cooling.
21f.	Convection Forced K	W/(m²·K) or coefficient	Forced convection coefficient when active cooling (airflow or liquid) is applied.
21g.	Thermal Capacity	J/K or J/°C	Heat storage capacity of the battery pack; controls temperature rise rate under sustained load.
21h.	Max Temp	°C (Degrees Celsius)	Maximum safe operating temperature before power derating or thermal shutdown.
21i	Min Temp	°C (Degrees Celsius)	Minimum operating temperature; below this, efficiency loss or power limiting may apply.

D. Example data

I. Chosen Car Engine for Example

- Alpine A290 b (slug : ks_alpine_a290_b)
- Ferrari SF 25 (slug : ks_ferrari_sf_25)
- Chevrolet Camaro ZL1 (slug : ks_renault_5_gt_turbo)
- Datsun 240z Fairlady (slug : ks_datsun_240z_fairlady)

II. Example

Alpine A290 b

- | 1. Engine Type : motor
- | 2. Inertia : 0.06000
- | 3. Power Curve : content\cars\ks_alpine_a290_b\data\HEADER_POWER.curve

- 4. Coast Curve : content\cars\ks_alpine_a290_b\data\coast.curve
- 5. Maps : None
- 6. Minimum : 0
- 7. Limiter : 15200
- 8. Limiter Cycles : 20
- 9. Throttle Response Curve :
content\cars\ks_alpine_a290_b\data\throttle.curve
- 10. Throttle Lag Up : 0.00000
- 11. Throttle Lag Dn : 0.00000
- 12. Throttle Rev Chocking : 0.00000
- 13. Ignition Time S : 0.00000
- 14. Starter Engine Torque : 0.00000
- 15. Start E C U Assist
 - 15a. Rpm Range : 0.00000
 - 15b. Gain : 0.00000
 - 15c. Speed Range K H M : 0.00000
 - 15d. Rpm Limiter : 0.00000
 - 15e. Limiter Cycles : 0
 - 15f. Use Clutch : false
 - 15g. Slip Ratio Target : 0.00000
 - 15h. Clutch Gain : 0.00000
 - 15i. Gas Gain : 0.00000
- 16. Turbo Controllers : None
- 17. Waste Gate Controllers : None
- 18. Max Turbo Boost : 0.00000
- 19. Bov Threshold : 0.00000
- 20. Turbos To Load : None
- 21. Battery Data
 - 21a. Capacity Kwh : 52.000
 - 21b. Dischage Efficiency : 0.800
 - 21c. Charge Efficiency : 0.600
 - 21d. Temp Eff Loss Per Deg : 0.01000
 - 21e. Convection K : 0.00100
 - 21f. Convection Forced K : 0.00050
 - 21g. Thermal Capacity : 100.000
 - 21h. Max Temp : 35.000
 - 21i. Min Temp : 15.000

Ferrari SF 25

- 1. Engine Type : combustion
- 2. Inertia : None
- 3. Power Curve : content\cars\ks_ferrari_sf_25\data\HEADER_POWER.curve
- 4. Coast Curve : content\cars\ks_ferrari_sf_25\data\coast_smooth.curve
- 5. Maps 1
 - 5a. Type : gamma
 - 5b. Power Mult : 1.00000
 - 5c. Consumption Mult : 0.60000
 - 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
 - 5e. Throttle Gain K R P M : 0.00000
 - 5f. Throttle Ref R P M Move : 0.00000
 - 5g. Throttle Lag Up : 0.00000
 - 5h. Throttle Lag Dn : 0.00000

```

| L 5i. Throttling Factor : 0.00000
- 5. Maps 2
| L 5a. Type : gamma
| L 5b. Power Mult : 1.00000
| L 5c. Consumption Mult : 0.60000
| L 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
| L 5e. Throttle Gain K R P M : 0.00000
| L 5f. Throttle Ref R P M Move : 0.00000
| L 5g. Throttle Lag Up : 0.00000
| L 5h. Throttle Lag Dn : 0.00000
| L 5i. Throttling Factor : 0.00000
- 5. Maps 3
| L 5a. Type : gamma
| L 5b. Power Mult : 0.89000
| L 5c. Consumption Mult : 0.56000
| L 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
| L 5e. Throttle Gain K R P M : 0.00000
| L 5f. Throttle Ref R P M Move : 0.00000
| L 5g. Throttle Lag Up : 0.00000
| L 5h. Throttle Lag Dn : 0.00000
| L 5i. Throttling Factor : 0.00000
- 5. Maps 4
| L 5a. Type : gamma
| L 5b. Power Mult : 0.89000
| L 5c. Consumption Mult : 0.56000
| L 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
| L 5e. Throttle Gain K R P M : 0.00000
| L 5f. Throttle Ref R P M Move : 0.00000
| L 5g. Throttle Lag Up : 0.00000
| L 5h. Throttle Lag Dn : 0.00000
| L 5i. Throttling Factor : 0.00000
- 5. Maps 5
| L 5a. Type : gamma
| L 5b. Power Mult : 0.82000
| L 5c. Consumption Mult : 0.55000
| L 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
| L 5e. Throttle Gain K R P M : 0.00000
| L 5f. Throttle Ref R P M Move : 0.00000
| L 5g. Throttle Lag Up : 0.00000
| L 5h. Throttle Lag Dn : 0.00000
| L 5i. Throttling Factor : 0.00000
- 5. Maps 6
| L 5a. Type : gamma
| L 5b. Power Mult : 0.82000
| L 5c. Consumption Mult : 0.55000
| L 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
| L 5e. Throttle Gain K R P M : 0.00000
| L 5f. Throttle Ref R P M Move : 0.00000
| L 5g. Throttle Lag Up : 0.00000
| L 5h. Throttle Lag Dn : 0.00000
| L 5i. Throttling Factor : 0.00000

```



```
- 5. Maps 7
  | 5a. Type : gamma
  | 5b. Power Mult : 0.72000
  | 5c. Consumption Mult : 0.50000
  | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
  | 5e. Throttle Gain K R P M : 0.00000
  | 5f. Throttle Ref R P M Move : 0.00000
  | 5g. Throttle Lag Up : 0.00000
  | 5h. Throttle Lag Dn : 0.00000
  | 5i. Throttling Factor : 0.00000
- 5. Maps 8
  | 5a. Type : gamma
  | 5b. Power Mult : 0.72000
  | 5c. Consumption Mult : 0.50000
  | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
  | 5e. Throttle Gain K R P M : 0.00000
  | 5f. Throttle Ref R P M Move : 0.00000
  | 5g. Throttle Lag Up : 0.00000
  | 5h. Throttle Lag Dn : 0.00000
  | 5i. Throttling Factor : 0.00000
- 5. Maps 9
  | 5a. Type : gamma
  | 5b. Power Mult : 0.95000
  | 5c. Consumption Mult : 0.58500
  | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
  | 5e. Throttle Gain K R P M : 0.00000
  | 5f. Throttle Ref R P M Move : 0.00000
  | 5g. Throttle Lag Up : 0.00000
  | 5h. Throttle Lag Dn : 0.00000
  | 5i. Throttling Factor : 0.00000
- 5. Maps 10
  | 5a. Type : gamma
  | 5b. Power Mult : 0.95000
  | 5c. Consumption Mult : 0.58500
  | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
  | 5e. Throttle Gain K R P M : 0.00000
  | 5f. Throttle Ref R P M Move : 0.00000
  | 5g. Throttle Lag Up : 0.00000
  | 5h. Throttle Lag Dn : 0.00000
  | 5i. Throttling Factor : 0.00000
- 5. Maps 11
  | 5a. Type : gamma
  | 5b. Power Mult : 1.00000
  | 5c. Consumption Mult : 0.60000
  | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_progressive.curve
  | 5e. Throttle Gain K R P M : 0.00000
  | 5f. Throttle Ref R P M Move : 0.00000
  | 5g. Throttle Lag Up : 0.00000
  | 5h. Throttle Lag Dn : 0.00000
  | 5i. Throttling Factor : 0.00000
- 5. Maps 12
```

- 5a. Type : gamma
- 5b. Power Mult : 1.00000
- 5c. Consumption Mult : 0.60000
- 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_wet.curve
- 5e. Throttle Gain K R P M : 0.00000
- 5f. Throttle Ref R P M Move : 0.00000
- 5g. Throttle Lag Up : 0.00000
- 5h. Throttle Lag Dn : 0.00000
- 5i. Throttling Factor : 0.00000
- 6. Minimum : 2000
- 7. Limiter : 15000
- 8. Limiter Cycles : 60
- 9. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
- 10. Throttle Lag Up : 0.00000
- 11. Throttle Lag Dn : 0.00000
- 12. Throttle Rev Chocking : 0.00000
- 13. Ignition Time S : 0.00000
- 14. Starter Engine Torque : 40.00000
- 15. Start E C U Assist
 - 15a. Rpm Range : 0.00000
 - 15b. Gain : 0.00000
 - 15c. Speed Range K H M : 0.00000
 - 15d. Rpm Limiter : 0.00000
 - 15e. Limiter Cycles : 0
 - 15f. Use Clutch : false
 - 15g. Slip Ratio Target : 0.00000
 - 15h. Clutch Gain : 0.00000
 - 15i. Gas Gain : 0.00000
- 16. Turbo Controllers 1
 - 16a. Name : None
 - 16b. Stages 1
 - 16b1. Input Var : RPMS
 - 16b2. Combinator Mode : Add
 - 16b3. Lut :
content\cars\ks_ferrari_sf_25\data\ctrl_turbo0CONTROLLER_0.curve
 - 16b4. Filter Gain : 0.99000
 - 16b5. Up Limit : 10000.00000
 - 16b6. Down Limit : 0.00000
 - 16b7. Current Value : 0.00000
 - 16b8. Const Value : 0.00000
- 17. Waste Gate Controllers : None
- 18. Max Turbo Boost : 2.20000
- 19. Bov Threshold : 0.00000
- 20. Turbos To Load 1 :
content\cars\ks_ferrari_sf_25\data\ks_ferrari_sf_250.turbo
- 21. Battery Data
 - 21a. Capacity Kwh : 0.000
 - 21b. Dischage Efficiency : 0.000
 - 21c. Charge Efficiency : 0.000
 - 21d. Temp Eff Loss Per Deg : 0.00000
 - 21e. Convection K : 0.00000
 - 21f. Convection Forced K : 0.00000
 - 21g. Thermal Capacity : 0.000
 - 21h. Max Temp : 0.000

L L 21i. Min Temp : 0.000

Chevrolet Camaro ZL1

- 1. Engine Type : combustion
- 2. Inertia : 0.45000
- 3. Power Curve :
content\cars\ks_chevrolet_camaro_zl1\data\HEADER_POWER.curve
- 4. Coast Curve : content\cars\ks_chevrolet_camaro_zl1\data\coast.curve
- 5. Maps 1
 - 5a. Type : gamma
 - 5b. Power Mult : 1.00000
 - 5c. Consumption Mult : 1.00000
 - 5d. Throttle Response Curve :
content\cars\ks_chevrolet_camaro_zl1\data\throttlemap_response.curve
 - 5e. Throttle Gain K R P M : 0.00000
 - 5f. Throttle Ref R P M Move : 0.00000
 - 5g. Throttle Lag Up : 0.80000
 - 5h. Throttle Lag Dn : 0.92000
 - 5i. Throttling Factor : 0.00000
- 6. Minimum : 900
- 7. Limiter : 6600
- 8. Limiter Cycles : 30
- 9. Throttle Response Curve :
content\cars\ks_chevrolet_camaro_zl1\data\throttlemap_response.curve
- 10. Throttle Lag Up : 0.80000
- 11. Throttle Lag Dn : 0.90000
- 12. Throttle Rev Chocking : 0.00000
- 13. Ignition Time S : 0.00000
- 14. Starter Engine Torque : 60.00000
- 15. Start E C U Assist
 - 15a. Rpm Range : 1500.00000
 - 15b. Gain : 0.10000
 - 15c. Speed Range K H M : 120.00000
 - 15d. Rpm Limiter : 3800.00000
 - 15e. Limiter Cycles : 0
 - 15f. Use Clutch : true
 - 15g. Slip Ratio Target : 0.14000
 - 15h. Clutch Gain : 5.00000
 - 15i. Gas Gain : 1.75000
- 16. Turbo Controllers 1
 - 16a. Name : None
 - 16b. Stages 1
 - 16b1. Input Var : RPMS
 - 16b2. Combinator Mode : Add
 - 16b3. Lut :
content\cars\ks_chevrolet_camaro_zl1\data\rpms_boost.curve
 - 16b4. Filter Gain : 0.00000
 - 16b5. Up Limit : 1.20000
 - 16b6. Down Limit : 0.00000
 - 16b7. Current Value : 0.00000
 - 16b8. Const Value : 0.00000
- 17. Waste Gate Controllers : None
- 18. Max Turbo Boost : 0.60000

- 19. Bov Threshold : 0.00000
- 20. Turbos To Load 1 :
content\cars\ks_chevrolet_camaro_zl1\data\ks_chevrolet_camaro_zl1_compressor.turbo
- 21. Battery Data
 - 21a. Capacity Kwh : 0.000
 - 21b. Dischage Efficiency : 0.000
 - 21c. Charge Efficiency : 0.000
 - 21d. Temp Eff Loss Per Deg : 0.00000
 - 21e. Convection K : 0.00000
 - 21f. Convection Forced K : 0.00000
 - 21g. Thermal Capacity : 0.000
 - 21h. Max Temp : 0.000
 - 21i. Min Temp : 0.000

Datsun 240z Fairlady

- 1. Engine Type : combustion
- 2. Inertia : 0.18000
- 3. Power Curve :
content\cars\ks_datsun_240z_fairlady\data\HEADER_POWER.curve
- 4. Coast Curve :
content\cars\ks_datsun_240z_fairlady\data\ks_datsun_240z_fairlady_coast1.curve
- 5. Maps 1
 - 5a. Type : cervone
 - 5b. Power Mult : 1.00000
 - 5c. Consumption Mult : 1.00000
 - 5d. Throttle Response Curve :
content\cars\ks_datsun_240z_fairlady\data\throttle.curve
 - 5e. Throttle Gain K R P M : 0.00000
 - 5f. Throttle Ref R P M Move : 0.00000
 - 5g. Throttle Lag Up : 0.00000
 - 5h. Throttle Lag Dn : 0.00000
 - 5i. Throttling Factor : 0.00000
- 6. Minimum : 600
- 7. Limiter : 7000
- 8. Limiter Cycles : 8
- 9. Throttle Response Curve :
content\cars\ks_datsun_240z_fairlady\data\throttle.curve
- 10. Throttle Lag Up : 0.00000
- 11. Throttle Lag Dn : 0.00000
- 12. Throttle Rev Chocking : 0.00000
- 13. Ignition Time S : 0.00000
- 14. Starter Engine Torque : 30.00000
- 15. Start E C U Assist
 - 15a. Rpm Range : 0.00000
 - 15b. Gain : 0.00000
 - 15c. Speed Range K H M : 0.00000
 - 15d. Rpm Limiter : 0.00000
 - 15e. Limiter Cycles : 0
 - 15f. Use Clutch : false
 - 15g. Slip Ratio Target : 0.00000
 - 15h. Clutch Gain : 0.00000
 - 15i. Gas Gain : 0.00000

- 16. Turbo Controllers : None
- 17. Waste Gate Controllers : None
- 18. Max Turbo Boost : 0.00000
- 19. Bov Threshold : 0.00000
- 20. Turbos To Load : None
- 21. Battery Data
 - 21a. Capacity Kwh : 0.000
 - 21b. Dischage Efficiency : 0.000
 - 21c. Charge Efficiency : 0.000
 - 21d. Temp Eff Loss Per Deg : 0.00000
 - 21e. Convection K : 0.00000
 - 21f. Convection Forced K : 0.00000
 - 21g. Thermal Capacity : 0.000
 - 21h. Max Temp : 0.000
 - 21i. Min Temp : 0.000

5. Car Setup [.carsetup]

A. Description

I. General Description

The **Car Setup** asset is the primary adjustment interface that translates mechanical physics constants into highly customizable track settings. While other files define the immovable physical constraints of a vehicle (like the frame weight or engine potential), the `CAR SETUP` file acts as the pit garage dashboard.

It stores all adjustable parameters that engineers and drivers modify to adapt the vehicle to specific racetracks, weather conditions, or driving styles. It acts as a set of instructions applied on top of the base suspension, aerodynamics, and drivetrain geometries to maximize grip, control stability, and balance the car's handling.

II. Area of Influence / Impact on Vehicle Dynamics

The variables configured in the `CAR SETUP` asset have an immediate and dramatic impact on every dynamic phase of lap execution:

- **Mechanical Grip & Tire Patch Optimization:** By altering alignment and tire pressures, this asset controls how flat the tire stays against the tarmac during high-load cornering, directly maximizing lateral adhesion.
- **Aerodynamic Platform Control:** Ride heights and rake (the angle of the car's floor relative to the track) dictate how much downforce wings and underbody diffusers generate, dramatically shifting high-speed stability and drag.
- **Transient Phase Handling (Weight Transfer):** Springs and dampers regulate *how fast* and *how far* the body rolls when cornering, pitches under braking, or squats under acceleration, giving the driver a predictable or snappy platform.
- **Corner Exit Propulsion (Power Delivery):** Differential settings govern how power shifts between the drive wheels, dictating whether a car neatly clips an exit apex or snaps into wheelspin/snap-oversteer.

III. Key Architecture & Data Fields Explained

The parameters in a `CAR SETUP` file are traditionally compartmentalized into five core operational groups: **Tyres**, **Aerodynamics**, **Suspension (Springs/Alignment)**, **Dampers (Shock Absorbers)**, and **Drivetrain (Differential)**.

1 - TYRES (THE CONTACT PATCH)

- **Cold Pressure:** The initial inflating pressure of the tire before leaving the pitlane. It directly influences the hot operating pressure and optimal carcass temperature window.

- **Camber (Front/Rear):** The inward or outward tilt of the wheels when viewed from the front. Negative camber ensures that when the car rolls into a corner, the outer tire leans flat against the asphalt for maximum cornering grip.
- **Toe (Front/Rear):** The angle of the wheels relative to the vehicle's longitudinal centerline. Front *toe-out* increases turn-in agility, while rear *toe-in* stabilizes the rear end during straight-line acceleration and heavy braking.

2 - AERODYNAMICS (THE AIRFLOW PLATFORM)

- **Wing Angles (Front/Rear Spoiler):** The mechanical inclination of aero planes. Higher rear wing angles increase downforce (stability in fast corners) at the expense of higher straight-line drag (lower top speed).
- **Ride Height (Front/Rear):** The clearance between the car's floor and the ground. Lowering the car increases venturi/ground-effect downforce, but going too low can cause the floor to bottom out on bumps, destroying all grip instantly.

3 - SUSPENSION GEOMETRY & RATES

- **Wheel Rate / Spring Stiffness:** The stiffness of the main suspension coil springs. Stiffer springs stabilize the aero platform and stop body roll, but reduce compliance on bumpy circuits or kerbs.
- **Anti-Roll Bar (ARB - Front/Rear):** A torsion bar linking the left and right suspension sides. It purely resists body roll in corners. Adjusting the stiffness ratio between front and rear ARBs is the primary tool for shifting handling balance between understeer and oversteer.

4 - DAMPERS (TRANSIENT SHOCK ABSORPTION)

Dampers control the speed of suspension movement and are split into travel directions and shaft speeds:

- **Bump / Compression (Slow & Fast):** Resists the suspension compressing. *Slow Bump* manages body roll from driver inputs; *Fast Bump* absorbs high-speed impacts like hitting a track kerb or bump.
- **Rebound / Extension (Slow & Fast):** Resists the suspension extending back out. *Slow Rebound* regulates how the car settles as it exits a corner or releases the brakes; *Fast Rebound* controls how fast the tire snaps back down to touch the track after a kerb strike.

5 - DRIVETRAIN & DIFFERENTIAL

- **Diff Preload:** The static friction locking the differential when zero throttle is applied. Higher preload stabilizes the car during snappy, mid-corner lift-offs.
- **Diff Power (Lock):** The locking percentage under acceleration. High lock forces both tires to spin together for maximum exit drive, but can cause power-understeer or sudden power-oversteer if traction breaks.
- **Diff Coast (Lock):** The locking percentage under deceleration/braking. High coast lock keeps the rear stable under heavy braking into a corner but resists initial turn-in agility.

IV. Interpretation of Setup Configuration Strategies

When reading a car's exported .SETUP file, the values map out a very specific racing engineering intent based on the target circuit layout:

- **The High-Downforce Monza/Spa vs. Monaco Profile:** For high-speed tracks like Spa-Francorchamps or Monza, you will notice ultra-low wing angles to minimize drag, stiffer springs to support the high aero load compressing the suspension, and lower ride heights. For Monaco, the wings are cranked to the absolute maximum, springs are softened to absorb bumpy street surfaces, and toe angles are aggressive to force the car around tight hairpins.
- **The Endurance Balance Management:** In high-tier setups, engineers deliberately leave a margin in the ride heights and tire pressures. This accommodates changing ambient temperatures over a multi-hour race stint and accounts for structural components wearing out, ensuring the aerodynamic floor platform stays stable regardless of fuel burn or tyre degradation.

B. Schema

```
1. Import Setup : string - path
2. Mechanical Balance : object
  | 2a. Arbs [x] : float | can have multiple Arbs
  | 2b. Steer Ratio : float
  | 2c. Brakes : object
  |   | 2c1. Front Bias : float
  |   | 2c2. Torque Multiplier : float
  |   | 2c3. Brake Ducts [x] : float | can have multiple Brake Ducts
  | 2d. Differential : object
  |   | 2d1. Power : float
  |   | 2d2. Coast : float
  |   | 2d3. Preload : float
3. Suspensions [x] : object
  | 3a. Wheel Rate : float
  | 3b. Bump Stop Up : object
  |   | 3b1. Range : float
  |   | 3b2. Rate : float
  | 3c. Bump Stop Down : object
  |   | 3c1. Range : float
  |   | 3c2. Rate : float
  | 3d. Helper Rate : float
  | 3e. Helper Range : float
4. Dampers [x] : object | can have multiple Dampers
  | 4a. Slow Bump : float
  | 4b. Fast Bump : float
  | 4c. Slow Rebound : float
  | 4e. Fast Rebound : float
5. Alignments [x] : object | can have multiple Alignments
  | 5a. Pressure : float
  | 5b. Camber : float
  | 5c. Toe : float
  | 5d. Caster : float
  | 5e. Static Camber : float
```


- | 5f. Toe Out Linear : float
- | 5g. Compound : float
- 6. Electronics : object
 - | 6a. Tc1 : float
 - | 6b. Tc2 : float
 - | 6c. Abs : float
 - | 6d. Esc : float
 - | 6e. Ebb : float
 - | 6f. Engine Map : float
 - | 6g. Telemetry Laps to Record : float
 - | 6h. Turbo Boost Lv : float
 - | 6i. Ers Deployment Map : float
 - | 6j. Ers Recharge Lv : float
 - | 6k. Ers Heat Charging : float
- 7. Aero : object
 - | 7a. Collar Positions Mm [x] : float | can have multiple Collar Positions Mm
 - | 7b. Front Target Height : float
 - | 7c. Rear Target Height : float
 - | 7d. Front Wing Angle : float
 - | 7e. Rear Wing Angle : float
- 8. Fuel Strategy : object
 - | 8a. Fuel : float
- 9. Final State Name : string - path
- 10. Version : integer
- 11. Is Setup Shared : boolean

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Import Setup	None (File path)	Optional path to a base .CARSETUP preset from which this setup inherits or overlays values.
2.	Mechanical Balance	None (Object)	Parent block grouping chassis balance adjustments: anti-roll bars, steering ratio, brakes, and differential.
2a.	Arbs	Nm/deg or N/m (Anti-roll bar stiffness)	Anti-roll bar stiffness values; index 1 = front axle, index 2 = rear axle. Primary tool for shifting roll stiffness balance.
2b.	Steer Ratio	Ratio (:1) or clicks	Adjustable steering rack ratio override applied at setup level; modifies low-speed agility and steering wheel sensitivity.
2c.	Brakes	None (Object)	Brake setup sub-block within Mechanical Balance; contains bias, torque scaling, and duct settings.
2c1.	Front Bias	% (Percentage)	Brake balance percentage sent to the front axle (e.g., 72 = 72% front / 28% rear).

ID	Name	Unit of Measurement	Description
2c2.	Torque Multiplier	% or dimensionless multiplier	Global braking torque scaling factor for this setup (100 = 100% of base brake system torque).
2c3.	Brake Ducts	Ratio (0.0 - 1.0) or clicks	Brake cooling duct opening level(s); multiple entries typically map to front/rear duct positions.
2d.	Differential	None (Object)	Differential setup sub-block within Mechanical Balance; governs power, coast, and preload lock.
2d1.	Power	% or Nm (Differential lock)	Differential locking intensity under acceleration (on-throttle); higher values increase drive-wheel coupling on corner exit.
2d2.	Coast	% or Nm (Differential lock)	Differential locking intensity on coast/deceleration; stabilizes the rear under braking and lift-off.
2d3.	Preload	Nm (Newton-meters)	Static preload torque locking the differential plates at zero throttle; resists free differential action mid-corner.
3.	Suspensions	None (Object array, per wheel)	Per-corner suspension block; indices 1–4 map to LF, RF, LR, RR wheels respectively.
3a.	Wheel Rate	N/m (Newtons per meter)	Effective wheel-rate spring stiffness at this corner; controls ride frequency, body roll, and aero platform stability.
3b.	Bump Stop Up	None (Object)	Upper bump-stop definition within each Suspensions entry; limits compression travel beyond spring range.
3b1.	Range	m (Meters)	Travel range before the upper bump-stop engages (Suspensions > Bump Stop Up).
3b2.	Rate	N/m or N (Bump-stop stiffness)	Stiffness/force rate of the upper bump-stop resisting compression beyond the range threshold.
3c.	Bump Stop Down	None (Object)	Lower bump-stop definition within each Suspensions entry; limits droop/extension travel.
3c1.	Range	m (Meters)	Travel range before the lower (rebound) bump-stop engages (Suspensions > Bump Stop Down).
3c2.	Rate	N/m or N (Bump-stop stiffness)	Stiffness/force rate of the lower bump-stop resisting droop/extension.

ID	Name	Unit of Measurement	Description
3d.	Helper Rate	N/m (Spring rate)	Stiffness of the helper/tender spring assisting the main coil at low compression.
3e.	Helper Range	m (Meters)	Travel range over which the helper spring is active before the main spring fully carries load.
4.	Dampers	None (Object array, per wheel)	Per-corner damper settings; indices 1–4 map to LF, RF, LR, RR. Values may be raw N·s/m or UI clicks depending on carsetupunits.
4a.	Slow Bump	N·s/m or clicks	Slow-shaft-speed compression damping; controls body roll and pitch from driver inputs (braking, cornering).
4b.	Fast Bump	N·s/m or clicks	Fast-shaft-speed compression damping; absorbs kerb strikes, bumps, and high-frequency track irregularities.
4c.	Slow Rebound	N·s/m or clicks	Slow-shaft-speed extension damping; regulates how the chassis settles after weight transfer releases.
4e.	Fast Rebound	N·s/m or clicks	Fast-shaft-speed extension damping; controls how quickly the tyre returns to the track surface after a kerb hit.
5.	Alignments	None (Object array, per wheel)	Per-corner tyre pressure and alignment geometry; indices 1–4 map to LF, RF, LR, RR.
5a.	Pressure	PSI or bar (Cold tyre pressure)	Cold inflation pressure before leaving the pits; determines hot operating pressure and tyre carcass temperature window.
5b.	Camber	deg (Degrees)	Adjustable camber angle (negative = top of wheel leans inward); maximizes contact patch during body roll in corners.
5c.	Toe	deg (Degrees)	Toe angle relative to vehicle centerline; toe-out front increases turn-in, toe-in rear stabilizes straight-line stability.
5d.	Caster	deg (Degrees)	Steering axis caster angle; influences straight-line stability, self-centering torque, and camber gain during steering.
5e.	Static Camber	deg (Degrees)	Computed/static camber at ride height including kinematic offset; read-only or derived alignment reference.

ID	Name	Unit of Measurement	Description
5f.	Toe Out Linear	deg/m or rad/m (Toe curve)	Linear toe change rate as a function of suspension travel or steering; models compliance and kinematic toe curve.
5g.	Compound	None (Compound index)	Tyre compound selection index referencing the available front/rear tyre compound paths in Car Data.
6.	Electronics	None (Object)	Driver-aid and powertrain map presets stored in this setup (TC, ABS, ESC, engine map, ERS, turbo).
6a.	Tc1	None (Map level / index)	Traction Control map level (primary TC); references TC settings defined in Car Data electronics.
6b.	Tc2	None (Map level / index)	Secondary Traction Control map level for dual-TC systems.
6c.	Abs	None (Map level / index)	ABS intervention map level; controls slip thresholds and pulsing aggressiveness.
6d.	Esc	None (Map level / index)	Electronic Stability Program map level; governs yaw/brake intervention authority.
6e.	Ebb	None (Map level / index)	Electronic Brake Balance mode or level for dynamic bias adjustment.
6f.	Engine Map	None (Map index)	Engine power map selection referencing engine map entries in the .CARENGINE asset.
6g.	Telemetry Laps To Record	None (Integer)	Number of laps of onboard telemetry to retain when this setup is active.
6h.	Turbo Boost Lv	bar or level index	Turbo boost level preset for forced-induction engines.
6i.	Ers Deployment Map	None (Map index)	Hybrid ERS power deployment map selection.
6j.	Ers Recharge Lv	None (Level index)	ERS regenerative recovery aggressiveness level.
6k.	Ers Heat Charging	None (Level index)	MGU-H heat-to-battery charging level for hybrid systems.
7.	Aero	None (Object)	Aerodynamic setup block: ride heights, spring collar positions, and adjustable wing angles.
7a.	Collar Positions Mm	mm (millimeters)	Spring perch/collar position per corner controlling static ride height; indices 1–4 = LF, RF, LR, RR.

ID	Name	Unit of Measurement	Description
7b.	Front Target Height	mm (millimeters)	Target front ride height reference for aero platform setup and regulatory ride-height checks.
7c.	Rear Target Height	mm (millimeters)	Target rear ride height; combined with front height defines rake (nose-up/nose-down aero attitude).
7d.	Front Wing Angle	deg (Degrees)	Mechanical angle of the front aero device (splitter/wing); increases downforce and drag when raised.
7e.	Rear Wing Angle	deg (Degrees)	Mechanical angle of the rear wing element; primary high-speed cornering stability vs. straight-line speed trade-off.
8.	Fuel Strategy	None (Object)	Fuel load planning block for this setup preset.
8a.	Fuel	L (Liters)	Fuel load strategy for this setup preset; affects weight, CG position, and stint length planning.
9.	Final State Name	None (String / preset identifier)	Composite preset identifier linking mechanical and visual setup states (e.g., mech + visual preset names).
10.	Version	None (Integer)	Setup file format version for backward compatibility with the editor and simulation loader.
11.	Is Setup Shared	None (Boolean : True / False)	Indicates whether this setup preset is shared/exportable across sessions or locked to a single vehicle profile.

D. Example data

I. Chosen Car Engine for Example

- Audi Sport Quattro (slug : ks_audi_sport_quattro)
- Alfa Romeo Junior (slug : ks_alfa_romeo_junior)
- Ferrari 488 Challenge Evo (slug : ks_ferrari_488_challenge_evo [preset : safe_1])

II. Example

Audi Sport Quattro

- | 1. Import Setup : None
- | 2. Mechanical Balance
 - | 2a. Arbs 1 : 25000.00000
 - | 2a. Arbs 2 : 20000.00000

- 2b. Steer Ratio : 18.00000
- 2c. Brakes
 - 2c1. Front Bias : 72.00000
 - 2c2. Torque Multiplier : 100.00000
 - 2c3. Brake Ducts : None
- 2d. Differential
 - 2d1. Power : 0.00000
 - 2d2. Coast : 0.00000
 - 2d3. Preload : 0.00000
- 3. Suspensions 1
 - 3a. Wheel Rate : 50000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.02645
 - 3b2. Rate : 650.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.11855
 - 3c2. Rate : 650.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 2
 - 3a. Wheel Rate : 50000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.02645
 - 3b2. Rate : 650.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.11855
 - 3c2. Rate : 650.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 3
 - 3a. Wheel Rate : 42750.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.04487
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.04513
 - 3c2. Rate : 400.00000
- 3. Suspensions 4
 - 3a. Wheel Rate : 42750.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.04487
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.04513
 - 3c2. Rate : 400.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 4. Dampers 1
 - 4a. Slow Bump : 6000.00000
 - 4b. Fast Bump : 1500.00000
 - 4c. Slow Rebound : 8000.00000
 - 4e. Fast Rebound : 2300.00000
- 4. Dampers 2
 - 4a. Slow Bump : 6000.00000
 - 4b. Fast Bump : 1500.00000
 - 4c. Slow Rebound : 8000.00000

- L 4e. Fast Rebound : 2300.00000
- 4. Dampers 3
 - | 4a. Slow Bump : 5000.00000
 - | 4b. Fast Bump : 1200.00000
 - | 4c. Slow Rebound : 7000.00000
 - | 4e. Fast Rebound : 1900.00000
- 4. Dampers 4
 - | 4a. Slow Bump : 5000.00000
 - | 4b. Fast Bump : 1200.00000
 - | 4c. Slow Rebound : 7000.00000
 - | 4e. Fast Rebound : 1900.00000
- 5. Alignements 1
 - | 5a. Pressure : 28.00000
 - | 5b. Camber : -1.00000
 - | 5c. Toe : -0.03000
 - | 5d. Caster : -0.04510
 - | 5e. Static Camber : -1.69155
 - | 5f. Toe Out Linear : 0.00098
 - | 5g. Compound : 0.00000
- 5. Alignements 2
 - | 5a. Pressure : 28.00000
 - | 5b. Camber : -1.00000
 - | 5c. Toe : -0.03000
 - | 5d. Caster : -0.04510
 - | 5e. Static Camber : -1.76592
 - | 5f. Toe Out Linear : 0.00095
 - | 5g. Compound : 0.00000
- 5. Alignements 3
 - | 5a. Pressure : 28.00000
 - | 5b. Camber : -0.60000
 - | 5c. Toe : 0.05000
 - | 5d. Caster : 0.00000
 - | 5e. Static Camber : -0.66797
 - | 5f. Toe Out Linear : 0.00022
 - | 5g. Compound : 0.00000
- 5. Alignements 4
 - | 5a. Pressure : 28.00000
 - | 5b. Camber : -0.60000
 - | 5c. Toe : 0.05000
 - | 5d. Caster : 0.00000
 - | 5e. Static Camber : -0.75145
 - | 5f. Toe Out Linear : 0.00022
 - | 5g. Compound : 0.00000
- 6. Electronics
 - | 6a. Tc1 : 0.00000
 - | 6b. Tc2 : 0.00000
 - | 6c. Abs : 1.00000
 - | 6d. Esc : 0.00000
 - | 6e. Ebb : 0.000
 - | 6f. Engine Map : 0.00000
 - | 6g. Telemetry Laps to Record : 0.00000
 - | 6h. Turbo Boost Lv : 0.00000
 - | 6i. Ers Deployment Map : 0.000
 - | 6j. Ers Recharge Lv : 0.000
 - | 6k. Ers Heat Charging : 0.000
- 7. Aero

- 7a. Collar Positions Mm 1 : 112.59116
- 7a. Collar Positions Mm 2 : 107.20738
- 7a. Collar Positions Mm 3 : 76.21446
- 7a. Collar Positions Mm 4 : 75.33346
- 7b. Front Target Height : 200.00000
- 7c. Rear Target Height : 200.00000
- 7d. Front Wing Angle : 0.00000
- 7e. Rear Wing Angle : 0.00000
- 8. Fuel Strategy
 - 8a. Fuel : 30.00000
- 9. Final State Name :
- ks_audi_sport_quattro_preset_sq_mech_1_preset_sq_visual_1
- 10. Version : None
- 11. Is Setup Shared : false

Alfa Romeo Junior

- 1. Import Setup
- 2. Mechanical Balance
 - 2a. Arbs 1 : 40000.00000
 - 2a. Arbs 1 : 38000.00000
 - 2b. Steer Ratio : -14.60000
 - 2c. Brakes
 - 2c1. Front Bias : 80.00000
 - 2c2. Torque Multiplier : 100.00000
 - 2c3. Brake Ducts : None
 - 2d. Differential
 - 2d1. Power : 0.00000
 - 2d2. Coast : 0.00000
 - 2d3. Preload : 0.00000
- 3. Suspensions 1
 - 3a. Wheel Rate : 49000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.01194
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.07306
 - 3c2. Rate : 400.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 2
 - 3a. Wheel Rate : 49000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.01194
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.07306
 - 3c2. Rate : 400.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 3
 - 3a. Wheel Rate : 45000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : -0.00872

- └ 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - └ 3c1. Range : 0.0372
 - └ 3c2. Rate : 500.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 4
 - 3a. Wheel Rate : 45000.00000
 - 3b. Bump Stop Up
 - └ 3b1. Range : -0.00872
 - └ 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - └ 3c1. Range : 0.09372
 - └ 3c2. Rate : 500.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 4. Dampers 1
 - 4a. Slow Bump : 8200.00000
 - 4b. Fast Bump : 3000.00000
 - 4c. Slow Rebound : 10500.00000
 - 4e. Fast Rebound : 3300.00000
- 4. Dampers 2
 - 4a. Slow Bump : 8200.00000
 - 4b. Fast Bump : 3000.00000
 - 4c. Slow Rebound : 10500.00000
 - 4e. Fast Rebound : 3300.00000
- 4. Dampers 3
 - 4a. Slow Bump : 5000.00000
 - 4b. Fast Bump : 3500.00000
 - 4c. Slow Rebound : 6500.00000
 - 4e. Fast Rebound : 3300.00000
- 4. Dampers 4
 - 4a. Slow Bump : 5000.00000
 - 4b. Fast Bump : 3500.00000
 - 4c. Slow Rebound : 6500.00000
 - 4e. Fast Rebound : 3300.00000
- 5. Alignements 1
 - 5a. Pressure : 26.00000
 - 5b. Camber : -1.30000
 - 5c. Toe : -0.05000
 - 5d. Caster : -0.04980
 - 5e. Static Camber : -1.57347
 - 5f. Toe Out Linear : 0.00018
 - 5g. Compound : 0.00000
- 5. Alignements 2
 - 5a. Pressure : 26.00000
 - 5b. Camber : -1.30000
 - 5c. Toe : -0.05000
 - 5d. Caster : -0.04980
 - 5e. Static Camber : -1.56779
 - 5f. Toe Out Linear : 0.00016
 - 5g. Compound : 0.00000
- 5. Alignements 3
 - 5a. Pressure : 27.00000
 - 5b. Camber : -1.40000
 - 5c. Toe : 0.05000

- 5d. Caster : -0.12000
- 5e. Static Camber : -1.39969
- 5f. Toe Out Linear : -0.00142
- 5g. Compound : 0.00000
- 5. Alignements 4
 - 5a. Pressure : 27.00000
 - 5b. Camber : -1.40000
 - 5c. Toe : 0.05000
 - 5d. Caster : -0.12000
 - 5e. Static Camber : -1.39560
 - 5f. Toe Out Linear : -0.00143
 - 5g. Compound : 0.00000
- 6. Electronics
 - 6a. Tc1 : 1.00000
 - 6b. Tc2 : 1.00000
 - 6c. Abs : 1.00000
 - 6d. Esc : 0.00000
 - 6e. Ebb : 0.000
 - 6f. Engine Map : 0.00000
 - 6g. Telemetry Laps to Record : 0.00000
 - 6h. Turbo Boost Lv : 0.00000
 - 6i. Ers Deployment Map : 0.000
 - 6j. Ers Recharge Lv : 0.000
 - 6k. Ers Heat Charging : 0.000
- 7. Aero
 - 7a. Collar Positions Mm 1 : 102.44676
 - 7a. Collar Positions Mm 2 : 102.44756
 - 7a. Collar Positions Mm 3 : 105.31527
 - 7a. Collar Positions Mm 4 : 105.31618
 - 7b. Front Target Height : 185.00000
 - 7c. Rear Target Height : 190.00000
 - 7d. Front Wing Angle : 0.00000
 - 7e. Rear Wing Angle : 0.00000
- 8. Fuel Strategy
 - 8a. Fuel : 3.00000
- 9. Final State Name : ks_alfa_romeo_junior_preset_mln_mech_1_preset_mln_visual_1
- 10. Version : 0
- 11. Is Setup Shared : false

Ferrari 488 Challenge Evo [preset : safe_1]

- 1. Import Setup : None
- 2. Mechanical Balance
 - 2a. Arbs 1 : 34000.00000
 - 2a. Arbs 2 : 17000.00000
 - 2b. Steer Ratio : 15.00000
 - 2c. Brakes
 - 2c1. Front Bias : 64.00000
 - 2c2. Torque Multiplier : 100.00000
 - 2c3. Brake Ducts : None
 - 2d. Differential
 - 2d1. Power : 0.00000

- | | 2d2. Coast : 0.30000
- | | 2d3. Preload : 10.00000
- 3. Suspensions 1
 - | 3a. Wheel Rate : 160000.00000
 - | 3b. Bump Stop Up
 - | 3b1. Range : 0.02748
 - | 3b2. Rate : 500.00000
 - | 3c. Bump Stop Down
 - | 3c1. Range : 0.01252
 - | 3c2. Rate : 300.00000
 - | 3d. Helper Rate : 0.00000
 - | 3e. Helper Range : 0.00000
- 3. Suspensions 2
 - | 3a. Wheel Rate : 160000.00000
 - | 3b. Bump Stop Up
 - | 3b1. Range : 0.02748
 - | 3b2. Rate : 500.00000
 - | 3c. Bump Stop Down
 - | 3c1. Range : 0.01252
 - | 3c2. Rate : 300.00000
 - | 3d. Helper Rate : 0.00000
 - | 3e. Helper Range : 0.00000
- 3. Suspensions 3
 - | 3a. Wheel Rate : 150000.00000
 - | 3b. Bump Stop Up
 - | 3b1. Range : 0.01958
 - | 3b2. Rate : 300.00000
 - | 3c. Bump Stop Down
 - | 3c1. Range : 0.04042
 - | 3c2. Rate : 300.00000
 - | 3d. Helper Rate : 0.00000
 - | 3e. Helper Range : 0.00000
- 3. Suspensions 4
 - | 3a. Wheel Rate : 150000.00000
 - | 3b. Bump Stop Up
 - | 3b1. Range : 0.01958
 - | 3b2. Rate : 300.00000
 - | 3c. Bump Stop Down
 - | 3c1. Range : 0.04042
 - | 3c2. Rate : 300.00000
 - | 3d. Helper Rate : 0.00000
 - | 3e. Helper Range : 0.00000
- 4. Dampers 1
 - | 4a. Slow Bump : 8.00000
 - | 4b. Fast Bump : 5000.00000
 - | 4c. Slow Rebound : 9.00000
 - | 4e. Fast Rebound : 5000.00000
- 4. Dampers 2
 - | 4a. Slow Bump : 8.00000
 - | 4b. Fast Bump : 5000.00000
 - | 4c. Slow Rebound : 9.00000
 - | 4e. Fast Rebound : 5000.00000
- 4. Dampers 3
 - | 4a. Slow Bump : 9.00000
 - | 4b. Fast Bump : 5000.00000
 - | 4c. Slow Rebound : 10.00000

- L 4e. Fast Rebound : 5000.00000
- 4. Dampers 4
 - 4a. Slow Bump : 9.00000
 - 4b. Fast Bump : 5000.00000
 - 4c. Slow Rebound : 10.00000
 - 4e. Fast Rebound : 5000.00000
- 5. Alignements 1
 - 5a. Pressure : 24.00000
 - 5b. Camber : -3.50000
 - 5c. Toe : -0.10000
 - 5d. Caster : -0.07000
 - 5e. Static Camber : -3.09347
 - 5f. Toe Out Linear : 0.00055
 - 5g. Compound : 0.00000
- 5. Alignements 2
 - 5a. Pressure : 24.00000
 - 5b. Camber : -3.50000
 - 5c. Toe : -0.10000
 - 5d. Caster : -0.07000
 - 5e. Static Camber : -3.09532
 - 5f. Toe Out Linear : 0.00056
 - 5g. Compound : 0.00000
- 5. Alignements 3
 - 5a. Pressure : 23.50000
 - 5b. Camber : -3.00000
 - 5c. Toe : 0.10000
 - 5d. Caster : 0.00000
 - 5e. Static Camber : -3.21468
 - 5f. Toe Out Linear : -0.00023
 - 5g. Compound : 0.00000
- 5. Alignements 4
 - 5a. Pressure : 23.50000
 - 5b. Camber : -3.00000
 - 5c. Toe : 0.10000
 - 5d. Caster : 0.00000
 - 5e. Static Camber : -3.21772
 - 5f. Toe Out Linear : -0.00023
 - 5g. Compound : 0.00000
- 6. Electronics
 - 6a. Tc1 : 3.00000
 - 6b. Tc2 : 3.00000
 - 6c. Abs : 3.00000
 - 6d. Esc : 0.00000
 - 6e. Ebb : 0.000
 - 6f. Engine Map : 0.00000
 - 6g. Telemetry Laps to Record : 5.00000
 - 6h. Turbo Boost Lv : 1.00000
 - 6i. Ers Deployment Map : 0.000
 - 6j. Ers Recharge Lv : 0.000
 - 6k. Ers Heat Charging : 0.000
- 7. Aero
 - 7a. Collar Positions Mm 1 : 0.88076
 - 7a. Collar Positions Mm 2 : 0.88185
 - 7a. Collar Positions Mm 3 : 29.68013
 - 7a. Collar Positions Mm 4 : 29.68015
 - 7b. Front Target Height : 75.00000

```
| | 7c. Rear Target Height : 90.00000
| | 7d. Front Wing Angle : 0.00000
| | 7e. Rear Wing Angle : 12.00000
| 8. Fuel Strategy
| | 8a. Fuel : 30.00000
| 9. Final State Name :
ks_ferrari_488_challenge_evo_preset_f488ce_mech_1_preset_f488ce_visual_1
| 10. Version : 0
| 11. Is Setup Shared : false
```

6. Car Setup Limits [.carsetuplimits]

A. Description

I. General Description

The `CAR SETUP LIMITS` asset is the structural and regulatory counterpart to the vehicle configuration file (Car Setup). While a setup file dictates the specific mechanical settings active on a vehicle at any given moment, the `CAR SETUP LIMITS` file defines the rigid boundaries, rules, and user interface properties governing those adjustments.

It serves as the master blueprint for the garage interface and simulation engine. It specifies which mechanical components can be altered by the user, the absolute minimum and maximum allowable physics values, the precise increments (steps) for tuning, and how data fields should be formatted, displayed, converted, or masked within the user interface.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters configured within the `CAR SETUP LIMITS` file act as a critical control filter over a vehicle's performance envelope and user accessibility:

- **Preservation of Vehicle Identity (Road vs. Race):** It prevents unrealistic modifications, ensuring a standard production street car cannot be fitted with fully adjustable racing spoilers, bespoke motorsport suspension systems, or pure open-wheel steering ratios
- **Exploration of the Performance Envelope:** It defines how far a race engineer can push tuning boundaries. If the minimum front ride height threshold is restricted too high, the vehicle will be fundamentally blocked from exploiting ground-effect underbody aerodynamics.
- **Garage User Interface and User Experience:** By setting precise incremental adjustments, it determines whether a value changes smoothly millimeter by millimeter or clicks through coarse mechanical steps. It also translates raw backend physics numbers into legible cockpit metrics (e.g., clicks, degrees, bars, or PSI).
- **Performance Balancing & Fair Play (BOP):** In competitive multiplayer and esports environments, locking down or narrowing these limits is the primary method used to standardize hardware components across different vehicle models, ensuring a balanced grid without rewriting core chassis physics meshes..

III. Key Architecture & Data Fields Explained

The schema for this asset mirrors the organizational layout of the standard setup file (Tyres, Aerodynamics, Suspension, Dampers, Drivetrain). However, instead of storing a single static setting, every individual parameter is transformed into a **nested object** containing regulatory and interface variables:

1 - THE ANATOMY OF A PARAMETER LIMIT OBJECT

Every adjustable mechanical setting utilizes the following structural parameters:

- **Min / Max:** The absolute lower and upper physical boundaries allowed by the simulation engine for this component.
- **Step:** The exact mechanical increment or decrement applied per single "click" inside the garage menu.
- **Is Modifiable:** A boolean flag (`TRUE/FALSE`). When set to `FALSE`, the adjustment slider is locked and greyed out in the user interface.
- **Hide Value:** A toggle allowing developers to obscure the raw physical value from the user screen (often used in high-level motorsport to simulate proprietary team data secrecy).
- **Unit:** The textual unit label rendered on screen to contextualize the value (e.g., `MM`, `NM`, `Hz`, `PSI`, `DEG`).
- **Fractional Digits:** Specifies the decimal point precision displayed on the user interface screen.
- **Lut (Look-Up Table):** Maps the adjustment steps to an external curve, allowing linear menu clicks to translate into non-linear physical changes.

2 - CORE CATEGORIES OF APPLICATION

- **Tyres:** Enforces minimum safe cold inflation pressures to prevent tire carcass deflation/delamination, alongside maximum ranges for wheel alignment geometry (camber and toe).
- **Aerodynamics:** Locks down the operational angles of front splitters and rear wings (often constrained to 1-degree or 0.5-degree steps) and limits the safe compression clearance of the underbody floor.
- **Suspension Geometry & Rates:** Dictates the stiffest and softest allowable coil spring options (`WHEEL RATE`) and limits the structural thickness options for front and rear anti-roll bars (`ARBS`).
- **Dampers:** Defines the total number of valving adjustments ("clicks") available for shock absorber damping across low-speed and high-speed compression (`BUMP`) and extension (`REBOUND`) cycles.
- **Drivetrain & Differential:** Sets boundaries on differential clutch locking percentages under acceleration (`POWER`), deceleration (`COAST`), and static resistance (`PRELOAD`).

IV. Interpretation of Setup Limits Strategies

By reading a vehicle's specific limits file, developers and players can immediately identify its structural classification and engineering intent:

- **The Stock Production Profile:** In standard road vehicles, the vast majority of `Is Modifiable` flags are hard-locked to `FALSE`. Parameters like spring rates, dampers, differential

behaviors, and aerodynamics are frozen to factory settings. Only basic tyre pressures and minor alignment tolerances remain adjustable for track days.

- **The Pure Motorsport Profile (GT3 / Prototypes / Open-Wheel):** Almost all mechanical and aerodynamic flags are set to `TRUE`. However, the window between `MIN` and `MAX` is deliberately narrow, engineered strictly to keep the car operating within its designed aerodynamic or suspension kinematic sweet spot. Increments match real-world race hardware (e.g., fixed damper click values).
- **The BOP (Balance of Performance) Regulation Strategy:** Rather than altering a vehicle's engine power or weight mesh directly, series organizers can adjust this file to restrict tuning freedom. For instance, raising the `MIN` value of a rear wing angle forces a car to run higher minimum downforce—intentionally increasing straight-line drag to slow it down on high-speed circuits.

B. Schema

```
1. Car Data : string - path
2. Mechanical Balance : object
  2a. Arbs [x] : object | can have multiple Arbs
    2a1. Step : float
    2a2. Min : float
    2a3. Max : float
    2a4. Lut : string - path
    2a5. Is Modifiable : boolean
    2a6. Hide Value : boolean
    2a7. Is Negative : boolean
    2a8. Unit : string
    2a9. Fractional Digits : integer
    2a10. Treat As Boolean : boolean
  2b. Steer Ratio : object
    2a1. Step : float
    2a2. Min : float
    2a3. Max : float
    2a4. Lut : string - path
    2a5. Is Modifiable : boolean
    2a6. Hide Value : boolean
    2a7. Is Negative : boolean
    2a8. Unit : string
    2a9. Fractional Digits : integer
    2a10. Treat As Boolean : boolean
  2c. Brakes : object
    2c1. Front Bias : object
      2a1. Step : float
      2a2. Min : float
      2a3. Max : float
      2a4. Lut : string - path
      2a5. Is Modifiable : boolean
      2a6. Hide Value : boolean
      2a7. Is Negative : boolean
      2a8. Unit : string
      2a9. Fractional Digits : integer
      2a10. Treat As Boolean : boolean
    2c2. Torque Multiplier : object
```


- 2a1. Step : float
- 2a2. Min : float
- 2a3. Max : float
- 2a4. Lut : string – path
- 2a5. Is Modifiable : boolean
- 2a6. Hide Value : boolean
- 2a7. Is Negative : boolean
- 2a8. Unit : string
- 2a9. Fractional Digits : integer
- 2a10. Treat As Boolean : boolean
- 2c3. Brake Ducts [x] : object | can have multiple Brake Ducts
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string – path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 2d. Differential : object
 - 2d1. Power : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string – path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 2d2. Coast : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string – path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 2d3. Preload : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string – path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 3. Suspensions [x] : object | can have multiple Suspensions

- 3a. Wheel Rate : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 3b. Bump Stop Up : object
 - 3b1. Range : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 3b2. Rate : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 3c. Bump Stop Down : object
 - 3b1. Range : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 3b2. Range : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer

- 2a10. Treat As Boolean : **boolean**
- 3d. Helper Rate : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string - path**
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**
 - 2a10. Treat As Boolean : **boolean**
- 3e. Helper Range : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string - path**
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**
 - 2a10. Treat As Boolean : **boolean**
- 4. Dampers [x] : **object** | can have multiple Dampers
 - 4a. Slow Bump : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string - path**
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**
 - 2a10. Treat As Boolean : **boolean**
 - 4b. Fast Bump : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string - path**
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**
 - 2a10. Treat As Boolean : **boolean**
 - 4c. Slow Rebound : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string - path**
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**

- └ 2a10. Treat As Boolean : **boolean**
- 4d. Fast Rebound : **object**
 - └ 2a1. Step : **float**
 - └ 2a2. Min : **float**
 - └ 2a3. Max : **float**
 - └ 2a4. Lut : **string - path**
 - └ 2a5. Is Modifiable : **boolean**
 - └ 2a6. Hide Value : **boolean**
 - └ 2a7. Is Negative : **boolean**
 - └ 2a8. Unit : **string**
 - └ 2a9. Fractional Digits : **integer**
 - └ 2a10. Treat As Boolean : **boolean**
- 5. Alignments **[x]** : **object** | can have multiple Alignments
 - 5a. Pressure : **object**
 - └ 2a1. Step : **float**
 - └ 2a2. Min : **float**
 - └ 2a3. Max : **float**
 - └ 2a4. Lut : **string - path**
 - └ 2a5. Is Modifiable : **boolean**
 - └ 2a6. Hide Value : **boolean**
 - └ 2a7. Is Negative : **boolean**
 - └ 2a8. Unit : **string**
 - └ 2a9. Fractional Digits : **integer**
 - └ 2a10. Treat As Boolean : **boolean**
 - 5b. Camber : **object**
 - └ 2a1. Step : **float**
 - └ 2a2. Min : **float**
 - └ 2a3. Max : **float**
 - └ 2a4. Lut : **string - path**
 - └ 2a5. Is Modifiable : **boolean**
 - └ 2a6. Hide Value : **boolean**
 - └ 2a7. Is Negative : **boolean**
 - └ 2a8. Unit : **string**
 - └ 2a9. Fractional Digits : **integer**
 - └ 2a10. Treat As Boolean : **boolean**
 - 5c. Toe : **object**
 - └ 2a1. Step : **float**
 - └ 2a2. Min : **float**
 - └ 2a3. Max : **float**
 - └ 2a4. Lut : **string - path**
 - └ 2a5. Is Modifiable : **boolean**
 - └ 2a6. Hide Value : **boolean**
 - └ 2a7. Is Negative : **boolean**
 - └ 2a8. Unit : **string**
 - └ 2a9. Fractional Digits : **integer**
 - └ 2a10. Treat As Boolean : **boolean**
 - 5d. Caster : **object**
 - └ 2a1. Step : **float**
 - └ 2a2. Min : **float**
 - └ 2a3. Max : **float**
 - └ 2a4. Lut : **string - path**
 - └ 2a5. Is Modifiable : **boolean**
 - └ 2a6. Hide Value : **boolean**
 - └ 2a7. Is Negative : **boolean**
 - └ 2a8. Unit : **string**
 - └ 2a9. Fractional Digits : **integer**

- └─ 2a10. Treat As Boolean : **boolean**
- 5e. Static Camber : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string** - path
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 5f. Toe Out Linear : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string** - path
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 5g. Compound : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string** - path
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 6. Electronics : **object**
 - 6a. Tc1 : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string** - path
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
 - 6b. Tc2 : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string** - path
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**

- └─ 2a10. Treat As Boolean : **boolean**
- 6c. Abs : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 6d. Esc : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 6e. Ebb : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 6f. Engine Map : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 6g. Telemetry Laps To Record : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**

- 6h. Turbo Boost Lv : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean

- 6i. Ers Deployment Map : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean

- 6j. Ers Recharge Lv : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean

- 6k. Ers Heat Charging : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean

- 7. Aero : object

- 7a. Collar Positions [x] : object | can have multiple Collar

Positions

- 2a1. Step : float
- 2a2. Min : float
- 2a3. Max : float
- 2a4. Lut : string - path
- 2a5. Is Modifiable : boolean
- 2a6. Hide Value : boolean
- 2a7. Is Negative : boolean
- 2a8. Unit : string
- 2a9. Fractional Digits : integer

- └─ 2a10. Treat As Boolean : **boolean**
- 7b. Front Target Height : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 7c. Rear Target Height : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 7d. Front Wing Angle : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 7e. Rear Wing Angle : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**
 - └─ 2a10. Treat As Boolean : **boolean**
- 8. Fuel Strategy : **object**
 - 8a. Fuel : **object**
 - └─ 2a1. Step : **float**
 - └─ 2a2. Min : **float**
 - └─ 2a3. Max : **float**
 - └─ 2a4. Lut : **string - path**
 - └─ 2a5. Is Modifiable : **boolean**
 - └─ 2a6. Hide Value : **boolean**
 - └─ 2a7. Is Negative : **boolean**
 - └─ 2a8. Unit : **string**
 - └─ 2a9. Fractional Digits : **integer**

- └┐ 2a10. Treat As Boolean : **boolean**
- └ 9. Use Single Compound : **boolean**

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Car Data	None (File path)	Path to the parent .car asset this limits file is bound to; links regulatory bounds to the correct vehicle.
2.	Mechanical Balance	None (Object)	Parent block for chassis balance limit definitions: ARBs, steering, brakes, and differential.
2a.	Arbs	None (Limit object array)	Per-axle anti-roll bar limit blocks; index 1 = front, 2 = rear. Each entry contains fields 2a1–2a10.
2a1.	Step	Depends on parameter (physics unit or clicks)	Garage increment applied per adjustment click; defines the resolution of slider/stepper controls.
2a2.	Min	Depends on parameter	Absolute minimum allowable physics value for the linked setup parameter.
2a3.	Max	Depends on parameter	Absolute maximum allowable physics value for the linked setup parameter.
2a4.	Lut	None (.CURVE file path)	Optional look-up table mapping linear UI steps to non-linear physical output values.
2a5.	Is Modifiable	None (Boolean : True / False)	When false, the garage slider is locked/greyed out; parameter cannot be changed by the user.
2a6.	Hide Value	None (Boolean : True / False)	When true, obscures the raw numeric value in the UI (proprietary team data simulation).
2a7.	Is Negative	None (Boolean : True / False)	UI sign convention flag; allows negative values to display/be edited correctly (camber, toe).
2a8.	Unit	None (String)	Display unit label rendered in the garage UI (e.g., mm, deg, PSI, bar, Nm, clicks).
2a9.	Fractional Digits	None (Integer)	Number of decimal places shown in the garage UI for this parameter.
2a10.	Treat As Boolean	None (Boolean : True / False)	Renders the parameter as an on/off toggle instead of a numeric slider when true.

ID	Name	Unit of Measurement	Description
2b.	Steer Ratio	None (Limit object)	Limit definition for the adjustable steering ratio parameter in Car Setup.
2c.	Brakes	None (Object)	Brake setup limits sub-block within Mechanical Balance.
2c1.	Front Bias	None (Limit object)	Limit definition for brake balance front percentage.
2c2.	Torque Multiplier	None (Limit object)	Limit definition for global braking torque scaling.
2c3.	Brake Ducts	None (Limit object array)	Per-duct brake cooling limit blocks; multiple entries for front/ rear duct positions.
2d.	Differential	None (Object)	Differential setup limits sub-block within Mechanical Balance.
2d1.	Power	None (Limit object)	Limit definition for on-throttle differential lock intensity.
2d2.	Coast	None (Limit object)	Limit definition for coast/off-throttle differential lock intensity.
2d3.	Preload	None (Limit object)	Limit definition for static differential preload torque.
3.	Suspensions	None (Object array, per wheel)	Per-corner suspension limit blocks; indices 1–4 = LF, RF, LR, RR.
3a.	Wheel Rate	None (Limit object)	Limit definition for per-corner wheel-rate spring stiffness.
3b.	Bump Stop Up	None (Object)	Upper bump-stop limit sub-block within each Suspensions entry.
3b1.	Range	None (Limit object)	Bump-stop travel range limit. Under 3b Bump Stop Up; ID reused under 3c Bump Stop Down.
3b2.	Rate	None (Limit object)	Bump-stop stiffness/force rate limit. Under 3b Bump Stop Up; ID reused under 3c Bump Stop Down.
3c.	Bump Stop Down	None (Object)	Lower bump-stop limit sub-block; contains 3b1 (Range) and 3b2 (Rate) limit objects per schema.
3d.	Helper Rate	None (Limit object)	Limit definition for helper/tender spring stiffness.
3e.	Helper Range	None (Limit object)	Limit definition for helper spring active travel range.
4.	Dampers	None (Object array, per wheel)	Per-corner damper limit blocks; indices 1–4 = LF, RF, LR, RR.

ID	Name	Unit of Measurement	Description
4a.	Slow Bump	None (Limit object)	Limit definition for slow-speed compression damper clicks/values.
4b.	Fast Bump	None (Limit object)	Limit definition for fast-speed compression damper clicks/values.
4c.	Slow Rebound	None (Limit object)	Limit definition for slow-speed rebound damper clicks/values.
4d.	Fast Rebound	None (Limit object)	Limit definition for fast-speed rebound damper clicks/values (4d here; Car Setup uses 4e).
5.	Alignments	None (Object array, per wheel)	Per-corner alignment and tyre limit blocks; indices 1–4 = LF, RF, LR, RR.
5a.	Pressure	None (Limit object)	Limit definition for cold tyre inflation pressure.
5b.	Camber	None (Limit object)	Limit definition for adjustable camber angle.
5c.	Toe	None (Limit object)	Limit definition for toe angle.
5d.	Caster	None (Limit object)	Limit definition for caster angle.
5e.	Static Camber	None (Limit object)	Limit definition for computed/static camber display or override bounds.
5f.	Toe Out Linear	None (Limit object)	Limit definition for linear toe compliance curve coefficient.
5g.	Compound	None (Limit object)	Limit definition for tyre compound selection index bounds.
6.	Electronics	None (Object)	Driver-aid and powertrain map limit definitions stored in the garage UI.
6a.	Tc1	None (Limit object)	Limit definition for primary Traction Control map level selection.
6b.	Tc2	None (Limit object)	Limit definition for secondary TC map level.
6c.	Abs	None (Limit object)	Limit definition for ABS map level selection.
6d.	Esc	None (Limit object)	Limit definition for ESP/stability control map level.
6e.	Ebb	None (Limit object)	Limit definition for Electronic Brake Balance level.
6f.	Engine Map	None (Limit object)	Limit definition for engine power map index selection.
6g.	Telemetry Laps To Record	None (Limit object)	Limit definition for onboard telemetry lap recording count.

ID	Name	Unit of Measurement	Description
6h.	Turbo Boost Lv	None (Limit object)	Limit definition for turbo boost level preset.
6i.	Ers Deployment Map	None (Limit object)	Limit definition for ERS deployment map index.
6j.	Ers Recharge Lv	None (Limit object)	Limit definition for ERS recharge aggressiveness level.
6k.	Ers Heat Charging	None (Limit object)	Limit definition for MGU-H heat charging level.
7.	Aero	None (Object)	Aerodynamic setup limit block: ride heights, collar positions, and wing angles.
7a.	Collar Positions	None (Limit object array)	Per-corner spring collar/ride-height limit blocks under 7. Aero; indices 1–4 = LF, RF, LR, RR.
7b.	Front Target Height	None (Limit object)	Limit definition for front target ride height.
7c.	Rear Target Height	None (Limit object)	Limit definition for rear target ride height.
7d.	Front Wing Angle	None (Limit object)	Limit definition for front aero device angle adjustment range.
7e.	Rear Wing Angle	None (Limit object)	Limit definition for rear wing angle adjustment range.
8.	Fuel Strategy	None (Object)	Fuel load planning limit block.
8a.	Fuel	None (Limit object)	Fuel load volume limit under 8. Fuel Strategy.
9.	Use Single Compound	None (Boolean : True / False)	Forces all four corners to share a single tyre compound selection when true.

D. Example data

I. Chosen Car Engine for Example

- BMW M4 CSL (slug : ks_bmw_m4_csl)
- Lamborghini Countach (slug : ks_lamborghini_countach)

II. Example

BMW M4 CSL

```

1. Car Data : None
2. Mechanical Balance
  2a. Arbs 1
    2a1. Step : 1.00000

```

- 2a2. Min : 30000.00000
- 2a3. Max : 30000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 2a. Arbs 2
 - 2a1. Step : 1.00000
 - 2a2. Min : 11000.00000
 - 2a3. Max : 11000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 2b. Steer Ratio
 - 2a1. Step : 1.00000
 - 2a2. Min : 15.00000
 - 2a3. Max : 15.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : None
 - 2a10. Treat As Boolean : 0
- 2c. Brakes
 - 2c1. Front Bias
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 100.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : None
 - 2a10. Treat As Boolean : false
 - 2c2. Torque Multiplier
 - 2a1. Step : 1.00000
 - 2a2. Min : 100.00000
 - 2a3. Max : 100.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 2d. Differential
 - 2d1. Power

- 2a1. Step : 1.00000
- 2a2. Min : 0.25000
- 2a3. Max : 0.25000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

2d2. Coast

- 2a1. Step : 1.00000
- 2a2. Min : 0.55000
- 2a3. Max : 0.55000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

2d3. Preload

- 2a1. Step : 1.00000
- 2a2. Min : 1.00000
- 2a3. Max : 1.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3. Suspensions 1

3a. Wheel Rate

- 2a1. Step : 1.00000
- 2a2. Min : 47000.00000
- 2a3. Max : 47000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3b. Bump Stop Up

3b1. Range

- 2a1. Step : 0.01000
- 2a2. Min : -0.09036
- 2a3. Max : 0.15964
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 600.00000
 - 2a3. Max : 600.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.07036
 - 2a3. Max : 0.07036
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 1000.00000
 - 2a3. Max : 1000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 5000.00000
 - 2a3. Max : 5000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3e. Helper Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.03000
 - 2a3. Max : 0.03000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false

3. Suspensions 2

3a. Wheel Rate

- 2a1. Step : 1.00000
- 2a2. Min : 47000.00000
- 2a3. Max : 47000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3b. Bump Stop Up

3b1. Range

- 2a1. Step : 0.01000
- 2a2. Min : -0.09036
- 2a3. Max : 0.15964
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3b2. Rate

- 2a1. Step : 1.00000
- 2a2. Min : 600.00000
- 2a3. Max : 600.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3c. Bump Stop Down

3b1. Range

- 2a1. Step : 1.00000
- 2a2. Min : 0.07036
- 2a3. Max : 0.07036
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3b2. Rate

- 2a1. Step : 1.00000
- 2a2. Min : 1000.00000
- 2a3. Max : 1000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None

- | | 2a9. Fractional Digits : 0
- | | 2a10. Treat As Boolean : false

- 3d. Helper Rate

- | 2a1. Step : 1.00000
- | 2a2. Min : 5000.00000
- | 2a3. Max : 5000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3e. Helper Range

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.03000
- | 2a3. Max : 0.03000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3. Suspensions 3

- 3a. Wheel Rate

- | 2a1. Step : 1.00000
- | 2a2. Min : 33000.00000
- | 2a3. Max : 33000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3b. Bump Stop Up

- 3b1. Range

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.13069
- | 2a3. Max : 0.13069
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3b2. Rate

- | 2a1. Step : 1.00000
- | 2a2. Min : 500.00000
- | 2a3. Max : 500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false

- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3c. Bump Stop Down

- 3b1. Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.10431
 - 2a3. Max : 0.10431
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false

- 3b2. Rate

- 2a1. Step : 1.00000
- 2a2. Min : 1000.00000
- 2a3. Max : 1000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3d. Helper Rate

- 2a1. Step : 1.00000
- 2a2. Min : 5000.00000
- 2a3. Max : 5000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3e. Helper Range

- 2a1. Step : 1.00000
- 2a2. Min : 0.03000
- 2a3. Max : 0.03000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3. Suspensions 4

- 3a. Wheel Rate

- 2a1. Step : 1.00000
- 2a2. Min : 33000.00000
- 2a3. Max : 33000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false

- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3b. Bump Stop Up
 - 3b1. Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.13069
 - 2a3. Max : 0.13069
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 500.00000
 - 2a3. Max : 500.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.10431
 - 2a3. Max : 0.10431
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 1000.00000
 - 2a3. Max : 1000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 5000.00000
 - 2a3. Max : 5000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false

- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3e. Helper Range

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.03000
- | 2a3. Max : 0.03000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 1

- | 4a. Slow Bump

- | 2a1. Step : 1.00000
- | 2a2. Min : 5500.00000
- | 2a3. Max : 5500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 4b. Fast Bump

- | 2a1. Step : 1.00000
- | 2a2. Min : 1500.00000
- | 2a3. Max : 1500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 4c. Slow Rebound

- | 2a1. Step : 1.00000
- | 2a2. Min : 7500.00000
- | 2a3. Max : 7500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 4d. Fast Rebound

- | 2a1. Step : 1.00000
- | 2a2. Min : 4750.00000
- | 2a3. Max : 4750.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false

- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 2

- 4a. Slow Bump

- | 2a1. Step : 1.00000
- | 2a2. Min : 5500.00000
- | 2a3. Max : 5500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 1.00000
- | 2a2. Min : 1500.00000
- | 2a3. Max : 1500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1.00000
- | 2a2. Min : 7500.00000
- | 2a3. Max : 7500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 1.00000
- | 2a2. Min : 4750.00000
- | 2a3. Max : 4750.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 3

- 4a. Slow Bump

- | 2a1. Step : 1.00000
- | 2a2. Min : 4500.00000
- | 2a3. Max : 4500.00000
- | 2a4. Lut : None

- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 1.00000
- | 2a2. Min : 2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1.00000
- | 2a2. Min : 6500.00000
- | 2a3. Max : 6500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 1.00000
- | 2a2. Min : 4250.00000
- | 2a3. Max : 4250.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 4

- 4a. Slow Bump

- | 2a1. Step : 1.00000
- | 2a2. Min : 4500.00000
- | 2a3. Max : 4500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 1.00000
- | 2a2. Min : 2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None

- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1.00000
- | 2a2. Min : 6500.00000
- | 2a3. Max : 6500.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 1.00000
- | 2a2. Min : 4250.00000
- | 2a3. Max : 4250.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 1

- 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -2.50000
- | 2a3. Max : -1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.15000
- | 2a3. Max : 0.20000
- | 2a4. Lut : None

- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5d. Caster

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.09195
- | 2a3. Max : 0.02195
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5e. Static Camber

- | 2a1. Step : 1.00000
- | 2a2. Min : -1.50000
- | 2a3. Max : -1.50000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00050
- | 2a3. Max : 0.00050
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 2.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 2

- 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None

- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -2.50000
- | 2a3. Max : -1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.15000
- | 2a3. Max : 0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5d. Caster

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.09195
- | 2a3. Max : 0.02195
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5e. Static Camber

- | 2a1. Step : 1.00000
- | 2a2. Min : -1.50000
- | 2a3. Max : -1.50000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00050
- | 2a3. Max : 0.00050
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false

- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 2.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 3

- | 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -3.00000
- | 2a3. Max : -0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.15000
- | 2a3. Max : 0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5d. Caster

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false

- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5e. Static Camber

- | 2a1. Step : 1.00000
- | 2a2. Min : -2.00000
- | 2a3. Max : -2.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00020
- | 2a3. Max : 0.00020
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 2.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 4

- | 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -3.00000
- | 2a3. Max : -0.50000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true

- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.15000
- | 2a3. Max : -0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5d. Caster

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5e. Static Camber

- | 2a1. Step : 1.00000
- | 2a2. Min : -2.00000
- | 2a3. Max : -2.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5f. Toe Out Linear

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00020
- | 2a3. Max : 0.00020
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 2.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false

- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6. Electronics

6a. Tc1

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 10.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6b. Tc2

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6c. Abs

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6d. Esc

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 2.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6e. Ebb

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false

- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6f. Engine Map

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6g. Telemetry Laps To Record

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 99.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6h. Turbo Boost Lv

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6i. Ers Deployment Map

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6j. Ers Recharge Lv

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false

- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6k. Ers Heat Charging

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 7. Aero

- | 7a. Collar Positions 1

- | 2a1. Step : 1.00000
- | 2a2. Min : 140.00000
- | 2a3. Max : 140.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 7a. Collar Positions 2

- | 2a1. Step : 1.00000
- | 2a2. Min : 140.00000
- | 2a3. Max : 140.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 7a. Collar Positions 3

- | 2a1. Step : 1.00000
- | 2a2. Min : 70.00000
- | 2a3. Max : 70.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 7a. Collar Positions 4

- | 2a1. Step : 1.00000
- | 2a2. Min : 70.00000
- | 2a3. Max : 70.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false

- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false
- 7b. Front Target Height
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 50.00000
 - | 2a3. Max : 200.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7c. Rear Target Height
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 50.00000
 - | 2a3. Max : 200.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7d. Front Wing Angle
 - | 2a1. Step : 0.00000
 - | 2a2. Min : 0.00000
 - | 2a3. Max : 0.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7e. Rear Wing Angle
 - | 2a1. Step : 0.00000
 - | 2a2. Min : 0.00000
 - | 2a3. Max : 0.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 8. Fuel Strategy
 - 8a. Fuel
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 2.00000
 - | 2a3. Max : 59.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false

- | | | 2a8. Unit : None
- | | | 2a9. Fractional Digits : 0
- | | | 2a10. Treat As Boolean : false
- | 9. Use Single Compound : false

Lamborghini Countach

- | 1. Car Data : None
- | 2. Mechanical Balance
 - | 2a. Arbs 1
 - | 2a1. Step : 4000.00000
 - | 2a2. Min : 28000.00000
 - | 2a3. Max : 68000.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
 - | 2a. Arbs 2
 - | 2a1. Step : 5000.00000
 - | 2a2. Min : 5000.00000
 - | 2a3. Max : 50000.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
 - | 2b. Steer Ratio
 - | 2a1. Step : 1.45000
 - | 2a2. Min : -14.50000
 - | 2a3. Max : -14.50000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : None
 - | 2a10. Treat As Boolean : 0
 - | 2c. Brakes
 - | 2c1. Front Bias
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 0.00000
 - | 2a3. Max : 100.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : None

- └ 2a10. Treat As Boolean : false
- 2c2. Torque Multiplier
 - └ 2a1. Step : 10.00000
 - └ 2a2. Min : 100.00000
 - └ 2a3. Max : 100.00000
 - └ 2a4. Lut : None
 - └ 2a5. Is Modifiable : false
 - └ 2a6. Hide Value : false
 - └ 2a7. Is Negative : false
 - └ 2a8. Unit : None
 - └ 2a9. Fractional Digits : 0
 - └ 2a10. Treat As Boolean : false

- 2d. Differential

- └ 2d1. Power
 - └ 2a1. Step : 0.02000
 - └ 2a2. Min : 0.20000
 - └ 2a3. Max : 0.20000
 - └ 2a4. Lut : None
 - └ 2a5. Is Modifiable : false
 - └ 2a6. Hide Value : false
 - └ 2a7. Is Negative : false
 - └ 2a8. Unit : None
 - └ 2a9. Fractional Digits : 0
 - └ 2a10. Treat As Boolean : false

- └ 2d2. Coast
 - └ 2a1. Step : 0.04000
 - └ 2a2. Min : 0.40000
 - └ 2a3. Max : 0.40000
 - └ 2a4. Lut : None
 - └ 2a5. Is Modifiable : false
 - └ 2a6. Hide Value : false
 - └ 2a7. Is Negative : false
 - └ 2a8. Unit : None
 - └ 2a9. Fractional Digits : 0
 - └ 2a10. Treat As Boolean : false

- └ 2d3. Preload
 - └ 2a1. Step : 1.00000
 - └ 2a2. Min : 10.00000
 - └ 2a3. Max : 10.00000
 - └ 2a4. Lut : None
 - └ 2a5. Is Modifiable : false
 - └ 2a6. Hide Value : false
 - └ 2a7. Is Negative : false
 - └ 2a8. Unit : None
 - └ 2a9. Fractional Digits : 0
 - └ 2a10. Treat As Boolean : false

- 3. Suspensions 1

- └ 3a. Wheel Rate
 - └ 2a1. Step : 4000.00000
 - └ 2a2. Min : 50000.00000
 - └ 2a3. Max : 100000.00000
 - └ 2a4. Lut : None
 - └ 2a5. Is Modifiable : false
 - └ 2a6. Hide Value : false
 - └ 2a7. Is Negative : false
 - └ 2a8. Unit : None

- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3b. Bump Stop Up
 - 3b1. Range
 - 2a1. Step : 0.01200
 - 2a2. Min : 0.02933
 - 2a3. Max : 0.02933
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 35.00000
 - 2a2. Min : 350.00000
 - 2a3. Max : 350.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 0.00150
 - 2a2. Min : 0.10567
 - 2a3. Max : 0.010567
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 35.00000
 - 2a2. Min : 350.00000
 - 2a3. Max : 350.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false

- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3e. Helper Range

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3. Suspensions 2

- 3a. Wheel Rate

- 2a1. Step : 4000.00000
- 2a2. Min : 50000.00000
- 2a3. Max : 100000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b. Bump Stop Up

- 3b1. Range

- 2a1. Step : 0.01200
- 2a2. Min : 0.02933
- 2a3. Max : 0.02933
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b2. Rate

- 2a1. Step : 35.00000
- 2a2. Min : 350.00000
- 2a3. Max : 350.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3c. Bump Stop Down

- 3b1. Range

- 2a1. Step : 0.00150
- 2a2. Min : 0.10567
- 2a3. Max : 0.010567
- 2a4. Lut : None
- 2a5. Is Modifiable : false

- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3b2. Rate
 - 2a1. Step : 35.00000
 - 2a2. Min : 350.00000
 - 2a3. Max : 350.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3e. Helper Range
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3. Suspensions 3
 - 3a. Wheel Rate
 - 2a1. Step : 4000.00000
 - 2a2. Min : 50000.00000
 - 2a3. Max : 100000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b. Bump Stop Up
 - 3b1. Range
 - 2a1. Step : 0.01200
 - 2a2. Min : 0.02911
 - 2a3. Max : 0.02911
 - 2a4. Lut : None

- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3b2. Rate
 - 2a1. Step : 35.00000
 - 2a2. Min : 350.00000
 - 2a3. Max : 350.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 0.00150
 - 2a2. Min : 0.10589
 - 2a3. Max : 0.10589
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 35.00000
 - 2a2. Min : 350.00000
 - 2a3. Max : 350.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3e. Helper Range
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None

- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3. Suspensions 4

- 3a. Wheel Rate

- 2a1. Step : 4000.00000
- 2a2. Min : 50000.00000
- 2a3. Max : 100000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b. Bump Stop Up

- 3b1. Range

- 2a1. Step : 0.01200
- 2a2. Min : 0.02911
- 2a3. Max : 0.02911
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b2. Rate

- 2a1. Step : 35.00000
- 2a2. Min : 350.00000
- 2a3. Max : 350.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3c. Bump Stop Down

- 3b1. Range

- 2a1. Step : 0.00150
- 2a2. Min : 0.10589
- 2a3. Max : 0.10589
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b2. Rate

- 2a1. Step : 35.00000
- 2a2. Min : 350.00000

- 2a3. Max : 350.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3d. Helper Rate

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3e. Helper Range

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 4. Dampers 1

- 4a. Slow Bump

- 2a1. Step : 1000.00000
- 2a2. Min : 10000.00000
- 2a3. Max : 10000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 4b. Fast Bump

- 2a1. Step : 100.00000
- 2a2. Min : -2000.00000
- 2a3. Max : 2000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- 2a1. Step : 1000.00000
- 2a2. Min : 10000.00000

- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 2

- 4a. Slow Bump

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000

- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 3

| 4a. Slow Bump

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

| 4b. Fast Bump

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

| 4c. Slow Rebound

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

| 4d. Fast Rebound

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 4

| 4a. Slow Bump

- | 2a1. Step : 1000.00000

- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 1

- 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : psi
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5b. Camber

- | 2a1. Step : 0.10000

- 2a2. Min : -2.50000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : Deg
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5c. Toe

- 2a1. Step : 0.01000
- 2a2. Min : -0.20000
- 2a3. Max : 0.20000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : Deg
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5d. Caster

- 2a1. Step : 0.00166
- 2a2. Min : -0.02000
- 2a3. Max : -0.02000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5e. Static Camber

- 2a1. Step : 0.18000
- 2a2. Min : -1.80000
- 2a3. Max : -1.80000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5f. Toe Out Linear

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5g. Compound

- 2a1. Step : 1.00000
- 2a2. Min : 0.00000

- | 2a3. Max : 1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5. Alignments 2

5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : psi
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -2.50000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.20000
- | 2a3. Max : 0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5d. Caster

- | 2a1. Step : 0.00166
- | 2a2. Min : -0.02000
- | 2a3. Max : -0.02000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

5e. Static Camber

- | 2a1. Step : 0.18000
- | 2a2. Min : -1.80000

- 2a3. Max : -1.80000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5g. Compound

- 2a1. Step : 1.00000
- 2a2. Min : 0.00000
- 2a3. Max : 1.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5. Alignments 3

- 5a. Pressure

- 2a1. Step : 1.00000
- 2a2. Min : 20.00000
- 2a3. Max : 35.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : psi
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5b. Camber

- 2a1. Step : 0.10000
- 2a2. Min : -2.50000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : Deg
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5c. Toe

- 2a1. Step : 0.01000
- 2a2. Min : -0.20000

- | 2a3. Max : 0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5d. Caster

- | 2a1. Step : 0.00166
- | 2a2. Min : -0.02000
- | 2a3. Max : -0.02000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5e. Static Camber

- | 2a1. Step : 0.18000
- | 2a2. Min : -1.80000
- | 2a3. Max : -1.80000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 4

| 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000

- 2a3. Max : 35.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : psi
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5b. Camber

- 2a1. Step : 0.10000
- 2a2. Min : -2.50000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : Deg
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5c. Toe

- 2a1. Step : 0.01000
- 2a2. Min : -0.20000
- 2a3. Max : 0.20000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : Deg
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5d. Caster

- 2a1. Step : 0.00166
- 2a2. Min : -0.02000
- 2a3. Max : -0.02000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5e. Static Camber

- 2a1. Step : 0.18000
- 2a2. Min : -1.80000
- 2a3. Max : -1.80000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5f. Toe Out Linear

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000

- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6. Electronics

- 6a. Tc1

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6b. Tc2

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6c. Abs

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6d. Esc

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000

- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6e. Ebb

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6f. Engine Map

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6g. Telemetry Laps To Record

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 99.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6h. Turbo Boost Lv

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6i. Ers Deployment Map

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None

- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6j. Ers Recharge Lv

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6k. Ers Heat Charging

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 7. Aero

- | 7a. Collar Positions 1

- | 2a1. Step : 7.00000
- | 2a2. Min : 70.00000
- | 2a3. Max : 70.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 7a. Collar Positions 2

- | 2a1. Step : 7.00000
- | 2a2. Min : 70.00000
- | 2a3. Max : 70.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 7a. Collar Positions 3

- | 2a1. Step : 5.00000
- | 2a2. Min : 50.00000
- | 2a3. Max : 50.00000
- | 2a4. Lut : None

- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false
- 7a. Collar Positions 4
 - | 2a1. Step : 5.00000
 - | 2a2. Min : 50.00000
 - | 2a3. Max : 50.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7b. Front Target Height
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 20.00000
 - | 2a3. Max : 180.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7c. Rear Target Height
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 20.00000
 - | 2a3. Max : 180.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7d. Front Wing Angle
 - | 2a1. Step : 0.00000
 - | 2a2. Min : 0.00000
 - | 2a3. Max : 0.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7e. Rear Wing Angle
 - | 2a1. Step : 0.00000
 - | 2a2. Min : 0.00000
 - | 2a3. Max : 0.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false

- | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
 - 8. Fuel Strategy
 - 8a. Fuel
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 1.00000
 - | 2a3. Max : 120.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 9. Use Single Compound : false

7. Car Setup Units [.carsetupunits]

A. Description

I. General Description

The **Car Setup Units** asset is the primary translation matrix that standardizes how physical metrics are rendered visually within the simulation's user interface. While other physics files handle raw mathematical processing (often utilizing raw SI metrics like Newtons, meters, or Kelvin internally), the `CAR SETUP UNITS` file dictates the exact localization string, data labels, and scaling rules for what the player sees on screen.

It ensures that complex mechanics—such as hydraulic pressure, kinetic stiffness, or geometric angles—are mapped out into readable, intuitive measurements that real-world mechanics and simulation drivers can interact with seamlessly across different regions.

II. Area of Influence / Impact on Vehicle Dynamics

While the `CAR SETUP UNITS` asset does not modify the vehicle's physics engine computations directly, its architecture heavily influences the configuration pipeline and user experience:

- **Tuning Accuracy and Readability:** It prevents telemetry translation errors. By mapping clear unit tokens (e.g., matching a dampening constant to a specific text string), it guides modders to configure logical tuning increments.
- **Localization & Community Usability:** It serves as the primary system allowing global simulation players to quickly comprehend the vehicle state. It dictates whether a driver works with Metric systems, Imperial defaults, or hardware-specific indicators (like "valving clicks").
- **Telemetry Data Alignment:** It simplifies the correlation between in-game setup adjustments and exported engineering telemetry graphs (e.g., matching MoTeC worksheets to the exact garage presentation units).

III. Key Architecture & Data Fields Explained

The file structure maps directly over the variable names found in the global tuning schemas. Instead of assigning thresholds or steps, it assigns text strings, dimensional values, and contextual formatting markers.

1 - WHEEL AND TYRE TELEMETRY ALIGNMENT

Every adjustable mechanical setting utilizes the following structural parameters:

- **Pressure Units:** Dictates the presentation of pneumatic pressure vectors. Typically configured to display as PSI (Pounds per Square Inch) or `BAR`, ensuring drivers can balance the thermal expanding footprint accurately.

- **Alignment Angles:** Sets the formatting label for spatial geometric alignment. Variables like Camber, Toe, and Caster are universally assigned the degrees symbol (°) or millimeter targets (MM) to track wheel placement relative to the contact patch.

2 - SUSPENSION COMPONENTS & KINEMATICS

- **Spring Stiffness / Wheel Rates:** Controls the representation of pure compression resistance. These strings translate internal structural forces into readable loads, standardizing on metrics like N/MM (Newtons per millimeter), KG/MM, or specialized frequency descriptors like Hz (Hertz)
- **Damper Damping Coefficients:** Maps out energy dissipation factors. Because raw internal forces use complex scientific units like Ns/M (Newton-seconds per meter), this asset often simplifies the interface display to strings like Ns/M or redirects the interface to parse numbers as absolute hardware steps labeled as CLICS OR CLICKS.

3 - AERODYNAMICS & AEROSTATIC CLEARANCE

- **Ride Heights / Ground Clearance:** Dictates the unit configuration for the aerostatic floor platform tracking. It handles structural frame clearances, mapping them cleanly to localized metric length units, universally defined as MM (millimeters) to allow precise platform rake tracking.
- **Aero Wing Inclinometers:** Configures the visual representation of adjustable wing blades, setting the display string to absolute angular units (such as degrees °) or fixed geometric positions.

IV. Interpretation of Setup Units Strategies

When interpreting a vehicle's exported architecture, the units configuration immediately establishes the target simulation profile and intended engineering depth:

- **The Scientific Engineering Profile:** Characterized by pure, unmodified physical engineering notations. Damper configurations read strictly as Ns/M, and spring configurations read as N/MM. This gives advanced race engineers absolute mathematical transparency when calculating real-time dynamic weight transfer or suspension frequencies.
- **The Driver-Centric / Click Profile:** Simplifies high-tier internal metrics to match active physical components. Instead of displaying a confusing velocity value like 1500 Ns/M, the unit system references the hardware's detent steps, displaying simply as integers. This replicates a real racing driver counting physical clicks on an adjustable Ohlins or Penske shock absorber matrix while sitting in the pitlane.

B. Schema

```

- 1. Mechanical Balance : object
  - 1a. Arbs [x] : string | can have multiple Arbs
  - 1b Steer Ratio : string
  - 1c. Brakes : object

```

- 1c1. Front Bias : `string`
 - 1c2. Torque Multiplier : `string`
 - 1c3. Brake Ducts `[x]` : `string` | can have multiple Brake Ducts
- 1d. Differential : `object`
 - 1d1. Power : `string`
 - 1d2. Coast : `string`
 - 1d3. Preload : `string`
- 2. Suspensions `[x]` : `object` | can have multiple Suspensions
 - 2a. Wheel Rate : `string`
 - 2b. Bump Stop Up : `object`
 - 2b1. Range : `string`
 - 2b2. Rate : `string`
 - 2c. Bump Stop Down : `object`
 - 2c1. Range : `string`
 - 2c2. Rate : `string`
 - 2d. Helper Rate : `string`
 - 2e. Helper Range : `string`
- 3. Dampers `[x]` : `object` | can have multiple Dampers
 - 3a. Slow Bump : `string`
 - 3b. Fast Bump : `string`
 - 3c. Slow Rebound : `string`
 - 3d. Fast Rebound : `string`
- 4. Alignments `[x]` : `object` | can have multiple Alignments
 - 4a. Pressure : `string`
 - 4b. Camber : `string`
 - 4c. Toe : `string`
 - 4d. Caster : `string`
 - 4e. Static Camber : `string`
 - 4f. Toe Out Camber : `string`
 - 4g. Compound : `string`
- 5. Electronics : `object`
 - 5a. Tc1 : `string`
 - 5b. Tc2 : `string`
 - 5c. Abs : `string`
 - 5d. Esc : `string`
 - 5e. Ebb : `string`
 - 5f. Engine Map : `string`
 - 5g. Telemetry Laps To Record : `string`
 - 5h. Turbo Boost Lv : `string`
 - 5i. Ers Deployment Map : `string`
 - 5j. Ers Recharge Lv : `string`
 - 5k. Ers Heat Charging : `string`
- 6. Aero : `object`
 - 6a. Collar Positions `[x]` : `string` | can have multiple Collar Positions
 - 6b. Front Target Height : `string`
 - 6c. Rear Target Height : `string`
 - 6e. Front Wing Angle : `string`
 - 6f. Rear Wing Angle : `string`
- 7. Fuel Strategy : `object`
 - 7a. Fuel : `string`
- 8. Use Single Compound : `string`

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Mechanical Balance	None (Object)	Parent block defining UI unit strings for chassis balance parameters.
1a.	Arbs	None (String)	Display unit label for anti-roll bar stiffness; index 1 = front, 2 = rear (e.g., N/m).
1b.	Steer Ratio	None (String)	Display unit label for steering ratio (e.g., ratio value or None if dimensionless).
1c.	Brakes	None (Object)	Brake setup unit sub-block within Mechanical Balance.
1c1.	Front Bias	None (String)	Display unit label for brake balance front percentage (e.g., %).
1c2.	Torque Multiplier	None (String)	Display unit label for global braking torque multiplier (e.g., %).
1c3.	Brake Ducts	None (String)	Display unit label for brake duct opening level(s); multiple entries for front/rear.
1d.	Differential	None (Object)	Differential setup unit sub-block within Mechanical Balance.
1d1.	Power	None (String)	Display unit label for on-throttle differential lock (e.g., %, Nm, or None).
1d2.	Coast	None (String)	Display unit label for coast differential lock (e.g., %, Nm, or None).
1d3.	Preload	None (String)	Display unit label for differential preload torque (e.g., Nm).
2.	Suspensions	None (Object array, per wheel)	Per-corner suspension unit block; indices 1–4 (or more) map to wheel positions.
2a.	Wheel Rate	None (String)	Display unit label for wheel-rate spring stiffness (e.g., N/m, N/mm, kg/mm, Hz).
2b.	Bump Stop Up	None (Object)	Upper bump-stop unit sub-block within each Suspensions entry.
2b1.	Range	None (String)	Display unit label for upper bump-stop travel range under 2b Bump Stop Up (e.g., m).
2b2.	Rate	None (String)	Display unit label for upper bump-stop force rate under 2b Bump Stop Up (e.g., N).

ID	Name	Unit of Measurement	Description
2c.	Bump Stop Down	None (Object)	Lower bump-stop unit sub-block; child fields reuse IDs 2b1 and 2b2 per schema.
2d.	Helper Rate	None (String)	Display unit label for helper/tender spring stiffness (e.g., N/m).
2e.	Helper Range	None (String)	Display unit label for helper spring travel range (e.g., m).
3.	Dampers	None (Object array, per wheel)	Per-corner damper unit block; indices 1–4 (or more) map to wheel positions.
3a.	Slow Bump	None (String)	Display unit label for slow-speed compression damping (e.g., Ns/m or clicks).
3b.	Fast Bump	None (String)	Display unit label for fast-speed compression damping (e.g., Ns/m or clicks).
3c.	Slow Rebound	None (String)	Display unit label for slow-speed rebound damping (e.g., Ns/m or clicks).
3d.	Fast Rebound	None (String)	Display unit label for fast-speed rebound damping (e.g., Ns/m or clicks).
4.	Alignments	None (Object array, per wheel)	Per-corner alignment and tyre unit block; indices 1–4 map to wheel positions.
4a.	Pressure	None (String)	Display unit label for cold tyre pressure (e.g., PSI, bar).
4b.	Camber	None (String)	Display unit label for camber angle (e.g., °).
4c.	Toe	None (String)	Display unit label for toe angle (e.g., °).
4d.	Caster	None (String)	Display unit label for caster angle (e.g., ° or None).
4e.	Static Camber	None (String)	Display unit label for computed/ static camber (e.g., °).
4f.	Toe Out Camber	None (String)	Display unit label for toe-out linear/camber curve coefficient (e.g., °).
4g.	Compound	None (String)	Display unit label for tyre compound index (often None / dimensionless).
5.	Electronics	None (Object)	Driver-aid and powertrain map unit labels for garage UI.
5a.	Tc1	None (String)	Display unit label for primary Traction Control map level.

ID	Name	Unit of Measurement	Description
5b.	Tc2	None (String)	Display unit label for secondary TC map level.
5c.	Abs	None (String)	Display unit label for ABS map level.
5d.	Esc	None (String)	Display unit label for ESP/stability control map level.
5e.	Ebb	None (String)	Display unit label for Electronic Brake Balance level.
5f.	Engine Map	None (String)	Display unit label for engine power map index.
5g.	Telemetry Laps To Record	None (String)	Display unit label for telemetry lap recording count.
5h.	Turbo Boost Lv	None (String)	Display unit label for turbo boost level (e.g., bar).
5i.	Ers Deployment Map	None (String)	Display unit label for ERS deployment map index.
5j.	Ers Recharge Lv	None (String)	Display unit label for ERS recharge level.
5k.	Ers Heat Charging	None (String)	Display unit label for MGU-H heat charging level.
6.	Aero	None (Object)	Aerodynamic setup unit block: ride heights, collar positions, and wing angles.
6a.	Collar Positions	None (String)	Display unit label for per-corner spring collar position (e.g., m, mm); indices 1–4 = LF, RF, LR, RR.
6b.	Front Target Height	None (String)	Display unit label for front target ride height (e.g., mm).
6c.	Rear Target Height	None (String)	Display unit label for rear target ride height (e.g., mm).
6e.	Front Wing Angle	None (String)	Display unit label for front aero device angle (e.g., °). Schema skips 6d.
6f.	Rear Wing Angle	None (String)	Display unit label for rear wing angle (e.g., °).
7.	Fuel Strategy	None (Object)	Fuel load planning unit block.
7a.	Fuel	None (String)	Display unit label for fuel load volume (e.g., L).
8.	Use Single Compound	None (String)	Display unit label for the single-compound toggle (often None / boolean).

D. Example data

I. Chosen Car Engine for Example

- Setup Units (slug : setup_units) [common_phsx]

II. Example

Setup Units

- 1. Mechanical Balance
 - 1a. Arbs 1 : N/m
 - 1a. Arbs 2 : N/m
 - 1b Steer Ratio : None
 - 1c. Brakes
 - 1c1. Front Bias : %
 - 1c2. Torque Multiplier : %
 - 1c3. Brake Ducts : None
 - 1d. Differential
 - 1d1. Power : None
 - 1d2. Coast : None
 - 1d3. Preload : Nm
- 2. Suspensions 1
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 2
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 3
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m

- 2. Suspensions 4
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 5
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 6
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 3. Dampers 1
 - 3a. Slow Bump : Ns/m
 - 3b. Fast Bump : Ns/m
 - 3c. Slow Rebound : Ns/m
 - 3d. Fast Rebound : Ns/m
- 3. Dampers 2
 - 3a. Slow Bump : Ns/m
 - 3b. Fast Bump : Ns/m
 - 3c. Slow Rebound : Ns/m
 - 3d. Fast Rebound : Ns/m
- 3. Dampers 3
 - 3a. Slow Bump : Ns/m
 - 3b. Fast Bump : Ns/m
 - 3c. Slow Rebound : Ns/m
 - 3d. Fast Rebound : Ns/m
- 3. Dampers 4
 - 3a. Slow Bump : Ns/m
 - 3b. Fast Bump : Ns/m
 - 3c. Slow Rebound : Ns/m
 - 3d. Fast Rebound : Ns/m
- 3. Dampers 5
 - 3a. Slow Bump : Ns/m
 - 3b. Fast Bump : Ns/m
 - 3c. Slow Rebound : Ns/m
 - 3d. Fast Rebound : Ns/m
- 3. Dampers 6

- 3a. Slow Bump : Ns/m
- 3b. Fast Bump : Ns/m
- 3c. Slow Rebound : Ns/m
- 3d. Fast Rebound : Ns/m
- 4. Alignments 1
 - 4a. Pressure : PSI
 - 4b. Camber : °
 - 4c. Toe : °
 - 4d. Caster : None
 - 4e. Static Camber : °
 - 4f. Toe Out Camber : °
 - 4g. Compound : None
- 4. Alignments 2
 - 4a. Pressure : PSI
 - 4b. Camber : °
 - 4c. Toe : °
 - 4d. Caster : None
 - 4e. Static Camber : °
 - 4f. Toe Out Camber : °
 - 4g. Compound : None
- 4. Alignments 3
 - 4a. Pressure : PSI
 - 4b. Camber : °
 - 4c. Toe : °
 - 4d. Caster : None
 - 4e. Static Camber : °
 - 4f. Toe Out Camber : °
 - 4g. Compound : None
- 4. Alignments 4
 - 4a. Pressure : PSI
 - 4b. Camber : °
 - 4c. Toe : °
 - 4d. Caster : None
 - 4e. Static Camber : °
 - 4f. Toe Out Camber : °
 - 4g. Compound : None
- 5. Electronics
 - 5a. Tc1 : None
 - 5b. Tc2 : None
 - 5c. Abs : None
 - 5d. Esc : None
 - 5e. Ebb : None
 - 5f. Engine Map : None
 - 5g. Telemetry Laps To Record : None
 - 5h. Turbo Boost Lv : None
 - 5i. Ers Deployment Map : None
 - 5j. Ers Recharge Lv : None
 - 5k. Ers Heat Charging : None
- 6. Aero
 - 6a. Collar Positions 1 : m
 - 6a. Collar Positions 2 : m
 - 6a. Collar Positions 3 : m
 - 6a. Collar Positions 4 : m
 - 6b. Front Target Height : mm
 - 6c. Rear Target Height : mm
 - 6e. Front Wing Angle : °

- | L 6f. Rear Wing Angle : °
- 7. Fuel Strategy
 - | L 7a. Fuel : L
- 8. Use Single Compound : None

8. Car Tuning Parts [.tuningpart]

A. Description

I. General Description

The **Car Tuning Parts** asset (typically using the `.TUNINGPART` extension) acts as the modular configuration switchboard for vehicle variants within the simulation engine. While other assets define the unyielding mechanical foundations of a baseline model, the `CAR TUNING PARTS` file defines optional component swaps, performance packages, upgrades, or regulatory restrictors.

It serves as an administrative routing layer that points directly to secondary sub-physics files (such as alternative engines, custom gearboxes, specialized aerodynamics profiles, or strict setup limits). When a specific tuning part is selected, the physics engine hot-swaps or overrides the default baseline references with the nested paths specified inside this module.

II. Area of Influence / Impact on Vehicle Dynamics

The configurations set within the `CAR TUNING PARTS` asset fundamentally alter a vehicle's mechanical identity, eligibility, and on-track physics behavior:

- **Variant and Package Management:** It dictates whether a car operates in a standard street trim, a modified track-day package, or a fully restricted competitive racing spec (e.g., converting a baseline road car into an upgraded GT-spec variant).
- **Dynamic Physics Overrides:** It intercepts default component behaviors. By changing structural paths, a single vehicle folder can dynamically simulate completely distinct power units, differential units, or aerodynamic maps depending on the activated part.
- **Electronic and System Level Toggles:** It regulates driver aids and system availability on a per-variant basis, allowing developers to cleanly toggle entire sub-systems—such as completely removing ABS and Traction Control—without altering the baseline vehicle architecture.
- **Targeted Balance of Performance (BOP):** It provides a clean vector for balancing competitive series by forcing specific tuning parts (like alternative engine limiters or forced ballast matrices) onto specific cars depending on the event configuration.

III. Key Architecture & Data Fields Explained

The internal hierarchy of a `CAR TUNING PARTS` object is built around system classification tags (`PHYSICS TUNING / CAR PART TYPE`) paired with explicit operational file pathways (`PATH`).

1 - TUNING PART CORE DEFINITIONS

- **Tuning Part Edit / Name:** The distinct identifier string labeling the specific package or modification option displayed inside the user selection menus.

- **Physics Tuning Type:** An enum declaration stating which core mechanical domain this modification controls or replaces (e.g., `DRIVETRAIN`, `ENGINE`, `SETUP LIMITS`, `<NONE>`).
- **Car Part Type:** Defines the structural categorization string for the modification package, standardizing on system nodes such as `MECHANICS_ENGINE`, `MECHANICS_DRIVETRAIN`, or `MECHANICS_ELECTRONICS`.

2 - COMPONENT PATH OVERRIDES

- **Engine Path Redirects:** Links directly to standalone, alternative propulsion blueprints (e.g., swapping a stock motor configuration path for a high-performance `.CARENGINE` profile).
- **Drivetrain Path Redirects:** Points to custom transmission or differential assets (e.g., swapping an open differential map for a highly specific Limited-Slip Differential `.DRIVETRAIN` asset).
- **Setup Limits Redirects:** Routes the vehicle garage interface to alternative slider constraints (e.g., enforcing an entirely unique `.CARSETUPLIMITS` file when a "No ABS/ No TC" motorsport variant is active).

IV. Interpretation of Tuning Part Strategies

Analyzing a vehicle's `.TUNINGPART` matrix reveals how developers structure component modularity across different racing series:

- **The Upgraded Component Profile:** Used to simulate aftermarket mechanical parts or step-by-step performance stages. For example, a vintage car might feature a tuning part named `LIMITED-SLIP DIFFERENTIAL`. Its internal fields explicitly route the `PHYSICS TUNING: DRIVETRAIN` node to a custom path containing tighter clutch plates and locking ramps, replacing the stock open differential behavior seamlessly when purchased or activated.
- **The Electronic Restrictor Profile (e.g., GT3 Cup / Monoregulation):** Used to strip consumer aids for pure motorsport specifications. A tuning part like `No ABS No TC` sets the `PHYSICS TUNING` pointer to `<NONE>` or redirects the hardware paths to lock out secondary algorithmic calculations, forcing the braking model to bypass anti-lock logic loops and rely purely on linear hydraulic pressure limits

B. Schema

```

1. Physics Tuning : enum
2. [physics_tuning] : object | object key take the value of 1. Physics
Tuning. Key-Value on this object will depend on 1. selection
  2a. (Engine, Gearbox, Drivetrain, Clutch, Brake System, Electronics,
Wing, Setup, Setup Limits) Path : string - path
  2b. (Engine Tune) Multiplier : float
  2c. (Engine Tune) Add : float
  2d. (Engine Tune) Cos Ampl : float
  2e. (Engine Tune) Cos Period : float
  2f. (Engine Tune) Cos Phase : float
  2g. (Brakes) Front Path : string - path
  2h. (Brakes) Rear Path : float

```

- 2i. (Suspensions) Coilover Path Front : string - path
- 2j. (Suspensions) Coilover Path Rear : string - path
- 2k. (Suspensions Geometry) Geometry Path Front : string - path
- 2l. (Suspensions Geometry) Geometry Path Rear : string - path
- 2m. (Wing) Wing Index Override : integer
- 2n. (Aero Package) Data : object
 - 2n1. Slip Gain Mult : float
 - 2n2. Speed Factor Mult : float
 - 2n3. Downforces [x] : object | can have multiple Downforces
 - 2n3a. Position : x, y, z float
 - 2n3b. Cl Gain : float
 - 2n3c. Cd Gain : float
 - 2n3d. Yaw Gain : float
 - 2n3e. Drag Per Cool Transfer : float
 - 2n3f. Damage C L [x] : float | can have multiple Damage C L
 - 2n3g. Damage C D [x] : float | can have multiple Damage C D
 - 2n3h. Downforce Controllers [x] : object | can have multiple

Downforce Controllers

- 2n3h1. Combinator Mode : enum
- 2n3h2. Input : enum
- 2n3h3. Filter : float
- 2n3h4. Up Limit : float
- 2n3h5. Down Limit : float
- 2n3h6. Lut : string - path
- 2n3i. Lift Per Front Angle : float
- 2n3j. Lift Per Rear Angle : float
- 2n3k. Drag Per Front Angle : float
- 2n3l. Drag Per Rear Angle : float
- 2n3m. Default Front Angle : float
- 2n3n. Default Rear Angle : float
- 2n4. Front Lift : string - path
- 2n5. Rear Lift : string - path
- 2n6. Drag : string - path
- 2n7. Wings Path [x] : string - path
- 2o. (Performance Modes Tuning) Performance Modes [x] : object | can

have multiple Performance Modes

- 2o1. Performance Mode Name : string
- 2o2. Electronics Settings : object
 - 2o2a. Tc1 : float
 - 2o2b. Tc2 : float
 - 2o2c. Abs : float
 - 2o2d. Esc : float
 - 2o2e. Ebb : float
 - 2o2f. Engine Map : float
 - 2o2g. Telemetry Laps To Record : float
 - 2o2h. Turbo Boost Lv : float
 - 2o2i. Ers Deployment Map : float
 - 2o2j. Ers Recharge Lv : float
 - 2o2k. Ers Heat Charging : float
- 2o3. Brake Settings : object
 - 2o3a. Front Bias : float
 - 2o3b. Torque Multiplier : float
 - 2o3c. Brake Ducts [x] : float | can have multiple Brake Ducts
- 2o4. Damper Settings [x] : object | can have multiple Damper

Settings

- 2o4a. Slow Bump : float

- 2o4b. Fast Bump : float
- 2o4c. Slow Rebound : float
- 2o4d. Fast Rebound : float
- 2o5. Differential Data : object
 - 2o5a. Type : enum
 - 2o5b. Power : float
 - 2o5c. Coast : float
 - 2o5d. Preload : float
 - 2o5e. Front Share : float
 - 2o5f. Torque Bias Ratio Power : float
 - 2o5g. Torque Bias Ratio Coast : float
 - 2o5h. Thermal Capacity : float
 - 2o5i. Surface : float
 - 2o5j. Heat Transfer Coeff : float
 - 2o5k. Wear Factor : float
 - 2o5l. Friction Reduction With T : float
 - 2o5m. Friction Ref T : float
- 2o6. Four W D Differentials : object
 - 2o6a. Front Diff : object
 - 2o6a1. Type : enum
 - 2o6a2. Power : float
 - 2o6a3. Coast : float
 - 2o6a4. Preload : float
 - 2o6a5. Front Share : float
 - 2o6a6. Torque Bias Ratio Power : float
 - 2o6a7. Torque Bias Ratio Coast : float
 - 2o6a8. Thermal Capacity : float
 - 2o6a9. Surface : float
 - 2o6a10. Heat Transfer Coeff : float
 - 2o6a11. Wear Factor : float
 - 2o6a12. Friction Reduction With T : float
 - 2o6a13. Friction Ref T : float
 - 2o6b. Center Diff : object
 - 2o6b1. Type : enum
 - 2o6b2. Power : float
 - 2o6b3. Coast : float
 - 2o6b4. Preload : float
 - 2o6b5. Front Share : float
 - 2o6b6. Torque Bias Ratio Power : float
 - 2o6b7. Torque Bias Ratio Coast : float
 - 2o6b8. Thermal Capacity : float
 - 2o6b9. Surface : float
 - 2o6b10. Heat Transfer Coeff : float
 - 2o6b11. Wear Factor : float
 - 2o6b12. Friction Reduction With T : float
 - 2o6b13. Friction Ref T : float
 - 2o6c. Rear Diff : object
 - 2o6c1. Type : enum
 - 2o6c2. Power : float
 - 2o6c3. Coast : float
 - 2o6c4. Preload : float
 - 2o6c5. Front Share : float
 - 2o6c6. Torque Bias Ratio Power : float
 - 2o6c7. Torque Bias Ratio Coast : float
 - 2o6c8. Thermal Capacity : float
 - 2o6c9. Surface : float

- 2o6c10. Heat Transfer Coeff : float
- 2o6c11. Wear Factor : float
- 2o6c12. Friction Reduction With T : float
- 2o6c13. Friction Ref T : float
- 2o7. Front Lock Controllers : object
 - 2o7a. Name : string
 - 2o7b. Stages [x] : object | can have multiple Stages
 - 2o7b1. Input Var : enum
 - 2o7b2. Combinator Mode : enum
 - 2o7b3. Lut : string – path
 - 2o7b4. Filter Gain : float
 - 2o7b5. Up Limit : float
 - 2o7b6. Down Limit : float
 - 2o7b7. Current Value : float
 - 2o7b8. Const Value : float
- 2o8. Center Lock Controllers : object
 - 2o8a. Name : string
 - 2o8b. Stages [x] : object
 - 2o8b1. Input Var : enum
 - 2o8b2. Combinator Mode : enum
 - 2o8b3. Lut : string – path
 - 2o8b4. Filter Gain : float
 - 2o8b5. Up Limit : float
 - 2o8b6. Down Limit : float
 - 2o8b7. Current Value : float
 - 2o8b8. Const Value : float
- 2o9. Rear Lock Controllers : object
 - 2o9a. Name : string
 - 2o9b. Stages [x] : object
 - 2o9b1. Input Var : enum
 - 2o9b2. Combinator Mode : enum
 - 2o9b3. Lut : string – path
 - 2o9b4. Filter Gain : float
 - 2o9b5. Up Limit : float
 - 2o9b6. Down Limit : float
 - 2o9b7. Current Value : float
 - 2o9b8. Const Value : float
- 2o10. Awd Clutches [x] : object
 - 2o10a. Position : integer
 - 2o10b. Preload : float
 - 2o10c. Controllers : object
 - 2o10c1. Name : string
 - 2o10c2. Stages [x] : object
 - 2o10c2a. Input Var : enum
 - 2o10c2b. Combinator Mode : enum
 - 2o10c2c. Lut : string – path
 - 2o10c2d. Filter Gain : float
 - 2o10c2e. Up Limit : float
 - 2o10c2f. Down Limit : float
 - 2o10c2g. Current Value : float
 - 2o10c2h. Const Value : float
- 2o11. Turbo Controllers [x] : object
 - 2o11a. Name : string
 - 2o11b. Stages [x] : object
 - 2o11b1. Input Var : enum
 - 2o11b2. Combinator Mode : enum

- 2o11b3. Lut : **string** – path
 - 2o11b4. Filter Gain : **float**
 - 2o11b5. Up Limit : **float**
 - 2o11b6. Down Limit : **float**
 - 2o11b7. Current Value : **float**
 - 2o11b8. Const Value : **float**
 - 2o12. Turbo Settings : **object**
 - 2o12a. Boost Lv : **float**
- 3. Car Part Type : **enum**

Enum - Car Tuning Pars

1	Physics Tuning	<None>, Engine Tune, Engine, Gearbox, Drivetrain, Clutch, Brakes, Brake System, Electronics, Suspensions, Suspensions Geometry, Wing, Aero Package, Setup, Setup Limits, Performance Modes Tuning
2n3h1	Combinator Mode	UndefinedMode, AddH, MultH, AddYawGain, MultYawGain, AddCIGain, MultCIGain
2n3h2	Input	UndefinedInput, Brake, Gas, Yaw, LatG, LonG, Steer, Speed, SusTravelLR, SusTravelRR
2o5a	Type	LSD, Spool, Torsen, EpicyclicTorsen, EpicyclicLSD, TorqueVectoring
2o7b1	Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
2o7b2	Combinator Mode	UndefinedMode, Add, Mult

3	Car Part Type	<None>, Exterior_Body, Exterior_FrontBumper, Exterior_FrontBumperLip, Exterior_FrontFogLights, Exterior_FrontBumperCanards, Exterior_FrontLights, Exterior_RearLights, Exterior_RearFogLights, Exterior_AuxLights, Exterior_FrontAirIntakeGrid, Exterior_FrontWindshield, Exterior_Hood, Exterior_HoodAirscoopes, Exterior_SlideFrontBody, Exterior_SlideMirrors, Exterior_LeftDoor, Exterior_RightDoor, Exterior_Skirts, Exterior_SlideRearBody, Exterior_RearDiffuser, Exterior_RearTailgateSpoiler, Exterior_RearSpoiler, Exterior_RoofAirScopes, Exterior_RearWindshield, Exterior_RimFL, Exterior_RimFR, Exterior_RimRL, Exterior_RimRR, Exterior_RimBlurFL, Exterior_RimBlurFR, Exterior_RimBlurRL, Exterior_RimBlurRR, Exterior_TyreFL, Exterior_TyreFR, Exterior_TyreRL, Exterior_TyreRR, Exterior_CaliperFL, Exterior_CaliperFR, Exterior_CaliperRL, Exterior_CaliperRR, Exterior_DiscFL, Exterior_DiscFR, Exterior_DiscRL, Exterior_DiscRR, Exterior_PlateFront, Exterior_PlateRear, Exterior_Exhaust, Exterior_Lumirank, Exterior_FrontSplitter, Exterior_FuelCap, Exterior_Decal, Exterior_Storage, Exterior_Top, Interior_Body, Interior_FrontSeats, Interior_RearSeats, Interior_FireExtinguisher, Interior_SteeringWheelAndHub, Interior_MirrorGadget, Interior_FloorCarpets, Interior_HandbrakeLever, Interior_GearShifterLever, Interior_RollbarOrCage, Interior_CameraDevice, Interior_CockpitLights, Interior_AnalogDashboard, Interior_DigitalDashboard, Interior_SeatbeltOn, Interior_SeatbeltOff, Interior_HarnessOn, Interior_HarnessOff, Interior_SafetyNetOn, Interior_SafetyNetOff, Interior_PedalAccelerator, Interior_PedalBrake, Interior_PedalClutch, Interior_Storage, Interior_CupHolder, Mechanics_SuspensionGeometry_Front, Mechanics_SuspensionGeometry_Rear, Mechanics_SuspensionCoilover_Front, Mechanics_SuspensionCoilover_Rear, Mechanics_Engine, Mechanics_Turbo, Mehanics_Ecu, Mechanics_Pistons, Mechanics_Rod, Mechanics_Cam_Shaft, Mechanics_Crank_Shaft, Mechanics_Head, Mechanics_Injection_System, Mechanics_Radiator, Mechanics_Oil_Radiator, Mechanics_Intercooler, Mechanics_Air_Filter, Mechanics_Intake_Manifold, Mechanics_Exhaust_Manifold, Mechanics_Catalitic_Converter, Mechanics_Center_Pipe, Mechanics_Muffler, Mechanics_Differential, Mechanics_Clutch, Mechanics_Gearbox, Mechanics_Electronics, Mechanics_Drivetrain, Mechanics_Aero_0, Mechanics_Aero_1, Mechanics_Aero_2, Mechanics_Aero_3, Mechanics_Aero_4, Mechanics_Aero_5, Mechanics_Aero_6, Mechanics_Aero_7, Mechanics_Aero_8, Mechanics_Aero_9, Mechanics_Downforce_0, Mechanics_Downforce_1, Mechanics_Downforce_2, Mechanics_Downforce_3, Mechanics_PadsFL, Mechanics_PadsFR, Mechanics_PadsRL, Mechanics_PadsRR, Mechanics_CaliperFL, Mechanics_CaliperFR, Mechanics_CaliperRL, Mechanics_CaliperRR, Mechanics_DiscFL, Mechanics_DiscFR, Mechanics_DiscRL, Mechanics_DiscRR
---	---------------	---

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Physics Tuning	None (Enum)	Primary enum selector declaring which mechanical domain this tuning part overrides (Engine, Gearbox, Drivetrain, Setup Limits, Aero Package, Performance Modes Tuning, etc.).
2.	[physics_tuning]	None (Object)	Dynamic object whose JSON key equals the Physics Tuning enum value; only the branch matching field 1 is populated at runtime.
2a.	Path	None (File path)	File path redirect for component swap tuning parts; active when Physics Tuning is Engine, Gearbox, Drivetrain, Clutch, Brake System, Electronics, Wing, Setup, or Setup Limits.
2b.	Multiplier	Dimensionless float	Engine tune torque multiplier scalar applied on top of the base power curve when Physics Tuning is Engine Tune.
2c.	Add	Nm (Newton-meters)	Flat torque offset added to the engine output curve when Physics Tuning is Engine Tune.
2d.	Cos Ampl	Dimensionless coefficient	Cosine-wave torque modulation amplitude for periodic engine tune effects (Engine Tune mode).
2e.	Cos Period	s (Seconds) or rad	Period of the cosine torque modulation in Engine Tune mode.
2f.	Cos Phase	rad (Radians) or deg	Phase offset of the cosine torque modulation in Engine Tune mode.
2g.	Front Path	None (File path)	Path to an alternative front brake asset (.brakes or related) when Physics Tuning is Brakes.
2h.	Rear Path	Depends on parameter (float)	Path or scalar reference for rear brake configuration when Physics Tuning is Brakes (schema types as float; verify against asset).
2i.	Coilover Path Front	None (File path)	Path to a replacement front .COILOVER asset when Physics Tuning is Suspensions.
2j.	Coilover Path Rear	None (File path)	Path to a replacement rear .COILOVER asset when Physics Tuning is Suspensions.
2k.	Geometry Path Front	None (File path)	Path to a replacement front .SUSPENSION geometry asset when Physics Tuning is Suspensions Geometry.

ID	Name	Unit of Measurement	Description
2l.	Geometry Path Rear	None (File path)	Path to a replacement rear .SUSPENSION geometry asset when Physics Tuning is Suspensions Geometry.
2m.	Wing Index Override	None (Integer)	Integer index selecting which wing aero profile slot to override when Physics Tuning is Wing.
2n.	Data	None (Object)	Root aero package data block when Physics Tuning is Aero Package; contains downforce modifiers and wing path overrides.
2n1.	Slip Gain Mult	Dimensionless multiplier	Global multiplier applied to tyre slip-driven aero gain within the aero package.
2n2.	Speed Factor Mult	Dimensionless multiplier	Multiplier scaling speed-dependent aero factor curves in the package.
2n3.	Downforces [x]	None (Object array)	Per-element downforce modifier block; multiple entries for distinct aero surfaces or zones.
2n3a.	Position	m (Meters, X / Y / Z)	3D position offset (X/Y/Z) of this downforce element relative to the vehicle reference frame.
2n3b.	Cl Gain	Dimensionless coefficient	Lift coefficient (CL) gain multiplier for this downforce element.
2n3c.	Cd Gain	Dimensionless coefficient	Drag coefficient (CD) gain multiplier for this downforce element.
2n3d.	Yaw Gain	Dimensionless coefficient	Yaw-angle sensitivity gain for this downforce element.
2n3e.	Drag Per Cool Transfer	Coefficient	Drag penalty per unit of cooling airflow transfer through this element.
2n3f.	Damage C L [x]	Dimensionless coefficient	Array of lift-damage coefficients reducing CL under aerodynamic damage states.
2n3g.	Damage C D [x]	Dimensionless coefficient	Array of drag-damage coefficients reducing CD efficiency under damage.
2n3h.	Downforce Controllers [x]	None (Object array)	Dynamic downforce controller block; multiple controllers can combine inputs (brake, steer, speed, etc.).
2n3h1.	Combinator Mode	None (Enum)	Enum selecting how this controller combines with others (AddH, MultH, AddClGain, MultClGain, etc.).

ID	Name	Unit of Measurement	Description
2n3h2.	Input	None (Enum)	Input signal source for the controller (Brake, Gas, Yaw, LatG, Speed, SusTravellR, etc.).
2n3h3.	Filter	Coefficient (Smoothing multiplier)	Low-pass filter time constant smoothing the controller input signal.
2n3h4.	Up Limit	Depends on input variable	Upper clamp limit for controller output contribution.
2n3h5.	Down Limit	Depends on input variable	Lower clamp limit for controller output contribution.
2n3h6.	Lut	None (.CURVE file path)	Optional .CURVE LUT mapping filtered input to aero output modifier.
2n3i.	Lift Per Front Angle	Coefficient per deg	Lift sensitivity per degree of front wing angle for this package.
2n3j.	Lift Per Rear Angle	Coefficient per deg	Lift sensitivity per degree of rear wing angle.
2n3k.	Drag Per Front Angle	Coefficient per deg	Drag sensitivity per degree of front wing angle.
2n3l.	Drag Per Rear Angle	Coefficient per deg	Drag sensitivity per degree of rear wing angle.
2n3m.	Default Front Angle	deg (Degrees)	Default front wing angle reference used when no setup override is active.
2n3n.	Default Rear Angle	deg (Degrees)	Default rear wing angle reference used when no setup override is active.
2n4.	Front Lift	None (File path)	Path to front lift aero curve/asset override within the aero package.
2n5.	Rear Lift	None (File path)	Path to rear lift aero curve/asset override within the aero package.
2n6.	Drag	None (File path)	Path to drag aero curve/asset override within the aero package.
2n7.	Wings Path [x]	None (File path)	Path(s) to alternative wing definition assets; multiple entries for multi-element wings.
2o.	Performance Modes [x]	None (Object array)	Performance mode preset block when Physics Tuning is Performance Modes Tuning; multiple modes (Wet, Quali, etc.).
2o1.	Performance Mode Name	None (String)	Display name string for this performance mode shown in UI selection menus.
2o2.	Electronics Settings	None (Object)	Electronics preset sub-block bundled into this performance mode.

ID	Name	Unit of Measurement	Description
2o2a.	Tc1	None (Map index / level)	Traction Control map level preset for this performance mode.
2o2b.	Tc2	None (Map index / level)	Secondary TC map level preset.
2o2c.	Abs	None (Map index / level)	ABS map level preset.
2o2d.	Esc	None (Map index / level)	ESC/ESP map level preset.
2o2e.	Ebb	None (Map index / level)	Electronic Brake Balance level preset.
2o2f.	Engine Map	None (Map index)	Engine power map index preset.
2o2g.	Telemetry Laps To Record	None (Integer)	Telemetry lap recording count preset.
2o2h.	Turbo Boost Lv	bar or level index	Turbo boost level preset.
2o2i.	Ers Deployment Map	None (Map index)	ERS deployment map preset.
2o2j.	Ers Recharge Lv	None (Level index)	ERS recharge level preset.
2o2k.	Ers Heat Charging	None (Level index)	MGU-H heat charging level preset.
2o3.	Brake Settings	None (Object)	Brake setup preset sub-block for this performance mode.
2o3a.	Front Bias	Ratio (0.0 - 1.0)	Brake balance front percentage preset
2o3b.	Torque Multiplier	Dimensionless multiplier	Global braking torque multiplier preset.
2o3c.	Brake Ducts [x]	Ratio (0.0 - 1.0)	Brake duct opening level preset(s); multiple entries for front/rear ducts.
2o4.	Damper Settings [x]	None (Object array)	Per-corner damper preset block; indices map to wheel positions.
2o4a.	Slow Bump	Clicks or N·s/m	Slow-speed compression damping preset for this corner.
2o4b.	Fast Bump	Clicks or N·s/m	Fast-speed compression damping preset.
2o4c.	Slow Rebound	Clicks or N·s/m	Slow-speed rebound damping preset.
2o4d.	Fast Rebound	Clicks or N·s/m	Fast-speed rebound damping preset.
2o5.	Differential Data	None (Object)	Single differential override block within a performance mode.
2o5a.	Type	None (Enum)	Differential type enum (LSD, Spool, Torsen, EpicyclicTorsen, TorqueVectoring, etc.).
2o5b.	Power	Nm or ratio	On-throttle differential lock intensity preset.

ID	Name	Unit of Measurement	Description
2o5c.	Coast	Nm or ratio	Coast/off-throttle differential lock intensity preset.
2o5d.	Preload	Nm (Newton-meters)	Static differential preload torque preset.
2o5e.	Front Share	Ratio (0.0 - 1.0)	Front torque share fraction for torque-vectoring or active diffs.
2o5f.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Torque bias ratio under power.
2o5g.	Torque Bias Ratio Coast	Ratio	Torque bias ratio on coast
2o5h.	Thermal Capacity	J/K or J/°C	Thermal mass of differential friction surfaces for heat model.
2o5i.	Surface	m ² (Square meters)	Effective friction surface area for differential heat generation.
2o5j.	Heat Transfer Coeff	W/(m ² ·K) (Coefficient)	Heat transfer coefficient to ambient/oil for differential cooling.
2o5k.	Wear Factor	Dimensionless coefficient	Wear accumulation rate factor for differential friction surfaces.
2o5l.	Friction Reduction With T	Ratio/°C	Friction reduction factor as a function of differential temperature.
2o5m.	Friction Ref T	°C (Degrees Celsius)	Reference temperature for friction reduction curve.
2o6.	Four W D Differentials	None (Object)	AWD/4WD triple-differential preset container (front, center, rear).
2o6a.	Front Diff	None (Object)	Front axle differential preset sub-block.
2o6a1.	Type	None (Enum)	Differential type enum for the front diff.
2o6a2.	Power	Nm or ratio	On-throttle lock intensity for the front diff.
2o6a3.	Coast	Nm or ratio	Coast lock intensity for the front diff.
2o6a4.	Preload	Nm (Newton-meters)	Preload torque for the front diff.
2o6a5.	Front Share	Ratio (0.0 - 1.0)	Front torque share for the front diff.
2o6a6.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Power torque bias ratio for the front diff.
2o6a7.	Torque Bias Ratio Coast	Ratio	Coast torque bias ratio for the front diff.
2o6a8.	Thermal Capacity	J/K or J/°C	Thermal capacity of the front diff friction pack.

ID	Name	Unit of Measurement	Description
2o6a9.	Surface	m ² (Square meters)	Friction surface area for the front diff.
2o6a10.	Heat Transfer Coeff	W/(m ² ·K) (Coefficient)	Heat transfer coefficient for the front diff.
2o6a11.	Wear Factor	Dimensionless coefficient	Wear rate factor for the front diff.
2o6a12.	Friction Reduction With T	Ratio/°C	Temperature-dependent friction reduction for the front diff.
2o6a13.	Friction Ref T	°C (Degrees Celsius)	Reference temperature for front diff friction model.
2o6b.	Center Diff	None (Object)	Center differential preset sub-block.
2o6b1.	Type	None (Enum)	Differential type enum for the center diff.
2o6b2.	Power	Nm or ratio	On-throttle lock intensity for the center diff.
2o6b3.	Coast	Nm or ratio	Coast lock intensity for the center diff.
2o6b4.	Preload	Nm (Newton-meters)	Preload torque for the center diff.
2o6b5.	Front Share	Ratio (0.0 - 1.0)	Front torque share for the center diff.
2o6b6.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Power torque bias ratio for the center diff.
2o6b7.	Torque Bias Ratio Coast	Ratio	Coast torque bias ratio for the center diff.
2o6b8.	Thermal Capacity	J/K or J/°C	Thermal capacity of the center diff friction pack.
2o6b9.	Surface	m ² (Square meters)	Friction surface area for the center diff.
2o6b10.	Heat Transfer Coeff	W/(m ² ·K) (Coefficient)	Heat transfer coefficient for the center diff.
2o6b11.	Wear Factor	Dimensionless coefficient	Wear rate factor for the center diff.
2o6b12.	Friction Reduction With T	Ratio/°C	Temperature-dependent friction reduction for the center diff.
2o6b13.	Friction Ref T	°C (Degrees Celsius)	Reference temperature for center diff friction model.
2o6c.	Rear Diff	None (Object)	Rear axle differential preset sub-block.
2o6c1.	Type	None (Enum)	Differential type enum for the rear diff.
2o6c2.	Power	Nm or ratio	On-throttle lock intensity for the rear diff.

ID	Name	Unit of Measurement	Description
2o6c3.	Coast	Nm or ratio	Coast lock intensity for the rear diff.
2o6c4.	Preload	Nm (Newton-meters)	Preload torque for the rear diff.
2o6c5.	Front Share	Ratio (0.0 - 1.0)	Front torque share for the rear diff.
2o6c6.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Power torque bias ratio for the rear diff.
2o6c7.	Torque Bias Ratio Coast	Ratio	Coast torque bias ratio for the rear diff.
2o6c8.	Thermal Capacity	J/K or J/°C	Thermal capacity of the rear diff friction pack.
2o6c9.	Surface	m² (Square meters)	Friction surface area for the rear diff.
2o6c10.	Heat Transfer Coeff	W/(m²·K) (Coefficient)	Heat transfer coefficient for the rear diff.
2o6c11.	Wear Factor	Dimensionless coefficient	Wear rate factor for the rear diff.
2o6c12.	Friction Reduction With T	Ratio/°C	Temperature-dependent friction reduction for the rear diff.
2o6c13.	Friction Ref T	°C (Degrees Celsius)	Reference temperature for rear diff friction model.
2o7.	Front Lock Controllers	None (Object)	Front axle lock controller definition for active/vectoring diffs.
2o7a.	Name	None (String)	Display/debug name for the front lock controller.
2o7b.	Stages [x]	None (Object array)	Multi-stage controller pipeline; each stage processes an input through a LUT and limits.
2o7b1.	Input Var	None (Telemetry enum)	Input variable enum for this stage (SlipRatio, Steer, LatG, ErsChargeLevel, etc.).
2o7b2.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode for merging stage output (Add or Mult).
2o7b3.	Lut	None (.curve file path)	Look-up table (.curve) mapping filtered input to lock fraction.
2o7b4.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain smoothing the stage input signal.
2o7b5.	Up Limit	Depends on input variable	Upper output limit for this stage.
2o7b6.	Down Limit	Depends on input variable	Lower output limit for this stage.
2o7b7.	Current Value	Depends on input variable	Runtime current value output of this stage (telemetry/state).
2o7b8.	Const Value	Depends on input variable	Constant fallback input value when Input Var is Const.

ID	Name	Unit of Measurement	Description
2o8.	Center Lock Controllers	None (Object)	Center differential lock controller definition.
2o8a.	Name	None (String)	Display/debug name for the center lock controller.
2o8b.	Stages [x]	None (Object array)	Multi-stage controller pipeline for center diff lock.
2o8b1.	Input Var	None (Telemetry enum)	Input variable enum for center lock stage.
2o8b2.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode (Add/Mult) for center lock stage.
2o8b3.	Lut	None (.curve file path)	LUT path for center lock stage.
2o8b4.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain for center lock stage input.
2o8b5.	Up Limit	Depends on input variable	Upper output limit for center lock stage.
2o8b6.	Down Limit	Depends on input variable	Lower output limit for center lock stage.
2o8b7.	Current Value	Depends on input variable	Current runtime value for center lock stage.
2o8b8.	Const Value	Depends on input variable	Constant input value for center lock stage.
2o9.	Rear Lock Controllers	None (Object)	Rear axle lock controller definition.
2o9a.	Name	None (String)	Display/debug name for the rear lock controller.
2o9b.	Stages [x]	None (Object array)	Multi-stage controller pipeline for rear diff lock.
2o9b1.	Input Var	None (Telemetry enum)	Input variable enum for rear lock stage.
2o9b2.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode for rear lock stage.
2o9b3.	Lut	None (.curve file path)	LUT path for rear lock stage.
2o9b4.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain for rear lock stage.
2o9b5.	Up Limit	Depends on input variable	Upper output limit for rear lock stage.
2o9b6.	Down Limit	Depends on input variable	Lower output limit for rear lock stage.
2o9b7.	Current Value	Depends on input variable	Current runtime value for rear lock stage.
2o9b8.	Const Value	Depends on input variable	Constant input value for rear lock stage.

ID	Name	Unit of Measurement	Description
2o10	Awd Clutches [x]	None (Object array)	AWD clutch pack definition; multiple clutches for front/rear axle coupling.
2o10a.	Position	None (Integer)	Integer position/index of this clutch in the AWD coupling matrix.
2o10b.	Preload	Nm (Newton-meters)	Static preload torque on this AWD clutch pack.
2o10c.	Controllers	None (Object)	Nested lock controller for this AWD clutch.
2o10c1.	Name	None (String)	Display name for the AWD clutch controller.
2o10c2.	Stages [x]	None (Object array)	Multi-stage pipeline controlling AWD clutch lock.
2o10c2a.	Input Var	None (Telemetry enum)	Input variable enum for AWD clutch stage.
2o10c2b.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode for AWD clutch stage.
2o10c2c.	Lut	None (.curve file path)	LUT path for AWD clutch stage.
2o10c2d.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain for AWD clutch stage.
2o10c2e.	Up Limit	Depends on input variable	Upper output limit for AWD clutch stage.
2o10c2f.	Down Limit	Depends on input variable	Lower output limit for AWD clutch stage.
2o10c2g.	Current Value	Depends on input variable	Current runtime value for AWD clutch stage.
2o10c2h.	Const Value	Depends on input variable	Constant input for AWD clutch stage.
2o11	Turbo Controllers [x]	None (Object array)	Turbo boost controller block within performance mode tuning.
2o11a.	Name	None (String)	Display name for turbo controller.
2o11b.	Stages [x]	None (Object array)	Multi-stage turbo boost control pipeline.
2o11b1.	Input Var	None (Telemetry enum)	Input variable enum for turbo stage.
2o11b2.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode for turbo stage.
2o11b3.	Lut	None (.curve file path)	LUT path mapping input to boost modifier.
2o11b4.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain for turbo stage input.
2o11b5.	Up Limit	Depends on input variable	Upper boost limit for this stage.

ID	Name	Unit of Measurement	Description
2o11b6.	Down Limit	Depends on input variable	Lower boost limit for this stage.
2o11b7.	Current Value	Depends on input variable	Current runtime boost modifier value.
2o11b8.	Const Value	Depends on input variable	Constant input for turbo stage.
2o12.	Turbo Settings	None (Object)	Turbo settings sub-block within performance mode.
2o12a.	Boost Lv	bar or level index	Target turbo boost level preset for this performance mode.
3.	Car Part Type	None (Enum)	Enum categorizing the visual/mechanical part slot this tuning part occupies in the parts menu (Mechanics_Engine, Mechanics_Gearbox, Exterior_*, etc.).

D. Example data

I. Chosen Cars for Example

- Toyota Supra MK IV (slug : ks_toyota_supra_mkiv) [7 tuning parts]
- Datsun 240z Fairlady (slug : ks_datsun_240z_fairlady) [8 tuning parts]
- Porsche 992 GT3 Cup (slug : ks_porsche_992_gt3_cup) [4 tuning parts]

II. Example

Toyota Supra MK IV

1. Drift Front Geometry (file :

ks_toyota_supra_geometry_front_drift_geometry.tuningpart)

```

| 1. Physics Tuning : Suspensions Geometry
| 2. Suspensions Geometry
|   | 2a. Geometry Path Front :
content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_front_drift.
suspension
|   | 2b. Geometry Path Rear :
content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_rear.suspens
ion
| 3. Car Part Type : Mechanics_SuspensionGeometry_Front

```

2. Drift Drivetrain (file : ks_toyota_supra_mkiv_drivetrain_drift.tuningpart)

```

| 1. Physics Tuning : Drivetrain
| 2. Drivetrain
|   | 2a. Path :
content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_drift.drivet
rain

```


└ 3. Car Part Type : Mechanics_Drivetrain

3. Drift Engine (file : ks_toyota_supra_mkiv_engine_drift.tuningpart)

└ 1. Physics Tuning : Engine

└ 2. Engine

└ └ 2a. Path :

content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_drift.carengine

└ 3. Car Part Type : Mechanics_Engine

4. Drift Front Suspension (file : ks_toyota_supra_mkiv_front_susp_drift.tuningpart)

└ 1. Physics Tuning : Suspensions

└ 2. Suspensions

└ └ 2a. Coilover Path Front :

content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_front.coilover

└ └ 2b. Coilover Path Rear :

content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_rear.coilover

└ 1c. Car Part Type : Mechanics_SuspensionCoilover_Front

5. Drift Gearbox (file : ks_toyota_supra_mkiv_gearbox_drift.tuningpart)

└ 1. Physics Tuning : Gearbox

└ 2. Gearbox

└ └ 2a. Path :

content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_drift.gearbox

└ 3. Car Part Type : Mechanics_Gearbox

6. Drift Setup (file : ks_toyota_supra_mkiv_setup_drift.tuningpart)

└ 1. Physics Tuning : Setup

└ 2. Setup

└ └ 2a. Path :

content\cars\ks_toyota_supra_mkiv\data\setup\supra_drift.carsetup

└ 3. Car Part Type : None

7. Drift Setup Limits (file : ks_toyota_supra_mkiv_setuplimits_drift.tuningpart)

└ 1. Physics Tuning : Setup Limits

└ 2. Setup Limits

└ └ 2a. Path :

content\cars\ks_toyota_supra_mkiv\data\setup\supra_mk_iv_limits_drift.carsetuplimits

└ 3. Car Part Type : None

Datsun 240z Fairlady

1. 5 Speed Gearbox (file : ks_datsun_240z_fairlady_5speed.tuningpart)

- | 1. Physics Tuning : Gearbox
- | 2. Gearbox
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_5speed.gearbox
- | 3. Car Part Type : Mechanics_Gearbox

2. Brake (file : ks_datsun_240z_fairlady_brake.tuningpart)

- | 1. Physics Tuning : Brake System
- | 2. Gearbox
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_brakes.brakesystem
- | 3. Car Part Type : Mechanics_CaliperFL

3. 5 Speed Clutch (file : ks_datsun_240z_fairlady_clutch_5speed.tuningpart)

- | 1. Physics Tuning : Clutch
- | 2. Clutch
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_clutch_5speed.clutch
- | 3. Car Part Type : Mechanics_Clutch

4. G Nose Coilovers (file : ks_datsun_240z_fairlady_g_nose_coilovers.tuningpart)

- | 1. Physics Tuning : Suspensions
- | 2. Suspensions
 - | 2a. Coilover Path Front :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_g_nose_front.coilover
 - | 2b. Coilover Path Rear :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_g_nose_rear.coilover
- | 3. Car Part Type : Mechanics_SuspensionCoilover_Front

5. Setup (file : ks_datsun_240z_fairlady_g_nose_setup.tuningpart)

- | 1. Physics Tuning : Setup
- | 2. Setup
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\setup\ks_datsun_240z_fairlady_g_nose.carsetup
- | 3. Car Part Type : None

6. Setup Limits (file : ks_datsun_240z_fairlady_g_nose_setuplimits.tuningpart)

- | 1. Physics Tuning : Setup Limits
- | 2. Setup Limits
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\setup\ks_datsun_240z_fairlady_g_nose.carsetuplimits
- | 3. Car Part Type : None

7. L28 Engine (file : ks_datsun_240z_fairlady_l28engine.tuningpart)

- | 1. Physics Tuning : Engine
- | 2. Engine
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_L28engine.carengine
- | 3. Car Part Type : Mechanics_Engine

8. Limited-Slip Differential (file : ks_datsun_240z_fairlady_lsd.tuningpart)

- | 1. Physics Tuning : Drivetrain
- | 2. Drivetrain
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_LSD.drivetrain
- | 3. Car Part Type : Mechanics_Drivetrain

Porsche 992 GT3 Cup

1. No ABS No TC (file : 992_gt3_no_abs_no_tc.tuningpart)

- | 1. Physics Tuning : <None>
- | 2. Car Part Type : Mechanics_Electronics

2. No ABS No TC Setup Limits (file : 992_gt3_no_abs_no_tc_setuplimits.tuningpart)

- | 1. Physics Tuning : Setup Limits
- | 2. Setup Limits
 - | 2a. Path :
content\cars\ks_porsche_992_gt3_cup\data\Setup\limitsporsche992cup_no_abs_no_tc.carsetuplimits
- | 3. Car Part Type : None

3. Only ABS (file : 992_gt3_only_abs.tuningpart)

- | 1. Physics Tuning : <None>
- | 2. Car Part Type : Mechanics_Electronics

4. Only ABS Setup Limits (file : 992_gt3_only_abs_setuplimits.tuningpart)

- | 1. Physics Tuning : Setup Limits
- | 2. Setup Limits
 - | 2a. Path :
content\cars\ks_porsche_992_gt3_cup\data\Setup\limitsporsche992cup_only_abs.carsetuplimits
- | 3. Car Part Type : None

9. Car Electronics [.carelectronics]

A. Description

I. General Description

The **Car Electronics** asset serves as the central processing unit for all algorithmic driver aids, active chassis controls, and safety sub-systems within the vehicle. While other files handle pure mechanical reactions (such as steel twisting or hydraulics compressing), this asset defines the software layer that intercepts driver inputs and wheel telemetry in real time.

It processes raw kinematic metrics—such as wheel speed deltas, steering wheel rotation, and longitudinal/lateral slip ratios—and calculates targeted mitigation interventions. This file maps the operational behavior of high-level motorsports electronic suites, most notably **ABS (Anti-lock Braking System)** and **TC (Traction Control)**.

II. Area of Influence / Impact on Vehicle Dynamics

The variables configured within the `CAR ELECTRONICS` asset dictate how aggressively the car's digital systems manage performance and stability thresholds:

- **Traction Optimization (Corner Exit Propulsion):** Traction Control systems actively monitor drive-wheel spin. When the slip exceeds predefined thresholds, the system limits engine torque output to preserve lateral stability and maximize forward bite without spinning the car.
- **Braking Deceleration Stability:** The ABS module continuously samples wheel lockup parameters during heavy braking. By dynamically modulating hydraulic line pressure, it prevents tires from skidding, preserving steering authority and optimal braking distances.
- **Transient Phase Control and Safety Windows:** It maps out the exact operating window between a skilled driver sliding the car for lap time versus a catastrophic loss of control, acting as a electronic safety net customized across various preset dashboard maps

III. Key Architecture & Data Fields Explained

The internal framework of a `CAR ELECTRONICS` file is highly structured around multi-stage **Settings Matrices** (e.g., Maps 1 through 12) to allow drivers to toggle intervention intensity on the fly from the cockpit.

1 - TRACTION CONTROL (TC) LOGIC MATRIX

For each step or map preset, the system evaluates the following core parameters:

- **Min Slip Ratio / Max Slip Ratio:** Defines the target allowable longitudinal tire slippage window (e.g., 0.04 to 0.06). Slippage below the minimum is ignored; slippage exceeding the maximum triggers full system suppression.

- **Ref Slip Angle (Degrees):** Establishes the baseline lateral slide tolerance. This prevents the system from prematurely cutting power during controlled, high-load cornering yaw angles.
- **Cut Level:** Dictates the severity of engine torque reduction when wheelspin occurs, utilizing ignition cuts, fuel cuts, or electronic throttle body modulation to choke power.
- **Max Torque Variation:** Governs the dampening and smoothing factor of the engine power re-engagement, preventing harsh shockwaves from upsetting the chassis as grip is recovered.

2 - ANTI-LOCK BRAKING SYSTEM (ABS) MAP SETTINGS

- **Target Slip Sliders:** Defines how much a tire is allowed to slip relative to vehicle ground speed under heavy braking before the ABS hydraulic valves cycle open to release line pressure.
- **Frequency and Response Cycles:** Sets the operational speed (measured in Hz) at which the digital controller samples data and updates valve adjustments, determining how smooth or violent the pedal feedback pulses.

IV. Interpretation of Tuning Part Strategies

By examining the .CARELECTRONICS maps, developers can identify the exact racing category and setup complexity built into the vehicle:

- **The Multi-Map Pro-Motorsport Profile (e.g., GT3 / GT4):** Features an expansive matrix spanning up to 10 or 12 selectable map configurations. Lower-numbered maps (e.g., Setting 1 or 2) feature loose slip parameters and subtle cut levels optimized for professional drivers on fresh slick tyres. Higher-numbered maps dramatically narrow the allowed slip ratios and maximize cut levels, configured specifically to stabilize the vehicle on worn tyres or during heavy rain.
- **The Simplified Production / Historic Profile:** Often completely empty, hard-locked to a single rigid mapping with non-adjustable entries, or disabled entirely via modular tuning parts. This accurately reflects vehicles running vintage mechanical setups or baseline road cars with binary, non-adjustable factory safety systems.

B. Schema

```

1. T C : object
  1a. Has T C2 : boolean
  1b. Frequency Hz : float
  1c. Min Speed Kmh : float
  1d. Gear Change Time : float
  1e. Min Cut Level : float
  1f. Max Cut Level : float
  1g. Settings [x] : object | can have multiple Settings
    1g1. Min Slip Ratio : float
    1g2. Max Slip Ratio : float
    1g3. Ref Slip Angle Deg : float
    1g4. Engine Cut Level : float
    1g5. Angular A C Cgain : float

```

- 1g6. Oversteer Gain : float
 - 1g7. Slip Angle Activation Deg : float
- 2. A B S : object
 - 2a. Settings [x] : object | can have multiple Settings
 - 2a1. Min Slip Ratio : float
 - 2a2. Max Slip Ratio : float
 - 2a3. Ref Slip Angledeg : float
 - 2a4. Cut Level : float
 - 2a5. Max Torque Variation : float
 - 2b. Frequency : float
 - 2c. Channels : integer
 - 2d. Min Speed Kmh : float
- 3. E D L : object
 - 3a. Active : boolean
 - 3b. Braketorquepower : float
 - 3c. Braketorquecoast : float
 - 3d. Deadzonecoast : float
 - 3e. Deadzonepower : float
 - 3f. Maxspinpower : float
 - 3g. Maxspincoast : float
 - 3h. Minspeed : float
- 4. E S P : object
 - 4a. Frequency Hz : float
 - 4b. Min Speed Kmh : float
 - 4c. Settings [x] : object | can have multiple Settings
 - 4c1. Gain : float
 - 4c2. Steer Gain : float
 - 4c3. Min Steer Gain : float
 - 4c4. Steer Gain Max Speed : float
 - 4c5. Oversteer Gain : float
 - 4c6. Understeer Gain : float
 - 4c7. Max Slip Ratio : float
 - 4c8. Dead Zone : float
 - 4c9. Filter Gain : float
 - 4c10. Brake Perc : float
 - 4c11. Brake Perc Activation : float

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	TC	None (Object)	Traction Control system block; processes wheel slip and yaw to modulate engine torque.
1a.	Has TC2	None (Boolean : True / False)	Enables a secondary traction-control map (TC2) for dual-stage intervention logic.
1b.	Frequency Hz	Hz (Hertz)	Control loop update frequency of the traction-control system (e.g., 333 Hz in Huracán ST Evo 2).
1c.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum vehicle speed below which traction control deactivates (e.g., 40 km/h).

ID	Name	Unit of Measurement	Description
1d.	Gear Change Time	s (Seconds)	Temporary TC suppression window duration during gear changes to avoid false interventions.
1e.	Min Cut Level	Ratio (0.0 - 1.0)	Minimum engine torque cut level applied at the lowest TC intervention step.
1f.	Max Cut Level	Ratio (0.0 - 1.0)	Maximum engine torque cut level at the most aggressive TC step.
1g.	Settings	None (Object array)	Per-map TC settings block; multiple entries (Settings 1–12+) selectable from the cockpit.
1g1.	Min Slip Ratio	Dimensionless ratio	Lower slip-ratio threshold at which TC begins monitoring wheel spin for this settings map.
1g2.	Max Slip Ratio	Dimensionless ratio	Upper slip-ratio threshold triggering full TC intervention for this settings map.
1g3.	Ref Slip Angle Deg	deg (Degrees)	Reference rear slip angle used to scale TC aggression relative to yaw behavior.
1g4.	Engine Cut Level	Ratio (0.0 - 1.0)	Engine torque reduction factor applied when slip thresholds are exceeded.
1g5.	Angular ACC Gain	Dimensionless gain	Yaw-rate / angular acceleration gain for TC correction authority (schema: Angular A C Cgain).
1g6.	Oversteer Gain	Dimensionless gain	Additional TC sensitivity multiplier when oversteer is detected.
1g7.	Slip Angle Activation Deg	deg (Degrees)	Minimum slip angle required before this TC map becomes active.
2.	ABS	None (Object)	Anti-lock Braking System block; modulates brake line pressure to prevent wheel lockup.
2a.	Settings	None (Object array)	Per-map ABS settings block; multiple entries (Settings 1–12+) for intervention intensity.
2a1.	Min Slip Ratio	Dimensionless ratio	Lower wheel slip threshold for ABS monitoring on this settings map.
2a2.	Max Slip Ratio	Dimensionless ratio	Upper slip threshold triggering maximum ABS pressure modulation.
2a3.	Ref Slip Angle Deg	deg (Degrees)	Reference slip angle for ABS yaw-stability cross-correlation (schema: Ref Slip Angledeg).

ID	Name	Unit of Measurement	Description
2a4.	Cut Level	Ratio (0.0 - 1.0)	Brake pressure reduction factor applied during ABS pulsing.
2a5.	Max Torque Variation	Nm or ratio	Maximum allowed brake torque fluctuation per ABS cycle on this map.
2b.	Frequency	Hz (Hertz)	ABS control loop pulsing frequency (e.g., 200 Hz in Huracán ST Evo 2).
2c.	Channels	None (Integer)	Number of independent ABS control channels (e.g., 4 = one per wheel).
2d.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum speed below which ABS intervention is disabled (e.g., 20 km/h).
3.	EDL	None (Object)	Electronic Differential Lock block; uses brake torque to emulate a locking differential (may be None/ disabled).
3a.	Active	None (Boolean : True / False)	Master enable for Electronic Differential Lock brake-based torque transfer.
3b.	Brake Torque Power	Nm (Newton-meters)	Brake torque applied to the spinning wheel under power to emulate diff lock (schema: Braketorquepower).
3c.	Brake Torque Coast	Nm (Newton-meters)	Brake torque applied during coasting to synchronize wheel speeds (schema: Braketorquecoast).
3d.	Dead Zone Coast	Ratio or rad/s	Speed-difference deadband below which coast EDL does not intervene (schema: Deadzonecoast).
3e.	Dead Zone Power	Ratio or rad/s	Speed-difference deadband below which power EDL does not intervene (schema: Deadzonepower).
3f.	Max Spin Power	rad/s or ratio	Maximum allowed wheel spin differential under power before EDL fully engages (schema: Maxspinpower).
3g.	Max Spin Coast	rad/s or ratio	Maximum allowed wheel spin differential on coast before EDL engages (schema: Maxspincoast).
3h.	Min Speed	M/s or km/h (Speed threshold	Minimum vehicle speed required for EDL operation (schema: Minspeed).

ID	Name	Unit of Measurement	Description
4.	ESP	None (Object)	Electronic Stability Program block; applies selective braking and torque reduction to control yaw.
4a.	Frequency Hz	Hz (Hertz)	ESP control loop update frequency (0 = disabled in some example configs).
4b.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum speed below which ESP deactivates.
4c.	Settings	None (Object array)	Per-map ESP settings block; multiple entries for intervention authority presets.
4c1.	Gain	Dimensionless gain	Global ESP intervention authority multiplier for this settings map.
4c2.	Steer Gain	Dimensionless gain	Steering-angle sensitivity gain for ESP yaw correction.
4c3.	Min Steer Gain	Dimensionless gain	Minimum steering gain applied at low speeds within ESP logic.
4c4.	Steer Gain Max Speed	km/h (Kilometers per hour)	Speed at which steering gain reaches its maximum ESP authority.
4c5.	Oversteer Gain	Dimensionless gain	Corrective gain when rear slip (oversteer) is detected.
4c6.	Understeer Gain	Dimensionless gain	Corrective gain when front slip (understeer) is detected.
4c7.	Max Slip Ratio	Dimensionless ratio	Maximum wheel slip ratio before ESP applies full brake/engine intervention.
4c8.	Dead Zone	deg or ratio	Yaw/steer error deadband where ESP remains passive.
4c9.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter on ESP sensor inputs to prevent oscillatory corrections.
4c10.	Brake Perc	Ratio (0.0 - 1.0)	Percentage of available brake pressure ESP may apply for stability correction.
4c11.	Brake Perc Activation	Ratio (0.0 - 1.0)	Slip/yaw threshold fraction at which ESP begins applying brake intervention.

D. Example data

I. Chosen Cars for Example

- Lamborghini Huracan ST Evo 2 (slug : ks_lamborghini_huracan_st_evo2)
- Maserati MC20 GT2 (slug : ks_maserati_mc20_gt2)
- Porsche 992 GT3 Cup (slug : ks_porsche_992_gt3_cup) [2 variants]

II. Example

Lamborghini Huracan ST Evo 2

- 1. T C
 - 1a. Has T C2 : false
 - 1b. Frequency Hz : 333.00000
 - 1c. Min Speed Kmh : 40.00000
 - 1d. Gear Change Time : 0.04500
 - 1e. Min Cut Level : 1000.00000
 - 1f. Max Cut Level : 1.00000
 - 1g. Settings 1
 - 1g1. Min Slip Ratio : 0.00000
 - 1g2. Max Slip Ratio : 0.00000
 - 1g3. Ref Slip Angle Deg : 0.00000
 - 1g4. Engine Cut Level : 0.00000
 - 1g5. Angular A C Cgain : 0.00000
 - 1g6. Oversteer Gain : 0.00000
 - 1g7. Slip Angle Activation Deg : 100.00000
 - 1g. Settings 2
 - 1g1. Min Slip Ratio : 0.18000
 - 1g2. Max Slip Ratio : 0.45000
 - 1g3. Ref Slip Angle Deg : 10.00000
 - 1g4. Engine Cut Level : 1.50000
 - 1g5. Angular A C Cgain : 3.00000
 - 1g6. Oversteer Gain : 3.00000
 - 1g7. Slip Angle Activation Deg : 6.00000
 - 1g. Settings 3
 - 1g1. Min Slip Ratio : 0.15000
 - 1g2. Max Slip Ratio : 0.40000
 - 1g3. Ref Slip Angle Deg : 9.50000
 - 1g4. Engine Cut Level : 1.50000
 - 1g5. Angular A C Cgain : 4.00000
 - 1g6. Oversteer Gain : 4.00000
 - 1g7. Slip Angle Activation Deg : 5.25000
 - 1g. Settings 4
 - 1g1. Min Slip Ratio : 0.12000
 - 1g2. Max Slip Ratio : 0.35000
 - 1g3. Ref Slip Angle Deg : 9.00000
 - 1g4. Engine Cut Level : 1.50000
 - 1g5. Angular A C Cgain : 5.00000
 - 1g6. Oversteer Gain : 5.00000
 - 1g7. Slip Angle Activation Deg : 4.75000
 - 1g. Settings 5
 - 1g1. Min Slip Ratio : 0.10000
 - 1g2. Max Slip Ratio : 0.30000
 - 1g3. Ref Slip Angle Deg : 9.00000
 - 1g4. Engine Cut Level : 1.50000
 - 1g5. Angular A C Cgain : 6.50000
 - 1g6. Oversteer Gain : 6.50000
 - 1g7. Slip Angle Activation Deg : 4.50000
 - 1g. Settings 6
 - 1g1. Min Slip Ratio : 0.09000

		-	1g2. Max Slip Ratio : 0.25000
		-	1g3. Ref Slip Angle Deg : 8.50000
		-	1g4. Engine Cut Level : 1.50000
		-	1g5. Angular A C Cgain : 7.50000
		-	1g6. Oversteer Gain : 7.50000

- 1g7. Slip Angle Activation Deg : 4.50000
 - 1g. Settings 7
 - 1g1. Min Slip Ratio : 0.08000
 - 1g2. Max Slip Ratio : 0.25000
 - 1g3. Ref Slip Angle Deg : 8.00000
 - 1g4. Engine Cut Level : 1.25000
 - 1g5. Angular A C Cgain : 8.50000
 - 1g6. Oversteer Gain : 7.50000
 - 1g7. Slip Angle Activation Deg : 4.50000
 - 1g. Settings 8
 - 1g1. Min Slip Ratio : 0.07000
 - 1g2. Max Slip Ratio : 0.20000
 - 1g3. Ref Slip Angle Deg : 8.00000
 - 1g4. Engine Cut Level : 1.25000
 - 1g5. Angular A C Cgain : 8.50000
 - 1g6. Oversteer Gain : 7.50000
 - 1g7. Slip Angle Activation Deg : 4.50000
 - 1g. Settings 9
 - 1g1. Min Slip Ratio : 0.06000
 - 1g2. Max Slip Ratio : 0.20000
 - 1g3. Ref Slip Angle Deg : 7.50000
 - 1g4. Engine Cut Level : 1.25000
 - 1g5. Angular A C Cgain : 8.50000
 - 1g6. Oversteer Gain : 7.50000
 - 1g7. Slip Angle Activation Deg : 4.00000
 - 1g. Settings 10
 - 1g1. Min Slip Ratio : 0.05000
 - 1g2. Max Slip Ratio : 0.15000
 - 1g3. Ref Slip Angle Deg : 7.00000
 - 1g4. Engine Cut Level : 1.00000
 - 1g5. Angular A C Cgain : 9.00000
 - 1g6. Oversteer Gain : 7.50000
 - 1g7. Slip Angle Activation Deg : 3.50000
- 2. A B S
 - 2a. Settings 1
 - 2a1. Min Slip Ratio : -1.00000
 - 2a2. Max Slip Ratio : -1.00000
 - 2a3. Ref Slip Angledeg : 0.00000
 - 2a4. Cut Level : 0.00000
 - 2a5. Max Torque Variation : 0.00000
 - 2a. Settings 2
 - 2a1. Min Slip Ratio : 0.08000
 - 2a2. Max Slip Ratio : 0.12000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.10000
 - 2a5. Max Torque Variation : 0.70000
 - 2a. Settings 3
 - 2a1. Min Slip Ratio : 0.07000
 - 2a2. Max Slip Ratio : 0.11000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.00000
 - 2a5. Max Torque Variation : 0.70000
 - 2a. Settings 4
 - 2a1. Min Slip Ratio : 0.06000

- 2a2. Max Slip Ratio : 0.08000
- 2a3. Ref Slip Angledeg : 5.00000
- 2a4. Cut Level : 0.10000
- 2a5. Max Torque Variation : 0.70000
- 2a. Settings 5
 - 2a1. Min Slip Ratio : 0.05000
 - 2a2. Max Slip Ratio : 0.07000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.10000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 6
 - 2a1. Min Slip Ratio : 0.04000
 - 2a2. Max Slip Ratio : 0.06000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.10000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 7
 - 2a1. Min Slip Ratio : 0.03200
 - 2a2. Max Slip Ratio : 0.07500
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.10000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 8
 - 2a1. Min Slip Ratio : 0.03000
 - 2a2. Max Slip Ratio : 0.07000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.10000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 9
 - 2a1. Min Slip Ratio : 0.03000
 - 2a2. Max Slip Ratio : 0.07000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.10000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 10
 - 2a1. Min Slip Ratio : 0.02900
 - 2a2. Max Slip Ratio : 0.06500
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.10000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 11
 - 2a1. Min Slip Ratio : 0.02900
 - 2a2. Max Slip Ratio : 0.06500
 - 2a3. Ref Slip Angledeg : 4.50000
 - 2a4. Cut Level : 0.10000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 12
 - 2a1. Min Slip Ratio : 0.03000
 - 2a2. Max Slip Ratio : 0.06500
 - 2a3. Ref Slip Angledeg : 4.25000
 - 2a4. Cut Level : 0.00000
 - 2a5. Max Torque Variation : 0.70000
- 2b. Frequency : 200.00000
- 2c. Channels : 4

- L 2d. Min Speed Kmh : 20.00000
- 3. E D L : None
- 4. E S P
 - 4a. Frequency Hz : 0.00000
 - 4b. Min Speed Kmh : 0.0

Maserati MC20 GT2

- 1. T C
 - 1a. Has T C2 : false
 - 1b. Frequency Hz : 333.00000
 - 1c. Min Speed Kmh : 40.00000
 - 1d. Gear Change Time : 0.02500
 - 1e. Min Cut Level : 1.50000
 - 1f. Max Cut Level : 0.10000
 - 1g. Settings 1
 - 1g1. Min Slip Ratio : 0.00000
 - 1g2. Max Slip Ratio : 0.00000
 - 1g3. Ref Slip Angle Deg : 0.00000
 - 1g4. Engine Cut Level : 0.00000
 - 1g5. Angular A C Cgain : 0.20000
 - 1g6. Oversteer Gain : 0.50000
 - 1g7. Slip Angle Activation Deg : 0.20000
 - 1g. Settings 2
 - 1g1. Min Slip Ratio : 0.25000
 - 1g2. Max Slip Ratio : 0.50000
 - 1g3. Ref Slip Angle Deg : 18.00000
 - 1g4. Engine Cut Level : 1.50000
 - 1g5. Angular A C Cgain : 2.00000
 - 1g6. Oversteer Gain : 2.00000
 - 1g7. Slip Angle Activation Deg : 4.00000
 - 1g. Settings 3
 - 1g1. Min Slip Ratio : 0.23000
 - 1g2. Max Slip Ratio : 0.31000
 - 1g3. Ref Slip Angle Deg : 14.80000
 - 1g4. Engine Cut Level : 1.25000
 - 1g5. Angular A C Cgain : 3.50000
 - 1g6. Oversteer Gain : 4.00000
 - 1g7. Slip Angle Activation Deg : 3.50000
 - 1g. Settings 4
 - 1g1. Min Slip Ratio : 0.20000
 - 1g2. Max Slip Ratio : 0.30000
 - 1g3. Ref Slip Angle Deg : 13.00000
 - 1g4. Engine Cut Level : 1.10000
 - 1g5. Angular A C Cgain : 5.00000
 - 1g6. Oversteer Gain : 6.00000
 - 1g7. Slip Angle Activation Deg : 3.00000
 - 1g. Settings 5
 - 1g1. Min Slip Ratio : 0.18000
 - 1g2. Max Slip Ratio : 0.30000
 - 1g3. Ref Slip Angle Deg : 11.00000
 - 1g4. Engine Cut Level : 0.95000
 - 1g5. Angular A C Cgain : 4.00000
 - 1g6. Oversteer Gain : 10.00000

L 1g7. Slip Angle Activation Deg : 2.50000

- 1g. Settings 6

- 1g1. Min Slip Ratio : 0.17000

- 1g2. Max Slip Ratio : 0.28000

- 1g3. Ref Slip Angle Deg : 9.00000

- 1g4. Engine Cut Level : 0.80000

- 1g5. Angular A C Cgain : 13.00000

- 1g6. Oversteer Gain : 11.00000

- 1g7. Slip Angle Activation Deg : 2.25000

- 1g. Settings 7

- 1g1. Min Slip Ratio : 0.15000

- 1g2. Max Slip Ratio : 0.24000

- 1g3. Ref Slip Angle Deg : 8.90000

- 1g4. Engine Cut Level : 0.65000

- 1g5. Angular A C Cgain : 20.00000

- 1g6. Oversteer Gain : 14.00000

- 1g7. Slip Angle Activation Deg : 2.00000

- 1g. Settings 8

- 1g1. Min Slip Ratio : 0.15000

- 1g2. Max Slip Ratio : 0.24000

- 1g3. Ref Slip Angle Deg : 8.90000

- 1g4. Engine Cut Level : 0.65000

- 1g5. Angular A C Cgain : 20.00000

- 1g6. Oversteer Gain : 14.00000

- 1g7. Slip Angle Activation Deg : 1.80000

- 1g. Settings 9

- 1g1. Min Slip Ratio : 0.12000

- 1g2. Max Slip Ratio : 0.25000

- 1g3. Ref Slip Angle Deg : 8.90000

- 1g4. Engine Cut Level : 0.65000

- 1g5. Angular A C Cgain : 20.00000

- 1g6. Oversteer Gain : 14.00000

- 1g7. Slip Angle Activation Deg : 1.80000

- 1g. Settings 10

- 1g1. Min Slip Ratio : 0.10000

- 1g2. Max Slip Ratio : 0.24000

- 1g3. Ref Slip Angle Deg : 8.90000

- 1g4. Engine Cut Level : 0.65000

- 1g5. Angular A C Cgain : 20.00000

- 1g6. Oversteer Gain : 14.00000

- 1g7. Slip Angle Activation Deg : 1.70000

- 1g. Settings 11

- 1g1. Min Slip Ratio : 0.08000

- 1g2. Max Slip Ratio : 0.18000

- 1g3. Ref Slip Angle Deg : 8.00000

- 1g4. Engine Cut Level : 0.10000

- 1g5. Angular A C Cgain : 0.00000

- 1g6. Oversteer Gain : 1.00000

- 1g7. Slip Angle Activation Deg : 1.60000

- 1g. Settings 12

- 1g1. Min Slip Ratio : 0.08000

- 1g2. Max Slip Ratio : 0.18000

- 1g3. Ref Slip Angle Deg : 7.00000

- 1g4. Engine Cut Level : 0.10000

- 1g5. Angular A C Cgain : 0.00000
- 1g6. Oversteer Gain : 1.00000
- 1g7. Slip Angle Activation Deg : 1.60000
- 1g. Settings 13
 - 1g1. Min Slip Ratio : 0.06000
 - 1g2. Max Slip Ratio : 0.18000
 - 1g3. Ref Slip Angle Deg : 5.00000
 - 1g4. Engine Cut Level : 0.10000
 - 1g5. Angular A C Cgain : 0.00000
 - 1g6. Oversteer Gain : 1.00000
 - 1g7. Slip Angle Activation Deg : 1.50000

- 2. A B S

- 2a. Settings 1
 - 2a1. Min Slip Ratio : -1.00000
 - 2a2. Max Slip Ratio : -1.00000
 - 2a3. Ref Slip Angledeg : 0.00000
 - 2a4. Cut Level : 0.00000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 2
 - 2a1. Min Slip Ratio : 0.02000
 - 2a2. Max Slip Ratio : 0.30000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.60000
- 2a. Settings 3
 - 2a1. Min Slip Ratio : 0.03800
 - 2a2. Max Slip Ratio : 0.09000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.80000
- 2a. Settings 4
 - 2a1. Min Slip Ratio : 0.03300
 - 2a2. Max Slip Ratio : 0.08400
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.80000
- 2a. Settings 5
 - 2a1. Min Slip Ratio : 0.03100
 - 2a2. Max Slip Ratio : 0.08200
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.80000
- 2a. Settings 6
 - 2a1. Min Slip Ratio : 0.03000
 - 2a2. Max Slip Ratio : 0.07900
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.80000
- 2a. Settings 7
 - 2a1. Min Slip Ratio : 0.02000
 - 2a2. Max Slip Ratio : 0.07500
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.80000

- 2a. Settings 8
 - 2a1. Min Slip Ratio : 0.04000
 - 2a2. Max Slip Ratio : 0.07000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 9
 - 2a1. Min Slip Ratio : 0.03700
 - 2a2. Max Slip Ratio : 0.07000
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 10
 - 2a1. Min Slip Ratio : 0.03000
 - 2a2. Max Slip Ratio : 0.06500
 - 2a3. Ref Slip Angledeg : 5.00000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 11
 - 2a1. Min Slip Ratio : 0.03000
 - 2a2. Max Slip Ratio : 0.06500
 - 2a3. Ref Slip Angledeg : 4.50000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 12
 - 2a1. Min Slip Ratio : 0.03000
 - 2a2. Max Slip Ratio : 0.06500
 - 2a3. Ref Slip Angledeg : 4.50000
 - 2a4. Cut Level : 0.15000
 - 2a5. Max Torque Variation : 0.70000
- 2b. Frequency : 200.00000
- 2c. Channels : 4
- 2d. Min Speed Kmh : 20.00000
- 3. E D L : None
- 4. E S P
 - 4a. Frequency Hz : 0.00000
 - 4b. Min Speed Kmh : 0.00000
- 4c. Settings : None

Porsche 992 GT3 Cup

1. No ABS No TC (file : 992_gt3_no_abs_no_tc.carelectronics)

- 1. T C : None
- 2. A B S : None
- 3. E D L : None
- 4. E S P
 - 4a. Frequency Hz : 0.00000
 - 4b. Min Speed Kmh : 0.00000
- 4c. Settings : None

2. Only ABS (file : 992_gt3_only_abs.carelectronics)

- 1. T C : None

2. A B S

2a. Settings 1

- 2a1. Min Slip Ratio : -1.00000 | | 2a2. Max Slip Ratio : -1.00000
- 2a3. Ref Slip Angledeg : 0.00000
- 2a4. Cut Level : 0.00000
- 2a5. Max Torque Variation : 0.00000

2a. Settings 2

- 2a1. Min Slip Ratio : 0.12000
- 2a2. Max Slip Ratio : 0.14000
- 2a3. Ref Slip Angledeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

2a. Settings 3

- 2a1. Min Slip Ratio : 0.11000
- 2a2. Max Slip Ratio : 0.14000
- 2a3. Ref Slip Angledeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

2a. Settings 4

- 2a1. Min Slip Ratio : 0.10000
- 2a2. Max Slip Ratio : 0.12000
- 2a3. Ref Slip Angledeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

2a. Settings 5

- 2a1. Min Slip Ratio : 0.08000
- 2a2. Max Slip Ratio : 0.10000
- 2a3. Ref Slip Angledeg : 7.00000
- 2a4. Cut Level : 0.02000
- 2a5. Max Torque Variation : 1.00000

2a. Settings 6

- 2a1. Min Slip Ratio : 0.07000
- 2a2. Max Slip Ratio : 0.08000
- 2a3. Ref Slip Angledeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

2a. Settings 7

- 2a1. Min Slip Ratio : 0.06000
- 2a2. Max Slip Ratio : 0.08000
- 2a3. Ref Slip Angledeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

2a. Settings 8

- 2a1. Min Slip Ratio : 0.05000
- 2a2. Max Slip Ratio : 0.07000
- 2a3. Ref Slip Angledeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

2a. Settings 9

- 2a1. Min Slip Ratio : 0.05000
- 2a2. Max Slip Ratio : 0.06000
- 2a3. Ref Slip Angledeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

2a. Settings 10

- 2a1. Min Slip Ratio : 0.04000

- 2a2. Max Slip Ratio : 0.05000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000
- 2a. Settings 11
 - 2a1. Min Slip Ratio : 0.02500
 - 2a2. Max Slip Ratio : 0.03500
 - 2a3. Ref Slip Angleddeg : 7.00000
 - 2a4. Cut Level : 0.20000
 - 2a5. Max Torque Variation : 1.00000
- 2a. Settings 12
 - 2a1. Min Slip Ratio : 0.01000
 - 2a2. Max Slip Ratio : 0.02000
 - 2a3. Ref Slip Angleddeg : 7.00000
 - 2a4. Cut Level : 0.20000
 - 2a5. Max Torque Variation : 1.00000
- 2b. Frequency : 200.00000
- 2c. Channels : 4
- 2d. Min Speed Kmh : 20.00000
- 3. E D L : None
- 4. E S P
 - 4a. Frequency Hz : 0.00000
 - 4b. Min Speed Kmh : 0.00000
 - 4c. Settings : None

10. Clutch [.clutch]

A. Description

I. General Description

The **Clutches** asset defines the mechanical and operational physics of the vehicle's decoupling mechanism between the engine's crankshaft and the transmission input shaft. It acts as the primary thermal, friction, and kinetic bridge during gear synchronization and stationary launches.

This asset governs how engine torque is progressively transferred to the drivetrain, defining both the physical properties of the clutch assembly (such as mass and maximum holding capacity) and the automated behaviors or electronic overrides that assist with shifting and stall prevention.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters configured within the `CLUTCHS` asset have a direct and critical impact on the vehicle's powertrain response and drivability:

- **Torque Transfer Efficiency:** It dictates the maximum power threshold the transmission can receive before slipping. If engine modifications increase torque beyond the clutch's mechanical limit, the asset causes the plates to slip, wasting power and slowing acceleration.
- **Shift Performance & Transition Smoothness:** During shifting phases, this asset regulates the speed and characteristics of the clamp engagement. It determines whether a gear change creates a sharp kinetic shockwave that can break rear-wheel traction (shift-lock) or slurs smoothly through a progressive friction zone.
- **Launch Control and Standing Starts:** It defines the bite-point characteristics. The structural settings determine how effectively a vehicle can launch from a dead stop without bogging down the engine RPM or inducing uncontrollable wheelspin.
- **Driver Assistance and Accessibility:** Through its automated profile parameters, it controls the electronic logic that bridges the gap between raw manual three-pedal mechanics and automated, paddle-activated, or software-assisted shifting maneuvers.

III. Key Architecture & Data Fields Explained

The internal structure of the `CLUTCHS` asset isolates core physical constants from algorithmic shift profiles.

1 - BASE MECHANICAL & INERTIAL PARAMETERS

- **Clutch Inertia:** The rotational mass moment of inertia of the clutch pressure plates and disc assembly. Lower inertia allows the engine to rev up and drop RPMs faster when decoupled, while higher inertia stores kinetic energy, easing standing starts on road cars.

- **Clutch Max Torque:** The maximum torque holding threshold (measured in Nm) that the friction plates can withstand when fully clamped. This acts as a physical fuse in the powertrain mesh.
- **Clutch Curve:** A look-up table or reference curve mapping the clamping force and friction coefficient relative to pedal travel or electronic actuator engagement, determining the progressive nature of the bite point.

2 - AUTOCLUTCH PROFILE PARAMETERS

When physical assistances or automated transmissions are active, the system routes logic through specific profile fields:

- **Upshift Profile / Down Shift Profile:** Explicit file paths pointing to external curve blueprints (.CURVE). These specify the exact engagement/disengagement timing, speed, and dampening curves applied to the clutch actuator during gear changes.
- **Forced On:** A boolean flag (TRUE/FALSE). When forced on, the simulation completely automates clutch operation, rendering a manual physical clutch pedal or binding obsolete.
- **Min Rpm's / Max Rpm's:** Defines the operational threshold window for automated anti-stall and launch logic. If engine speed drops below the minimum RPM, the automated actuator slips or completely pulls the clutch in to prevent an engine stall.
- **Use On Changes:** Dictates whether the automated clutch actuator should intervene only during active gear shifts or remain dynamically involved during low-speed maneuvers..

IV. Interpretation of Tuning Part Strategies

By cross-referencing these fields across various vehicle archetypes, specific mechanical and competitive design philosophies become apparent:

- **The Analog Historic Profile (e.g., Caterham 485 CSR):** Features a balanced CLUTCH INERTIA and a realistic CLUTCH MAX TORQUE overhead. In this profile, FORCED ON is set to FALSE and USE ON CHANGES is bypassed. The physics model relies entirely on the driver's manual execution of the friction zone, leaving no electronic safety net against stalls if the car spins or launches poorly.
- **The High-Performance Motorsport Profile (e.g., Formula Cars / Modern GT):** Designed with ultra-low CLUTCH INERTIA (e.g., 0.00500) to maximize engine responsiveness. The CLUTCH MAX TORQUE is tightly calculated to handle massive peak launch forces. Advanced shifting profiles (UPSHIFT/DOWNSHIFT .CURVE) are meticulously tuned to match the rapid millimeter-precision cuts of pneumatic or hydraulic actuators, completing engagements in milliseconds without upsetting the vehicle's balance.

B. Schema

| 1. Clutch Inertia : float

- 2. Clutch Max Torque : float
- 3. Autoclutch : object
 - 3a. Upshift Profile : string – path
 - 3b. Down Shift Profile : string – path
 - 3c. Forced On : boolean
 - 3d. Min Rpms : float
 - 3e. Max Rpms : float
 - 3f. Use On Changes : boolean
- 4. Clutch Curve : string – path

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Clutch Inertia	kg·m ² (Kilogram square meters)	Rotational inertia of the clutch pressure plates and disc assembly; lower values allow faster RPM changes when decoupled (e.g., 0.005 F2004, 0.017 Caterham 485 CSR).
2.	Clutch Max Torque	Nm (Newton-meters)	Maximum torque the friction plates can transmit when fully clamped before slip; acts as a powertrain fuse (e.g., 500 Nm Caterham, 700 Nm F2004).
3.	Autoclutch	None (Object)	Automated clutch assist profile block for shift timing, anti-stall, and launch logic.
3a.	Upshift Profile	None (.curve file path)	Engagement/disengagement curve applied during upshifts (e.g., upShiftProfile.curve); may be None if unused.
3b.	Down Shift Profile	None (.curve file path)	Engagement curve applied during downshifts (e.g., downShiftProfile.curve); may be None if unused.
3c.	Forced On	None (Boolean : True / False)	When true, fully automates clutch operation (no manual pedal required; e.g., Golf GTI mk8).
3d.	Min Rpms	RPM (Revolutions per minute)	Lower RPM threshold for automated anti-stall logic; clutch slips or disengages below this speed (e.g., 1200 Caterham, 4300 F2004).
3e.	Max Rpms	RPM (Revolutions per minute)	Upper RPM window bound for automated clutch/launch assist logic (e.g., 2000 Caterham, 4900 F2004).

ID	Name	Unit of Measurement	Description
3f.	Use On Changes	None (Boolean : True / False)	When true, autoclutch intervenes during gear changes; when false, driver controls friction zone manually (Caterham: false).
4.	Clutch Curve	None (.curve file path)	Look-up table mapping pedal travel or actuator position to clamping force / friction coefficient; often None when autoclutch profiles are used.

D. Example data

I. Chosen Cars for Example

- Caterham 485 CSR (slug : ks_caterham_485_csr)
- Ferrari F2004 (slug : ks_ferrari_f2004)
- Volkswagen Golf GTI mk8 (slug : ks_volkswagen_golf_gti_mk8)

II. Example

Caterham 485 CSR

```

- 1. Clutch Inertia : 0.01700
- 2. Clutch Max Torque : 500.00000
- 3. Autoclutch
  | 3a. Upshift Profile :
content\cars\ks_caterham_485_csr\data\upShiftProfile.curve
  | 3b. Down Shift Profile :
content\cars\ks_caterham_485_csr\data\downShiftProfile.curve
  | 3c. Forced On : false
  | 3d. Min Rpms : 1200.00000
  | 3e. Max Rpms : 2000.00000
  | 3f. Use On Changes : false
- 4. Clutch Curve : None

```

Ferrari F2004

```

- 1. Clutch Inertia : 0.00500
- 2. Clutch Max Torque : 700.00000
- 3. Autoclutch
  | 3a. Upshift Profile : None
  | 3b. Down Shift Profile :
content\cars\ks_ferrari_f2004\data\downShiftProfile.curve
  | 3c. Forced On : false
  | 3d. Min Rpms : 4300.00000
  | 3e. Max Rpms : 4900.00000
  | 3f. Use On Changes : true
- 4. Clutch Curve : None

```

Volkswagen Golf GTI mk8

- 1. Clutch Inertia : 0.01000
- 2. Clutch Max Torque : 450.00000
- 3. Autoclutch
 - 3a. Upshift Profile : None
 - 3b. Down Shift Profile : None
 - 3c. Forced On : true
 - 3d. Min Rpms : 1500.00000
 - 3e. Max Rpms : 2400.00000
 - 3f. Use On Changes : true
- 4. Clutch Curve : None

11. Coilover [.coilover]

A. Description

I. General Description

The **Coilovers** asset defines the mechanical, damping, and geometric physics of the vehicle's primary suspension assemblies. It acts as the central structural, kinetic, and dampening bridge between the unsprung mass (wheels, hubs, brakes) and the sprung mass (chassis and bodywork).

This asset governs how vertical, lateral, and longitudinal forces generated by track irregularities and aerodynamic loads are managed, defining both the physical properties of the suspension units (such as spring rates, helper springs, and bump stops) and the precise hydraulic damping profiles (bump and rebound) that control the velocity of suspension travel.

Additionally, it integrates wheel alignment parameters to establish the static and dynamic geometry of the tire contact patch.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters configured within the **COILOVERS** asset have a direct and critical impact on the vehicle's mechanical grip, aerodynamic stability, and overall handling balance:

- **Mechanical Grip and Tire Compliance:** It dictates how effectively the tires maintain contact with the track surface over bumps, kerbs, and undulations. Optimizing spring and damper rates prevents the tires from overloading or losing contact, directly impacting traction and braking efficiency.
- **Platform Control and Aerodynamic Stability:** On downforce-heavy vehicles, this asset regulates pitch, roll, and ride height variations. By controlling chassis movement under braking, acceleration, and cornering, it ensures the aerodynamic undertray and wings operate within their optimal performance window.
- **Transient Handling and Response:** During corner entry, apex, and exit phases, the dampers regulate the speed of weight transfer. This asset determines whether the vehicle responds sharply to driver inputs or exhibits sluggish transitions, directly influencing understeer and oversteer characteristics.
- **Endurance and Structural Management:** Through the configuration of progressive bump stops and helper springs, it defines how the vehicle absorbs severe impacts (such as high-speed kerb strikes) without destabilizing the chassis or causing harsh mechanical bottoming-out.

III. Key Architecture & Data Fields Explained

The internal structure of the **COILOVERS** asset isolates core elastic elements, hydraulic damping forces, and static alignment geometry.

1 - WHEEL RATE & SPRING PARAMETERS

- **Wheel Rate:** The effective stiffness of the suspension spring measured at the wheel. It determines the basic mechanical resistance to vertical loads and establishes the natural frequency of the chassis.
- **Bump Stop Up (Range & Rate):** Defines the clearance distance and progressive stiffness of the compression bump stop. It acts as a secondary, highly stiff spring to prevent the suspension from bottoming out mechanically against the chassis.
- **Bump Stop Down (Range & Rate):** Defines the travel limit and stiffness of the rebound stop, restricting the maximum extension of the suspension to prevent unsettled chassis behavior during severe unloading or airborne phases.
- **Helper Rate & Range:** Configures the characteristics of the helper (or tender) spring. It maintains tension on the main spring when the suspension is fully extended, ensuring predictable tire re-engagement under heavy droop conditions.

2 - DAMPER PROFILE PARAMETERS

The hydraulic dampening logic isolates low-speed chassis control from high-speed surface absorption:

- **Slow Bump / Slow Rebound:** Controls low-velocity suspension movements typically induced by driver inputs, such as body roll during cornering, pitching under braking, and squatting under acceleration..
- **Fast Bump / Fast Rebound:** Controls high-velocity suspension movements triggered by track inputs, such as hitting kerbs, ripples, or sudden track compression changes.
- **Damping Strategy Profiles:** Determines whether the asset operates on a Scientific Profile (exporting absolute raw forces like N_s/m for telemetry analysis in software like MoTeC) or a Driver-Centric Profile (mapping adjustments to discrete, tactile clicks on the damper valves).

3 - ALIGNMENTS & GEOMETRY FIELDS

- **Pressure:** Establishes the cold and hot tire inflation targets, which directly interact with the vertical spring rate of the tire sidewall.
- **Camber & Static Camber:** The angular inclination of the wheel relative to the ground. It optimizes the tire's contact patch under heavy cornering loads to maximize lateral grip.
- **Toe / Toe Out Linear:** The static angle of the wheels relative to the vehicle's longitudinal centerline, critical for tuning straight-line stability, turn-in response, and tire temperature generation.
- **Caster:** The forward or backward tilt of the steering axis, dictating the self-centering steering force and dynamic camber gain when turning the wheel.
- **Compound:** Links the alignment geometry to the specific tire construction metadata, ensuring the suspension mechanics align with the thermal and grip characteristics of the selected rubber.

IV. Interpretation of Tuning Part Strategies

By cross-referencing these fields across various vehicle archetypes, specific mechanical and competitive design philosophies become apparent:

- **The Compliant Road / Historic Profile (e.g., Caterham 485 CSR):** Features lower `WHEEL RATES` and generous `BUMP STOP RANGES` to allow significant suspension travel, prioritizing mechanical compliance and driver feedback over flat platform control. `DAMPERS` are tuned with softer `SLOW BUMP` settings to absorb road imperfections, and tires utilize higher profiles with conservative camber angles to maintain a forgiving, predictable slip angle progression.
- **The High-Downforce Motorsport Profile (e.g., Formula Cars / Modern GT):** Designed with ultra-stiff `WHEEL RATES` and minimal ride heights to lock down the aerodynamic platform. `BUMP STOPS` are short and aggressive, engaging almost immediately under high aero loads to maintain a stable splitter and diffuser height. `DAMPERS` utilize highly sophisticated fast/slow blow-off valves, tuned with millimeter precision to absorb track kerbs rapidly without letting the aerodynamic floor stall or upsetting the car's high-speed stability.

B. Schema

- 1. Spring Rate : float
- 2. Progressive Spring Rate : float
- 3. Bump Stop Up : object
 - 3a. Range : float
 - 3b. Reference : float
 - 3c. Force : float
 - 3d. Gamma : float
 - 3e. Length : float
 - 3f. Damping : float
- 4. Bump Stop Down : object
 - 4a. Range : float
 - 4b. Reference : float
 - 4c. Force : float
 - 4d. Gamma : float
 - 4e. Length : float
 - 4f. Damping : float
- 5. Collar Position : float
- 6. Damper : object
 - 6a. Fast : object
 - 6a1. Bump : float
 - 6a2. Rebound : float
 - 6b. Slow : object
 - 6b1. Bump : float
 - 6b2. Rebound : float
 - 6c. Fast Threshold Bump : float
 - 6d. Fast Threshold Rebound : float
 - 6e. Cooling Surface : float
 - 6f. Nominal Force : float
 - 6g. Min Stress Fatigue : float
 - 6h. Max Stress Fatigue : float
 - 6i. Thermal Capacity : float
 - 6j. Heat Transfer Coef : float

```

| 6k. Lut List : string - path
| 6l. Damper Lut Scale : float
| 7. Helper K : float
| 8. Helper Range : float
| 9. Rod Controllers : object
|   | 9a. Name : string
|   | 9b. Stages [x] : object | can have multiple Stages
|   |   | 9b1. Input Var : enum
|   |   | 9b2. Combinator Mode : enum
|   |   | 9b3. Lut : string - path
|   |   | 9b4. Filter Gain : float
|   |   | 9b5. Up Limit : float
|   |   | 9b6. Down Limit : float
|   |   | 9b7. Current Value : float
|   |   | 9b8. Const Value : float

```

Enum - Car Coilover

9b2	Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
9b2	Combinator Mode	UndefinedMode, Add, Mult

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Spring Rate	N/m (Newtons per meter)	Primary linear spring stiffness at the wheel (schema label Sprint Rate; likely Spring Rate). Caterham front: 14000 N/m, Alpine rear: 40000 N/m.
2.	Progressive Spring Rate	N/m ² or N/m (Progressive rate)	Secondary progressive spring rate; stiffness rises non-linearly with compression (0 = disabled on Caterham).
3.	Bump Stop Up	None (Object)	Upper bump-stop block limiting compression travel beyond main spring range.
3a.	Range	m (Meters)	Travel range before upper bump-stop engages (Bump Stop Up).
3b.	Reference	m (Meters)	Reference suspension position where upper bump-stop force scaling begins.

ID	Name	Unit of Measurement	Description
3c.	Force	N (Newtons)	Peak resistive force of the upper bump-stop at maximum compression.
3d.	Gamma	Dimensionless exponent	Non-linear shaping exponent for upper bump-stop force vs. compression.
3e.	Length	m (Meters)	Physical contact length of the upper bump-stop element.
3f.	Damping	N·s/m (Damping coefficient)	Damping applied during upper bump-stop engagement.
4.	Bump Stop Down	None (Object)	Lower bump-stop block limiting droop/rebound travel.
4a.	Range	m (Meters)	Travel range before lower bump-stop engages (Bump Stop Down).
4b.	Reference	m (Meters)	Reference position where lower bump-stop force model activates.
4c.	Force	N (Newtons)	Peak resistive force of the lower bump-stop at maximum extension.
4d.	Gamma	Dimensionless exponent	Non-linear shaping exponent for lower bump-stop force curve.
4e.	Length	m (Meters)	Physical contact length of the lower bump-stop element.
4f.	Damping	N·s/m (Damping coefficient)	Damping applied during lower bump-stop engagement on rebound.
5.	Collar Position	m (Meters)	Spring perch/collar position controlling static ride height and preload (e.g., 0.04961 m Caterham front).
6.	Damper	None (Object)	Hydraulic damper block: fast/slow circuits, thresholds, thermal model, and optional LUT.
6a.	Fast	None (Object)	Fast damper circuit sub-block for high-velocity wheel movements (kerbs, bumps).
6a1.	Bump	N·s/m (Damping rate)	Fast-speed compression damping under 6a Fast (e.g., 700 Caterham, 1500 Alpine).
6a2.	Rebound	N·s/m (Damping rate)	Fast-speed extension damping under 6a Fast (e.g., 1700 Caterham front).
6b.	Slow	None (Object)	Slow damper circuit sub-block for body roll, pitch, and weight transfer.

ID	Name	Unit of Measurement	Description
6a1.	Bump	N·s/m (Damping rate)	Slow-speed compression damping under 6b Slow; schema reuses ID 6a1 (examples may label 6b1).
6a2.	Rebound	N·s/m (Damping rate)	Slow-speed extension damping under 6b Slow; schema reuses ID 6a2 (examples may label 6b2).
6c.	Fast Threshold Bump	m/s (Velocity threshold)	Compression velocity above which the fast bump circuit engages (e.g., 0.15 Caterham, 0.50 Alpine).
6d.	Fast Threshold Rebound	m/s (Velocity threshold)	Extension velocity above which the fast rebound circuit engages.
6e.	Cooling Surface	m ² (Square meters)	Exposed damper body surface area for thermal dissipation modeling.
6f.	Nominal Force	N (Newtons)	Reference force level for damper nominal operating point and fatigue calculations.
6g.	Min Stress Fatigue	Pa or N (Stress threshold)	Lower bound stress below which damper fatigue accumulation is negligible.
6h.	Max Stress Fatigue	Pa or N (Stress threshold)	Upper bound stress at which damper fatigue reaches maximum degradation rate.
6i.	Thermal Capacity	J/K or J/°C	Heat storage capacity of the damper assembly under repeated high-energy cycles.
6j.	Heat Transfer Coef	W/(m ² ·K) (Coefficient)	Convective heat transfer rate from damper body to ambient airflow.
6k.	Lut List	None (File path)	Path to damper force-velocity LUT list; often None when using scalar rates only.
6i.	Damper Lut Scale	Dimensionless multiplier	Scaling factor applied to LUT damper output (0 = unused when Lut List is None).
7.	Helper K	N/m (Spring Rate)	Stiffness of helper/tender spring assisting the main coil at low compression.
8.	Helper Range	m (Meters)	Travel range over which the helper spring is active before the main spring carries full load.
9.	Rod Controllers	None (Object)	Dynamic suspension rod controller block; optional active/heave control pipelines.
9a.	Name	None (String)	Internal identifier for the rod controller; often None when inactive.

ID	Name	Unit of Measurement	Description
9b.	Stages	None (Object array)	Multi-stage controller pipeline; each stage maps telemetry input through a LUT (often None).
9b1.	Input Var	None (Telemetry enum)	Telemetry input channel (Speed, SusTravelLR, SteerDEG, SlipRatio, ErsChargeLevel, etc.).
9b2.	Combinator Mode	None (Math enum : Add / Mult)	How stage output combines with prior stages: additive or multiplicative.
9b3.	Lut	None (.curve file path)	Look-up table mapping filtered input to rod/damper modifier output.
9b4.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter coefficient smoothing controller input spikes.
9b5.	Up Limit	Depends on input variable	Upper clamp on processed input before LUT evaluation.
9b6.	Down Limit	Depends on input variable	Lower clamp on processed input before LUT evaluation.
9b7.	Current Value	Depends on input variable	Runtime controller stage output during simulation (telemetry/ debug).
9b8.	Const Value	Depends on input variable	Constant fallback output when Input Var is Const or LUT is inactive.

D. Example data

I. Chosen Cars for Example

- Caterham 485 CSR (slug : ks_caterham_485_csr)
- Alpine A110s (slug : ks_alpine_a110_s)
- Dallara EXP (slug : ks_dallara_exp)

II. Example

Caterham 485 CSR

1. Front Coilover (file : caterham_485_csr_fr_coil.coilover)

```

1. Spring Rate : 14000.00000
2. Progressive Spring Rate : 0.00000
3. Bump Stop Up : object
  3a. Range : 0.04539
  3b. Reference : 0.01000
  3c. Force : 100.00000
  3d. Gamma : 2.00000

```

- | 3e. Length : 0.02000
- | 3f. Damping : 300.00000
- 4. Bump Stop Down : object
 - | 4a. Range : 0.06961
 - | 4b. Reference : 0.01000
 - | 4c. Force : 80.00000
 - | 4d. Gamma : 2.00000
 - | 4e. Length : 0.02000
 - | 4f. Damping : 100.00000
- 5. Collar Position : 0.04961
- 6. Damper : object
 - | 6a. Fast : object
 - | 6a1. Bump : 700.00000
 - | 6a2. Rebound : 1700.00000
 - | 6b. Slow : object
 - | 6b1. Bump : 2000.00000
 - | 6b2. Rebound : 3000.00000
 - | 6c. Fast Threshold Bump : 0.15000
 - | 6d. Fast Threshold Rebound : 0.15000
 - | 6e. Cooling Surface : 0.00000
 - | 6f. Nominal Force : 0.00000
 - | 6g. Min Stress Fatigue : 0.00000
 - | 6h. Max Stress Fatigue : 0.00000
 - | 6i. Thermal Capacity : 0.00000
 - | 6j. Heat Transfer Coef : 0.00000
 - | 6k. Lut List : None
 - | 6l. Damper Lut Scale : 0.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
 - | 9a. Name : None
 - | 9b. Stages [x] : None

2. Rear Coilover (file : caterham_485_csr_r_coil.coilover)

- 1. Spring Rate : 16000.00000
- 2. Progressive Spring Rate : 0.00000
- 3. Bump Stop Up : object
 - | 3a. Range : 0.04031
 - | 3b. Reference : 0.01000
 - | 3c. Force : 100.00000
 - | 3d. Gamma : 2.00000
 - | 3e. Length : 0.02000
 - | 3f. Damping : 300.00000
- 4. Bump Stop Down : object
 - | 4a. Range : 0.04469
 - | 4b. Reference : 0.01000
 - | 4c. Force : 80.00000
 - | 4d. Gamma : 2.00000
 - | 4e. Length : 0.02000
 - | 4f. Damping : 100.00000
- 5. Collar Position : 0.06969
- 6. Damper : object
 - | 6a. Fast : object
 - | 6a1. Bump : 700.00000

- └─┬ 6a2. Rebound : 1700.00000
- └─6b. Slow : object
 - └─6b1. Bump : 2000.00000
 - └─6b2. Rebound : 3000.00000
- └─6c. Fast Threshold Bump : 0.15000
- └─6d. Fast Threshold Rebound : 0.15000
- └─6e. Cooling Surface : 0.00000
- └─6f. Nominal Force : 0.00000
- └─6g. Min Stress Fatigue : 0.00000
- └─6h. Max Stress Fatigue : 0.00000
- └─6i. Thermal Capacity : 0.00000
- └─6j. Heat Transfer Coef : 0.00000
- └─6k. Lut List : None
- └─6l. Damper Lut Scale : 0.00000
- └─7. Helper K : 0.00000
- └─8. Helper Range : 0.00000
- └─9. Rod Controllers : object
 - └─9a. Name : None
 - └─9b. Stages [x] : None

Alpine A110s

1. Front Coilover (file : ks_alpine_a110_s_front.coilover)

- └─1. Spring Rate : 31000.00000
- └─2. Progressive Spring Rate : 30000.00000
- └─3. Bump Stop Up : object
 - └─3a. Range : 0.01616
 - └─3b. Reference : 0.00600
 - └─3c. Force : 300.00000
 - └─3d. Gamma : 3.00000
 - └─3e. Length : 0.02000
 - └─3f. Damping : 400.00000
- └─4. Bump Stop Down : object
 - └─4a. Range : 0.05184
 - └─4b. Reference : 0.01000
 - └─4c. Force : 300.00000
 - └─4d. Gamma : 1.50000
 - └─4e. Length : 0.04500
 - └─4f. Damping : 400.00000
- └─5. Collar Position : 0.07184
- └─6. Damper : object
 - └─6a. Fast : object
 - └─6a1. Bump : 1500.00000
 - └─6a2. Rebound : 3500.00000
 - └─6b. Slow : object
 - └─6b1. Bump : 3000.00000
 - └─6b2. Rebound : 7000.00000
 - └─6c. Fast Threshold Bump : 0.50000
 - └─6d. Fast Threshold Rebound : 0.80000
 - └─6e. Cooling Surface : 0.00000
 - └─6f. Nominal Force : 0.00000
 - └─6g. Min Stress Fatigue : 0.00000
 - └─6h. Max Stress Fatigue : 0.00000
 - └─6i. Thermal Capacity : 0.00000

- | 6j. Heat Transfer Coef : 0.00000
- | 6k. Lut List : None
- | 6l. Damper Lut Scale : 0.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
 - | 9a. Name : None
 - | 9b. Stages [x] : None

2. Rear Coilover (file : ks_alpine_a110_s_rear.coilover)

- 1. Spring Rate : 40000.00000
- 2. Progressive Spring Rate : 40000.00000
- 3. Bump Stop Up : object
 - | 3a. Range : 0.06087
 - | 3b. Reference : 0.00600
 - | 3c. Force : 400.00000
 - | 3d. Gamma : 2.00000
 - | 3e. Length : 0.02000
 - | 3f. Damping : 500.00000
- 4. Bump Stop Down : object
 - | 4a. Range : -0.00887
 - | 4b. Reference : 0.01000
 - | 4c. Force : 400.00000
 - | 4d. Gamma : 1.50000
 - | 4e. Length : 0.02000
 - | 4f. Damping : 500.00000
- 5. Collar Position : 0.00913
- 6. Damper : object
 - | 6a. Fast : object
 - | 6a1. Bump : 1500.00000
 - | 6a2. Rebound : 2300.00000
 - | 6b. Slow : object
 - | 6b1. Bump : 4200.00000
 - | 6b2. Rebound : 7500.00000
 - | 6c. Fast Threshold Bump : 0.50000
 - | 6d. Fast Threshold Rebound : 0.80000
 - | 6e. Cooling Surface : 0.00000
 - | 6f. Nominal Force : 0.00000
 - | 6g. Min Stress Fatigue : 0.00000
 - | 6h. Max Stress Fatigue : 0.00000
 - | 6i. Thermal Capacity : 0.00000
 - | 6j. Heat Transfer Coef : 0.00000
 - | 6k. Lut List : None
 - | 6l. Damper Lut Scale : 0.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
 - | 9a. Name : None
 - | 9b. Stages [x] : None

Dallara EXP

1. Front Coilover (file : ks_dallara_exp_front_coil.coilover)

- 1. Spring Rate : 50000.00000
- 2. Progressive Spring Rate : 0.00000
- 3. Bump Stop Up : object
 - 3a. Range : -0.01135
 - 3b. Reference : 0.08000
 - 3c. Force : 350.00000
 - 3d. Gamma : 3.00000
 - 3e. Length : 0.01500
 - 3f. Damping : 1000.00000
- 4. Bump Stop Down : object
 - 4a. Range : 0.05635
 - 4b. Reference : 0.01000
 - 4c. Force : 350.00000
 - 4d. Gamma : 1.50000
 - 4e. Length : 0.02000
 - 4f. Damping : 500.00000
- 5. Collar Position : 0.05135
- 6. Damper : object
 - 6a. Fast : object
 - 6a1. Bump : 1234.00000
 - 6a2. Rebound : 2889.00000
 - 6b. Slow : object
 - 6b1. Bump : 5500.00000
 - 6b2. Rebound : 8200.00000
 - 6c. Fast Threshold Bump : 0.80000
 - 6d. Fast Threshold Rebound : 0.80000
 - 6e. Cooling Surface : 0.00000
 - 6f. Nominal Force : 0.00000
 - 6g. Min Stress Fatigue : 0.00000
 - 6h. Max Stress Fatigue : 0.00000
 - 6i. Thermal Capacity : 0.00000
 - 6j. Heat Transfer Coef : 0.00000
 - 6k. Lut List :
 - content\cars\common_phsx\dampers\penske\damper_penske.dampercurves
 - 6l. Damper Lut Scale : 1.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
 - 9a. Name : None
 - 9b. Stages [x] : None

2. Rear Coilover (file : ks_dallara_exp_rear_coil.coilover)

- 1. Spring Rate : 80000.00000
- 2. Progressive Spring Rate : 0.00000
- 3. Bump Stop Up : object
 - 3a. Range : 0.01310
 - 3b. Reference : 0.08000
 - 3c. Force : 400.00000
 - 3d. Gamma : 3.00000
 - 3e. Length : 0.01500
 - 3f. Damping : 1000.00000
- 4. Bump Stop Down : object
 - 4a. Range : 0.03490
 - 4b. Reference : 0.01000

- 4c. Force : 400.00000
- 4d. Gamma : 1.50000
- 4e. Length : 0.02000
- 4f. Damping : 1000.00000
- 5. Collar Position : 0.02690
- 6. Damper : object
 - 6a. Fast : object
 - 6a1. Bump : 2042.00000
 - 6a2. Rebound : 6889.00000
 - 6b. Slow : object
 - 6b1. Bump : 7056.00000
 - 6b2. Rebound : 12800.00000
 - 6c. Fast Threshold Bump : 0.80000
 - 6d. Fast Threshold Rebound : 0.80000
 - 6e. Cooling Surface : 0.00000
 - 6f. Nominal Force : 0.00000
 - 6g. Min Stress Fatigue : 0.00000
 - 6h. Max Stress Fatigue : 0.00000
 - 6i. Thermal Capacity : 0.00000
 - 6j. Heat Transfer Coef : 0.00000
 - 6k. Lut List :
content\cars\common_phsx\dampers\penske\damper_penske.dampercurves
 - 6l. Damper Lut Scale : 1.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
 - 9a. Name : None
 - 9b. Stages [x] : None

12. Damper Curves [.dampercurves]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx
-

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxx

B. Schema

[Data in a "Damper Curves List Edit" object]

- 1. Damper Lut [x] : string – path | can have multiple Damper Lut

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Damper Lut	None (.curve file path)	Indexed list of damper force–velocity LUT curves (.curve); each entry defines non-linear bump/rebound damping for a discrete damper click or valve map. Referenced from Coilover 6k Lut List (.DAMPERCURVES asset). Ford GT3: 5 curves per axle; Penske kit: 60 curves; Porsche Cayman GT4: 12 per axle.

D. Example data

I. Chosen Cars for Example

- Ford - GT3 Dampers (common_phsx) [2 dampers]
- Penske Dampers (common_phsx)
- Porsche Cayman Dampers (common_phsx) [2 dampers]

II. Example

Ford - GT3 Dampers

1. Front Damper (file : ford_gt3_front_damper.dampercurves)

```

└ 1. Damper Lut 1 :
content\cars\common_phsx\dampers\ford\ford_damper_front_1.curve
└ 1. Damper Lut 2 :
content\cars\common_phsx\dampers\ford\ford_damper_front_2.curve
└ 1. Damper Lut 3 :
content\cars\common_phsx\dampers\ford\ford_damper_front_3.curve
└ 1. Damper Lut 4 :
content\cars\common_phsx\dampers\ford\ford_damper_front_4.curve
└ 1. Damper Lut 5 :
content\cars\common_phsx\dampers\ford\ford_damper_front_5.curve

```

2. Rear Damper (file : ford_gt3_rear_damper.dampercurves)

```

└ 1. Damper Lut 1 :
content\cars\common_phsx\dampers\ford\ford_damper_rear_1.curve
└ 1. Damper Lut 2 :
content\cars\common_phsx\dampers\ford\ford_damper_rear_2.curve
└ 1. Damper Lut 3 :
content\cars\common_phsx\dampers\ford\ford_damper_rear_3.curve
└ 1. Damper Lut 4 :
content\cars\common_phsx\dampers\ford\ford_damper_rear_4.curve
└ 1. Damper Lut 5 :
content\cars\common_phsx\dampers\ford\ford_damper_rear_5.curve

```

Penske

└ 1. Damper Lut 1 :
content\cars\common_phsx\dampers\penske\damper_1.curve
└ 1. Damper Lut 2 :
content\cars\common_phsx\dampers\penske\damper_2.curve
└ 1. Damper Lut 3 :
content\cars\common_phsx\dampers\penske\damper_3.curve
└ 1. Damper Lut 4 :
content\cars\common_phsx\dampers\penske\damper_4.curve
└ 1. Damper Lut 5 :
content\cars\common_phsx\dampers\penske\damper_5.curve
└ 1. Damper Lut 6 :
content\cars\common_phsx\dampers\penske\damper_6.curve
└ 1. Damper Lut 7 :
content\cars\common_phsx\dampers\penske\damper_7.curve
└ 1. Damper Lut 8 :
content\cars\common_phsx\dampers\penske\damper_8.curve
└ 1. Damper Lut 9 :
content\cars\common_phsx\dampers\penske\damper_9.curve
└ 1. Damper Lut 10 :
content\cars\common_phsx\dampers\penske\damper_10.curve
└ 1. Damper Lut 11 :
content\cars\common_phsx\dampers\penske\damper_11.curve
└ 1. Damper Lut 12 :
content\cars\common_phsx\dampers\penske\damper_12.curve
└ 1. Damper Lut 13 :
content\cars\common_phsx\dampers\penske\damper_13.curve
└ 1. Damper Lut 14 :
content\cars\common_phsx\dampers\penske\damper_14.curve
└ 1. Damper Lut 15 :
content\cars\common_phsx\dampers\penske\damper_15.curve
└ 1. Damper Lut 16 :
content\cars\common_phsx\dampers\penske\damper_16.curve
└ 1. Damper Lut 17 :
content\cars\common_phsx\dampers\penske\damper_17.curve
└ 1. Damper Lut 18 :
content\cars\common_phsx\dampers\penske\damper_18.curve
└ 1. Damper Lut 19 :
content\cars\common_phsx\dampers\penske\damper_19.curve
└ 1. Damper Lut 20 :
content\cars\common_phsx\dampers\penske\damper_20.curve
└ 1. Damper Lut 21 :
content\cars\common_phsx\dampers\penske\damper_21.curve
└ 1. Damper Lut 22 :
content\cars\common_phsx\dampers\penske\damper_22.curve
└ 1. Damper Lut 23 :
content\cars\common_phsx\dampers\penske\damper_23.curve
└ 1. Damper Lut 24 :
content\cars\common_phsx\dampers\penske\damper_24.curve
└ 1. Damper Lut 25 :
content\cars\common_phsx\dampers\penske\damper_25.curve
└ 1. Damper Lut 26 :
content\cars\common_phsx\dampers\penske\damper_26.curve

└ 1. Damper Lut 27 :
content\cars\common_phsx\dampers\penske\damper_27.curve
└ 1. Damper Lut 28 :
content\cars\common_phsx\dampers\penske\damper_28.curve
└ 1. Damper Lut 29 :
content\cars\common_phsx\dampers\penske\damper_29.curve
└ 1. Damper Lut 30 :
content\cars\common_phsx\dampers\penske\damper_30.curve
└ 1. Damper Lut 31 :
content\cars\common_phsx\dampers\penske\damper_31.curve
└ 1. Damper Lut 32 :
content\cars\common_phsx\dampers\penske\damper_32.curve
└ 1. Damper Lut 33 :
content\cars\common_phsx\dampers\penske\damper_33.curve
└ 1. Damper Lut 34 :
content\cars\common_phsx\dampers\penske\damper_34.curve
└ 1. Damper Lut 35 :
content\cars\common_phsx\dampers\penske\damper_35.curve
└ 1. Damper Lut 36 :
content\cars\common_phsx\dampers\penske\damper_36.curve
└ 1. Damper Lut 37 :
content\cars\common_phsx\dampers\penske\damper_37.curve
└ 1. Damper Lut 38 :
content\cars\common_phsx\dampers\penske\damper_38.curve
└ 1. Damper Lut 39 :
content\cars\common_phsx\dampers\penske\damper_39.curve
└ 1. Damper Lut 40 :
content\cars\common_phsx\dampers\penske\damper_40.curve
└ 1. Damper Lut 41 :
content\cars\common_phsx\dampers\penske\damper_41.curve
└ 1. Damper Lut 42 :
content\cars\common_phsx\dampers\penske\damper_42.curve
└ 1. Damper Lut 43 :
content\cars\common_phsx\dampers\penske\damper_43.curve
└ 1. Damper Lut 44 :
content\cars\common_phsx\dampers\penske\damper_44.curve
└ 1. Damper Lut 45 :
content\cars\common_phsx\dampers\penske\damper_45.curve
└ 1. Damper Lut 46 :
content\cars\common_phsx\dampers\penske\damper_46.curve
└ 1. Damper Lut 47 :
content\cars\common_phsx\dampers\penske\damper_47.curve
└ 1. Damper Lut 48 :
content\cars\common_phsx\dampers\penske\damper_48.curve
└ 1. Damper Lut 49 :
content\cars\common_phsx\dampers\penske\damper_49.curve
└ 1. Damper Lut 50 :
content\cars\common_phsx\dampers\penske\damper_50.curve
└ 1. Damper Lut 51 :
content\cars\common_phsx\dampers\penske\damper_51.curve
└ 1. Damper Lut 52 :
content\cars\common_phsx\dampers\penske\damper_52.curve
└ 1. Damper Lut 53 :
content\cars\common_phsx\dampers\penske\damper_53.curve
└ 1. Damper Lut 54 :
content\cars\common_phsx\dampers\penske\damper_54.curve


```
└ 1. Damper Lut 55 :  
content\cars\common_phsx\dampers\penske\damper_55.curve  
└ 1. Damper Lut 56 :  
content\cars\common_phsx\dampers\penske\damper_56.curve  
└ 1. Damper Lut 57 :  
content\cars\common_phsx\dampers\penske\damper_57.curve  
└ 1. Damper Lut 58 :  
content\cars\common_phsx\dampers\penske\damper_58.curve  
└ 1. Damper Lut 59 :  
content\cars\common_phsx\dampers\penske\damper_59.curve  
└ 1. Damper Lut 60 :  
content\cars\common_phsx\dampers\penske\damper_60.curve
```

Porsche Cayman Dampers

1. Front Damper (file : cayman_gt4_front.dampercurves)

```
└ 1. Damper Lut 1 :  
content\cars\common_phsx\dampers\porsche\cayman\front1.curve  
└ 1. Damper Lut 2 :  
content\cars\common_phsx\dampers\porsche\cayman\front2.curve  
└ 1. Damper Lut 3 :  
content\cars\common_phsx\dampers\porsche\cayman\front3.curve  
└ 1. Damper Lut 4 :  
content\cars\common_phsx\dampers\porsche\cayman\front4.curve  
└ 1. Damper Lut 5 :  
content\cars\common_phsx\dampers\porsche\cayman\front5.curve  
└ 1. Damper Lut 6 :  
content\cars\common_phsx\dampers\porsche\cayman\front6.curve  
└ 1. Damper Lut 7 :  
content\cars\common_phsx\dampers\porsche\cayman\front7.curve  
└ 1. Damper Lut 8 :  
content\cars\common_phsx\dampers\porsche\cayman\front8.curve  
└ 1. Damper Lut 9 :  
content\cars\common_phsx\dampers\porsche\cayman\front9.curve  
└ 1. Damper Lut 10 :  
content\cars\common_phsx\dampers\porsche\cayman\front10.curve  
└ 1. Damper Lut 11 :  
content\cars\common_phsx\dampers\porsche\cayman\front11.curve  
└ 1. Damper Lut 12 :  
content\cars\common_phsx\dampers\porsche\cayman\front12.curve
```

2. Rear Damper (file : cayman_gt4_rear.dampercurves)

```
└ 1. Damper Lut 1 :  
content\cars\common_phsx\dampers\porsche\cayman\rear1.curve  
└ 1. Damper Lut 2 :  
content\cars\common_phsx\dampers\porsche\cayman\rear2.curve  
└ 1. Damper Lut 3 :  
content\cars\common_phsx\dampers\porsche\cayman\rear3.curve  
└ 1. Damper Lut 4 :  
content\cars\common_phsx\dampers\porsche\cayman\rear4.curve
```

└ 1. Damper Lut 5 :
content\cars\common_phsx\dampers\porsche\cayman\rear5.curve
└ 1. Damper Lut 6 :
content\cars\common_phsx\dampers\porsche\cayman\rear6.curve
└ 1. Damper Lut 7 :
content\cars\common_phsx\dampers\porsche\cayman\rear7.curve
└ 1. Damper Lut 8 :
content\cars\common_phsx\dampers\porsche\cayman\rear8.curve
└ 1. Damper Lut 9 :
content\cars\common_phsx\dampers\porsche\cayman\rear9.curve
└ 1. Damper Lut 10 :
content\cars\common_phsx\dampers\porsche\cayman\rear10.curve
└ 1. Damper Lut 11 :
content\cars\common_phsx\dampers\porsche\cayman\rear11.curve
└ 1. Damper Lut 12 :
content\cars\common_phsx\dampers\porsche\cayman\rear12.curve

13. Drivetrain [.drivetrain]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxxx

B. Schema

- 1. Traction Type : enum
- 2. Differential Data : object
 - 2a. Type : enum
 - 2b. Power : float
 - 2c. Coast : float
 - 2d. Preload : float
 - 2e. Front Share : float
 - 2f. Torque Bias Ratio Power : float
 - 2g. Torque Bias Ratio Coast : float
 - 2h. Thermal Capacity : float

- 2i. Surface : float
- 2j. Heat Transfer Coeff : float
- 2k. Wear Factor : float
- 2l. Friction Reduction With T : float
- 2m. Friction Ref T : float
- 3. Four W D Differentials : object
 - 3a. Front Diff : object
 - 2a. Type : float
 - 2b. Power : float
 - 2c. Coast : float
 - 2d. Preload : float
 - 2e. Front Share : float
 - 2f. Torque Bias Ratio Power : float
 - 2g. Torque Bias Ratio Coast : float
 - 2h. Thermal Capacity : float
 - 2i. Surface : float
 - 2j. Heat Transfer Coeff : float
 - 2k. Wear Factor : float
 - 2l. Friction Reduction With T : float
 - 2m. Friction Ref T : float
 - 3b. Center Diff : object
 - 2a. Type : enum
 - 2b. Power : float
 - 2c. Coast : float
 - 2d. Preload : float
 - 2e. Front Share : float
 - 2f. Torque Bias Ratio Power : float
 - 2g. Torque Bias Ratio Coast : float
 - 2h. Thermal Capacity : float
 - 2i. Surface : float
 - 2j. Heat Transfer Coeff : float
 - 2k. Wear Factor : float
 - 2l. Friction Reduction With T : float
 - 2m. Friction Ref T : float
 - 3c. Rear Diff : object
 - 2a. Type : enum
 - 2b. Power : float
 - 2c. Coast : float
 - 2d. Preload : float
 - 2e. Front Share : float
 - 2f. Torque Bias Ratio Power : float
 - 2g. Torque Bias Ratio Coast : float
 - 2h. Thermal Capacity : float
 - 2i. Surface : float
 - 2j. Heat Transfer Coeff : float
 - 2k. Wear Factor : float
 - 2l. Friction Reduction With T : float
 - 2m. Friction Ref T : float
- 4. Stiffness : float
- 5. Stiffness Mult : float
- 6. Damping Ratio : float
- 7. Max Torsion Deg : float
- 8. Non Linear Model : boolean
- 9. Max Between Lsd And Elsd : boolean
- 10. Has Cockpit Controls : boolean
- 11. Front Lock Controllers : object

- 11a. Name : string
- 11b. Stages [x] : object | can have multiple Stages
 - 11b1. Input Var : enum
 - 11b2. Combinator Mode : enum
 - 11b3. Lut : string - path
 - 11b4. Filter Gain : float
 - 11b5. Up Limit : float
 - 11b6. Down Limit : float
 - 11b7. Current Value : float
 - 11b8. Const Value : float
- 12. Center Lock Controllers : object
 - 11a. Name : string
 - 11b. Stages [x] : object | can have multiple Stages
 - 11b1. Input Var : enum
 - 11b2. Combinator Mode : enum
 - 11b3. Lut : string - path
 - 11b4. Filter Gain : float
 - 11b5. Up Limit : float
 - 11b6. Down Limit : float
 - 11b7. Current Value : float
 - 11b8. Const Value : float
- 13. Rear Lock Controllers : object
 - 11a. Name : string
 - 11b. Stages [x] : object | can have multiple Stages
 - 11b1. Input Var : enum
 - 11b2. Combinator Mode : enum
 - 11b3. Lut : string - path
 - 11b4. Filter Gain : float
 - 11b5. Up Limit : float
 - 11b6. Down Limit : float
 - 11b7. Current Value : float
 - 11b8. Const Value : float
- 14. Left Lock Controllers : object
 - 11a. Name : string
 - 11b. Stages [x] : object | can have multiple Stages
 - 11b1. Input Var : enum
 - 11b2. Combinator Mode : enum
 - 11b3. Lut : string - path
 - 11b4. Filter Gain : float
 - 11b5. Up Limit : float
 - 11b6. Down Limit : float
 - 11b7. Current Value : float
 - 11b8. Const Value : float
- 15. Right Lock Controllers : object
 - 11a. Name : string
 - 11b. Stages [x] : object | can have multiple Stages
 - 11b1. Input Var : enum
 - 11b2. Combinator Mode : enum
 - 11b3. Lut : string - path
 - 11b4. Filter Gain : float
 - 11b5. Up Limit : float
 - 11b6. Down Limit : float
 - 11b7. Current Value : float
 - 11b8. Const Value : float
- 16. Awd Clutches [x] : object | can have multiple Awd Clutches
 - 16a. Position : integer

- 16b. Preload : float
- 16c. Controllers : object
 - 11a. Name : string
 - 11b. Stages [x] : object | can have multiple Stages
 - 11b1. Input Var : enum
 - 11b2. Combinator Mode : enum
 - 11b3. Lut : string – path
 - 11b4. Filter Gain : float
 - 11b5. Up Limit : float
 - 11b6. Down Limit : float
 - 11b7. Current Value : float
 - 11b8. Const Value : float

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Traction Type	None (Enum : FWD / RWD / AWDF / ...)	Global driveline layout enum (e.g., AWDF Audi RS3, RWD Ferrari F40 LM, FWD Abarth 695).
2.	Differential Data	None (Object)	Primary differential parameter block for single-diff or default axle configuration.
2a.	Type	None (Enum : LSD / Spool / Torsen / EpicyclicTorsen / EpicyclicLSD / TorqueVectoring)	Differential mechanism type governing torque split and lock behavior.
2b.	Power	Nm or ratio	Lock/acceleration sensitivity under power (on-throttle); higher values increase drive-wheel coupling on corner exit.
2c.	Coast	Nm or ratio	Lock sensitivity on coast/off-throttle; stabilizes the axle under braking and lift-off.
2d.	Preload	Nm (Newton-meters)	Static preload torque before differential plates begin slipping.
2e.	Front Share	Ratio (0.0 - 1.0)	Fraction of total driveline torque directed to the front axle (AWD/ center diff).
2f.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Torque bias ratio between sides or axles under power.
2g.	Torque Bias Ratio Coast	Ratio	Torque bias ratio between sides or axles on coast.
2h.	Thermal Capacity	J/K or J/°C	Heat storage capacity of differential friction components for thermal wear modeling.
2i.	Surface	m ² (Square meters)	Exposed friction surface area of the differential housing for cooling calculations.

ID	Name	Unit of Measurement	Description
2j.	Heat Transfer Coeff	W/(m ² ·K) (Coefficient)	Convective heat dissipation rate from differential to ambient/oil.
2k.	Wear Factor	Dimensionless coefficient	Rate at which differential friction surfaces degrade under load and temperature.
2l.	Friction Reduction With T	Ratio/°C	Linear friction reduction per degree of differential temperature rise.
2m.	Friction Ref T	°C (Degrees Celsius)	Reference temperature at which nominal differential friction is defined.
3.	Four W D Differentials	None (Object)	AWD/4WD triple-differential container: front, center, and rear axle blocks.
3a.	Front Diff	None (Object)	Front Diff sub-block within Four W D Differentials (3).
3b.	Center Diff	None (Object)	Center Diff sub-block within Four W D Differentials (3).
3c.	Rear Diff	None (Object)	Rear Diff sub-block within Four W D Differentials (3).
4.	Stiffness	Nm/deg (Newton-meters per degree)	Torsional stiffness of the differential coupling model (e.g., 4500 Nm/deg Ferrari F40 LM).
5.	Stiffness Mult	Dimensionless multiplier	Scalar multiplier applied to base stiffness for non-linear or epicyclic models (e.g., 2.0 F40 LM).
6.	Damping Ratio	Dimensionless ratio (ζ)	Torsional damping ratio of the differential dynamics model (e.g., 0.05 F40 LM).
7.	Max Torsion Deg	deg (Degrees)	Maximum torsion angle before the differential model saturates (e.g., 40° F40 LM).
8.	Non Linear Model	None (Boolean : True / False)	When true, enables non-linear torsional differential physics (true on F40 LM).
9.	Max Between Lsd And Elsa	None (Boolean : True / False)	When true, caps combined LSD and eLSD lock contribution to a maximum blended value.
10.	Has Cockpit Controls	None (Boolean : True / False)	When true, exposes in-cockpit differential lock or bias adjustment to the driver.
11.	Front Lock Controllers	None (Object)	Dynamic controller modulating front lock controllers under telemetry conditions.

ID	Name	Unit of Measurement	Description
11a.	Name	None (String)	Internal identifier for the lock controller; often None when inactive. Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b.	Stages	None (Object array)	Multi-stage controller pipeline mapping telemetry input through LUTs (often None on simple LSD setups). Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b1.	Input Var	None (Telemetry enum)	Telemetry input channel (Gas, SlipRatioFrontAVG, LatG, Speed, ErsChargeLevel, etc.). Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b2.	Combinator Mode	None (Math enum : Add / Mult)	How stage output combines with prior stages: additive or multiplicative. Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b3.	Lut	None (.curve file path)	Look-up table mapping filtered input to differential/clutch lock modifier. Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b4.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter coefficient smoothing controller input spikes. Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b5.	Up Limit	Depends on input variable	Upper clamp on processed input before LUT evaluation. Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b6.	Down Limit	Depends on input variable	Lower clamp on processed input before LUT evaluation. Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b7.	Current Value	Depends on input variable	Runtime controller stage output during simulation (telemetry/ debug). Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.
11b8.	Const Value	Depends on input variable	Constant fallback output when Input Var is Const or LUT is inactive. Under Front Lock Controllers ; schema reuses IDs 11a/11b/11b1–11b8.

ID	Name	Unit of Measurement	Description
12.	Center Lock Controllers	None (Object)	Dynamic controller modulating center lock controllers under telemetry conditions.
13.	Rear Lock Controllers	None (Object)	Dynamic controller modulating rear lock controllers under telemetry conditions.
14.	Left Lock Controllers	None (Object)	Dynamic controller modulating left lock controllers under telemetry conditions.
15.	Right Lock Controllers	None (Object)	Dynamic controller modulating right lock controllers under telemetry conditions.
16.	Awd Clutches	None (Object array)	AWD clutch pack array; each entry couples axles with preload and optional active controllers (Audi RS3).
16a.	Position	None (Integer index)	Clutch pack index/position in the AWD coupling matrix (e.g., 2 on Audi RS3).
16b.	Preload	Nm (Newton-meters)	Static preload torque on this AWD clutch before slip occurs.
16c.	Controllers	None (Object)	Nested lock controller block for this AWD clutch pack.

D. Example data

I. Chosen Cars for Example

- Audi RS3 Sportback (slug : ks_audi_rs_3_sportback)
- Ferrari F40 LM (slug : ks_ferrari_f40_lm)
- Abarth 695 Biposto (slug : ks_abarth_695_biposto)

II. Example

Audi RS3 Sportback

- 1. Traction Type : AWDF
- 2. Differential Data
 - 2a. Type : LSD
 - 2b. Power : 0.00000
 - 2c. Coast : 0.00000
 - 2d. Preload : 0.00000
 - 2e. Front Share : 0.00000
 - 2f. Torque Bias Ratio Power : 0.00000
 - 2g. Torque Bias Ratio Coast : 0.00000
 - 2h. Thermal Capacity : 0.00000
 - 2i. Surface : 0.00000

- 2j. Heat Transfer Coeff : 0.00000
- 2k. Wear Factor : 0.00000
- 2l. Friction Reduction With T : 0.00000
- 2m. Friction Ref T : 0.00000
- 3. Four W D Differentials
 - 3a. Front Diff
 - 2a. Type : LSD
 - 2b. Power : 0.50000
 - 2c. Coast : 0.10000
 - 2d. Preload : 30.00000
 - 2e. Front Share : 0.00000
 - 2f. Torque Bias Ratio Power : 0.00000
 - 2g. Torque Bias Ratio Coast : 0.00000
 - 2h. Thermal Capacity : 0.00000
 - 2i. Surface : 0.00000
 - 2j. Heat Transfer Coeff : 0.00000
 - 2k. Wear Factor : 0.00000
 - 2l. Friction Reduction With T : 0.00000
 - 2m. Friction Ref T : 0.00000
 - 3b. Center Diff
 - 2a. Type : EpicyclicLSD
 - 2b. Power : 0.03000
 - 2c. Coast : 0.03000
 - 2d. Preload : 1.00000
 - 2e. Front Share : 0.40000
 - 2f. Torque Bias Ratio Power : 0.00000
 - 2g. Torque Bias Ratio Coast : 0.00000
 - 2h. Thermal Capacity : 0.00000
 - 2i. Surface : 0.00000
 - 2j. Heat Transfer Coeff : 0.00000
 - 2k. Wear Factor : 0.00000
 - 2l. Friction Reduction With T : 0.00000
 - 2m. Friction Ref T : 0.00000
 - 3c. Rear Diff
 - 2a. Type : Spool
 - 2b. Power : 0.03000
 - 2c. Coast : 0.03000
 - 2d. Preload : 0.00000
 - 2e. Front Share : 0.00000
 - 2f. Torque Bias Ratio Power : 2.50000
 - 2g. Torque Bias Ratio Coast : 2.00000
 - 2h. Thermal Capacity : 0.00000
 - 2i. Surface : 0.00000
 - 2j. Heat Transfer Coeff : 0.00000
 - 2k. Wear Factor : 0.00000
 - 2l. Friction Reduction With T : 0.00000
 - 2m. Friction Ref T : 0.00000
- 4. Stiffness : 0.00000
- 5. Stiffness Mult : 0.00000
- 6. Damping Ratio : 0.00000
- 7. Max Torsion Deg : 0.00000
- 8. Non Linear Model : false
- 9. Max Between Lsd And Elsd : false
- 10. Has Cockpit Controls : false
- 11. Front Lock Controllers
 - 11a. Name : None

```

| 11b. Stages : None
- 12. Center Lock Controllers
| 11a. Name : None
| 11b. Stages : None
- 13. Rear Lock Controllers
| 11a. Name : None
| 11b. Stages : None
- 14. Left Lock Controllers
| 11a. Name : None
| 11b. Stages : None
- 15. Right Lock Controllers
| 11a. Name : None
| 11b. Stages : None
- 16. Awd Clutches 1
| 16a. Position : 2
| 16b. Preload : 0.000
| 16c. Controllers
| 11a. Name : None
| 11b. Stages 1
| 11b1. Input Var : Gas
| 11b2. Combinator Mode : Add
| 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_awd_gas_add.curve
| 11b4. Filter Gain : 0.00000
| 11b5. Up Limit : 3000.00000
| 11b6. Down Limit : 0.00000
| 11b7. Current Value : 0.00000
| 11b8. Const Value : 0.00000
| 11b. Stages 2
| 11b1. Input Var : SlipRatioFrontAVG
| 11b2. Combinator Mode : Add
| 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_awd_clutch_SRavg.curve
| 11b4. Filter Gain : 0.00000
| 11b5. Up Limit : 3000.00000
| 11b6. Down Limit : 0.00000
| 11b7. Current Value : 0.00000
| 11b8. Const Value : 0.00000
| 11b. Stages 3
| 11b1. Input Var : LatG
| 11b2. Combinator Mode : Mult
| 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_steer_right_clutch_Perf1.curve
| 11b4. Filter Gain : 0.00000
| 11b5. Up Limit : 3000.00000
| 11b6. Down Limit : 0.00000
| 11b7. Current Value : 0.00000
| 11b8. Const Value : 0.00000
| 11b. Stages 4
| 11b1. Input Var : OversteerFactor
| 11b2. Combinator Mode : Mult

```

```

| | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_awd_clutch_oversteer.curve
| | | | 11b4. Filter Gain : 0.00000
| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000
| | | | 11b8. Const Value : 0.00000
| | | | 11b. Stages 5
| | | | 11b1. Input Var : Speed
| | | | 11b2. Combinator Mode : Mult
| | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_awd_clutch_speed_power.curve
| | | | 11b4. Filter Gain : 0.00000
| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000
| | | | 11b8. Const Value : 0.00000
| | | | 16. Awd Clutches 2
| | | | 16a. Position : 3
| | | | 16b. Preload : 0.000
| | | | 16c. Controllers
| | | | 11a. Name : None
| | | | 11b. Stages 1
| | | | 11b1. Input Var : Gas
| | | | 11b2. Combinator Mode : Add
| | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_awd_gas_add.curve
| | | | 11b4. Filter Gain : 0.00000
| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000
| | | | 11b8. Const Value : 0.00000
| | | | 11b. Stages 2
| | | | 11b1. Input Var : SlipRatioFrontAVG
| | | | 11b2. Combinator Mode : Add
| | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_awd_clutch_SRavg.curve
| | | | 11b4. Filter Gain : 0.00000
| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000
| | | | 11b8. Const Value : 0.00000
| | | | 11b. Stages 3
| | | | 11b1. Input Var : LatG
| | | | 11b2. Combinator Mode : Mult
| | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_steer_right_clutch_Perf1.curve
| | | | 11b4. Filter Gain : 0.00000
| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000

```

```

| | | L 11b8. Const Value : 0.00000
| | | - 11b. Stages 4
| | | | 11b1. Input Var : OversteerFactor
| | | | 11b2. Combinator Mode : Mult
| | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_awd_clutch_oversteer.curve
| | | | 11b4. Filter Gain : 0.00000
| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000
| | | | 11b8. Const Value : 0.00000
| | | - 11b. Stages 5
| | | | 11b1. Input Var : Speed
| | | | 11b2. Combinator Mode : Mult
| | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_awd_clutch_speed_power.curve
| | | | 11b4. Filter Gain : 0.00000
| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000
| | | | 11b8. Const Value : 0.00000

```

Ferrari F40 LM

```

- 1. Traction Type : RWD
- 2. Differential Data
| 2a. Type : LSD
| 2b. Power : 0.00000
| 2c. Coast : 0.00000
| 2d. Preload : 0.00000
| 2e. Front Share : 0.00000
| 2f. Torque Bias Ratio Power : 0.00000
| 2g. Torque Bias Ratio Coast : 0.00000
| 2h. Thermal Capacity : 0.00000
| 2i. Surface : 0.00000
| 2j. Heat Transfer Coeff : 0.00000
| 2k. Wear Factor : 0.00000
| 2l. Friction Reduction With T : 0.00000
| 2m. Friction Ref T : 0.00000
- 3. Four W D Differentials
| 3a. Front Diff
| | 2a. Type : LSF
| | 2b. Power : 0.00000
| | 2c. Coast : 0.00000
| | 2d. Preload : 0.00000
| | 2e. Front Share : 0.00000
| | 2f. Torque Bias Ratio Power : 0.00000
| | 2g. Torque Bias Ratio Coast : 0.00000
| | 2h. Thermal Capacity : 0.00000
| | 2i. Surface : 0.00000
| | 2j. Heat Transfer Coeff : 0.00000
| | 2k. Wear Factor : 0.00000

```

- | 2l. Friction Reduction With T : 0.00000
 - | 2m. Friction Ref T : 0.00000
- 3b. Center Diff
 - | 2a. Type : LSD
 - | 2b. Power : 0.00000
 - | 2c. Coast : 0.00000
 - | 2d. Preload : 0.00000
 - | 2e. Front Share : 0.00000
 - | 2f. Torque Bias Ratio Power : 0.00000
 - | 2g. Torque Bias Ratio Coast : 0.00000
 - | 2h. Thermal Capacity : 0.00000
 - | 2i. Surface : 0.00000
 - | 2j. Heat Transfer Coeff : 0.00000
 - | 2k. Wear Factor : 0.00000
 - | 2l. Friction Reduction With T : 0.00000
 - | 2m. Friction Ref T : 0.00000
- 3c. Rear Diff
 - | 2a. Type : LSD
 - | 2b. Power : 0.00000
 - | 2c. Coast : 0.00000
 - | 2d. Preload : 0.00000
 - | 2e. Front Share : 0.00000
 - | 2f. Torque Bias Ratio Power : 0.00000
 - | 2g. Torque Bias Ratio Coast : 0.00000
 - | 2h. Thermal Capacity : 0.00000
 - | 2i. Surface : 0.00000
 - | 2j. Heat Transfer Coeff : 0.00000
 - | 2k. Wear Factor : 0.00000
 - | 2l. Friction Reduction With T : 0.00000
 - | 2m. Friction Ref T : 0.00000
- 4. Stiffness : 4500.00000
- 5. Stiffness Mult : 2.00000
- 6. Damping Ratio : 0.05000
- 7. Max Torsion Deg : 40.00000
- 8. Non Linear Model : true
- 9. Max Between Lsd And Elsd : false
- 10. Has Cockpit Controls : false
- 11. Front Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 12. Center Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 13. Rear Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 14. Left Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 15. Right Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 16. Awd Clutches : None

Abarth 695 Biposto

- 1. Traction Type : FWD
- 2. Differential Data
 - 2a. Type : LSD
 - 2b. Power : 0.25000
 - 2c. Coast : 0.25000
 - 2d. Preload : 20.00000
 - 2e. Front Share : 0.00000
 - 2f. Torque Bias Ratio Power : 0.00000
 - 2g. Torque Bias Ratio Coast : 0.00000
 - 2h. Thermal Capacity : 0.00000
 - 2i. Surface : 0.00000
 - 2j. Heat Transfer Coeff : 0.00000
 - 2k. Wear Factor : 0.00000
 - 2l. Friction Reduction With T : 0.00000
 - 2m. Friction Ref T : 0.00000
- 3. Four W D Differentials
 - 3a. Front Diff
 - 2a. Type : LSD
 - 2b. Power : 0.00000
 - 2c. Coast : 0.00000
 - 2d. Preload : 0.00000
 - 2e. Front Share : 0.00000
 - 2f. Torque Bias Ratio Power : 0.00000
 - 2g. Torque Bias Ratio Coast : 0.00000
 - 2h. Thermal Capacity : 0.00000
 - 2i. Surface : 0.00000
 - 2j. Heat Transfer Coeff : 0.00000
 - 2k. Wear Factor : 0.00000
 - 2l. Friction Reduction With T : 0.00000
 - 2m. Friction Ref T : 0.00000
 - 3b. Center Diff
 - 2a. Type : LSD
 - 2b. Power : 0.00000
 - 2c. Coast : 0.00000
 - 2d. Preload : 0.00000
 - 2e. Front Share : 0.00000
 - 2f. Torque Bias Ratio Power : 0.00000
 - 2g. Torque Bias Ratio Coast : 0.00000
 - 2h. Thermal Capacity : 0.00000
 - 2i. Surface : 0.00000
 - 2j. Heat Transfer Coeff : 0.00000
 - 2k. Wear Factor : 0.00000
 - 2l. Friction Reduction With T : 0.00000
 - 2m. Friction Ref T : 0.00000
 - 3c. Rear Diff
 - 2a. Type : LSD
 - 2b. Power : 0.00000
 - 2c. Coast : 0.00000
 - 2d. Preload : 0.00000
 - 2e. Front Share : 0.00000
 - 2f. Torque Bias Ratio Power : 0.00000
 - 2g. Torque Bias Ratio Coast : 0.00000
 - 2h. Thermal Capacity : 0.00000

- | 2i. Surface : 0.00000
 - | 2j. Heat Transfer Coeff : 0.00000
 - | 2k. Wear Factor : 0.00000
 - | 2l. Friction Reduction With T : 0.00000
 - | 2m. Friction Ref T : 0.00000
- 4. Stiffness : 0.00000
- 5. Stiffness Mult : 0.00000
- 6. Damping Ratio : 0.00000
- 7. Max Torsion Deg : 0.00000
- 8. Non Linear Model : false
- 9. Max Between Lsd And Elsd : false
- 10. Has Cockpit Controls : false
- 11. Front Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 12. Center Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 13. Rear Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 14. Left Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 15. Right Lock Controllers
 - | 11a. Name : None
 - | 11b. Stages : None
- 16. Awd Clutches : None

14. Gearbox [.gearbox]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxxx

B. Schema

```
| 1. Gear Count : integer
| 2. Gears [x] : object | can have multiple Gears
|   | 2a. Name : string
|   | 2b. Ratio : float
| 3. Final Ratio : float
| 4. Gear Up Time : float
| 5. Gear Dn Time : float
| 6. Auto Cut Off Time : float
| 7. Gas Cut Off Level : float
| 8. Has Dual Clutch : boolean
```

- 9. Is Shifter Supported : **boolean**
- 10. Downshift Protection : **object**
 - 10a. Is Active : **boolean**
 - 10b. Is Debug : **boolean**
 - 10c. Overrev : **integer**
 - 10d. Lock N : **boolean**
- 11. Damage Rpm Window : **float**
- 12. Valid Shift Rpm Window : **float**
- 13. Controls Window Gain : **float**
- 14. Gearbox Inertia : **float**
- 15. Autoblip : **object**
 - 15a. Profile : **float**
 - 15b. Is Electronic : **boolean**
- 16. Autoshifter : **object**
 - 16a. Up : **integer**
 - 16b. Down Rpm Threshold : **integer**
 - 16c. Slip Threshold : **float**
 - 16d. Gas Cutoff Time : **float**
- 17. Gears Fatigue Log10 A : **float**
- 18. Gears Fatigue Nominal Torque : **float**
- 19. Gears Fatigue Max Stress : **float**
- 20. Gears Fatigue Min Stress : **float**

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Gear Count	None (Integer count)	Number of forward/reverse gear ratios configured (excludes R/N slots in Gear Count; Porsche GT4: 6).
2.	Gears	None (Object array)	Per-gear entries including reverse (R), neutral (N), and numbered forward ratios; each slot repeats IDs 2a/2b.
2a.	Name	None (String)	Gear label (R, N, 1, 2, ...); identifies reverse, neutral, and forward gears in the ratio table.
2b.	Ratio	Ratio (dimensionless)	Internal gear ratio; negative for reverse (e.g., -3.55 R), 0 for neutral, positive for forward (3.909 1st Porsche GT4).
3.	Final Ratio	Ratio (dimensionless)	Final-drive/differential ring-and-pinion multiplier applied after the selected gear (e.g., 3.72461 Porsche GT4).
4.	Gear Up Time	ms (Milliseconds)	Upshift engagement duration; lower values yield faster shifts (30 ms DCT Porsche, 250 ms Renault 5 manual).

ID	Name	Unit of Measurement	Description
5.	Gear Dn Time	ms (Milliseconds)	Downshift engagement duration; often longer than upshift for rev-matching (80 ms Porsche, 16 ms Alpine).
6.	Auto Cut Off Time	ms (Milliseconds)	Engine fuel/torque cut duration during automated or semi-automated shifts.
7.	Gas Cut Off Level	Ratio (0.0 - 1.0)	Fraction of throttle cut during shift fuel interruption (0.7 Porsche DCT; 0 on single-speed Alpine).
8.	Has Dual Clutch	None (Boolean : True / False)	When true, gearbox uses a dual-clutch layout with overlapping shift phases (Porsche GT4: true).
9.	Is Shifter Supported	None (Boolean : True / False)	When true, H-pattern or sequential shifter hardware input is supported for manual shifts.
10.	Downshift Protection	None (Object)	Anti-over-rev downshift protection block preventing destructive downshift RPM excursions.
10a.	Is Active	None (Boolean : True / False)	Master enable for downshift over-rev protection logic.
10b.	Is Debug	None (Boolean : True / False)	When true, exposes debug telemetry for downshift protection triggers.
10c.	Overrev	RPM (Revolutions per minute)	Allowed RPM overshoot margin before downshift is blocked (300 Porsche, 100 Alpine).
10d.	Lock N	None (Boolean : True / False)	When true, prevents accidental departure from neutral without clutch/shift validation.
11.	Damage Rpm Window	RPM (Revolutions per minute)	RPM band width used when evaluating gearbox/drivetrain damage from over-rev events (100 in examples).
12.	Valid Shift Rpm Window	RPM (Revolutions per minute)	RPM tolerance window within which a shift request is considered valid (800 in examples).
13.	Controls Window Gain	Dimensionless coefficient	Gain scaling driver shift-input timing tolerance (0.4 across example cars).
14.	Gearbox Intertia	kg·m ² (Kilogram square meters)	Rotational inertia of gears and shafts; affects rev hang and shift feel (0.012 Porsche, 0.017 Renault 5).
15.	Autoblip	None (Object)	Throttle-blip profile block for downshift rev-matching on manual/semi-auto setups.

ID	Name	Unit of Measurement	Description
15a.	Profile	None (.curve file path)	Autoblip throttle envelope curve (e.g., 718_gt4_autoblip.curve); schema types float but examples use path.
15b.	Is Electronic	None (Boolean : True / False)	When true, autoblip is applied electronically via ECU/throttle actuation rather than pedal coaching.
16.	Autoshifter	None (Object)	Automated upshift/downshift assist thresholds for sequential or auto modes.
16a.	Up	RPM (Revolutions per minute)	Engine RPM threshold triggering upshift in autoshifter mode (7700 Porsche, 6850 Alpine).
16b.	Down Rpm Threshold	RPM (Revolutions per minute)	RPM threshold below which downshift is requested (0 = disabled in examples).
16c.	Slip Threshold	Ratio (0.0 - 1.0)	Wheel-slip ratio threshold influencing autoshifter shift decisions (0.96 Porsche, 0.95 Alpine).
16d.	Gas Cutoff Time	s (Seconds)	Duration of throttle cut during autoshifter shifts (0.063 s Porsche, 0.15 s Alpine).
17.	Gears Fatigue Log10 A	Dimensionless coefficient (log10 coefficient)	Log10 fatigue curve coefficient A for gear tooth stress accumulation (0 = disabled in examples).
18.	Gears Fatigue Nominal Torque	Nm (Newton-meters)	Reference torque level for gear fatigue life calculations.
19.	Gears Fatigue Max Stress	Pa or N (Stress threshold)	Upper stress bound for gear fatigue degradation modeling.
20.	Gears Fatigue Min Stress	Pa or N (Stress threshold)	Lower stress bound below which gear fatigue accumulation is negligible.

D. Example data

I. Chosen Cars for Example

- Porsche 718 Cayman GT4 CS MR (slug : ks_porsche_718_cayman_gt4_cs_mr)
- Alpine A290 b (slug : ks_alpine_a290_b)
- Renault 5 GT Turbo (slug : ks_renault_5_gt_turbo)

II. Example

Porsche 718 Cayman GT4 CS MR

- 1. Gear Count : 6
- 2. Gears 1
 - 2a. Name : R
 - 2b. Ratio : -3.55000
- 2. Gears 2
 - 2a. Name : N
 - 2b. Ratio : 0.00000
- 2. Gears 3
 - 2a. Name : 1
 - 2b. Ratio : 3.90900
- 2. Gears 4
 - 2a. Name : 2
 - 2b. Ratio : 2.29167
- 2. Gears 5
 - 2a. Name : 3
 - 2b. Ratio : 1.65384
- 2. Gears 6
 - 2a. Name : 4
 - 2b. Ratio : 1.30303
- 2. Gears 7
 - 2a. Name : 5
 - 2b. Ratio : 1.08108
- 2. Gears 8
 - 2a. Name : 6
 - 2b. Ratio : 0.88095
- 3. Final Ratio : 3.72461
- 4. Gear Up Time : 30.00000
- 5. Gear Dn Time : 80.00000
- 6. Auto Cut Off Time : 30.00000
- 7. Gas Cut Off Level : 0.70000
- 8. Has Dual Clutch : true
- 9. Is Shifter Supported : false
- 10. Downshift Protection
 - 10a. Is Active : true
 - 10b. Is Debug : false
 - 10c. Overrev : 300
 - 10d. Lock N : true
- 11. Damage Rpm Window : 100.00000
- 12. Valid Shift Rpm Window : 800.00000
- 13. Controls Window Gain : 0.40000
- 14. Gearbox Inertia : 0.01200
- 15. Autoblip
 - 15a. Profile :
 - content\cars\ks_porsche_718_cayman_gt4_cs_mr\data\curves\718_gt4_autoblip.curve
 - 15b. Is Electronic : true
- 16. Autosifter
 - 16a. Up : 7700
 - 16b. Down Rpm Threshold : 0
 - 16c. Slip Threshold : 0.96000
 - 16d. Gas Cutoff Time : 0.06300
- 17. Gears Fatigue Log10 A : 0.00000
- 18. Gears Fatigue Nominal Torque : 0.00000
- 19. Gears Fatigue Max Stress : 0.00000

L 20. Gears Fatigue Min Stress : 0.00000

Alpine A290 b

```
- 1. Gear Count : 1
- 2. Gears 1
  | 2a. Name : R
  | 2b. Ratio : -9.00000
- 2. Gears 2
  | 2a. Name : N
  | 2b. Ratio : 0.00000
- 2. Gears 3
  | 2a. Name : 1
  | 2b. Ratio : 9.00000
- 3. Final Ratio : 1.00000
- 4. Gear Up Time : 10.00000
- 5. Gear Dn Time : 16.00000
- 6. Auto Cut Off Time : 11.00000
- 7. Gas Cut Off Level : 0.00000
- 8. Has Dual Clutch : false
- 9. Is Shifter Supported : false
- 10. Downshift Protection
  | 10a. Is Active : true
  | 10b. Is Debug : false
  | 10c. Overrev : 100
  | 10d. Lock N : true
- 11. Damage Rpm Window : 100.00000
- 12. Valid Shift Rpm Window : 800.00000
- 13. Controls Window Gain : 0.40000
- 14. Gearbox Inertia : 0.00800
- 15. Autoblip
  | 15a. Profile :
content\cars\ks_alpine_a290_b\data\autoBlipProfile.curve
  | 15b. Is Electronic : true
- 16. Autoshibter
  | 16a. Up : 6850
  | 16b. Down Rpm Threshold : 0
  | 16c. Slip Threshold : 0.95000
  | 16d. Gas Cutoff Time : 0.15000
- 17. Gears Fatigue Log10 A : 0.00000
- 18. Gears Fatigue Nominal Torque : 0.00000
- 19. Gears Fatigue Max Stress : 0.00000
- 20. Gears Fatigue Min Stress : 0.00000
```

Renault 5 GT Turbo

```
- 1. Gear Count : 5
- 2. Gears 1
  | 2a. Name : R
  | 2b. Ratio : -3.54500
- 2. Gears 2
  | 2a. Name : N
  | 2b. Ratio : 0.00000
```

- 2. Gears 3
 - 2a. Name : 1
 - 2b. Ratio : 3.09100
- 2. Gears 4
 - 2a. Name : 2
 - 2b. Ratio : 1.84200
- 2. Gears 5
 - 2a. Name : 3
 - 2b. Ratio : 1.32100
- 2. Gears 6
 - 2a. Name : 4
 - 2b. Ratio : 0.96700
- 2. Gears 7
 - 2a. Name : 5
 - 2b. Ratio : 0.75800
- 3. Final Ratio : 3.73300
- 4. Gear Up Time : 250.00000
- 5. Gear Dn Time : 270.00000
- 6. Auto Cut Off Time : 0.00000
- 7. Gas Cut Off Level : 0.00000
- 8. Has Dual Clutch : false
- 9. Is Shifter Supported : true
- 10. Downshift Protection
 - 10a. Is Active : false
 - 10b. Is Debug : false
 - 10c. Overrev : 0
 - 10d. Lock N : false
- 11. Damage Rpm Window : 100.00000
- 12. Valid Shift Rpm Window : 1000.00000
- 13. Controls Window Gain : 0.40000
- 14. Gearbox Inertia : 0.01700
- 15. Autoblip
 - 15a. Profile :
content\cars\ks_renault_5_gt_turbo\data\autoBlipProfile.curve
 - 15b. Is Electronic : false
- 16. Autosifter
 - 16a. Up : 6000
 - 16b. Down Rpm Threshold : 0
 - 16c. Slip Threshold : 0.95000
 - 16d. Gas Cutoff Time : 0.30000
- 17. Gears Fatigue Log10 A : 0.00000
- 18. Gears Fatigue Nominal Torque : 0.00000
- 19. Gears Fatigue Max Stress : 0.00000
- 20. Gears Fatigue Min Stress : 0.00000

15. General [.generalcar]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxxx

B. Schema

- 1. Screen Name : string
- 2. Total Mass : float
- 3. Tank Position : x, y, z float
- 4. Fuel : float
- 5. Max Fuel : float
- 6. Efficiency : float
- 7. Kg Per Liter : float
- 8. Body Box Sizes : x, y, z float
- 9. Pickup Front Height : float
- 10. Pickup Rear Height : float

- 11. Check Rules : **boolean**
- 12. Minimum Height : **float**
- 13. Torsional Stiffness : **float**
- 14. Torsional Damping : **float**
- 15. Body Mesh Offset : **object**
 - 15a. Position : **x, y, z float**
 - 15b. Rotation : **x, y, z float**
 - 15c. Scale : **x, y, z float**

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Screen Name	None (String)	Human-readable label for this .GENERALCAR profile in editor or UI tooling; does not affect physics.
2.	Total Mass	Kg (Kilograms)	Baseline vehicle mass including driver and fluids; fundamental input for weight transfer and inertia (Car Data 2b).
3.	Tank Position	m (Meters, X / Y / Z)	3D coordinates of the fuel tank centroid relative to the vehicle reference frame (Car Data 2c).
4.	Fuel	L (Liters)	Starting fuel volume at session load; reduces mass and shifts CG as fuel burns (Car Data 2d).
5.	Max Fuel	L (Liters)	Maximum fuel tank capacity; upper bound for fuel load and endurance calculations (Car Data 2e).
6.	Efficiency	Dimensionless ratio	Fuel consumption efficiency modifier applied to engine burn calculations (Car Data 2f).
7.	Kg Per Liter	kg/L (Kilograms per liter)	Fuel density factor converting volumetric fuel into mass for physics updates (Car Data 2g).
8.	Body Box Sizes	m (Meters, X / Y / Z)	Axis-aligned bounding box dimensions of the chassis body envelope for regulatory/collision checks (Car Data 2h).
9.	Pickup Front Height	m (Meters)	Vertical pickup/reference height at the front axle; aligns suspension hardpoints (Car Data 2i).
10.	Pickup Rear Height	m (Meters)	Vertical pickup/reference height at the rear axle (Car Data 2j).
11.	Check Rules	None (Boolean : True / False)	When true, enables automated regulatory validation such as minimum ride height (Car Data 2k).

ID	Name	Unit of Measurement	Description
12.	Minimum Height	m (Meters)	Regulatory minimum ride-height threshold enforced when Check Rules is active (Car Data 2l).
13.	Torsional Stiffness	Nm/rad or N/m (Torsional spring rate)	Chassis torsional spring rate resisting twist between front and rear axles (Car Data 2m).
14.	Torsional Damping	Nm·s/rad (Torsional damping)	Chassis torsional damping coefficient dissipating body twist oscillations (Car Data 2n).
15.	Body Mesh Offset	None (Object)	Visual/collision body mesh alignment block relative to the physics origin (Car Data 2o).
15a.	Position	m (Meters, X / Y / Z)	Translation offset of the body mesh from the physics origin (Car Data 2o1).
15b.	Rotation	deg or rad (X / Y / Z)	Euler rotation offset applied to the body mesh for model alignment (Car Data 2o2).
15c.	Scale	Dimensionless (X / Y / Z)	Non-uniform scale factor on the body mesh offset block (Car Data 2o3).

D. Example data

I. Chosen Cars for Example

No cars or common assets use an asset of the “GENERALCAR” type.

16. Surface 3D [.surface3d]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx
-

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxxx

B. Schema

```
| 1. Downforce h mm : string - path
| 2. Downforce dh mm : string - path
| 3. size rear : range integer ( 0 - 30 )
| 4. size front : range integer ( 0 - 30 )
| 5. min rear : float
| 6. max rear : float
| 7. min front : float
| 8. max front : float
| (9). Interpolation Table : table of size "3. size rear" * "4. size
front"
```

- 10. interpolation done : **boolean**
- 11. Interp Map : **object**
 - 11a. min val : **float**
 - 11b. max val : **float**
 - 11c. resolution front : **range integer (3 - 90)**
 - 11d. resolution rear : **range integer (3 - 90)**
- 12. Import front : **boolean**
- 13. Import range x y : **boolean**
- 14. CSV import path : **string - path**
- 15. CSV export name : **string - path**

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Downforce h mm	None (File path)	Optional source height map defining absolute ride-height reference for aero lookup (often None; Mercedes drag map).
2.	Downforce dh mm	None (File path)	Optional delta ride-height map layered on Downforce h mm for dynamic aero offset (often None).
3.	Size rear	Integer count (0 - 30)	Rear-axis grid resolution of the 2D interpolation table (rows; e.g., 8 on Mercedes/Audi GT3 maps).
4.	Size front	Integer count (0 - 30)	Front-axis grid resolution of the interpolation table (columns; e.g., 6-8).
5.	Min rear	mm (Millimeters)	Minimum rear ride-height sample bound for table axis (e.g., 65.338 mm Mercedes drag, 30 mm Audi Cx).
6.	Max rear	mm (Millimeters)	Maximum rear ride-height sample bound for table axis (e.g., 160.227 mm Mercedes drag).
7.	Min front	mm (Millimeters)	Minimum front ride-height sample bound for table axis.
8.	Max front	mm (Millimeters)	Maximum front ride-height sample bound for table axis.
9.	Interpolation Table	Dimensionless coefficient (2D matrix)	size rear × size front lookup of aero coefficients (Cd, Cl, Cz) vs. front/rear ride height; schema ID shown as (9). in examples.
10.	Interpolation done	None (Boolean : True / False)	When true, raw table data has been processed into the Interp Map representation (true on downforce maps).
11.	Interp Map	None (Object)	Baked interpolation map metadata block derived from the table (None when using raw matrix only).

ID	Name	Unit of Measurement	Description
11a.	Min val	Dimensionless coefficient	Lower clamp of interpolated aero coefficient output (e.g., -0.500).
11b.	Max val	Dimensionless coefficient	Upper clamp of interpolated aero coefficient output (e.g., 2.000).
11c.	Resolution front	Integer count (3 - 90)	Front-axis resolution of the baked interpolation map (18 on processed GT3/GT2 maps).
11d.	Resolution rear	Integer count (3 - 90)	Rear-axis resolution of the baked interpolation map (18 in examples).
12.	Import front	None (Boolean : True / False)	When true, front-axis data is imported from an external source rather than edited inline.
13.	Import range x y	None (Boolean : True / False)	When true, imports axis range bounds from external CSV/tooling rather than manual min/max fields.
14.	CSV Import path	None (File path)	Path to a CSV file for bulk table import; often None when table is authored in-editor.
15.	CSV export name	None (File path or string)	Target path or filename for CSV export of the interpolation table; often None.

D. Example data

I. Chosen Cars for Example

- Mercedes AMG GT2 (slug : ks_mercedes_amg_gt2) [3 surface 3d]
- Audi R8 LMS GT3 Evo 2 (slug : ks_audi_r8_lms_gt3_evo_2) [3 surface 3d]
- Dallara Stradale Coupe (slug : ks_dallara_stradale_coupe) [3 surface 3d]

II. Example

Mercedes AMG GT2

1. Drag map (file : drag_map.surface3d)

```

1. Downforce h mm : None
2. Downforce dh mm : None
3. size rear : 8
4. size front : 8
5. min rear : 65.338
6. max rear : 160.227
7. min front : 65.338
8. max front : 160.227
(9). Interpolation Table

```

Front v / Rear	65.3	78.9	92.4	106.0	119.6	133.1	146.7	160.2
65.3	0.454	0.475	0.487	0.498	0.509	0.539	0.579	0.671
78.9	0.487	0.456	0.476	0.489	0.506	0.522	0.545	0.582
92.4	0.515	0.489	0.457	0.478	0.497	0.519	0.528	0.548
106.0	0.540	0.516	0.490	0.459	0.486	0.510	0.525	0.531
119.6	0.556	0.541	0.518	0.492	0.467	0.499	0.516	0.528
133.1	0.572	0.557	0.542	0.520	0.500	0.480	0.505	0.519
146.7	0.597	0.573	0.558	0.545	0.528	0.513	0.486	0.508
160.2	0.642	0.599	0.574	0.561	0.553	0.541	0.519	0.489

```

| 10. interpolation done : false
| 11. Interp Map : None
| 12. Import front : false
| 13. Import range x y : false
| 14. CSV import path : None
| 15. CSV export name : None

```

2. Front downforce (file : front_downforce.surface3d)

```

| 1. Downforce h mm : None
| 2. Downforce dh mm : None
| 3. size rear : 8
| 4. size front : 8
| 5. min rear : 20.039
| 6. max rear : 154.236
| 7. min front : 20.039
| 8. max front : 154.236
| (9). Interpolation Table

```

Front v / Rear	20.0	39.2	58.4	77.6	96.7	115.9	135.1	154.2
20.0	0.210	0.273	0.288	0.282	0.272	0.260	0.255	0.255
39.2	0.176	0.211	0.273	0.288	0.283	0.272	0.261	0.256
58.4	0.172	0.177	0.212	0.275	0.289	0.284	0.273	0.262
77.6	0.175	0.175	0.180	0.215	0.277	0.292	0.287	0.276
96.7	0.181	0.175	0.176	0.181	0.215	0.278	0.293	0.287
115.9	0.184	0.174	0.169	0.169	0.174	0.209	0.271	0.286
135.1	0.189	0.180	0.170	0.164	0.164	0.169	0.204	0.266
154.2	0.187	0.187	0.178	0.168	0.162	0.163	0.167	0.202

```

| 10. interpolation done : true

```

- 11. Interp Map
 - 11a. min val : -0.500
 - 11b. max val : 2.000
 - 11c. resolution front : 18
 - 11d. resolution rear : 18
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

3. Rear downforce (file : rear_downforce.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 8
- 4. size front : 8
- 5. min rear : 1.732
- 6. max rear : 288.196
- 7. min front : 1.732
- 8. max front : 288.196
- (9). Interpolation Table

Front v / Rear	1.7	42.7	83.6	124.5	165.4	206.3	247.3	288.2
1.7	0.444	0.431	0.453	0.439	0.440	0.422	0.384	0.346
42.7	0.445	0.477	0.452	0.439	0.440	0.422	0.384	0.346
83.6	0.443	0.477	0.499	0.438	0.440	0.422	0.384	0.346
124.5	0.443	0.475	0.499	0.484	0.439	0.422	0.384	0.346
165.4	0.443	0.475	0.497	0.484	0.485	0.421	0.384	0.346
206.3	0.443	0.475	0.497	0.482	0.486	0.467	0.383	0.346
247.3	0.443	0.475	0.497	0.482	0.484	0.467	0.429	0.45
288.2	0.443	0.475	0.497	0.482	0.484	0.465	0.430	0.391

- 10. interpolation done : true
- 11. Interp Map
 - 11a. min val : -0.500
 - 11b. max val : 2.000
 - 11c. resolution front : 18
 - 11d. resolution rear : 18
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

Audi R8 LMS GT3 Evo 2

1. CX Map (file : cxmap.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 8
- 4. size front : 6
- 5. min rear : 30.000
- 6. max rear : 100.000
- 7. min front : 30.000
- 8. max front : 80.000
- (9). Interpolation Table

Front v / Rear	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
30.0	1.041	1.049	1.051	1.054	1.057	1.061	1.065	1.068
40.0	1.076	1.080	1.080	1.078	1.080	1.085	1.089	1.092
50.0	1.092	1.104	1.108	1.113	1.121	1.119	1.123	1.127
60.0	1.098	1.110	1.119	1.127	1.129	1.137	1.141	1.145
70.0	1.110	1.123	1.131	1.140	1.141	1.145	1.152	1.153
80.0	1.120	1.134	1.141	1.146	1.149	1.152	1.158	1.164

- 10. interpolation done : true
- 11. Interp Map
 - 11a. min val : -0.500
 - 11b. max val : 2.000
 - 11c. resolution front : 18
 - 11d. resolution rear : 18
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

2. Front CZ Map (file : frontczmap.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 8
- 4. size front : 6
- 5. min rear : 30.000
- 6. max rear : 100.000
- 7. min front : 30.000
- 8. max front : 80.000
- (9). Interpolation Table

Front v / Rear	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
30.0	0.913	1.056	1.197	1.323	1.428	1.516	1.581	1.633

Front v / Rear	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
40.0	0.784	0.911	1.022	1.120	1.215	1.300	1.384	1.455
50.0	0.616	0.726	0.836	0.946	1.051	1.138	1.214	1.281
60.0	0.415	0.508	0.618	0.740	0.854	0.954	1.035	1.100
70.0	0.214	0.311	0.415	0.519	0.621	0.721	0.816	0.898
80.0	0.011	0.116	0.223	0.333	0.431	0.516	0.605	0.689

```

10. interpolation done : true
11. Interp Map
  11a. min val : -0.500
  11b. max val : 2.000
  11c. resolution front : 18
  11d. resolution rear : 18
12. Import front : false
13. Import range x y : false
14. CSV import path : None
15. CSV export name : None

```

3. Rear CZ Map (file : rearczmap.surface3d)

```

1. Downforce h mm : None
2. Downforce dh mm : None
3. size rear : 8
4. size front : 6
5. min rear : 30.000
6. max rear : 100.000
7. min front : 30.000
8. max front : 80.000
(9). Interpolation Table

```

Front v / Rear	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
30.0	1.927	2.011	2.026	1.953	1.905	1.892	1.859	1.823
40.0	2.085	2.122	2.125	2.074	2.015	1.965	1.917	1.871
50.0	2.267	2.262	2.240	2.191	2.124	2.053	1.992	1.934
60.0	2.428	2.407	2.365	2.299	2.227	2.151	2.080	2.005
70.0	2.542	2.513	2.464	2.395	2.316	2.232	2.156	2.079
80.0	2.633	2.591	2.539	2.481	2.401	2.305	2.217	2.138

```

10. interpolation done : true
11. Interp Map
  11a. min val : -0.500
  11b. max val : 2.000
  11c. resolution front : 18

```

```

| L 11d. resolution rear : 18
| 12. Import front : false
| 13. Import range x y : false
| 14. CSV import path : None
| 15. CSV export name : None

```

Dallara Stradale Coupe

1. Drag map (file : drag_map.surface3d)

```

| 1. Downforce h mm : None
| 2. Downforce dh mm : None
| 3. size rear : 7
| 4. size front : 4
| 5. min rear : 60.000
| 6. max rear : 120.000
| 7. min front : 60.000
| 8. max front : 120.000
| (9). Interpolation Table

```

Front v / Rear	60.0	70.0	80.0	90.0	100.0	110.0	120.0
60.0	0.930	0.943	0.950	0.960	0.986	0.994	1.010
80.0	0.945	0.960	0.965	0.980	1.000	1.016	1.040
100.0	0.960	0.970	0.974	0.990	1.001	1.015	1.023
120.0	0.970	0.972	0.976	0.995	1.003	1.010	1.022

```

| 10. interpolation done : false
| 11. Interp Map : None
| 12. Import front : false
| 13. Import range x y : false
| 14. CSV import path : None
| 15. CSV export name : None

```

2. Front lift map (file : front_lift_map.surface3d)

```

| 1. Downforce h mm : None
| 2. Downforce dh mm : None
| 3. size rear : 7
| 4. size front : 4
| 5. min rear : 60.000
| 6. max rear : 120.000
| 7. min front : 60.000
| 8. max front : 120.000
| (9). Interpolation Table

```

Front v / Rear	60.0	70.0	80.0	90.0	100.0	110.0	120.0
60.0	0.930	0.943	0.950	0.960	0.986	0.994	1.010

Front v / Rear	60.0	70.0	80.0	90.0	100.0	110.0	120.0
80.0	0.945	0.960	0.965	0.980	1.000	1.016	1.040
100.0	0.960	0.970	0.974	0.990	1.001	1.015	1.023
120.0	0.970	0.972	0.976	0.995	1.003	1.010	1.022

```

10. interpolation done : false
11. Interp Map : None
12. Import front : false
13. Import range x y : false
14. CSV import path : None
15. CSV export name : None

```

3. Rear lift map (file : rear_lift_map.surface3d)

```

1. Downforce h mm : None
2. Downforce dh mm : None
3. size rear : 7
4. size front : 4
5. min rear : 60.000
6. max rear : 120.000
7. min front : 60.000
8. max front : 120.000
(9). Interpolation Table

```

Front v / Rear	60.0	70.0	80.0	90.0	100.0	110.0	120.0
60.0	1.619	1.562	1.482	1.414	1.402	1.365	1.396
80.0	1.766	1.751	1.664	1.575	1.517	1.474	1.428
100.0	1.716	1.739	1.754	2.096	1.645	1.551	1.534
120.0	1.833	1.837	1.835	1.754	1.711	1.613	1.597

```

10. interpolation done : false
11. Interp Map : None
12. Import front : false
13. Import range x y : false
14. CSV import path : None
15. CSV export name : None

```

17. Suspension [.suspension]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxxx

B. Schema

- 1. Basic Data : object
 - 1a. Hub Mass : float
 - 1b. Toe Out Linear : float
 - 1c. Static Camber : float
 - 1d. Rim Offset : float
- 2. D W Data : object
 - 2a. Car Top Front : x, y, z float
 - 2b. Car Top Rear : x, y, z float
 - 2c. Tyre Top : x, y, z float
 - 2d. Car Bottom Front : x, y, z float

- 2e. Car Bottom Rear : x, y, z float
- 2f. Tyre Bottom : x, y, z float
- 2g. Car Steer : x, y, z float
- 2h. Tyre Steer : x, y, z float
- 3. Strut : object
 - 3a. Car Strut : x, y, z float
 - 3b. Tyre Strut : x, y, z float
 - 3c. Car Bottom W B F : x, y, z float
 - 3d. Car Bottom W B R : x, y, z float
 - 3e. Tyre Bottom W B : x, y, z float
 - 3f. Car Steer : x, y, z float
 - 3g. Tyre Steer : x, y, z float
- 4. Strut Ml : object
 - 4a. Car Strut : x, y, z float
 - 4b. Tyre Strut : x, y, z float
 - 4c. Car Bottom W B F : x, y, z float
 - 4d. Car Bottom W B R : x, y, z float
 - 4e. Tyre Bottom W B : x, y, z float
 - 4f. Car Thrust Ball1 : x, y, z float
 - 4g. Car Thrust Ball2 : x, y, z float
 - 4h. Tyre Thrust Arm : x, y, z float
 - 4i. Car Steer : x, y, z float
 - 4j. Tyre Steer : x, y, z float
- 5. Axle : object
 - 5a. Attach Relative Pos : float
 - 5b. Link Count : integer
 - 5c. Car Side [x] : x, y, z float | can have multiple Car Side
 - 5d. Axle Side [x] : x, y, z float | can have multiple Axle Side
 - 5e. Hub Mass : float
 - 5f. Leaf Spring Lat K : float
 - 5g. Torquereaction : float
- 6. Multi Link Data : object
 - 6a. Joint Car [x] : x, y, z float | can have multiple Joint Car
 - 6b. Joint Type [x] : x, y, z float | can have multiple Joint Type
- 7. Trailing Arm Data : object
 - 7a. Car Hinge Int : x, y, z float
 - 7b. Car Hinge Ext : x, y, z float
 - 7c. Tyre Top : x, y, z float
 - 7d. Tyre Bottom : x, y, z float
 - 7e. Car Steer : x, y, z float
 - 7f. Tyre Steer : x, y, z float
 - 7g. Steer Link To Body : boolean
- 8. Multi Link New Data : object
 - 8a. Links [x] : object | can have multiple Links
 - 8a1. Name : string
 - 8a2. Ball Car : x, y, z float
 - 8a3. Ball Tyre : x, y, z float
 - 8a4. Has Coilover Attached : boolean
 - 8a5. Is Toe : boolean
 - 8b. Arms [x] : object | can have multiple Arms
 - 8b1. Name : string
 - 8b2. Ball Car 1 : x, y, z float
 - 8b3. Ball Car 2 : x, y, z float
 - 8b4. Ball Tyre : x, y, z float
 - 8b5. Has Coilover Attached : boolean
 - 8c. Car Coilover : x, y, z float

- └ 8d. Bottom Coilover : **x, y, z float**
- └ 9. Dw Coil Data : **object**
 - └ 9a. Car Top Front : **x, y, z float**
 - └ 9b. Car Top Rear : **x, y, z float**
 - └ 9c. Tyre Top : **x, y, z float**
 - └ 9d. Car Bottom Front : **x, y, z float**
 - └ 9e. Car Bottom Rear : **x, y, z float**
 - └ 9f. Tyre Bottom : **x, y, z float**
 - └ 9g. Car Steer : **x, y, z float**
 - └ 9h. Tyre Steer : **x, y, z float**
 - └ 9i. Car Coilover : **x, y, z float**
 - └ 9j. Bottom Coilover : **x, y, z float**

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Basic Data	None (Object)	Per-corner suspension asset root; holds unsprung mass, static alignment, and exactly one kinematic block (2–9). Referenced from Car Data front/rear suspension paths (.suspension).
1a.	Hub Mass	kg (Kilograms)	Unsprung mass at the hub: wheel, tire, brake rotor/caliper, hub, and upright share. Typical range ~37–51 kg in examples (S2000 light front 37 kg, Golf 50 kg, GT3 R 51 kg). Lower mass improves bump compliance and tire contact response.
1b.	Toe Out Linear	rad/m (Radians per meter)	Linear kinematic toe gradient vs. suspension displacement or stroke (aligned with Car Setup 5f). Small signed values (e.g. ± 0.0002 – 0.0011) model bump-steer / compliance toe; positive often means increasing toe-out with travel depending on axis convention.
1c.	Static Camber	deg (Degrees)	Fixed geometric camber at reference ride height before live setup offsets. Negative = top of wheel leans inward (-0.3° street Golf, -1.6° S2000 rear, -5.7° GT3 R race front). Sets baseline contact patch and camber gain reference.
1d.	Rim Offset	m (Meters)	Wheel rim lateral offset (ET equivalent) from hub mounting face; shifts scrub radius and track (e.g. 0.045 m S2000 front, 0.037 m rear). Affects steering feel, kingpin offset, and fender clearance.

ID	Name	Unit of Measurement	Description
2.	D W Data	None (Object)	Double-wishbone (DWB) kinematic hardpoints; used when Strut / Axle / Trailing Arm / Multi Link blocks are None (e.g. Honda S2000 front). Defines upper/lower arm pivots and steer tie-rod points in the car body frame.
2a.	Car Top Front	m (Meters, X / Y / Z)	Chassis-side pivot of the upper wishbone, forward inner bushing (subframe / body mount). With 2b defines upper arm axis and camber / caster kinematics during jounce and rebound.
2b.	Car Top Rear	m (Meters, X / Y / Z)	Chassis-side pivot of the upper wishbone, rear inner bushing. Pair with 2a sets upper control arm length, inclination, and roll-center contribution.
2c.	Tyre Top	m (Meters, X / Y / Z)	Upright-side ball joint for the upper wishbone (outer upper pivot). Primary camber-change link as the wheel moves vertically and steers.
2d.	Car Bottom Front	m (Meters, X / Y / Z)	Chassis-side pivot of the lower wishbone, forward inner bushing. Sets anti-dive geometry with 2e and influences roll center height.
2e.	Car Bottom Rear	m (Meters, X / Y / Z)	Chassis-side pivot of the lower wishbone, rear inner bushing. With 2d defines lower arm plane and longitudinal compliance.
2f.	Tyre Bottom	m (Meters, X / Y / Z)	Upright-side ball joint for the lower wishbone (outer lower pivot). Carries vertical load into the lower arm; strongly affects scrub radius and kingpin axis.
2g.	Car Steer	m (Meters, X / Y / Z)	Steering tie-rod inner attachment on rack or steering arm (body side). With 2h defines bump-steer and Ackermann geometry for DWB layouts.
2h.	Tyre Steer	m (Meters, X / Y / Z)	Steering tie-rod outer ball joint on the upright/knuckle. Motion relative to 2g produces steer angle change through travel.
3.	Strut	None (Object)	McPherson strut kinematic set; lower arm + telescopic strut define wheel motion (e.g. Golf GTI Mk1 front). Mutually exclusive with other geometry blocks when set to None elsewhere.

ID	Name	Unit of Measurement	Description
3a.	Car Strut	m (Meters, X / Y / Z)	Strut top mount on the body (shock tower / dome). Defines primary vertical load path and strut axis inclination (caster/camber offset from strut angle).
3b.	Tyre Strut	m (Meters, X / Y / Z)	Strut lower joint at the steering knuckle (wheel-side strut pin). Sliding joint along strut axis; sets kingpin inclination with 3a.
3c.	Car Bottom W B F	m (Meters, X / Y / Z)	Lower control arm (wishbone) inner forward bushing on the subframe. "W B" = wishbone; front pivot of the LCA.
3d.	Car Bottom W B R	m (Meters, X / Y / Z)	Lower control arm inner rear bushing on the subframe. With 3c sets LCA swing axis, anti-roll, and longitudinal location.
3e.	Tyre Bottom W B	m (Meters, X / Y / Z)	Lower control arm outer ball joint at the knuckle. Vertical wheel motion rotates about the LCA line 3c-3d-3e.
3f.	Car Steer	m (Meters, X / Y / Z)	Steering tie-rod inner point (rack or steering arm on body) for McPherson layout.
3g.	Tyre Steer	m (Meters, X / Y / Z)	Tie-rod outer joint on the knuckle; combined with 3f defines steer angle and McPherson bump-steer.
4.	Strut MI	None (Object)	McPherson-strut variant with additional locating links (multi-link McPherson / strut-plus-links). Adds thrust links and extra steer points beyond block 3.
4a.	Car Strut	m (Meters, X / Y / Z)	Strut top mount on the body for Strut MI layout (same role as 3a).
4b.	Tyre Strut	m (Meters, X / Y / Z)	Strut-to-knuckle lower joint for Strut MI (same role as 3b).
4c.	Car Bottom W B F	m (Meters, X / Y / Z)	Lower wishbone forward inner pivot (Strut MI).
4d.	Car Bottom W B R	m (Meters, X / Y / Z)	Lower wishbone rear inner pivot (Strut MI).
4e.	Tyre Bottom W B	m (Meters, X / Y / Z)	Lower wishbone outer ball joint at the upright (Strut MI).
4f.	Car Thrust Ball1	m (Meters, X / Y / Z)	First chassis-side pivot of a thrust / radius rod limiting knuckle rotation about the strut axis; improves camber control under lateral load.

ID	Name	Unit of Measurement	Description
4g.	Car Thrust Ball2	m (Meters, X / Y / Z)	Second chassis-side pivot paired with 4f (dual-link or split bush axis) for the thrust linkage.
4h.	Tyre Thrust Arm	m (Meters, X / Y / Z)	Upright-side joint for the thrust / radius arm connecting to 4f/4g; resists fore-aft knuckle rotation and brake steer.
4i.	Car Steer	m (Meters, X / Y / Z)	Steering tie-rod inner attachment (Strut MI geometry).
4j.	Tyre Steer	m (Meters, X / Y / Z)	Steering tie-rod outer joint on the upright (Strut MI).
5.	Axle	None (Object)	Live / solid axle kinematic block for dependent rear (or front) suspension: locating links, leaf spring, and torque reaction. Used on trucks and classic live-axle cars when DW/Strut blocks are None
5a.	Attach Relative Pos	m (Meters) or dimensionless ratio (0 -1)	Longitudinal or normalized position along the axle/leaf where the spring or link attaches relative to axle center; sets load transfer and wrap-up geometry.
5b.	Link Count	Integer count	Number of axle locating links modeled (Panhard rod, trailing links, Watt's link, etc.); pairs with indexed 5c/5d entries.
5c.	Car Side	m (Meters, X / Y / Z)	Chassis-side attachment point for an axle locating link (indexed [x] per link). Defines roll center and lateral axle location.
5d.	Axle Side	m (Meters, X / Y / Z)	Axle-side attachment for the corresponding locating link [x]. With 5c sets link length and lateral constraint stiffness.
5e.	Hub Mass	kg (Kilograms)	Unsprung mass per corner for the axle layout (same physical meaning as 1a but authored inside the Axle block for live-axle assets).
5f.	Leaf Spring Lat K	N/m (Newtons per meter)	Lateral stiffness of the leaf spring stack resisting sideways axle shift (leaf-as-lateral-locator). Higher values reduce axle walk under cornering and braking.
5g.	Torquereaction	Nm/rad (Newton-meters per radian)	Axle torque-reaction stiffness against driveline/brake wrap-up (rotation of axle about its lateral axis). Models anti-squat / anti-lift from link geometry and compliant mounts.

ID	Name	Unit of Measurement	Description
6.	Multi Link Data	None (Object)	Legacy multi-link hardpoint list (indexed joints) predating Multi Link New Data ; simpler joint-car enumeration for older ACE suspension templates.
6a.	Joint Car	m (Meters, X / Y / Z)	Chassis-side pivot for link index [x] in the legacy multi-link solver (one point per link).
6b.	Joint Type	m (Meters, X / Y / Z)	Per-link constraint data for index [x]; schema types as x,y,z float (likely encoded joint class / axis flags rather than a literal position—treat as solver metadata vector in meters or normalized codes).
7.	Trailing Arm Data	None (Object)	Semi-trailing or trailing-arm rear geometry (e.g. Golf GTI Mk1 rear). Arm hinges on body and carries hub at outer end.
7a.	Car Hinge Int	m (Meters, X / Y / Z)	Inner trailing-arm bushing on the body (inboard pivot). Defines primary arm axis with 7b.
7b.	Car Hinge Ext	m (Meters, X / Y / Z)	Secondary inner hinge point on the body (outboard bushing of the arm inner pair). Two-point inner mount sets bushing compliance axis and toe change under load.
7c.	Tyre Top	m (Meters, X / Y / Z)	Hub/upright attachment at the upper outer end of the trailing arm.
7d.	Tyre Bottom	m (Meters, X / Y / Z)	Hub/upright attachment at the lower outer end; with 7c fixes wheel plane orientation on the arm.
7e.	Car Steer	m (Meters, X / Y / Z)	Steering or toe-link inner point on body (often inactive on driven rear arms).
7f.	Tyre Steer	m (Meters, X / Y / Z)	Steering / toe-link outer point on the hub for rear steer or toe compliance.
7g.	Steer Link To Body	None (Boolean : True / False)	When true, the steer/toe link inner mount is fixed to the body; when false, it may attach to the arm, changing toe compliance under traction/braking.
8.	Multi Link New Data	None (Object)	Explicit graph-based multi-link: indexed Links (two-point struts) and Arms (three-point control arms) plus optional coilover mounts (e.g. Porsche 992 GT3 R rear).

ID	Name	Unit of Measurement	Description
8a.	Links	None (Object array [x])	Repeatable link elements (tie rods, toe links, strut rods); each index defines a rigid link between Ball Car and Ball Tyre.
8a1.	Name	None (String)	Optional label for the link (e.g. "steer" on GT3 R toe link); aids authoring/debug, not used in physics directly.
8a2.	Ball Car	m (Meters, X / Y / Z)	Chassis-side spherical joint for link [x].
8a3.	Ball Tyre	m (Meters, X / Y / Z)	Upright-side spherical joint for link [x]; distance to 8a2 sets link length and constraint direction.
8a4.	Has Coilover Attached	None (Boolean : True / False)	When true, the coilover damper unit (.coilover asset) acts on this link (force along link or at pickup on link geometry).
8a5.	Is Toe	None (Boolean : True / False)	When true, this link is flagged as the primary toe-control member (toe link / steering compliance arm); GT3 R example sets true on link named steer.
8b.	Arms	None (Object array [x])	Repeatable control arms with two body pivots and one upright pivot (classic multi-link arm).
8b1.	Name	None (String)	Optional arm identifier for authoring (camper arm, upper lateral, etc.).
8b2.	Ball Car 1	m (Meters, X / Y / Z)	First chassis-side pivot of arm [x] (inner bushing pair or front link point).
8b3.	Ball Car 2	m (Meters, X / Y / Z)	Second chassis-side pivot of arm [x]; with 8b2 defines arm plane and bush axis.
8b4.	Ball Tyre	m (Meters, X / Y / Z)	Upright-side ball joint for arm [x]; primary load path from wheel into the arm.
8b5.	Has Coilover Attached	None (Boolean : True / False)	When true, coilover spring/damper attaches to this arm (common on lower or virtual swing arm).
8c.	Car Coilover	m (Meters, X / Y / Z)	Coilover upper mount on the body when not carried by a flagged link/ arm (shock tower pickup).
8d.	Bottom Coilover	m (Meters, X / Y / Z)	Coilover lower mount on the upright or link; with 8c defines damper line and motion ratio.

ID	Name	Unit of Measurement	Description
9.	Dw Coil Data	None (Object)	Double wishbone geometry with integrated coilover pick-ups (e.g. Porsche 992 GT3 R front). Combines DW hardpoints with explicit damper mount points 9i/9j.
9a.	Car Top Front	m (Meters, X / Y / Z)	Upper wishbone forward inner pivot (Dw Coil layout; same role as 2a).
9b.	Car Top Rear	m (Meters, X / Y / Z)	Upper wishbone rear inner pivot (Dw Coil; same role as 2b).
9c.	Tyre Top	m (Meters, X / Y / Z)	Upper wishbone outer ball joint on upright (Dw Coil; same role as 2c).
9d.	Car Bottom Front	m (Meters, X / Y / Z)	Lower wishbone forward inner pivot (Dw Coil; same role as 2d).
9e.	Car Bottom Rear	m (Meters, X / Y / Z)	Lower wishbone rear inner pivot (Dw Coil; same role as 2e).
9f.	Tyre Bottom	m (Meters, X / Y / Z)	Lower wishbone outer ball joint (Dw Coil; same role as 2f).
9g.	Car Steer	m (Meters, X / Y / Z)	Steering tie-rod inner point for Dw Coil geometry.
9h.	Tyre Steer	m (Meters, X / Y / Z)	Steering tie-rod outer joint for Dw Coil geometry.
9i.	Car Coilover	m (Meters, X / Y / Z)	Coilover upper mount on chassis/ subframe; defines damper axis relative to DW kinematics.
9j.	Bottom Coilover	m (Meters, X / Y / Z)	Coilover lower mount on upright or lower arm; with 9i sets spring motion ratio and anti-dive line.

D. Example data

I. Chosen Cars for Example

- Volkswagen Golf GTI Mk1 (slug : ks_volkswagen_golf_gti_mk1) [2 suspensions]
- Honda S2000 AP1 (slug : ks_honda_s2000_ap1) [3 suspensions]
- Porsche 992 GT3 R Rennsport (slug : ks_porsche_992_gt3_r_rennsport) [2 suspensions]

II. Example

Volkswagen Golf GTI Mk1

1. Front Suspension (file : ks_volkswagen_golf_gti_mk1_front.suspension)

```

1. Basic Data
| 1a. Hub Mass : 50.00000

```

- 1b. Toe Out Linear : 0.00030
- 1c. Static Camber : -0.30000
- 1d. Rim Offset : 0.00000
- 2. D W Data : None
- 3. Strut
 - 3a. Car Strut : 0.13200, 0.38800, -0.01050
 - 3b. Tyre Strut : 0.05200, -0.05000, 0.01500
 - 3c. Car Bottom W B F : 0.41000, -0.05700, 0.32500
 - 3d. Car Bottom W B R : 0.39000, -0.05000, -0.04000
 - 3e. Tyre Bottom W B : 0.05200, -0.05000, 0.01500
 - 3f. Car Steer : 0.39000, -0.05250, 0.06000
 - 3g. Tyre Steer : 0.05200, -0.05000, 0.15000
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

2. Rear Suspension (file : ks_volkswagen_golf_gti_mk1_rear.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 45.00000
 - 1b. Toe Out Linear : 0.00025
 - 1c. Static Camber : -1.00000
 - 1d. Rim Offset : 0.00000
- 2. D W Data : None
- 3. Strut : None
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data
 - 7a. Car Hinge Int : 1.40000, 0.00000, 0.35000
 - 7b. Car Hinge Ext : 1.30000, 0.00000, 0.40000
 - 7c. Tyre Top : 0.09622, 0.49070, -0.01650
 - 7d. Tyre Bottom : 0.04602, 0.03294, 0.00870
 - 7e. Car Steer : 0.48002, -0.07906, -0.10540
 - 7f. Tyre Steer : 0.04602, -0.07506, -0.04540
 - 7g. Steer Link To Body : false
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

Honda S2000 AP1

1. Front Suspension (file : ks_honda_s2000_ap1_front.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 37.00000
 - 1b. Toe Out Linear : -0.00020
 - 1c. Static Camber : -0.40000
 - 1d. Rim Offset : 0.04500
- 2. D W Data
 - 2a. Car Top Front : 0.34203, 0.12311, 0.08000
 - 2b. Car Top Rear : 0.34302, 0.12451, -0.08000

- 2c. Tyre Top : 0.12479, 0.10951, -0.02210
- 2d. Car Bottom Front : 0.45520, -0.06960, -0.26000
- 2e. Car Bottom Rear : 0.42590, -0.08110, 0.03000
- 2f. Tyre Bottom : 0.09050, -0.13502, 0.01250
- 2g. Car Steer : 0.45520, -0.02000, 0.05890
- 2h. Tyre Steer : 0.09480, -0.06012, 0.11691
- 3. Strut : None
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

2. Front Drift Suspension (file : ks_honda_s2000_ap1_front_drift.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 37.00000
 - 1b. Toe Out Linear : -0.00020
 - 1c. Static Camber : -0.40000
 - 1d. Rim Offset : 0.04500
- 2. D W Data
 - 2a. Car Top Front : 0.34302, 0.08311, 0.08000
 - 2b. Car Top Rear : 0.34302, 0.08451, -0.08000
 - 2c. Tyre Top : 0.11479, 0.10951, -0.02210
 - 2d. Car Bottom Front : 0.45520, -0.06960, -0.26000
 - 2e. Car Bottom Rear : 0.42590, -0.08110, 0.03000
 - 2f. Tyre Bottom : 0.08050, -0.10502, 0.01250
 - 2g. Car Steer : 0.45520, -0.05000, 0.05890
 - 2h. Tyre Steer : 0.09480, -0.06012, 0.11691
- 3. Strut : None
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

3. Rear Suspension (file : ks_honda_s2000_ap1_front_drift.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 44.50000
 - 1b. Toe Out Linear : -0.00030
 - 1c. Static Camber : -1.60000
 - 1d. Rim Offset : 0.03700
- 2. D W Data
 - 2a. Car Top Front : 0.39022, 0.08701, 0.04129
 - 2b. Car Top Rear : 0.42022, 0.09469, -0.24111
 - 2c. Tyre Top : 0.11000, 0.11220, 0.00000
 - 2d. Car Bottom Front : 0.44100, -0.07010, 0.22113
 - 2e. Car Bottom Rear : 0.44042, -0.08989, -0.10935
 - 2f. Tyre Bottom : 0.09130, -0.10511, -0.01000
 - 2g. Car Steer : 0.38990, -0.04540, 0.16020
 - 2h. Tyre Steer : 0.09130, -0.05342, 0.14988
- 3. Strut : None

- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

Porsche 992 GT3 R Rennport

1. Front Suspension (file : ks_porsche_992_gt3_r_rennsport_front.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 51.00000
 - 1b. Toe Out Linear : 0.00044
 - 1c. Static Camber : -5.70000
 - 1d. Rim Offset : 0.00000
- 2. D W Data : None
- 3. Strut : None
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data : None
- 9. Dw Coil Data
 - 9a. Car Top Front : 0.36128, 0.06671, 0.10834
 - 9b. Car Top Rear : 0.36128, 0.03131, -0.14089
 - 9c. Tyre Top : 0.08582, 0.10517, -0.01868
 - 9d. Car Bottom Front : 0.45261, -0.15879, 0.21226
 - 9e. Car Bottom Rear : 0.45095, -0.14676, -0.08504
 - 9f. Tyre Bottom : 0.03941, -0.11288, 0.01898
 - 9g. Car Steer : 0.49910, -0.11521, 0.13421
 - 9h. Tyre Steer : 0.03338, -0.06370, 0.13886
 - 9i. Car Coilover : 0.40708, 0.21593, -0.04583
 - 9j. Bottom Coilover : 0.10508, -0.11831, 0.00237

2. Rear Suspension (file : ks_porsche_992_gt3_r_rennsport_rear.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 46.00000
 - 1b. Toe Out Linear : -0.00110
 - 1c. Static Camber : -4.40000
 - 1d. Rim Offset : 0.00000
- 2. D W Data : None
- 3. Strut : None
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data
 - 8a. Links 1
 - 8a1. Name : None
 - 8a2. Ball Car : 0.45733, -0.21779, -0.03274
 - 8a3. Ball Tyre : 0.05940, -0.15917, -0.05147
 - 8a4. Has Coilover Attached : true

- └ 8a5. Is Toe : false
- 8a. Links 2
 - └ 8a1. Name : None
 - └ 8a2. Ball Car : 0.33837, -0.13867, 0.26058
 - └ 8a3. Ball Tyre : 0.07196, -0.16195, 0.00295
 - └ 8a4. Has Coilover Attached : false
 - └ 8a5. Is Toe : false
- 8a. Links 3
 - └ 8a1. Name : None
 - └ 8a2. Ball Car : 0.29399, -0.00924, 0.18603
 - └ 8a3. Ball Tyre : 0.08416, 0.09013, 0.15594
 - └ 8a4. Has Coilover Attached : false
 - └ 8a5. Is Toe : false
- 8a. Links 4
 - └ 8a1. Name : None
 - └ 8a2. Ball Car : 0.31754, 0.04240, 0.18574
 - └ 8a3. Ball Tyre : 0.06361, 0.16582, -0.00746
 - └ 8a4. Has Coilover Attached : false
 - └ 8a5. Is Toe : false
- 8a. Links 5
 - └ 8a1. Name : steer
 - └ 8a2. Ball Car : 0.32123, -0.00156, -0.12150
 - └ 8a3. Ball Tyre : 0.05699, 0.10790, -0.12715
 - └ 8a4. Has Coilover Attached : false
 - └ 8a5. Is Toe : true
- 8b. Arms : None
- 8c. Car Coilover : 0.32123, -0.00156, -0.12150
- 8d. Bottom Coilover : 0.05699, 0.10790, -0.12715
- 9. Dw Coil Data : None

18. Turbo [.turbo]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxxx

B. Schema

- 1. Max Boost : float
- 2. Lag U P : float
- 3. Lag D N : float
- 4. Rpm Ref : float
- 5. Gamma Rpm : float
- 6. Gamma Gas : float
- 7. Wastegate : float
- 8. Is Adjustable : boolean
- 9. CarData : string – path
- 10. Turbo no : integer

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Max Boost	bar (Bar gauge pressure)	Peak compressor boost relative to ambient (gauge). Typical street/ rally ~0.7–1.2 bar (205 T16), race/ drift up to ~2.0 bar; 0.0 disables forced-induction output on that stage. Aligns with Car Engine \$18 Max Turbo Boost ceiling.
2.	Lag U P	Dimensionless lag coefficient (0 - 1)	Exponential spool-up filter toward target boost under rising demand (UP = up). Values near 1.0 (e.g. 0.995–0.9975) mean heavy turbo lag; 0.0 = instantaneous rise (Camaro ZL1 compressor / supercharger-style response).
3.	Lag D N	Dimensionless lag coefficient (0 - 1)	Exponential decay filter when boost demand falls (DN = down): throttle lift, gear change, or RPM drop. Slightly below Lag UP in examples (0.988–0.990) so boost bleeds off a bit faster than it builds; 0.0 with Lag UP = no lag (ZL1).
4.	Rpm Ref	RPM (Revolutions per minute)	Reference engine speed for the RPM-shaped boost curve (with Gamma Rpm). Below this region spool is weak; around/above it boost builds toward Max Boost. Examples: 2300/4100 (205 staged twin), 6000 (ZL1), 3200–4400 (Supra drift).
5.	Gamma Rpm	Dimensionless exponent (power-law shape)	Exponent shaping how boost scales with RPM relative to Rpm Ref. Higher gamma (>1, e.g. 2.5–3.0) = abrupt high-RPM spool; lower (0.5–1.0) = progressive/early response (Supra stock / ZL1).
6.	Gamma Gas	Dimensionless exponent (power-law shape)	Exponent shaping boost vs. throttle/gas pedal. Higher = needs more pedal to spool; 0.0 = boost independent of throttle position (ZL1 compressor model); ~0.5–2.5 typical turbo response.
7.	Wastegate	bar (Bar gauge pressure)	Wastegate setpoint. Caps regulated boost; often equals Max Boost (205 T16) or slightly below (ZL1 0.9 vs 1.0; Supra drift 1.5 vs 2.0 Max). Models mechanical/ electronic wastegate opening pressure.

ID	Name	Unit of Measurement	Description
8.	Is Adjustable	None (Boolean : True / False)	When true, boost level is exposed to setup / performance-mode adjustment (Turbo Boost Lv); false = fixed factory map (205 T16, ZL1, Supra stock). Supra drift turbos set true.
9.	CarData	None (File path)	Optional path linking this .turbo asset to a Car Data context; None in all documented examples (boost controlled via Car Engine Turbos To Load + controllers).
10.	Turbo no	Integer index (0-based)	Ordinal index of this turbocharger in a multi-turbo / sequential stack (Turbo 0 = primary/low-RPM, Turbo 1 = secondary). Examples list file Turbo 0/1 but store index 0 in-asset—verify against loaded stage order.

D. Example data

I. Chosen Cars for Example

- Peugeot 205 T16 (slug : ks_peugeot_205_t16) [2 turbos]
- Chevrolet Camaro ZL1 (slug : ks_chevrolet_camaro_zl1) [compressor]
- Toyota Supra MKIV (slug : ks_toyota_supra_mkiv) [2 turbos / 2 types]

II. Example

Peugeot 205 T16

1. Turbo 0 (file : ks_peugeot_205_t15.turbo)

```

1. Max Boost : 0.70000
2. Lag U P : 0.99500
3. Lag D N : 0.99000
4. Rpm Ref : 2300.00000
5. Gamma Rpm : 2.50000
6. Gamma Gas : 2.50000
7. Wastegate : 0.70000
8. Is Adjustable : false
9. CarData : None
10. Turbo no : 0

```

2. Turbo 1 (file : ks_peugeot_205_t15_turbo0.turbo)

```

1. Max Boost : 1.20000
2. Lag U P : 0.99500
3. Lag D N : 0.99000

```

- | 4. Rpm Ref : 4100.00000
- | 5. Gamma Rpm : 2.50000
- | 6. Gamma Gas : 2.50000
- | 7. Wastegate : 1.20000
- | 8. Is Adjustable : false
- | 9. CarData : None
- | 10. Turbo no : 0

Chevrolet Camaro ZL1

- | 1. Max Boost : 1.00000
- | 2. Lag U P : 0.00000
- | 3. Lag D N : 0.00000
- | 4. Rpm Ref : 6000.00000
- | 5. Gamma Rpm : 1.00000
- | 6. Gamma Gas : 0.00000
- | 7. Wastegate : 0.90000
- | 8. Is Adjustable : false
- | 9. CarData : None
- | 10. Turbo no : 0

Toyota Supra MKIV

1. Turbo 0 (file : ks_toyota_supra_mkiv0.turbo)

- | 1. Max Boost : 0.00000
- | 2. Lag U P : 0.99500
- | 3. Lag D N : 0.99000
- | 4. Rpm Ref : 3500.00000
- | 5. Gamma Rpm : 0.50000
- | 6. Gamma Gas : 0.50000
- | 7. Wastegate : 0.00000
- | 8. Is Adjustable : false
- | 9. CarData : None
- | 10. Turbo no : 0

2. Turbo 1 (file : ks_toyota_supra_mkiv1.turbo)

- | 1. Max Boost : 0.00000
- | 2. Lag U P : 0.99100
- | 3. Lag D N : 0.98800
- | 4. Rpm Ref : 4000.00000
- | 5. Gamma Rpm : 0.50000
- | 6. Gamma Gas : 0.50000
- | 7. Wastegate : 0.00000
- | 8. Is Adjustable : false
- | 9. CarData : None
- | 10. Turbo no : 0

3. Drift Turbo 0 (file : ks_toyota_supra_mkiv_drift0.turbo)

- | 1. Max Boost : 0.30000
- | 2. Lag U P : 0.99750

- | 3. Lag D N : 0.98800
- | 4. Rpm Ref : 3200.00000
- | 5. Gamma Rpm : 1.00000
- | 6. Gamma Gas : 1.00000
- | 7. Wastegate : 0.30000
- | 8. Is Adjustable : true
- | 9. CarData : None
- | 10. Turbo no : 0

4. Drift Turbo 1 (file : ks_toyota_supra_mkiv_drift1.turbo)

- | 1. Max Boost : 2.00000
- | 2. Lag U P : 0.99650
- | 3. Lag D N : 0.99700
- | 4. Rpm Ref : 4400.00000
- | 5. Gamma Rpm : 3.00000
- | 6. Gamma Gas : 1.00000
- | 7. Wastegate : 1.50000
- | 8. Is Adjustable : true
- | 9. CarData : None
- | 10. Turbo no : 0

19. Tyre [.tyre]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxxx

B. Schema

```
| 1. Name : string
| 2. Short Name : string
| 3. Tyre Compound : enum
| 4. Tyre Data : object
|   | 4a. Width : float
|   | 4b. Radius : float
|   | 4c. Rate : float
|   | 4d. Progressive Rate : float
|   | 4e. Damping : float
```

- 4f. Angular Inertia : float
- 4g. Rim Radius : float
- 4h. Radius Raise K : float
- 4i. Tread Height M M : float
- 4j. Tread Consumption K : float
- 4k. Mass : float
- 4l. Lateral Flex K : float
- 4m. Lateral Flex C : float
- 4n. Longitudinal Flex K : float
- 4o. Longitudinal Flex C : float
- 4p. Explosion Temperature : float
- 4q. Blanket Temperature : float
- 4r. Flat Spot K : float
- 4s. Normal To Flex Ratio : float
- 4t. Contact Camber : float
- 4u. Contact Flex : float
- 4v. Contact Vertical Flex : float
- 4w. Damping Mode : enum
- 4x. Maxwell Damping Peak Frequency : float
- 4y. Maxwell Stiffening Percent : float
- 4z. Damping Threshold Speed Ms : float
- 4aa. Speed Damping Factor : float
- 4ab. Deflection Damping Factor : float
- 5. Model Data : object
 - 5a. Dy0 : float
 - 5b. Dx0 : float
 - 5c. Ls Exp Y : float
 - 5d. Ls Exp X : float
 - 5e. Fz0 : float
 - 5f. Friction Limit Angle : float
 - 5g. Flex Gain : float
 - 5h. Cf Xmult : float
 - 5i. Brake D X Mod : float
 - 5j. Combined Factor : float
 - 5k. Grip Slip Factor : float
 - 5l. Wear Curve : string – path
 - 5m. Grain Factor : float
 - 5n. Contact Wear I M 0 : float
 - 5o. Relaxation Length Y : float
 - 5p. Relaxation Length X : float
 - 5q. Mz Tweak Mult : float
 - 5r. Mz Scale : float
 - 5s. Mz Trail Nd Slip Reduction : float
 - 5t. Mz Trail Remap : float
 - 5u. Brush Exponent : float
- 6. Thermal Data : object
 - 6a. Vertical Spring : float
 - 6b. Vertical Damp K : float
 - 6c. Wear Mult : float
 - 6d. Density : float
 - 6e. Specific Heat : float
 - 6f. Surface Core Mass Ratio : float
 - 6g. Cool Factor Rain : float
 - 6h. Cool Factor : float
 - 6i. Heat Partition Coeff : float
 - 6j. Road Conduction : float

- 6k. Thermal Conductivity : float
- 6l. Rolling Factor : float
- 6m. Grain Gain : float
- 6n. Grain Gamma : float
- 6o. Grain Slip Angle Gain : float
- 6p. Grain Slip Angle Gamma : float
- 6q. Grain Slip Angle Threshold : float
- 6r. Blister Gain : float
- 6s. Blister Gamma : float
- 6t. Practical Temp Source : float
- 6u. Brake Transfer Factor : float
- 6v. Thermal Performance Curve : string - path
- 6w. Contact I M 0 : float
- 6x. Tref Friction Limit Angle : float
- 6y. Friction Limit Angle T Sensitivity : float
- 7. Pressure Data : object
 - 7a. Pressure Flex Gain : float
 - 7b. Rolling Resistance Gain : float
 - 7c. Rolling Heat Gain : float
 - 7d. Gain D : float
 - 7e. Ideal Pressure : float
 - 7f. Pressure Reference : float
 - 7g. Pressure Spring Curve : string - path
 - 7h. Pressure Spring Gain : float
 - 7i. Curb Pressure Loss Must : float
 - 7j. Curb Pressure Loss Threshold : float
- 8. Camber Data : object
 - 8a. Camber Gain : float
 - 8b. Camber Vertical K Range Deg : float
 - 8c. Camber Vertical K Gain : float
 - 8d. Dcamber0 : float
 - 8e. Dcamber1 : float
- 9. Speed Sensitivity : object
 - 9a. Mu0 T : float
 - 9b. Ref Speed : float
 - 9c. Mu Speed Sensitivity : float
 - 9d. Mu T Sensitivity : float
 - 9e. Tref Mu T : float
 - 9f. Mu0 T X : float
 - 9g. Ref Speed X : float
 - 9h. Mu Speed Sensitivity X : float
 - 9i. Mu T Sensitivity X : float
 - 9j. Tref Mu T X : float
- 10. Rolling Resistance : object
 - 10a. Rr0 : float
 - 10b. Rr1 : float
 - 10c. Rr Slip : float
 - 10d. Rr Wear Mult : float
- 11. Groove Data : object
 - 11a. Groove Factor : float
 - 11b. Groove S A Factor : float
 - 11c. Groove S R Factor : float
- 12. Pressure : float
- 13. Init Data : object
 - 13a. Width : float
 - 13b. Aspect Ratio : float

- 13c. Diameter : float
- 13d. Load Index : float
- 13e. Pressure : float

Enum - Car Tyre

3	Tyre Compound	Eco, Road, SuperCar, HyperCar, Slick_Medium, Wet, Racing_Vintage, F1_Soft, F1_Medium, F1_Hard, F1_Wet, F1_Intermediate
4w	Damping Mode	Simple, hystereticMaxwell, hystereticNando

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Name	None (String)	Full display name of the compound (e.g. Eco (E), Racing Vintage (RV), Wet (W)).
2.	Short Name	None (String)	Abbreviated HUD / telemetry label (E, RV, W).
3.	Tyre Compound	None (Enum : Eco / Road / SuperCar / HyperCar / Slick_Medium / Wet / Racing_Vintage / F1_Soft / F1_Medium / F1_Hard / F1_Wet / F1_Intermediate)	Compound class selecting grip, thermal, and wear baselines (Eco street, Racing_Vintage, F1_Wet, etc.).
4.	Tyre Data	None (Object)	Structural carcass block: geometry, vertical spring/damper, flex, temperatures, and contact-patch parameters.
4a.	Width	m (Meters)	Contact-section width in SI (0.165 Eco = 165 mm, 0.305 F1 wet = 305 mm). Distinct from Init Data width in mm.
4b.	Radius	m (Meters)	Unloaded overall tyre radius (Eco 0.251 m, Vintage 0.308 m, F1 wet 0.360 m). Sets rolling circumference and gearing.
4c.	Rate	N/m (Newtons per meter)	Vertical carcass spring rate (~263–307 kN/m in examples). Primary tyre contribution to wheel rate and ride frequency.
4d.	Progressive Rate	N/m ² or N/m ³ (Progressive stiffening)	Additional rate that rises with deflection (progressive carcass). 0.0 in examples = linear spring.
4e.	Damping	N·s/m (Newton-seconds per meter)	Vertical viscous damping of the tyre spring (Eco ~549, F1 wet ~846). Controls carcass oscillation and curb impact.

ID	Name	Unit of Measurement	Description
4f.	Angular Inertia	kg·m ² (Kilogram square meters)	Rotational inertia of tyre + rim about spin axis (Eco 0.91, Vintage 2.27, F1 wet 4.67). Affects acceleration, braking, and ABS feel.
4g.	Rim Radius	m (Meters)	Wheel rim outer radius (12" → 0.1524 m, 15" → 0.1905 m, 18" → 0.2286 m). Used with tyre radius for sidewall height.
4h.	Radius Raise K	m/N or dimensionless gain	Radius growth coefficient under load/centrifugal effects (0.001 in examples); slight diameter increase with speed/load.
4i.	Tread Height M M	mm (Millimeters)	Usable tread depth (Eco/Vintage 8 mm, F1 wet 3 mm). Depletes via wear; wet grooves need remaining depth for water evacuation.
4j.	Tread Consumption K	Dimensionless wear rate	Scalar converting slip/energy work into tread-height loss (6.0 across examples).
4k.	Mass	kg (Kilograms)	Tyre (and often rim share) mass for unsprung dynamics (6 kg in examples — simplified vs. real F1 ~9–10 kg).
4l.	Lateral Flex K	N/m (Newtons per meter)	Lateral carcass stiffness (Eco 28 kN/m, F1 wet 30 kN/m). Negative values (Vintage –10 kN/m) can encode special flex models; governs sidewall lean under Fy.
4m.	Lateral Flex C	Dimensionless damping ratio or N·s/m	Lateral flex damping (typically ~2.0). Limits oscillation of the contact patch relative to the rim.
4n.	Longitudinal Flex K	N/m (Newtons per meter)	Fore-aft carcass stiffness under Fx (Eco 18 kN/m, F1 wet 20 kN/m). Affects traction/braking compliance feel.
4o.	Longitudinal Flex C	Dimensionless damping ratio or N·s/m	Longitudinal flex damping (~2.0–2.5).
4p.	Explosion Temperature	°C (Degrees Celsius)	Catastrophic overheat threshold (450 °C in examples) above which the tyre is considered failed/ exploded.
4q.	Blanket Temperature	°C (Degrees Celsius)	Tyre-warmer / pit-blanket target start temperature (40 °C in examples).
4r.	Flat Spot K	Dimensionless gain	Sensitivity of flat-spot damage under locked-wheel scrub (0.1 in examples).

ID	Name	Unit of Measurement	Description
4s.	Normal To Flex Ratio	Dimensionless ratio	Coupling of vertical load into carcass flex (0.0 = no extra coupling in examples).
4t.	Contact Camber	deg (Degrees) or model coefficient	Contact-patch camber sensitivity / effective patch camber shaping (Eco ~2.8, F1 wet ~11).
4u.	Contact Flex	m (Meters) or dimensionless	Lateral contact-patch flex magnitude under load (Eco 0.023, F1 wet 0.345).
4v.	Contact Vertical Flex	Dimensionless or mm-scale	Vertical contact-patch compliance shaping (Eco ~3.6, F1 wet ~16.6).
4w.	Damping Mode	None (Enum : Simple / hystereticMaxwell / hystereticNando)	Carcass damping model: Simple viscous, Maxwell hysteretic, or Nando hysteretic (examples use hystereticNando).
4x.	Maxwell Damping Peak Frequency	Hz (Hertz)	Peak frequency for Maxwell hysteretic damper (0 when unused / Nando mode).
4y.	Maxwell Stiffening Percent	% (Percent)	Extra dynamic stiffening in Maxwell mode (0 when unused).
4z.	Damping Threshold Speed Ms	m/s (Meters per second)	Vertical deflection-speed threshold where speed-dependent damping engages (1.0 m/s in examples).
4aa.	Speed Damping Factor	Dimensionless gain	Multiplier on damping above the speed threshold (0.5 in examples).
4ab.	Deflection Damping Factor	Dimensionless gain	Multiplier scaling damping with deflection amplitude (3.0 in examples).
5.	Model Data	None (Object)	Brush / friction model: peak μ , load sensitivity, slip limits, wear, relaxation lengths, and self-aligning torque (Mz).
5a.	Dy0	Dimensionless friction coefficient (μ_y)	Peak lateral friction coefficient at reference load Fz0 (Eco 0.96, Vintage 1.11, F1 wet 1.47).
5b.	Dx0	Dimensionless friction coefficient (μ_x)	Peak longitudinal friction coefficient at Fz0 (Eco 1.00, F1 wet 1.56). Braking/traction grip baseline.
5c.	Ls Exp Y	Dimensionless exponent	Lateral load-sensitivity exponent: how μ_y falls as Fz rises above Fz0 (~0.82–0.87).
5d.	Ls Exp X	Dimensionless exponent	Longitudinal load-sensitivity exponent for μ_x (~0.83–1.07).
5e.	Fz0	N (Newtons)	Reference vertical load for Dy0/ Dx0 (~3840–4230 N $\approx$ 390–430 kgf per tyre).

ID	Name	Unit of Measurement	Description
5f.	Friction Limit Angle	deg (Degrees)	Slip angle (or slip equivalent) at which peak friction is reached (~7.8–8.6°). Beyond this, grip falls toward sliding.
5g.	Flex Gain	Dimensionless gain	Scales carcass flex contribution into the friction model (0.0207 in examples).
5h.	Cf Xmult	Dimensionless multiplier	Multiplier on longitudinal cornering/stiffness factor Cf_x (1.8–2.0).
5i.	Brake D X Mod	Dimensionless gain	Modifier reducing Dx under heavy braking (Eco 0.05, F1 wet 0.00). Models brake-lock μ drop.
5j.	Combined Factor	Dimensionless coefficient	Combined-slip ellipse shaping (Pacejka-like): how Fx and Fy share friction budget (1.8–2.0).
5k.	Grip Slip Factor	Dimensionless coefficient	Post-peak sliding grip retention vs. peak (0.76–0.82). Higher = more forgiving slide.
5l.	Wear Curve	None (.curve file path)	LUT of wear state → grip/thermal modifiers (wear_eco.curve, wear_f12025_wet.curve, ...).
5m.	Grain Factor	Dimensionless gain	Base graining susceptibility (0.001). Raised in thermal grain gains for soft compounds.
5n.	Contact Wear I M O	Dimensionless I/M/O weights	Inner / middle / outer contact-wear distribution weight (4.0 in examples).
5o.	Relaxation Length Y	m (Meters)	Lateral force lag distance (~0.10 m). Shorter = snappier turn-in; longer = delayed Fy build-up.
5p.	Relaxation Length X	m (Meters)	Longitudinal force relaxation length (~0.10 m). Affects ABS and throttle traction transient.
5q.	Mz Tweak Mult	Dimensionless multiplier	Global scale on self-aligning torque Mz feel (0.6).
5r.	Mz Scale	Dimensionless multiplier	Base Mz magnitude scale (0.12). Steering weight / trail feedback.
5s.	Mz Trail Nd Slip Reduction	Dimensionless coefficient	How pneumatic trail (Mz) shrinks with normalized slip (0.5).
5t.	Mz Trail Remap	Dimensionless coefficient	Remapping curve intensity for trail vs. slip (0.2).
5u.	Brush Exponent	Dimensionless exponent	Brush-model stress distribution exponent (3.0). Shapes the force build-up before peak slip.

ID	Name	Unit of Measurement	Description
6.	Thermal Data	None (Object)	Heat capacity, cooling, conduction, grain/blister, brake heat soak, and temperature→grip curve.
6a.	Vertical Spring	N/m or thermal model scale	Thermal-node coupling spring (40 in examples) between surface/core temperature layers.
6b.	Vertical Damp K	Thermal damping coefficient	Damping of surface↔core temperature exchange (2.0).
6c.	Wear Mult	Dimensionless multiplier	Temperature-linked wear rate multiplier (0.2).
6d.	Density	kg/m ³ (Kilograms per cubic meter)	Rubber density (1100 kg/m ³ — typical elastomer). Used with volume for thermal mass.
6e.	Specific Heat	J/(kg·K) (Joules per kilogram-kelvin)	Specific heat capacity of the tread rubber (1600 J/(kg·K) in examples).
6f.	Surface Core Mass Ratio	Dimensionless ratio (0 -1)	Fraction of thermal mass in the surface skin vs. core (0.15). Low = surface heats/cool fast.
6g.	Cool Factor Rain	Dimensionless gain	Extra convective cooling in wet/ rain conditions (3.0).
6h.	Cool Factor	Dimensionless gain	Base air/road cooling rate (Eco 1.4, Vintage 0.7, F1 wet 0.9).
6i.	Heat Partition Coeff	Dimensionless coefficient	Split of generated heat into surface vs. bulk (0.5–1.4).
6j.	Road Conduction	W/(m ² ·K) or model units	Conductive heat transfer to asphalt (1000 in examples).
6k.	Thermal Conductivity	W/(m·K) (Watts per meter-kelvin)	Rubber conductivity (Eco 0.20, Vintage 0.28, F1 wet 0.35 W/(m·K)).
6l.	Rolling Factor	Dimensionless gain	Heat generation from rolling hysteresis (Eco 0.6, F1 wet 1.6).
6m.	Grain Gain	Dimensionless gain	Rate of surface graining under thermal/slip stress (Eco 0.017, F1 wet 0.47).
6n.	Grain Gamma	Dimensionless exponent	Non-linear shaping of grain growth vs. drivers (matches Grain Gain in examples).
6o.	Grain Slip Angle Gain	Dimensionless gain	Extra grain rate from slip angle (F1 wet 0.565 vs Eco 0.165).
6p.	Grain Slip Angle Gamma	Dimensionless exponent	Exponent on slip-angle grain contribution.
6q.	Grain Slip Angle Threshold	deg (Degrees)	Slip angle above which grain acceleration engages (~10.5–11.3°).

ID	Name	Unit of Measurement	Description
6r.	Blister Gain	Dimensionless gain	Blister formation rate (0 in these compounds; used on soft slicks).
6s.	Blister Gamma	Dimensionless exponent	Non-linear blister growth shaping (0 when blister unused).
6t.	Practical Temp Source	Dimensionless blend (0 - 1)	Blend between surface and core temperature for “practical” grip temp (0.8).
6u.	Brake Transfer Factor	Dimensionless gain	Fraction of brake rotor heat conducted into the tyre (0.0007).
6v.	Thermal Performance Curve	None (.curve file path)	LUT temperature (°C) → friction efficiency (tcurve_eco.curve, tcurve_f12025_wet.curve).
6w.	Contact I M O	Dimensionless I/M/O weights	Inner/middle/outer contact thermal weighting (1.6).
6x.	Tref Friction Limit Angle	°C (Degrees Celsius)	Reference temperature for friction-limit-angle thermal correction (75 °C).
6y.	Friction Limit Angle T Sensitivity	deg/°C or gain	How Friction Limit Angle shifts with temperature (0.06).
7.	Pressure Data	None (Object)	Inflation effects on flex, rolling resistance, heat, spring rate, and curb puncture loss.
7a.	Pressure Flex Gain	Dimensionless gain	How carcass flex changes with pressure offset from reference (0.25).
7b.	Rolling Resistance Gain	Dimensionless gain	Pressure→rolling-resistance coupling (1.2). Under-inflation raises RR.
7c.	Rolling Heat Gain	Dimensionless gain	Extra heat from pressure-related hysteresis (0.1).
7d.	Gain D	Dimensionless gain	Pressure influence on longitudinal grip/damping term (~0.0025–0.0026).
7e.	Ideal Pressure	psi (Pounds per square inch)	Target hot pressure for peak performance (Eco/Vintage 31 psi, F1 wet 26 psi).
7f.	Pressure Reference	psi (Pounds per square inch)	Cold/reference pressure for spring and flex tables (Eco 30, Vintage 28, F1 wet 17 psi).
7g.	Pressure Spring Curve	None (.curve file path)	Optional LUT of pressure → vertical spring rate; None when using Pressure Spring Gain only.
7h.	Pressure Spring Gain	N/m per psi (or N/m)	Vertical rate contribution from inflation (~9.9–11.1 kN/m scale in examples).

ID	Name	Unit of Measurement	Description
7i.	Curb Pressure Loss Must	psi or loss amount	Pressure drop applied when curb impact exceeds threshold (1.0).
7j.	Curb Pressure Loss Threshold	m/s or impact units	Impact severity before curb puncture loss (Eco/Vintage 10, F1 wet 30 — tougher wet carcass).
8.	Camber Data	None (Object)	Camber sensitivity of grip and vertical stiffness.
8a.	Camber Gain	Dimensionless gain	Grip change per degree of camber (0.2).
8b.	Camber Vertical K Range Deg	deg (Degrees)	Camber window over which vertical rate gain applies ($\pm 5^\circ$).
8c.	Camber Vertical K Gain	Dimensionless gain	Vertical stiffness change with camber (-0.1).
8d.	Dcamber0	Dimensionless coefficient	Camber-thrust / μ offset coefficient at small camber (0.6).
8e.	Dcamber1	Dimensionless coefficient	Higher-order camber sensitivity (-11). Shapes camber thrust curve.
9.	Speed Sensitivity	None (Object)	Speed- and temperature-dependent friction for lateral (T) and longitudinal (X) axes.
9a.	Mu0 T	Dimensionless friction coefficient (μ)	Base lateral μ at reference speed/temp (Eco 0.78, F1 wet 0.85).
9b.	Ref Speed	m/s (Meters per second)	Reference slip/ground speed for lateral μ speed law (4.5 m/s).
9c.	Mu Speed Sensitivity	Dimensionless gain	Lateral μ change with speed relative to Ref Speed (Eco 0.78, F1 wet 1.2).
9d.	Mu T Sensitivity	1/°C or gain	Lateral μ temperature sensitivity (0.025–0.035).
9e.	Tref Mu T	°C (Degrees Celsius)	Reference temperature for lateral μ thermal correction (70–80 °C).
9f.	Mu0 T X	Dimensionless friction coefficient (μ_x)	Base longitudinal μ at reference conditions (Eco 0.70, F1 wet 0.68).
9g.	Ref Speed X	m/s (Meters per second)	Reference speed for longitudinal μ law (2.0–3.5 m/s).
9h.	Mu Speed Sensitivity	Dimensionless gain	Longitudinal μ speed sensitivity (Eco 0.62, F1 wet 1.8 — wet more speed-critical).
9i.	Mu T Sensitivity X	1/°C or gain	Longitudinal μ temperature sensitivity (0.035–0.095).
9j.	Tref Mu T X	°C (Degrees Celsius)	Reference temperature for longitudinal μ thermal correction (70–80 °C).

ID	Name	Unit of Measurement	Description
10.	Rolling Resistance	None (Object)	Steady and slip-dependent rolling drag opposing wheel rotation.
10a.	Rr0	N (Newtons)	Constant rolling-resistance force (~17.8–20.1 N). Dominant at low speed.
10b.	Rr1	N·s/m or N per (m/s)	Speed-proportional RR term (0.002). Small linear rise with velocity.
10c.	Rr Slip	N or model units	Extra RR under slip/scrub (646). Raises drag when sliding.
10d.	Rr Wear Mult	Dimensionless multiplier	How wear state scales rolling resistance (0.14).
11.	Groove Data	None (Object)	Tread groove / void effects on wet grip and aquaplaning (critical for F1_Wet).
11a.	Groove Factor	Dimensionless or groove index	Overall groove/void intensity (street Eco 0.49 vs F1 wet 68 — deep rain channels).
11b.	Groove S A Factor	Dimensionless coefficient	Groove effect on slip-angle grip (0.45). SA = slip angle.
11c.	Groove S R Factor	Dimensionless coefficient	Groove effect on slip-ratio / longitudinal grip (0.45). SR = slip ratio.
12.	Pressure	psi (Pounds per square inch)	Default cold inflation for this compound asset (Eco 30, Vintage 28, F1 wet 17 psi). Overridden by setup pressures.
13.	Init Data	None (Object)	Human-readable tyre size designation (width–aspect–rim) and load/pressure ratings.
13a.	Width	mm (Millimeters)	Nominal section width (165 / 195 / 305). Matches commercial size marking.
13d.	Aspect Ratio	% (Percent of width)	Sidewall height as % of width (60 street, 43 F1 wet ≈ low-profile race).
13c.	Diameter	in (Inches)	Rim diameter (12 / 15 / 18). Must match Rim Radius = Diameter × 0.0254 / 2.
13d.	Load Index	None (Integer load index)	ISO load index (Eco 99, Vintage 88) — max load capacity class.
13e.	Pressure	psi (Pounds per square inch)	Rated / nominal inflation for the size designation (mirrors field 12 / Pressure Reference).

D. Example data

I. Chosen Tyres for Example

- Eco (slug : eco_165_60_12)
- Vintage (slug : vintage_195_60_15)
- F1 2025 Wet (slug : f12025_wet_305_720_18)

II. Example

Eco | Size : 165 - 60 - 12

- 1. Name : Eco (E)
- 2. Short Name : E
- 3. Tyre Compound : Eco
- 4. Tyre Data
 - 4a. Width : 0.16500
 - 4b. Radius : 0.25140
 - 4c. Rate : 306900.00000
 - 4d. Progressive Rate : 0.00000
 - 4e. Damping : 549.08398
 - 4f. Angular Inertia : 0.91034
 - 4g. Rim Radius : 0.15240
 - 4h. Radius Raise K : 0.00100
 - 4i. Tread Height M M : 8.00000
 - 4j. Tread Consumption K : 6.00000
 - 4k. Mass : 6.00000
 - 4l. Lateral Flex K : 28000.00000
 - 4m. Lateral Flex C : 2.00000
 - 4n. Longitudinal Flex K : 18000.00000
 - 4o. Longitudinal Flex C : 2.50000
 - 4p. Explosion Temperature : 450.00000
 - 4q. Blanket Temperature : 40.00000
 - 4r. Flat Spot K : 0.10000
 - 4s. Normal To Flex Ratio : 0.00000
 - 4t. Contact Camber : 2.79367
 - 4u. Contact Flex : 0.02293
 - 4v. Contact Vertical Flex : 3.58828
 - 4w. Damping Mode : hystereticNando
 - 4x. Maxwell Damping Peak Frequency : 0.00000
 - 4y. Maxwell Stiffening Percent : 0.00000
 - 4z. Damping Threshold Speed Ms : 1.00000
 - 4aa. Speed Damping Factor : 0.50000
 - 4ab. Deflection Damping Factor : 3.00000
- 5. Model Data
 - 5a. Dy0 : 0.96160
 - 5b. Dx0 : 1.00483
 - 5c. Ls Exp Y : 0.82088
 - 5d. Ls Exp X : 1.07088
 - 5e. Fz0 : 4170.00000
 - 5f. Friction Limit Angle : 8.56818
 - 5g. Flex Gain : 0.02070
 - 5h. Cf Xmult : 2.00000

- 5i. Brake D X Mod : 0.05000
- 5j. Combined Factor : 1.90000
- 5k. Grip Slip Factor : 0.78000
- 5l. Wear Curve : content\cars\common_phsx\tyres\eco\wear_eco.curve
- 5m. Grain Factor : 0.00100
- 5n. Contact Wear I M 0 : 4.00000
- 5o. Relaxation Length Y : 0.10051
- 5p. Relaxation Length X : 0.10051
- 5q. Mz Tweak Mult : 0.60000
- 5r. Mz Scale : 0.12000
- 5s. Mz Trail Nd Slip Reduction : 0.50000
- 5t. Mz Trail Remap : 0.20000
- 5u. Brush Exponent : 3.00000

- 6. Thermal Data

- 6a. Vertical Spring : 40.00000
- 6b. Vertical Damp K : 2.00000
- 6c. Wear Mult : 0.20000
- 6d. Density : 1100.00000
- 6e. Specific Heat : 1600.00000
- 6f. Surface Core Mass Ratio : 0.15000
- 6g. Cool Factor Rain : 3.00000
- 6h. Cool Factor : 1.40000
- 6i. Heat Partition Coeff : 0.50000
- 6j. Road Conduction : 1000.00000
- 6k. Thermal Conductivity : 0.20000
- 6l. Rolling Factor : 0.60000
- 6m. Grain Gain : 0.01700
- 6n. Grain Gamma : 0.01700
- 6o. Grain Slip Angle Gain : 0.16500
- 6p. Grain Slip Angle Gamma : 0.24500
- 6q. Grain Slip Angle Threshold : 11.26818
- 6r. Blister Gain : 0.00000
- 6s. Blister Gamma : 0.00000
- 6t. Practical Temp Source : 0.80000
- 6u. Brake Transfer Factor : 0.00070
- 6v. Thermal Performance Curve :

content\cars\common_phsx\tyres\eco\tcurve_eco.curve

- 6w. Contact I M 0 : 1.60000
- 6x. Tref Friction Limit Angle : 75.00000
- 6y. Friction Limit Angle T Sensitivity : 0.06000

- 7. Pressure Data

- 7a. Pressure Flex Gain : 0.25000
- 7b. Rolling Resistance Gain : 1.20000
- 7c. Rolling Heat Gain : 0.10000
- 7d. Gain D : 0.00260
- 7e. Ideal Pressure : 31.00000
- 7f. Pressure Reference : 30.00000
- 7g. Pressure Spring Curve : None
- 7h. Pressure Spring Gain : 10890.00000
- 7i. Curb Pressure Loss Must : 1.00000
- 7j. Curb Pressure Loss Threshold : 10.00000

- 8. Camber Data

- 8a. Camber Gain : 0.20000
- 8b. Camber Vertical K Range Deg : 5.00000
- 8c. Camber Vertical K Gain : -0.10000
- 8d. Dcamber0 : 0.60000

- L 8e. Dcamber1 : -11.00000
- 9. Speed Sensitivity
 - 9a. Mu0 T : 0.78000
 - 9b. Ref Speed : 4.50000
 - 9c. Mu Speed Sensitivity : 0.78000
 - 9d. Mu T Sensitivity : 0.02500
 - 9e. Tref Mu T : 70.00000
 - 9f. Mu0 T X : 0.70000
 - 9g. Ref Speed X : 3.50000
 - 9h. Mu Speed Sensitivity X : 0.62000
 - 9i. Mu T Sensitivity X : 0.09500
 - 9j. Tref Mu T X : 70.00000
- 10. Rolling Resistance
 - 10a. Rr0 : 17.79780
 - 10b. Rr1 : 0.00200
 - 10c. Rr Slip : 646.00000
 - 10d. Rr Wear Mult : 0.14000
- 11. Groove Data
 - 11a. Groove Factor : 0.49000
 - 11b. Groove S A Factor : 0.45000
 - 11c. Groove S R Factor : 0.45000
- 12. Pressure : 30.00000
- 13. Init Data
 - 13a. Width : 165.00000
 - 13b. Aspect Ratio : 60.00000
 - 13c. Diameter : 12.00000
 - 13d. Load Index : 99.00000
 - 13e. Pressure : 30.00000

Vintage | Size : 195 - 60 - 15

- 1. Name : Racing Vintage (RV)
- 2. Short Name : RV
- 3. Tyre Compound : Racing_Vintage
- 4. Tyre Data
 - 4a. Width : 0.19500
 - 4b. Radius : 0.30750
 - 4c. Rate : 272800.00000
 - 4d. Progressive Rate : 0.00000
 - 4e. Damping : 488.07468
 - 4f. Angular Inertia : 2.26550
 - 4g. Rim Radius : 0.19050
 - 4h. Radius Raise K : 0.00100
 - 4i. Tread Height M M : 8.00000
 - 4j. Tread Consumption K : 6.00000
 - 4k. Mass : 6.00000
 - 4l. Lateral Flex K : -10000.00000
 - 4m. Lateral Flex C : 2.00000
 - 4n. Longitudinal Flex K : -1800.00000
 - 4o. Longitudinal Flex C : 2.00000
 - 4p. Explosion Temperature : 450.00000
 - 4q. Blanket Temperature : 40.00000
 - 4r. Flat Spot K : 0.10000

- 4s. Normal To Flex Ratio : 0.00000
- 4t. Contact Camber : 9.33809
- 4u. Contact Flex : 0.16605
- 4v. Contact Vertical Flex : 14.45922
- 4w. Damping Mode : hystereticNando
- 4x. Maxwell Damping Peak Frequency : 0.00000
- 4y. Maxwell Stiffening Percent : 0.00000
- 4z. Damping Threshold Speed Ms : 1.00000
- 4aa. Speed Damping Factor : 0.50000
- 4ab. Deflection Damping Factor : 3.00000

- 5. Model Data

- 5a. Dy0 : 1.10844
- 5b. Dx0 : 1.14557
- 5c. Ls Exp Y : 0.85581
- 5d. Ls Exp X : 0.88381
- 5e. Fz0 : 3840.00000
- 5f. Friction Limit Angle : 8.30538
- 5g. Flex Gain : 0.02070
- 5h. Cf Xmult : 2.00000
- 5i. Brake D X Mod : 0.05000
- 5j. Combined Factor : 1.80000
- 5k. Grip Slip Factor : 0.76000
- 5l. Wear Curve :

content\cars\common_phsx\tyres\vintage\wear_vintage_racing.curve

- 5m. Grain Factor : 0.00100
- 5n. Contact Wear I M 0 : 4.00000
- 5o. Relaxation Length Y : 0.10051
- 5p. Relaxation Length X : 0.10051
- 5q. Mz Tweak Mult : 0.60000
- 5r. Mz Scale : 0.12000
- 5s. Mz Trail Nd Slip Reduction : 0.50000
- 5t. Mz Trail Remap : 0.20000
- 5u. Brush Exponent : 3.00000

- 6. Thermal Data

- 6a. Vertical Spring : 40.00000
- 6b. Vertical Damp K : 2.00000
- 6c. Wear Mult : 0.20000
- 6d. Density : 1100.00000
- 6e. Specific Heat : 1600.00000
- 6f. Surface Core Mass Ratio : 0.15000
- 6g. Cool Factor Rain : 3.00000
- 6h. Cool Factor : 0.70000
- 6i. Heat Partition Coeff : 0.70000
- 6j. Road Conduction : 1000.00000
- 6k. Thermal Conductivity : 0.28000
- 6l. Rolling Factor : 1.00000
- 6m. Grain Gain : 0.27000
- 6n. Grain Gamma : 0.27000
- 6o. Grain Slip Angle Gain : 0.26500
- 6p. Grain Slip Angle Gamma : 0.34500
- 6q. Grain Slip Angle Threshold : 11.00538
- 6r. Blister Gain : 0.00000
- 6s. Blister Gamma : 0.00000
- 6t. Practical Temp Source : 0.80000
- 6u. Brake Transfer Factor : 0.00070

```

| | 6v. Thermal Performance Curve :
content\cars\common_phsx\tyres\vintage\tcurve_vintage_racing.curve
| | 6w. Contact I M 0 : 1.60000
| | 6x. Tref Friction Limit Angle : 75.00000
| | 6y. Friction Limit Angle T Sensitivity : 0.06000
- 7. Pressure Data
| | 7a. Pressure Flex Gain : 0.25000
| | 7b. Rolling Resistance Gain : 1.20000
| | 7c. Rolling Heat Gain : 0.10000
| | 7d. Gain D : 0.00250
| | 7e. Ideal Pressure : 31.00000
| | 7f. Pressure Reference : 28.00000
| | 7g. Pressure Spring Curve : None
| | 7h. Pressure Spring Gain : 9860.00000
| | 7i. Curb Pressure Loss Must : 1.00000
| | 7j. Curb Pressure Loss Threshold : 10.00000
- 8. Camber Data
| | 8a. Camber Gain : 0.20000
| | 8b. Camber Vertical K Range Deg : 5.00000
| | 8c. Camber Vertical K Gain : -0.10000
| | 8d. Dcamber0 : 0.60000
| | 8e. Dcamber1 : -11.00000
- 9. Speed Sensitivity
| | 9a. Mu0 T : 0.78000
| | 9b. Ref Speed : 4.50000
| | 9c. Mu Speed Sensitivity : 0.78000
| | 9d. Mu T Sensitivity : 0.02500
| | 9e. Tref Mu T : 70.00000
| | 9f. Mu0 T X : 0.80000
| | 9g. Ref Speed X : 2.00000
| | 9h. Mu Speed Sensitivity X : 0.82000
| | 9i. Mu T Sensitivity X : 0.08500
| | 9j. Tref Mu T X : 70.00000
- 10. Rolling Resistance
| | 10a. Rr0 : 18.36000
| | 10b. Rr1 : 0.00200
| | 10c. Rr Slip : 646.00000
| | 10d. Rr Wear Mult : 0.14000
- 11. Groove Data
| | 11a. Groove Factor : 0.49000
| | 11b. Groove S A Factor : 0.45000
| | 11c. Groove S R Factor : 0.45000
- 12. Pressure : 28.00000
- 13. Init Data
| | 13a. Width : 195.00000
| | 13b. Aspect Ratio : 60.00000
| | 13c. Diameter : 15.00000
| | 13d. Load Index : 88.00000
| | 13e. Pressure : 28.00000

```

F1 2025 [Wet] | Size : 305 - 720 - 18

- 1. Name : Wet (W)
- 2. Short Name : W

- 3. Tyre Compound : F1_Wet
- 4. Tyre Data
 - 4a. Width : 0.30500
 - 4b. Radius : 0.35975
 - 4c. Rate : 262600.00000
 - 4d. Progressive Rate : 0.00000
 - 4e. Damping : 845.68597
 - 4f. Angular Inertia : 4.66531
 - 4g. Rim Radius : 0.22860
 - 4h. Radius Raise K : 0.00100
 - 4i. Tread Height M M : 3.00000
 - 4j. Tread Consumption K : 6.00000
 - 4k. Mass : 6.00000
 - 4l. Lateral Flex K : 30000.00000
 - 4m. Lateral Flex C : 2.00000
 - 4n. Longitudinal Flex K : 20000.00000
 - 4o. Longitudinal Flex C : 2.50000
 - 4p. Explosion Temperature : 450.00000
 - 4q. Blanket Temperature : 40.00000
 - 4r. Flat Spot K : 0.10000
 - 4s. Normal To Flex Ratio : 0.00000
 - 4t. Contact Camber : 10.98116
 - 4u. Contact Flex : 0.34464
 - 4v. Contact Vertical Flex : 16.61791
 - 4w. Damping Mode : hystereticNando
 - 4x. Maxwell Damping Peak Frequency : 0.00000
 - 4y. Maxwell Stiffening Percent : 0.00000
 - 4z. Damping Threshold Speed Ms : 1.00000
 - 4aa. Speed Damping Factor : 0.500000
 - 4ab. Deflection Damping Factor : 3.00000
- 5. Model Data
 - 5a. Dy0 : 1.46825
 - 5b. Dx0 : 1.56499
 - 5c. Ls Exp Y : 0.86528
 - 5d. Ls Exp X : 0.83328
 - 5e. Fz0 : 4230.00000
 - 5f. Friction Limit Angle : 7.81311
 - 5g. Flex Gain : 0.02070
 - 5h. Cf Xmult : 1.80000
 - 5i. Brake D X Mod : 0.00000
 - 5j. Combined Factor : 2.00000
 - 5k. Grip Slip Factor : 0.82000
 - 5l. Wear Curve :
content\cars\common_phsx\tyres\f1_2025\wear_f12025_wet.curve
 - 5m. Grain Factor : 0.00100
 - 5n. Contact Wear I M 0 : 4.00000
 - 5o. Relaxation Length Y : 0.10051
 - 5p. Relaxation Length X : 0.10051
 - 5q. Mz Tweak Mult : 0.60000
 - 5r. Mz Scale : 0.12000
 - 5s. Mz Trail Nd Slip Reduction : 0.50000
 - 5t. Mz Trail Remap : 0.20000
 - 5u. Brush Exponent : 3.00000
- 6. Thermal Data
 - 6a. Vertical Spring : 40.00000
 - 6b. Vertical Damp K : 2.00000

- 6c. Wear Mult : 0.20000
- 6d. Density : 1100.00000
- 6e. Specific Heat : 1600.00000
- 6f. Surface Core Mass Ratio : 0.15000
- 6g. Cool Factor Rain : 3.00000
- 6h. Cool Factor : 0.90000
- 6i. Heat Partition Coeff : 1.40000
- 6j. Road Conduction : 1000.00000
- 6k. Thermal Conductivity : 0.35000
- 6l. Rolling Factor : 1.60000
- 6m. Grain Gain : 0.47000
- 6n. Grain Gamma : 0.47000
- 6o. Grain Slip Angle Gain : 0.56500
- 6p. Grain Slip Angle Gamma : 0.84500
- 6q. Grain Slip Angle Threshold : 10.51311
- 6r. Blister Gain : 0.00000
- 6s. Blister Gamma : 0.00000
- 6t. Practical Temp Source : 0.80000
- 6u. Brake Transfer Factor : 0.00070
- 6v. Thermal Performance Curve :

content\cars\common_phsx\tyres\f1_2025\tcurve_f12025_wet.curve

- 6w. Contact I M 0 : 1.60000
- 6x. Tref Friction Limit Angle : 75.00000
- 6y. Friction Limit Angle T Sensitivity : 0.06000

- 7. Pressure Data

- 7a. Pressure Flex Gain : 0.25000
- 7b. Rolling Resistance Gain : 1.20000
- 7c. Rolling Heat Gain : 0.10000
- 7d. Gain D : 0.00250
- 7e. Ideal Pressure : 26.00000
- 7f. Pressure Reference : 17.00000
- 7g. Pressure Spring Curve : None
- 7h. Pressure Spring Gain : 11110.00000
- 7i. Curb Pressure Loss Must : 1.00000
- 7j. Curb Pressure Loss Threshold : 30.00000

- 8. Camber Data

- 8a. Camber Gain : 0.20000
- 8b. Camber Vertical K Range Deg : 5.00000
- 8c. Camber Vertical K Gain : -0.10000
- 8d. Dcamber0 : 0.60000
- 8e. Dcamber1 : -11.00000

- 9. Speed Sensitivity

- 9a. Mu0 T : 0.85000
- 9b. Ref Speed : 4.50000
- 9c. Mu Speed Sensitivity : 1.20000
- 9d. Mu T Sensitivity : 0.03500
- 9e. Tref Mu T : 80.00000
- 9f. Mu0 T X : 0.68000
- 9g. Ref Speed X : 2.60000
- 9h. Mu Speed Sensitivity X : 1.80000
- 9i. Mu T Sensitivity X : 0.03500
- 9j. Tref Mu T X : 80.00000

- 10. Rolling Resistance

- 10a. Rr0 : 20.11450
- 10b. Rr1 : 0.00200
- 10c. Rr Slip : 646.00000

- | L 10d. Rr Wear Mult : 0.14000
- 11. Groove Data
 - | L 11a. Groove Factor : 68.00000
 - | L 11b. Groove S A Factor : 0.45000
 - | L 11c. Groove S R Factor : 0.45000
- 12. Pressure : 17.00000
- 13. Init Data
 - | L 13a. Width : 305.00000
 - | L 13b. Aspect Ratio : 43.00000
 - | L 13c. Diameter : 18.00000
 - | L 13d. Load Index : 101.00000
 - | L 13e. Pressure : 17.00000

20. Wing [.wing]

A. Description

I. General Description

Xxxxxxx

II. Area of Influence / Impact on Vehicle Dynamics

Xxxxxxx

III. Key Architecture & Data Fields Explained

xxxxxxxxx

1 - WHEEL RATE & SPRING PARAMETERS

- Xxx

2 - DAMPER PROFILE PARAMETERS

XXxxxxx

3 - ALIGNMENTS & GEOMETRY FIELDS

- Pressure: EXxxxxxxx

IV. Interpretation of Tuning Part Strategies

By cross-Xxxxxxxxxxx

B. Schema

- 1. Vertical : boolean
- 2. Name : string
- 3. Chord : float
- 4. Span : float
- 5. Position : x, y, z float
- 6. Lut A 0 A C L : string - path
- 7. Lut A 0 A C D : string - path
- 8. Lut G H C L Mult : string - path
- 9. Lut G H C D Mult : string - path

- 10. Cd Gain : float
- 11. Cl Gain : float
- 12. Angle : float
- 13. Yaw Gain : float
- 14. Drag Per Cool Transfer : float
- 15. Has Fan : boolean
- 16. Fan Speed : float
- 17. Damage C D [x] : float | can have multiple Damage C D
- 18. Damage C L [x] : float | can have multiple Damage C L
- 19. Wing Controllers [x] : object | can have multiple Wing Controllers
 - 19a. Combinator Mode : enum
 - 19b. Input : enum
 - 19c. Filter : float
 - 19d. Up Limit : float
 - 19e. Down Limit : float
 - 19f. Lut : string - path

Enum - Car Wing

19a	Combinator Mode	UndefinedMode, Add, Mult
19b	Input	UndefinedInput, Brake, Gas, Yaw, LatG, LonG, Steer, Speed, SusTravelLR, SusTravelRR

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Vertical	None (Boolean : True / False)	When true, the element is a vertical aero surface (endplate / fin) producing mainly side force; false = horizontal wing, body, or diffuser plane (all documented examples are false).
2.	Name	None (String)	Element label used for authoring and DRS/setup mapping (BODY, FRONT, REAR, DIFFUSER_F, DIFFUSER_RL/RR).
3.	Chord	m (Meters)	Streamwise reference length of the aero element. Often 1.0 as a unit chord for LUT scaling; physical chords appear on race cars (SF-25 rear wing 0.45 m, front diffuser 2.5 m). With Span sets effective area $A \approx \text{Chord} \times \text{Span}$.
4.	Span	m (Meters)	Lateral width of the element (RS6/Emira body ~1.95–2.41 m; SF-25 split front wings 0.42 m each, rear wing 2.0 m, half-diffusers 0.683 m).

ID	Name	Unit of Measurement	Description
5.	Position	m (Meters, X / Y / Z)	Force application point in the car body frame (m). Sets pitch/heave moment arms: front wings at +Z, rear at -Z, body near CG (e.g. RS6 FRONT z=1.9, REAR z=-2.1).
6.	Lut A O A C L	None (.curve file path)	Look-up table: angle of attack (AoA, deg) → lift coefficient Cl (wing_body/front/rear_AOA_CL.curve). Core polar for downforce vs. incidence.
7.	Lut A O A C D	None (.curve file path)	Look-up table: AoA → drag coefficient Cd (..._AOA_CD.curve). Paired with Cl LUT for L/D trade-off.
8.	Lut G H C L Mult	None (.curve file path)	Ground-height (ride-height) multiplier LUT on Cl (height_front_CL.curve, height_diffuser_CL.curve). Models ground effect / diffuser sensitivity; often None on body/rear wing.
9.	Lut G H C D Mult	None (.curve file path)	Ground-height multiplier LUT on Cd. Used with GH Cl Mult for front wings and underbody elements.
10.	Cd Gain	Dimensionless coefficient	Scalar on Cd from LUTs (aligned with Car Data §17c3). Body often carries most drag (RS6 BODY 1.6, Emira BODY 1.0); wings may use 0–1.5.
11.	Cl Gain	Dimensionless coefficient	Scalar on Cl from LUTs (aligned with Car Data §17c2). BODY often 0 (pure drag blob); FRONT/REAR 1–2 (street) up to ~3.9 (SF-25 front).
12.	Angle	deg (Degrees)	Static geometric incidence / flap angle relative to the body (AoA offset into CL/CD LUTs). Setup-adjustable on race cars (Emira FRONT 2°, SF-25 FRONT 6°, REAR 12°).
13.	Yaw Gain	Dimensionless coefficient	Sensitivity of aero forces to vehicle yaw (aligned with Car Data §17c4). Negative values on F1 (SF-25 -0.2 to -1.5) model yaw-induced Cl/Cd change and side-force coupling.
14.	Drag Per Cool Transfer	Dimensionless coefficient	Drag penalty coupled to cooling-air / duct mass-flow (aligned with Car Data §17c5). 0.0 in all documented examples.

ID	Name	Unit of Measurement	Description
15.	Has Fan	None (Boolean : True / False)	Enables an active suction fan on this element (e.g. fan-car / blown diffuser). False on RS6, Emira, and SF-25 examples.
16.	Fan Speed	RPM or rad/s (Fan rotational speed)	Commanded fan speed when Has Fan is true; 0 when unused.
17.	Damage C D	Dimensionless ΔC_d per damage stage	Indexed [x] damage stages increasing Cd after aero damage (e.g. 0.005–0.015). Stages map to progressive wing/body damage levels.
18.	Damage C L	Dimensionless ΔC_d per damage stage	Indexed [x] damage stages reducing (or shifting) Cl after damage; often matched to Damage CD on the impacted side (front crash → stage 1, rear → stage 2).
19.	Wing Controllers	None (Object array [x])	Dynamic aero controllers (DRS, brake-duct, speed-sensitive flaps). None in documented examples; schema exposes per-controller fields 19a–19f.
19a.	Combinator Mode	None (Enum : UndefinedMode / Add / Mult)	How this controller's LUT output combines with base Angle/gains (Add = offset; Must likely Mult/override — schema spelling).
19b.	Input	None (Enum : UndefinedInput / Brake / Gas / Yaw / LatG / LonG / Steer / Speed / SusTravelLR / SusTravelRR)	Telemetry channel driving the controller (brake for DRS-off, speed, lateral G, suspension travel L/R, etc.).
19c.	Filter	Dimensionless filter coefficient (0 - 1)	Low-pass smoothing on the controller input to avoid aero chatter.
19d.	Up Limit	deg or gain (Depends on controller output)	Upper clamp on controller output (max flap angle / gain).
19e.	Down Limit	deg or gain (Depends on controller output)	Lower clamp on controller output (min flap angle / gain).
19f.	Lut	None (.curve file path)	Look-up table mapping Input → angle or Cl/Cd modifier for this wing controller.

D. Example data

I. Chosen Tyres for Example

- Audi RS6 Avant (slug : ks_audi_rs_6_avant) [3 wings]
- Lotus Emira (slug : ks_lotus_emira)
- Ferrari SF-25 (slug : ks_ferrari_sf_25)

II. Example

Audi RS6 Avant

1. Wing 0 [Body] (file : ks_audi_rs_6_avant0.wing)

```
| 1. Vertical : false
| 2. Name : BODY
| 3. Chord : 1.00000
| 4. Span : 2.41000
| 5. Position : 0.00000, 0.18000, -0.10000
| 6. Lut A 0 A C L :
content\cars\ks_audi_rs_6_avant\data\wing_body_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_audi_rs_6_avant\data\wing_body_A0A_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 1.60000
| 11. Cl Gain : 0.00000
| 12. Angle : 0.00000
| 13. Yaw Gain : 0.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00500
| 17. Damage C D 2 : 0.00500
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None
```

1. Wing 1 [Front] (file : ks_audi_rs_6_avant1.wing)

```
| 1. Vertical : false
| 2. Name : FRONT
| 3. Chord : 1.00000
| 4. Span : 2.41000
| 5. Position : 0.00000, -0.34000, 1.90000
| 6. Lut A 0 A C L :
content\cars\ks_audi_rs_6_avant\data\wing_front_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_audi_rs_6_avant\data\wing_front_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_audi_rs_6_avant\data\height_front_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_audi_rs_6_avant\data\height_front_CD.curve
| 10. Cd Gain : 0.00000
| 11. Cl Gain : 1.00000
| 12. Angle : 2.00000
```

| 13. Yaw Gain : 0.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01500
| 17. Damage C D 2 : 0.00500
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.01500
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None

1. Wing 2 [Rear] (file : ks_audi_rs_6_avant2.wing)

| 1. Vertical : false
| 2. Name : REAR
| 3. Chord : 1.00000
| 4. Span : 2.41000
| 5. Position : 0.00000, 0.80000, -2.10000
| 6. Lut A O A C L :
content\cars\ks_audi_rs_6_avant\data\wing_rear_AOA_CL.curve
| 7. Lut A O A C D :
content\cars\ks_audi_rs_6_avant\data\wing_rear_AOA_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 0.00000
| 11. Cl Gain : 1.00000
| 12. Angle : 0.00000
| 13. Yaw Gain : 0.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01500
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None

Lotus Emira

1. Wing 0 [Body] (file : ks_lotus_emira0.wing)

| 1. Vertical : false
| 2. Name : BODY
| 3. Chord : 1.00000
| 4. Span : 1.94500
| 5. Position : 0.00000, 0.20000, 0.50000

```

| 6. Lut A O A C L :
content\cars\ks_lotus_emira\data\wing_body_AOA_CL.curve
| 7. Lut A O A C D :
content\cars\ks_lotus_emira\data\wing_body_AOA_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 1.00000
| 11. Cl Gain : 0.00000
| 12. Angle : 0.00000
| 13. Yaw Gain : 0.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00500
| 17. Damage C D 2 : 0.00500
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None

```

1. Wing 1 [Front] (file : ks_lotus_emira1.wing)

```

| 1. Vertical : false
| 2. Name : FRONT
| 3. Chord : 1.00000
| 4. Span : 1.94500
| 5. Position : 0.00000, -0.30000, 1.48700
| 6. Lut A O A C L :
content\cars\ks_lotus_emira\data\wing_front_AOA_CL.curve
| 7. Lut A O A C D :
content\cars\ks_lotus_emira\data\wing_front_AOA_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 0.00000
| 11. Cl Gain : 2.00000
| 12. Angle : 2.00000
| 13. Yaw Gain : 0.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01500
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.01500
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None

```

1. Wing 2 [Rear] (file : ks_lotus_emira2.wing)

```

1. Vertical : false
2. Name : REAR
3. Chord : 1.00000
4. Span : 1.94500
5. Position : 0.00000, 0.10000, -1.00000
6. Lut A 0 A C L :
content\cars\ks_lotus_emira\data\wing_rear_A0A_CL.curve
7. Lut A 0 A C D :
content\cars\ks_lotus_emira\data\wing_rear_A0A_CD.curve
8. Lut G H C L Mult : None
9. Lut G H C D Mult : None
10. Cd Gain : 0.00000
11. Cl Gain : 2.20000
12. Angle : 2.00000
13. Yaw Gain : 0.00000
14. Drag Per Cool Transfer : 0.00000
15. Has Fan : false
16. Fan Speed : 0.00000
17. Damage C D 1 : 0.00000
17. Damage C D 2 : 0.01500
17. Damage C D 3 : 0.00000
17. Damage C D 4 : 0.00000
18. Damage C L 1 : 0.00000
18. Damage C L 2 : 0.01000
18. Damage C L 3 : 0.00000
18. Damage C L 4 : 0.00000
19. Wing Controllers : None

```

Ferrari SF-25

1. Wing 0 [Body] (file : ks_ferrari_sf_250.wing)

```

1. Vertical : false
2. Name : BODY
3. Chord : 1.00000
4. Span : 1.36600
5. Position : 0.00000, 0.15000, -0.08000
6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_body_A0A_CL.curve
7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_body_A0A_CD.curve
8. Lut G H C L Mult : None
9. Lut G H C D Mult : None
10. Cd Gain : 0.50000
11. Cl Gain : 1.00000
12. Angle : 3.50000
13. Yaw Gain : -0.20000
14. Drag Per Cool Transfer : 0.00000
15. Has Fan : false
16. Fan Speed : 0.00000
17. Damage C D 1 : 0.00000
17. Damage C D 2 : 0.00000
17. Damage C D 3 : 0.01000
17. Damage C D 4 : 0.01000

```


- | 18. Damage C L 1 : 0.00000
- | 18. Damage C L 2 : 0.00000
- | 18. Damage C L 3 : 0.00000
- | 18. Damage C L 4 : 0.00000
- | 19. Wing Controllers : None

2. Wing 1 [Front] (file : ks_ferrari_sf_251.wing)

- | 1. Vertical : false
- | 2. Name : FRONT
- | 3. Chord : 1.00000
- | 4. Span : 0.42000
- | 5. Position : -0.56000, -0.09000, 2.90000
- | 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_front_A0A_CL.curve
- | 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_front_A0A_CD.curve
- | 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_frontwing_CL.curve
- | 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_frontwing_CD.curve
- | 10. Cd Gain : 1.25000
- | 11. Cl Gain : 3.90000
- | 12. Angle : 6.00000
- | 13. Yaw Gain : -0.57100
- | 14. Drag Per Cool Transfer : 0.00000
- | 15. Has Fan : false
- | 16. Fan Speed : 0.00000
- | 17. Damage C D 1 : 0.01000
- | 17. Damage C D 2 : 0.00000
- | 17. Damage C D 3 : 0.00000
- | 17. Damage C D 4 : 0.00000
- | 18. Damage C L 1 : 0.01000
- | 18. Damage C L 2 : 0.00000
- | 18. Damage C L 3 : 0.00000
- | 18. Damage C L 4 : 0.00000
- | 19. Wing Controllers : None

3. Wing 2 [Front] (file : ks_ferrari_sf_252.wing)

- | 1. Vertical : false
- | 2. Name : FRONT
- | 3. Chord : 1.00000
- | 4. Span : 0.42000
- | 5. Position : 0.56000, -0.09000, 2.90000
- | 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_front_A0A_CL.curve
- | 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_front_A0A_CD.curve
- | 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_frontwing_CL.curve
- | 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_frontwing_CD.curve
- | 10. Cd Gain : 1.25000
- | 11. Cl Gain : 3.90000

```
| 12. Angle : 6.00000
| 13. Yaw Gain : -0.57100
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.01000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None
```

4. Wing 3 [Rear] (file : ks_ferrari_sf_253.wing)

```
| 1. Vertical : false
| 2. Name : REAR
| 3. Chord : 0.45000
| 4. Span : 2.00000
| 5. Position : 0.00000, 0.55000, -1.70000
| 6. Lut A O A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_rear_A0A_CL.curve
| 7. Lut A O A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_rear_A0A_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 1.50000
| 11. Cl Gain : 0.70000
| 12. Angle : 12.00000
| 13. Yaw Gain : -1.50000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01000
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None
```

5. Wing 4 [Diffuser Front] (file : ks_ferrari_sf_254.wing)

```
| 1. Vertical : false
| 2. Name : DIFFUSER_F
| 3. Chord : 2.50000
| 4. Span : 1.36600
| 5. Position : 0.00000, -0.16000, 0.00000
| 6. Lut A O A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
```

```

| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CD.curve
| 10. Cd Gain : 1.00000
| 11. Cl Gain : 0.80000
| 12. Angle : 3.50000
| 13. Yaw Gain : -0.07000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.01000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None

```

6. Wing 5 [Diffuser Rear Left] (file : ks_ferrari_sf_255.wing)

```

| 1. Vertical : false
| 2. Name : DIFFUSER_RL
| 3. Chord : 1.00000
| 4. Span : 0.68300
| 5. Position : -0.34230, -0.17500, -1.57000
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_rear_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_rear_CD.curve
| 10. Cd Gain : 0.55000
| 11. Cl Gain : 0.71000
| 12. Angle : 3.50000
| 13. Yaw Gain : -1.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None

```

7. Wing 6 [Diffuser Rear Right] (file : ks_ferrari_sf_256.wing)

```
| 1. Vertical : false
| 2. Name : DIFFUSER_RR
| 3. Chord : 1.00000
| 4. Span : 0.68300
| 5. Position : 0.34230, -0.17500, -1.57000
| 6. Lut A O A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
| 7. Lut A O A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_rear_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_rear_CD.curve
| 10. Cd Gain : 0.55000
| 11. Cl Gain : 0.71000
| 12. Angle : 3.50000
| 13. Yaw Gain : -1.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None
```

8. Wing 7 [DRS] (file : ks_ferrari_sf_257.wing)

```
| 1. Vertical : false
| 2. Name : DRS
| 3. Chord : 1.00000
| 4. Span : 1.36600
| 5. Position : 0.00000, 0.59000, -1.80000
| 6. Lut A O A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_rear2_A0A_CL.curve
| 7. Lut A O A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_rear2_A0A_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 1.70000
| 11. Cl Gain : 2.00000
| 12. Angle : 17.00000
| 13. Yaw Gain : -1.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01000
| 17. Damage C D 3 : 0.00000
```

| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None

9. Wing 8 [Monkeyseat] (file : ks_ferrari_sf_258.wing)

| 1. Vertical : false
| 2. Name : DRS
| 3. Chord : 1.00000
| 4. Span : 1.36600
| 5. Position : 0.00000, -0.10000, -1.80000
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_monkey_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_monkey_A0A_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 1.00000
| 11. Cl Gain : 1.00000
| 12. Angle : 10.00000
| 13. Yaw Gain : -0.15000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01000
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None

10. Wing 9 [Diffuser Front Left] (file : ks_ferrari_sf_diff_l.wing)

| 1. Vertical : false
| 2. Name : DIFFUSER_FL
| 3. Chord : 2.50000
| 4. Span : 0.35000
| 5. Position : 0.31000, -0.16000, 0.00000
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CD.curve
| 10. Cd Gain : 0.83000
| 11. Cl Gain : 1.70000
| 12. Angle : 3.50000

```
| 13. Yaw Gain : -0.07000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.01000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None
```

11. Wing 10 [Diffuser Front Left L] (file : ks_ferrari_sf_diff_ll.wing)

```
| 1. Vertical : false
| 2. Name : DIFFUSER_FLL
| 3. Chord : 2.50000
| 4. Span : 0.35000
| 5. Position : 0.66000, -0.16000, 0.00000
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CD.curve
| 10. Cd Gain : 0.83000
| 11. Cl Gain : 1.10000
| 12. Angle : 3.50000
| 13. Yaw Gain : -0.07000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.01000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None
```

12. Wing 11 [Diffuser Front Right] (file : ks_ferrari_sf_diff_r.wing)

```
| 1. Vertical : false
| 2. Name : DIFFUSER_FR
| 3. Chord : 2.50000
| 4. Span : 0.35000
| 5. Position : -0.31000, -0.16000, 0.00000
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
```

```

| 7. Lut A O A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_AOA_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CD.curve
| 10. Cd Gain : 0.83000
| 11. Cl Gain : 1.70000
| 12. Angle : 3.50000
| 13. Yaw Gain : -0.07000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.01000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None

```

13. Wing 12 [Diffuser Front Right R] (file : ks_ferrari_sf_diff_rr.wing)

```

| 1. Vertical : false
| 2. Name : DIFFUSER_FRR
| 3. Chord : 2.50000
| 4. Span : 0.35000
| 5. Position : -0.66000, -0.16000, 0.00000
| 6. Lut A O A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_AOA_CL.curve
| 7. Lut A O A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_AOA_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CD.curve
| 10. Cd Gain : 0.83000
| 11. Cl Gain : 1.10000
| 12. Angle : 3.50000
| 13. Yaw Gain : -0.07000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.01000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None

```

